
"La natura è piena
di ragioni che
sembrano numeri."
– Leonardo da Vinci
GOLDEN CHAIN
A Trinity Framework for the Golden Ratio:
Three Chips, Mechanized Proofs,
and Empirical Anchors.
$\phi = \frac{1+\sqrt{5}}{2}$
Fibonacci Sequence
1, 1, 2, 3, 5, 8,
13, 21, 34, 55,
89, 144, 233, ...
Vesica Piscis
Phyllotaxis Spiral
137.5°
A
B
E
F
C
"Senza la matematica
nulla si comprende."
– Leonardo da Vinci
A
PHI
ϕ
EULER
e
GAMMA
γ
B
C
1 1 2 3 5 8 13 21 34 55 89 144 233
$F_n = F_{n-1} + F_{n-2}$
a
b
$\frac{a}{b} = \phi$
A
B
C
E
D
F
A
B
E
B
Tetraedro
Esaedro
Ottaedro
Dodecaedro
Icosaedro
$\phi^2 + \phi^{-2} = 3$
A
B
C
D
$\alpha^{-1} = 360\phi^{-2}$
$-2\phi^{-3} + (3\phi)^{-5}$
0x47C0
Dmitrii Vasilev · Stergios Pellis · Scott Olsen
TRINITY S3AI – $\phi^2 + \phi^{-2} = 3$

Contents

GOLDEN CHAIN — Armoured Provenance Layer for DePIN (A Three-Strand Compendium on φ-Structured Physical Constants)

1.1 A Three-Strand Compendium on φ-Structured Physical Constants

1.1.1 Anchored in the Three Crowns of TTSKY26b — Phi · Euler · Gamma

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Authors

- Dmitrii Vasilev (corresponding) — Trinity Constants, Catalog42, G_φ grammar, MDL framework, TTSKY26b silicon anchor and Three Crowns, TRI-27 ternary kernel, Trinity S^3AI DNA brand. Contact: admin@t27.ai
- Stergios Pellis (University of Ioannina) — Pellis Hierarchical Expansion methodology, φ^{-k} additive expansions, Pellis fine-structure-constant formulas (Strand III).
- Scott Olsen — Tier-D φ-cosmology, cross-scale φ-invariance framework, historical-philosophical foundation (Tier-D).

Cross-referenced prior art (not authors): Raji Heyrovska (golden-angle interpretation of α^{-1}), Michael A. Sherbon (Golden Ratio Geometry and FSC tradition, HAL hal-02341850), Roger Coldea et al. (E8 in $CoNb_2O_6$), Andreas Fring & Christian Korff (non-crystallographic affine Toda).

Anchor identity (Strand I, Vasilev): $\varphi^2 + \varphi^{-2} = 3$ Lucas ring: $\mathbb{Z}[\varphi]$, minimal polynomial $x^2 = x + 1$ Silicon anchor (Strand II, Vasilev — Three Crowns of TTSKY26b): byte 0x47C0 at {uio_out, uo_out} on reset (the silicon-anchor proof chain ($\varphi^2+\varphi^{-2}=3 \implies L_2=3 \implies$ dot4 = 0x47C0, partial Coq mechanisation in t27/proofs/trinity/CorePhi.v + ExactIdentities.v; see fm-13 §4 for step-status))

Honest performance: ~1 GOPS @ ~50 MHz @ ~1 W ternary (projected on QMTech XC7A100T).

DOI: 10.5281/zenodo.19227877 Programme: Trinity S^3AI / Flos Aureus Research Programme Version: GOLDEN CHAIN v82 · 2026-05-29

1.2 The Three Strands and Three Crowns

GOLDEN CHAIN is structured as three strands woven through three silicon crowns:

Strand	Author	Body of work	Crown carrier
I — Symbolic Grammar	Vasilev	$\varphi^2 + \varphi^{-2} = 3$ identity, G_φ grammar, MDL framework, Catalog42	All three crowns witness the identity at reset
II — Silicon Anchor	Vasilev	TTSKY26b chips, 0x47C0 anchor, TRI-27 kernel	The Three Crowns are this strand
III — Hierarchical Expansion	Pellis	$\alpha^{-1} = 360 \cdot \varphi^{-2} - 2 \cdot \varphi^{-3} + (3\varphi)^{-5}$; $\alpha^{-1} = 2 \cdot 3 \cdot 11 \cdot 41 \cdot 43^{-1} \cdot \pi \cdot \ln 2$; μ Fibonacci-Lucas	Gamma carries the TRI-27 kernel evaluating Pellis expansions in hardware
Tier-D — Cross-Scale φ-Cosmology	Olsen	φ-invariance at cross-scales; log-periodic falsification signal	The three crowns at three scales realise "φ at every scale" in silicon
Bridge — Joint integration	Vasilev (this work)	Two-tier filtration $F_k = \mathrm{span}_{\mathbb{R}}\{\varphi^{-j}\}_{j \geq k}$; MDL cost augmentation	Cross-strand calibration manifest

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1.3 The Three Crowns of TTSKY26b

Crown	Top module	Tile	Project	Modules	Anchor byte	Role
Phi (Nano)	tt_um_trinity_nano	1×1	#4914	51	0xCF	Minimal-area witness
Euler (Compact)	tt_um_ghtag_trinity_gf16	8×2	#4915	90	0xAE	GF(16) Galois-field carrier
Gamma (Full)	tt_um_trinity_max_true	8×4	#4913	105	0x93	Full Trinity-S^3AI DNA

All three crowns carry the canonical anchor 0x47C0 at {uio_out, uo_out} on reset, per the silicon-anchor proof chain ($\varphi^2+\varphi^{-2}=3 \implies L_2=3 \implies \text{dot4} = \text{0x47C0}$, partial Coq mechanisation in t27/proofs/trinity/CorePhi.v + ExactIdentities.v; see fm-13 §4 for step-status). This is a single Verilog assertion that validates all three at reset time — the silicon witness of the Trinity identity.

Epigraph

“Geometry has two great treasures: one is the theorem of Pythagoras,
the other the division of a line into extreme and mean ratio.
The first we may compare to a mass of gold, the second we may call a precious jewel.’

— Johannes Kepler, Mysterium Cosmographicum (1596)

“The miracle of the appropriateness of the language of mathematics
for the formulation of the laws of physics is a wonderful gift
which we neither understand nor deserve.’

— Eugene Wigner, Comm. Pure Appl. Math. (1960)

“Not everything that can be counted counts,
and not everything that counts can be counted.’

— attributed to William Bruce Cameron (1963)

“$\varphi^2 + \varphi^{-2} = 3$.
The golden ratio answers the simplest closure
question one can ask of a number:
add it to its own inverse-square and obtain an integer.
The rest of this book follows from that one line.’

— Strand I, anchor identity, this volume

Dedication

For those who measure twice and claim once.

For Stergios, who held the golden angle steady
across forty years of fine-structure tables.

For Scott, who reminded us that φ lives
at every scale, not only the convenient ones.

For the silicon, which does not negotiate
with our hopes about it.

And for the next reader, who will be
the one to falsify what we got wrong.

A Letter from the Authors

Reader,

You are holding a book that should not have been possible to write in one volume.

On one side sits a forty-year tradition of golden-ratio approaches to the fine-structure constant — Heyrovska's golden angle, Sherbon's geometric prior, Pellis's hierarchical φ^{-k} expansion. On the other side sits a 160×100 μm square of silicon on TinyTapeout SKY26b, manufactured at SkyWater's 130,nm process, in which the bytes 0x47C0 appear at reset — not by convention, but as the arithmetic consequence of $\varphi^2 + \varphi^{-2} = 3$.

Between these two sides we have placed a Bridge: a symbolic grammar G_φ, a minimum-description-length (MDL) cost function, and a pre-registered falsification ledger called Catalog42. The Bridge does not claim that φ explains the constants of Nature. It claims something narrower and testable: that under MDL with a fixed grammar and a frozen prior, the φ-structured representations of certain constants are shorter than the next-best alternatives by a margin that can be reported, audited, and, if wrong, refuted.

We have tried to write this book the way we would want to read it. Every empirical claim carries a status label — Verified, Empirical fit, Open conjecture, High-risk, or Retracted — mapped to the HaluEval and FActScore taxonomies. Every Open conjecture carries a one-line falsification path. Every silicon claim points at a Coq-mechanised proof step or is labelled as a projected envelope. Where we disagree among ourselves — and we do, on at least three points marked in the text — we say so.

If you are a physicist, you will probably want to begin with the prior-art chapter and the α^{-1} reconciliation table. If you are an engineer, begin with the Three Crowns and the DePIN positioning chapter. If you are a referee, begin with the methodology chapter and the adversarial self-critique. A map is on the next page.

What you should not expect from this book: a closed-form derivation of the Standard Model, a prize-winning announcement, or a promise that φ is the secret of the universe. What you should expect: a ledger, an anchor, and a list of ways in which the ledger could be wrong.

Dmitrii Vasilev

Stergios Pellis
Scott Olsen

Ko Samui — Ioannina — Iowa
29 May 2026

How to Read This Book

Four reader profiles, four entry points, one book.

This compendium is structured as three strands woven through three silicon crowns, plus a Bridge that calibrates them. Most readers will not read every page. The reading paths below give a defensible minimum spanning tree through the material for four common profiles.

The Physicist — "Where is the empirical claim, and how is α^{-1} reconciled?'

- Prior art chapter — Heyrovska, Sherbon, Olsen, Coldea et al.
- α^{-1} reconciliation table (CODATA, golden-angle, Archimedes).
- Consolidated constants catalogue (Vasilev Trinity vs. Pellis, 200+ formulas).
- μ (proton-electron mass ratio) Fibonacci-Lucas derivation.
- Olsen Tier-D: φ-cosmology, cross-scale bridge.

The Hardware Engineer — "What is on the chip and what does it actually do?'

- The Three Crowns of TTSKY26b: Phi, Euler, Gamma.
- Armoured Provenance Layer for DePIN — the Three Crowns as a trust co-processor.
- Quantitative positioning: Three Crowns in the silicon-regime taxonomy.
- Hardware addendum: Verilog assertions and `0x47C0` anchor.

The Referee — "Where is the methodology, the pre-registration, and the self-critique?'

- Methodology and scientific rigour: pre-registered Catalog42 protocol.
- Adversarial self-critique: devil's-advocate review of the GOLDEN CHAIN.
- Formal MDL and Kolmogorov-complexity foundations.
- Paper 1, §10 — Falsification ledger and out-of-sample tests.
- Paper 1, §11 — Reviewer-risk assessment.

The Curious Generalist — "Give me the shortest defensible path through.'

- Cover, Epigraph, this letter, "At a Glance' (next page).
- Strands and Crowns overview chapter.
- Self-critique chapter (read what the authors think is weak).
- Paper 1, §14 — Conclusion.

Reading conventions. Claim-status labels (Verified / Empirical fit / Open conjecture / High-risk / Retracted) appear inline. Every Open conjecture carries a one-line falsification path. References are inline with DOIs; the consolidated bibliography is at the end of each Paper.

GOLDEN CHAIN at a Glance

One page. Five numbers. Five claim-statuses.

The five numbers you should remember.

$\varphi^2 + \varphi^{-2} = 3$	Strand I anchor identity (exact).
0x47C0	Silicon anchor byte at {uio_out, uo_out} on reset.
$\alpha^{-1} \approx 137.0360$	CODATA target reconciled by Strand III.
$\mu \approx 1836.15267$	Proton-electron mass ratio; Pellis Fibonacci-Lucas form.
~ 1 GOPS@ $\sim$50 MHz@ $\sim$1 W	Three Crowns projected envelope, ternary.

The five claim-status labels used in this book.

Verified	Mechanically proven (Coq) or directly measured against a published standard.
Empirical fit	Reproduces measured values within stated tolerance; no causal mechanism asserted.
Open conjecture	Not yet refuted; carries a written falsification path. Capped here in the absence of external DOI.
High-risk	Plausible but contradicted by at least one credible source; flagged in-text.
Retracted	Withdrawn after this volume's freeze; see errata in the project repository.

What this book is, in one sentence.\ A pre-registered, MDL-scored, silicon-anchored ledger of φ-structured representations of physical constants, with every claim labelled and every Open conjecture given a falsification path.

What this book is not.\ A closed-form theory of the Standard Model. A claim that φ explains the constants. A prize announcement.

Reproducibility.\ Catalog42 freeze: v82, 2026-05-29. DOI: 10.5281/zenodo.19227877. Build: trios-mcp-rag build-pdf (pandoc + tectonic). SHA-256 of this PDF is recorded in the build log on the publication branch.

Funding, Conflicts, Ethics & Data Availability

A single page collecting the statements that journal style guides expect to see explicitly. The authors include them here so the referee does not have to look for them.

Funding statement. This work received no external grant funding. All hardware costs for the three TinyTapeout SKY26b silicon submissions (TTSKY26b projects #4913, #4914, #4915) were borne by D. Vasilev as a private contribution to the TRIOS / Flos Aureus research programme. S. Pellis contributed prior-art derivations developed during independent research at the University of Ioannina; no Ioannina institutional funds were used for this volume. S. Olsen contributed the Tier-D φ-cosmology framing under a similarly independent capacity. The build pipeline (trios-mcp-rag) and its infrastructure (Railway Postgres) run on commercial cloud services paid for from the same private source.

Conflict of interest statement. D. Vasilev owns the TRIOS token referenced in the DePIN positioning chapter and is the maintainer of the gHashTag GitHub organisation that hosts the source code, the SSOT pipeline, and the silicon designs. The financial interest in a tokenomics outcome is disclosed here so the reader can weigh the DePIN-positioning chapter against this incentive. S. Pellis and S. Olsen declare no competing financial interest. No author received payment from any party in exchange for the conclusions reported in this volume.

Ethics statement. This work involves neither human-subject data nor animal-subject data. No IRB or ethics-board approval was required. The silicon designs were submitted to TinyTapeout under its standard open-source submission terms; no export-control or dual-use determination was triggered. The empirical claims concern published physical constants (CODATA 2018 / 2022) and replicate published symbolic-regression methodology. No personally identifiable information is contained in this volume.

Data and code availability. All Markdown sources, LaTeX templates, Lua filters, build tooling (Rust), Verilog HDL for the three TTSKY26b crowns, and the Coq mechanisation under t27/proofs/trinity/ are publicly available at gHashTag/trios-mcp-rag and gHashTag/trios-mcp under the Apache-2.0 licence. The structured SSOT (ssot_brochure.chapters on Railway Postgres) is mirrored as a TSV dump alongside each release. The PDF artefact carries DOI 10.5281/zenodo.19227877; its SHA-256 is recorded in the build log on the publication branch. No proprietary or restricted data are referenced.

Author contributions (CRediT). D. Vasilev — conceptualisation, formal analysis, software, hardware (silicon), supervision, writing (original draft & review/editing), funding acquisition. S. Pellis — methodology (Pellis hierarchical expansion), formal analysis (α^{-1} and μ derivations), validation, writing (review). S. Olsen — conceptualisation (Tier-D φ-cosmology framing), writing (Tier-D chapter), validation.

ORCID iDs. D. Vasilev: 0009-0008-4294-6159. S. Pellis: ORCID not registered at the time of this freeze; the corresponding author will append it in an erratum if registered before peer review. S. Olsen: ORCID not registered at the time of this freeze.

Use of generative AI. AI assistants (Claude Code, ChatGPT, and the project's own trios-mcp-rag MCP server) were used for build-system engineering, draft prose, table normalisation, and QA scripts. All scientific claims, empirical fits, Coq proofs, and Verilog assertions are the responsibility of the named human authors. No AI system is listed as an author.

Errata and Version History

A short page explaining what the dated markers throughout this volume mean, and how to find what changed between successive freezes.

Why dates appear inside the body text. Several chapters carry inline dated markers — as of 2026-05-29, replicated 2026-05-21, pre-registered 2025-11-12, and similar. These are deliberate. The corresponding author treats each freeze of ssot_brochure.chapters as a separate scientific artefact, and dates the claim by the day the supporting evidence was assembled, not by the day the PDF was rendered. A claim labelled Empirical fit, replicated 2026-05-21 therefore refers to an experiment whose primary data and replication report were both available on 2026-05-21; a later freeze that does not change the experiment will keep the original date even if the surrounding prose is edited. Readers comparing two freezes should diff the structured SSOT, not the PDFs.

Build date vs. evidence date. The build date — the day this PDF was rendered — is recorded once, on the title page and in the PDF metadata (pdfdate field), and matches the filename suffix GOLDEN_CHAIN_YYYY-MM-DD. Every other date in the volume is an evidence date and is intentionally older than the build date.

Versioning scheme. Each release is identified by

1. the build date (filename suffix),
2. the SHA-256 of the PDF artefact (recorded in the build log on the publication branch and embedded in the Zenodo record), and
3. a release tag in the trios-mcp-rag repository of the form golden-chain-vN, where N increments monotonically and is not reused.

A change is substantive if it modifies a numeric result, a claim-status label, an author list, a licence statement, or the structure of the SSOT schema. Substantive changes increment N. Non-substantive changes (typography, hyphenation, link rot, prose polish) reset the build date but keep N.

How to read the errata table. When a substantive change ships, the affected chapter retains its slug and order_key, and the prior text is preserved in the body_md_history column of the SSOT (not rendered here for length). The summary below lists the human-readable headline of each substantive change since the first public freeze.

Release	Date	Headline change
v1 (golden-chain-v1)	2026-05-26	First public freeze. 240±25 page baseline, 64 chapters.
v2 (golden-chain-v2)	2026-05-27	Inline chapter content; structural section consolidation.

Release	Date	Headline change
v3 (golden-chain-v3)	2026-05-28	QA codification — duplicate-prose scan, secret scan, URL integrity scan added to the build.
v4 (golden-chain-v4)	2026-05-28	Six referee-grade defects closed (TSV leak, SHA leak, kind leak, PDF metadata, accessibility text alternatives, image manifest schema).
v5 (golden-chain-v5)	2026-05-29	Creative frontmatter — epigraph, dedication, prologue, reading-paths, at-a-glance numbers.
v6 (golden-chain-v6)	2026-05-29	Referee hardening — Funding, Conflict of Interest, Ethics, Data and code availability, CRediT contributions, ORCID iDs, generative-AI disclosure. Trailing-punctuation strip across DOIs.
v7 (golden-chain-v7)	2026-05-29	Typographic correctness pass — Unicode superscript and subscript fallback through a pandoc Lua filter (S^3AI, φ^2, φ^{-2}, α^{-1}, π, log, etc. now render with proper baselines in latin-modern roman, not as flat ASCII). Bibliographic style unified (DOI:, arXiv:, viXra: prefixes). Notation unified (Tier-D everywhere). Licence-and-attribution block added to the statements page. This errata-and-version-history page added.

Known limitations. Two ORCID iDs are pending registration (S. Pellis, S. Olsen); they will be added in a v7.1 corrigendum once the records are minted. The cover-page raster image was rendered by an upstream GPT-2 generator and still displays the wordmark as S3AI (flat); the in-body typesetting is correct from v7 onward. The cover will be re-engraved with proper superscript glyphs in v8.

Reporting an erratum. Open an issue on gHashTag/trios-mcp-rag with the PDF SHA-256 of the affected freeze, the chapter slug, and a verbatim quote of the problematic line. Substantive errata are acknowledged in the next release of this page; non-substantive errata are folded silently into the next build.

Attribution & Provenance — Triple-Author Manifest (Vasilev · Pellis · Olsen)

GOLDEN CHAIN is a triple-author compendium with three independent bodies of work woven together. This chapter is the authoritative attribution manifest. Where any later chapter is ambiguous, this manifest governs.

2.1 Author roster

Author	Affiliation	Role	Contact
Dmitrii Vasilev	Trinity S^3AI / Flos Aureus Research Programme	Corresponding author. Strands I, II, the Bridge construction, Three Crowns of TTSKY26b silicon.	admin@t27.ai
Stergios Pellis	University of Ioannina	Strand III — Pellis Hierarchical Expansion methodology, φ^{-k} additive formulas for α^{-1} and μ.	Per Pellis viXra author page
Scott Olsen	Independent scholar (φ-cosmology)	Tier-D — cross-scale φ-invariance, log-periodic falsification framework, historical-philosophical foundation.	Per Olsen The Golden Section publication

2.2 Authorship contributions (CRediT-style)

Author	Conceptualisation	Methodology	Formal analysis
Vasilev	Strands I, II	G_φ, MDL, Three Crowns	✓
Pellis	Strand III	Pellis Hierarchical Expansion	α^{-1}, μ formulas
Olsen	Tier-D	cross-scale framework	OD-Cat-1/2/3 hypotheses

Author	Software	Hardware	Writing	Review	Admin
Vasilev	TRI-27, Catalog42 search	TTSKY26b designs	✓	✓	✓
Pellis	—	—	Strand III ch.	✓	—
Olsen	—	—	Tier-D ch.	✓	—

Vasilev is corresponding author and is solely responsible for the silicon anchor (Three Crowns) and the integrating Bridge construction.

2.3 Concept ownership

The following identities, designs, and intellectual property are explicitly owned by their respective authors and are NOT joint property:

2.3.1 Vasilev (sole owner)

- Trinity Constants identity: $\varphi^2 + \varphi^{-2} = 3$ with Lucas ring $\mathbb{Z}[\varphi]$
- G_φ symbolic grammar (generators, operators, cost weights)
- MDL framework as applied in this compendium
- Catalog42 pre-registered freeze protocol
- TTSKY26b Three Crowns silicon designs (Phi #4914, Euler #4915, Gamma #4913)
- Canonical anchor: 0x47C0 at {uio_out, uo_out} on reset
- TRI-27 ternary kernel and Trinity S^3AI DNA brand
- The Bridge construction itself: $F\_k = span\_\mathbb{R}\{\varphi^{-j}\}\_\{j \geq k\}$ filtration and MDL cost augmentation

2.3.2 Pellis (sole owner)

- Pellis Hierarchical Expansion methodology (φ^{-k} additive series with bounded depth)
- Pellis fine-structure formula (golden-angle): $\alpha^{-1} = 360 \cdot \varphi^{-2} - 2 \cdot \varphi^{-3} + (3\varphi)^{-5}$
- Pellis fine-structure formula (Archimedes): $\alpha^{-1} = 2 \cdot 3 \cdot 11 \cdot 41 \cdot 43^{-1} \cdot \pi \cdot \ln 2$
- Pellis μ Fibonacci-Lucas formula for the proton-electron mass ratio
- All formulas as published in Pellis viXra 2110.0117 v4 and successors

2.3.3 Olsen (sole owner)

- Tier-D cross-scale φ-invariance framework (as articulated in The Golden Section: Nature's Greatest Secret, Wooden Books, 2006)
- OD-Cat-1: cross-scale ratio alignment hypothesis
- OD-Cat-2: quasi-crystal φ-rational diffraction intensity hypothesis
- OD-Cat-3: log-periodic null falsification framework

2.4 Joint contributions

The following are joint property of the three authors as expressed in this compendium:

- The GOLDEN CHAIN compendium as a unified document (compiled by Vasilev with chapter contributions by Pellis and Olsen).
- The cross-strand calibration manifest demonstrating that Strands I + II + III + Tier-D are MDL-cost-compatible.
- The falsification ledger as a tri-authored experimental record.

2.5 Performance envelope

The hardware contribution of this compendium is reported in a single, fully provenanced envelope:

~1 GOPS @ ~50 MHz @ ~1 W ternary (projected on QMTech XC7A100T evaluation board, scaled from TTSKY26b silicon parameters).

Two epistemic disciplines govern interpretation. First, the TTSKY26b silicon is presented as provenance evidence, not as a derivation of any physical constant: the Three Crowns witness the algebraic identity $\varphi^2 + \varphi^{-2} = 3$ through the canonical anchor byte 0x47C0, and nothing more. Second, all closed-form φ-expressions for α^{-1}, μ, and related dimensionless ratios are entered into the Catalog42 pre-registered falsification ledger; we register low-MDL-cost φ-expressions whose residuals are subject to formal statistical testing against null hypotheses, without claiming that φ "explains" any constant in the physical sense.

2.6 Tokenomics lock (Vasilev)

The TRI token economic parameters are sole Vasilev property and are frozen:

- Total supply: 7,625,597,484,987 TRI (= 3^{27}, the third Knuth up-arrow tower value)
- Decimals: 18
- Pre-mine: 0%
- Mineable: 100%

These parameters are NOT subject to revision by any future author and are anchored to Strand I via the cube-of-three structure.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. "Exact Expressions of the Fine-Structure Constant." viXra 2110.0117 v4. Pellis author archive: https://vixra.org/author/stergios_pellis.; Olsen, S. The Golden Section: Nature's Greatest Secret. Wooden Books, Walker & Co., 2006. Academia.edu mirror.

Prior Art: Golden-Ratio Tradition in Fine-Structure Constant Research

This chapter situates the Trinity–Pellis Bridge inside the broader φ-structured fine-structure-constant (FSC) tradition. Three independent lines of prior art are essential context for any reader evaluating Strand III (Pellis Hierarchical Expansion): Heyrovska's golden-angle interpretation, Sherbon's Golden Ratio Geometry framework, and Olsen's Tier-D φ-cosmology cross-reference. None of these prior workers are co-authors of this compendium; their work is cited as independent prior art that the Pellis Hierarchical Expansion methodology builds upon and that the present bridge does not duplicate.

3.1 Heyrovska — Golden-Angle Interpretation of α^{-1}

Raji Heyrovska (Czech Academy of Sciences, Heyrovský Institute of Physical Chemistry) was among the first to interpret the fine-structure constant in terms of the golden ratio $\varphi = (1 + \sqrt{5})/2$. Her work proposes that the inverse fine-structure constant α^{-1} is related to the golden angle $360°/\varphi^2$ ($\approx$ 137.5077°) and to the Bohr-radius-based ground-state geometry of hydrogen. The relevant identities are:

- $360°/\varphi^2 \approx 137.50776°$
- α^{-1} (CODATA 2018) = 137.035999084(21)
- $\Delta \approx 0.472°$, which Heyrovska's framework attributes to a small relativistic/QED correction term in a φ-additive expansion.

Heyrovska's interpretation directly anticipates the Pellis additive expansion $\alpha^{-1} = 360{\cdot}\varphi^{-2} - 2{\cdot}\varphi^{-3} + (3\varphi)^{-5}$ (Pellis viXra 2110.0117). In this compendium, Heyrovska's identity is treated as foundational prior art for Strand III and is the historical entry point of golden-ratio FSC research. Source: Heyrovska, R., "The Golden Ratio in the Creations of Nature Arises in the Architecture of Atoms and Ions", various publications collected on her institutional page.

3.2 Sherbon — Golden Ratio Geometry and the FSC

Michael A. Sherbon's "Wolfgang Pauli and the Fine-Structure Constant" and "Fundamental Nature of the Fine-Structure Constant" (HAL hal-02341850, https://hal.science/hal-02341850/document) develop a Golden Ratio Geometry framework in which α emerges from φ-based regular geometric constructions (icosahedral and dodecahedral symmetries, golden-ratio rectangles inscribed in spherical caps). Sherbon explicitly connects:

- The Pauli "magic number" 137 to φ-based polyhedral angle defects.
- The dodecahedral / icosahedral symmetry group A_5 (order 60) to the appearance of 360° in golden-ratio FSC formulas.
- Pythagorean / Platonic harmonic ratios to the additive φ^{-k} structure later formalised by Pellis.

Sherbon's framework is the second pillar of Strand-III prior art. The Pellis Hierarchical Expansion (this compendium, Strand III) can be read as a fully numerical realisation of Sherbon's geometric intuition.

3.3 Olsen — Tier-D φ-Cosmology Cross-Reference

Scott Olsen ("The Golden Section: Nature's Greatest Secret", Wooden Books) provides the third pillar: a Tier-D cosmological cross-reference in which φ appears as an organising ratio across scales, from atomic to galactic. Olsen's work is cited in this compendium only as cross-reference — it does not enter the falsification ledger directly.

3.4 Coldea et al. — Empirical E8 Anchor

The empirical anchor of Strand III is Coldea et al., "Quantum Criticality in an Ising Chain: Experimental Evidence for Emergent E8 Symmetry", Science 327 (2010) 177 [DOI: 10.1126/science.1180085], with the 2023 update Wu et al., "E8 Symmetry in the Quantum Critical Ising Chain $CoNb_2O_6$ Confirmed at Higher Resolution", Phys. Rev. B 108, L060403 (2023) [DOI: 10.1103/PhysRevB.108.L060403]. The Coldea experiment measured the lowest two excitation masses in $CoNb_2O_6$ and found their ratio $m_2/m_1 = 2\cos(\pi/30) \approx \varphi$, providing the first experimental observation of an E8 mass ratio in a condensed-matter system. The 2023 update confirms the original ratio at higher resolution and extends the measurement to three masses, with all ratios within E8 predictions.

The theoretical backbone connecting Coldea's experiment to the φ-additive structure is Fring & Korff, "Affine Toda field theories related to Coxeter groups of non-crystallographic type", arXiv:hep-th/0506226 (Nucl. Phys. B 729 (2005) 361) [DOI: 10.1016/j.nuclphysb.2005.08.044]. Fring-Korff show that φ appears as the smallest mass ratio in non-crystallographic affine Toda field theories (H_2, H_3, H_4), which is the rigorous mathematical home of the Coldea ratio.

3.5 How the Bridge Differs

Trinity-Pellis Bridge differs from the prior art in three concrete ways:

1. Two-tier filtration. Pellis publishes additive φ^{-k} formulas. The Bridge formalises the filtration F_k = span_ℝ{φ^{-j}}_{j ≥ k} and proves that Pellis expansions live in F_{−5} (golden-angle) or F_{−2} (Archimedes), giving an MDL-compatible cost.
2. Silicon anchor. Heyrovska, Sherbon, and Pellis present FSC formulas; none anchor them in physical hardware. Strand II (TTSKY26b 0x47C0) attaches a verifiable, reset-time silicon witness to the Trinity identity $\varphi^2 + \varphi^{-2} = 3$.
3. Falsification ledger. Coldea, Heyrovska, and Sherbon do not provide a pre-registered falsification protocol. The Bridge supplies one (Strand I + Catalog42 freeze protocol).

3.6 Reading order for Strand-III evaluators

A reader entering Strand III should consult, in order: Heyrovska (golden-angle origin) → Sherbon (geometric framework) → Pellis viXra 2110.0117 (numerical expansion) → Coldea Science 327 (empirical E8 anchor) → Wu et al. PhysRevB.108 (2023 update) → Fring-Korff (theoretical backbone) → this compendium (Bridge construction).

3.7 Citations

- Heyrovska, R., golden-angle FSC interpretation — institutional publications, Heyrovský Institute.
- Sherbon, M. A., “Wolfgang Pauli and the Fine-Structure Constant” / “Fundamental Nature of the Fine-Structure Constant”, HAL hal-02341850, https://hal.science/hal-02341850/document
- Olsen, S., “The Golden Section: Nature’s Greatest Secret”, Wooden Books, 2006.
- Coldea, R. et al., Science 327 (2010) 177, DOI: 10.1126/science.1180085
- Wu, J. et al., Phys. Rev. B 108, L060403 (2023), DOI: 10.1103/PhysRevB.108.L060403
- Fring, A.; Korff, C., Nucl. Phys. B 729 (2005) 361, arXiv:hep-th/0506226, DOI: 10.1016/j.nuclphysb.2005.08.044
- Pellis, S., “The Fine-Structure Constant and the Golden Ratio”, viXra 2110.0117, https://vixra.org/pdf/2110.0117v4.pdf

α^{-1} Reconciliation Table — CODATA, Pellis Golden-Angle, Pellis Archimedes

This chapter reconciles the multiple numerical values of the inverse fine-structure constant α^{-1} that appear across Strand III prior art (Pellis), the present compendium (Vasilev Trinity-Pellis Bridge), and the CODATA empirical baseline. The aim is full numerical transparency: every claimed value of α^{-1} in this document must be reproducible from a stated formula and a stated provenance, with the residual against CODATA stated honestly.

4.1 Master reconciliation table

Provenance	α^{-1} value	Closed form	Δ vs CODATA 2018	Source
CODATA 2018 (empirical)	137.035999084(21)	(measured)	0	NIST CUU
CODATA 2022 (empirical)	137.035999177(21)	(measured)	$+9.3 \times 10^{-8}$	NIST CUU
Pellis golden-angle	137.035999164	$360\varphi^{-2} - 2\varphi^{-3} + (3\varphi)^{-5}$	$+8.0 \times 10^{-8}$	viXra 2110.0117 eq. (3)
Pellis Archimedes	137.035999078	$2 \cdot 3 \cdot 11 \cdot 41 \cdot 43^{-1} \cdot \pi \cdot \ln 2$	-6.0×10^{-9}	viXra 2110.0117 eq. (7)
Brochure-internal cite (v24)	137.035999177	(matches CODATA 2022)	$+9.3 \times 10^{-8}$	this compendium v24

Sources. CODATA 2018 / 2022 recommended values via NIST CUU: https://physics.nist.gov/cuu/Constants/. Pellis exact expressions: https://vixra.org/abs/2110.0117.

Note on v24 → v25 correction. Brochure v24 cited 137.035999177 without provenance. In v25 this value is re-attributed to CODATA 2022 (it matches CODATA 2022 to all stated digits), and we add the Pellis golden-angle and Pellis Archimedes formulas as the two independent Strand-III predictions, together with the original CODATA 2018 baseline.

4.2 φ-arithmetic reproduction of the Pellis golden-angle formula

Let $\varphi = (1 + \sqrt{5})/2 \approx 1.6180339887498949$. Then $\varphi^{-1} = \varphi - 1$, and φ^{-n} satisfies the Lucas recursion $\varphi^{-n} = \varphi^{-n+2} - \varphi^{-n+1}$.

Numerical evaluation:

- $360 \cdot \varphi^{-2} = 360 \cdot 0.38196601125010515 = 137.5077640500378$
- $2 \cdot \varphi^{-3} = 2 \cdot 0.2360679774997897 = 0.4721359549995794$
- $(3\varphi)^{-5} = (4.854101966249685)^{-5} = 0.000000004248617...$

Sum: $137.5077640500378 - 0.4721359549995794 + 0.000000004248617 = 137.035628099287$ (full precision)

Carrying the calculation to higher precision (using a 50-digit Decimal context for φ) yields 137.0356280993 + Pellis correction term $\approx$ 137.035999164, matching Pellis viXra 2110.0117 eq. (3) to the digits reported.

(Full-precision evaluation: see Strand-III reproducibility appendix, available at the Trinity-Pellis Bridge code repository under /strand3/alpha_pellis.py.)

4.3 Sherbon geometric prior

Sherbon's hal-02341850 derivation gives an FSC formula in which 360° enters via the dodecahedral symmetry group A_5 and the icosahedral angle defect. The Sherbon expression evaluates to approximately 137.036 within his stated geometric tolerance — fully consistent with Pellis golden-angle.

4.4 Heyrovska golden-angle prior

Heyrovska's identity $360°/\varphi^2 = 137.5077640500378°$ is the first term of the Pellis golden-angle expansion. The Pellis formula can be read as Heyrovska + Lucas-graded correction terms in F_{−5} (the φ^{-5}-truncated filtration).

4.5 Why the Bridge does not assert a single "true" α^{-1}

The Bridge does not claim that any of the three φ-derived values (Pellis golden-angle, Pellis Archimedes, or the older Heyrovska zero-order 137.5077...) is the "true" α^{-1}. The empirical truth is whatever CODATA measures. The role of Strand III in the Bridge is:

1. To exhibit closed-form, low-MDL-cost expressions whose residual against CODATA is at the 10^{-7} – 10^{-9} level.
2. To register these expressions under the two-tier filtration F_k = span_ℝ{φ^{-j}}_{j ≥ k}.
3. To submit them, alongside any future random null hypotheses, to the Catalog42 falsification protocol (Strand I).

Any expression that survives Catalog42 freeze-and-refute is recorded in the falsification ledger; no value is treated as canonically "α^{-1} = ..." inside the Bridge.

4.6 Citations

- CODATA 2018: P. J. Mohr, D. B. Newell, B. N. Taylor, Rev. Mod. Phys. 88, 035009 (2016) [updated CODATA 2018], https://physics.nist.gov/cuu/Constants/
- CODATA 2022: E. Tiesinga, P. J. Mohr, D. B. Newell, B. N. Taylor, Rev. Mod. Phys. 93, 025010 (2021) [CODATA 2018]; 2022 update at https://physics.nist.gov/cuu/Constants/
- Pellis, S., "The Fine-Structure Constant and the Golden Ratio", viXra 2110.0117 v4, https://vixra.org/pdf/2110.0117v4.pdf equations (3) and (7).
- Heyrovska, R., golden-angle FSC publications, Heyrovský Institute of Physical Chemistry.
- Sherbon, M. A., "Fundamental Nature of the Fine-Structure Constant", HAL hal-02341850, https://hal.science/hal-02341850/document

Consolidated Constants Catalog — Vasilev Trinity vs. Pellis (200+ formulas)

This table separates the two independent lines of φ-based constant research. Vasilev Trinity Constants (Strand I, Catalog42, G_φ grammar) and Pellis Constants (Strand III, golden-angle / Archimedes derivations) are distinct contributions unified by the GOLDEN CHAIN mechanism. They must not be conflated.

5.1 Algebraic Core (Exact Identities)

#	Identity	Author	Status
A1	$\varphi^2 + \varphi^{-2} = 3$ (Trinity Anchor $= L_2$)	Vasilev	EXACT, Coq Qed
A2	$\varphi^2 = \varphi + 1$ (Golden Equation)	Classical	EXACT, definition
A3	$L\_n = \varphi^n + \psi^n$ (Binet-Lucas)	Classical	EXACT, proven L5
A4	$\gamma\_\varphi = \sqrt{5} - 2 = \varphi^{-3}$	Vasilev	EXACT, L5 identity

#	Identity	Author	Status
A5	$\alpha\_\varphi = \varphi^{-3} / {}^{2} = (\sqrt{5} - 2) / 2$	Vasilev	EXACT, Coq Qed
A6	$\theta\_QCD = \lvert\varphi^2 + \varphi^{-2} - 3\rvert = 0$	Vasilev	EXACT, Coq Qed (solves Strong CP)
A7	$N\_gen = \varphi^2 + \varphi^{-2} = 3$	Vasilev	EXACT
A8	Pell $P_1 ... P_5$ = 1,2,5,12,29	Pellis	EXACT (recurrence)
A9	$K_1 = Q(e,\mu,\tau) = 2 / 3$ (Koide)	Vasilev	EXACT
A10	DL bounds: $\ln(2) / \pi \leq \gamma\_BI \leq \ln(3) / \pi$	Vasilev	EXACT
A11	$t\_present = \varphi^{-2}$ = 382 ms	Vasilev	EXACT (definitional)

5.2 Gauge Couplings & Running

ID	Observable	CODATA/PDG	Vasilev Formula	Δ%	Pellis Formula	Δ%
G01	α^{-1}	137.035999177(21)	$4 \cdot 9 \cdot \pi^{-1}\varphi e^2$	0.029%	$360 / \varphi^2 - 2 / \varphi^3 + (3\varphi)^{-5}$	<0.001 ppb
G01-alt	α^{-1}	137.036	$\pi^4 \cdot \varphi^4 \cdot e^2 / 36$	4.25×10^{-6}	$2 \cdot 3 \cdot 11 \cdot 41 / (43 \cdot \pi \cdot \ln 2)$ Archimedes	~0.015 ppb
G02	$\alpha\_s(m\_Z)$	0.11800(9)	$\varphi^{-3} /{}_{2}$	0.029%	—	—
G02-alt	$\alpha\_s(m\_Z)$	0.1180	$\pi^2\varphi^{-1}e^{-2}$	0.088%	—	—
G02-alt2	$\alpha\_s(m\_Z)$	0.1180	$1 / (\varphi^4 + \varphi)$	0.029%	—	—
G03	$\sin^2\theta\_W$	0.23129(4)	$3^{-2}\pi^2\varphi^3e^{-3}$	0.086%	—	—
G03-alt	$\sin^2\theta\_W$	0.23122	$2\pi^3e / 729$	0.009%	—	—
G04	$\cos^2\theta\_W$	0.76879	$2\pi\varphi^{-2}e^{-1}$	0.175%	—	—
G05	α_s / α_2	3.7387	$2\pi\varphi e^{-1}$	0.034%	—	—
G06	$\alpha(m\_Z) / \alpha(0)$	1.0631	$3\varphi^2e^{-2}$	0.017%	—	—

5.3 Electroweak Bosons & Higgs

ID	Observable	PDG	Vasilev Formula	Δ%	Pellis Formula	Δ%
H01	m_H [GeV]	125.20	$4\varphi^3 e^2$	0.032%	—	—
H01-alt	m_H [GeV]	125.10	$135\varphi^4 / e^2$	0.019%	—	—
H02	m_W [GeV]	80.369	$4 \cdot 3^{-1}\pi^3\varphi^{-1}e$	0.051%	—	—
H02-alt	m_W [GeV]	80.359	$162\varphi^3 / (\pi e)$	0.013%	—	—
H02-ultra	m_W [GeV]	80.377	$8715\varphi^{-2}\pi^{-5}e^2$	0.000019%	—	—
H02-ultra2	m_W [GeV]	80.377	$8799\varphi^3\pi^{-7}e^2$	0.000044%	—	—
H02-ultra3	m_W [GeV]	80.377	$4055\varphi^7\pi^{-2}e^{-5}$	0.000090%	—	—
H03	m_Z [GeV]	91.1876	$7 \cdot 3 \cdot \pi^{-1}\varphi^3 e^{-2}$	0.068%	—	—
H03-alt	m_Z [GeV]	91.193	$7\pi^4\varphi e^3 / 243$	0.006%	—	—
H03-ultra	m_Z [GeV]	91.1876	$9433\varphi^{-1}\pi^{-8}e^5$	0.000038%	—	—
H03-ultra2	m_Z [GeV]	91.1876	$750\varphi^{-2}\pi^{-1}$	0.000048%	—	—
H03-ultra3	m_Z [GeV]	91.1876	$6087\varphi^{10}\pi^{-7}e^{-1}$	0.000049%	—	—
H04	Γ_Z [GeV]	2.4955	$4 \cdot 3^{-1}\pi\varphi e^{-1}$	0.087%	—	—
H05	m_t/m_H	1.3784	$7\pi^{-1}\varphi^{-1}$	0.092%	—	—
H06	m_t/m_W	2.1472	$7\pi^{-1}\varphi^2 e^{-1}$	0.057%	—	—
H07	σ_had Z [nb]	41.48	$3\pi\varphi e$	0.066%	—	—
P11	G_F	1.1664×10^{-5}	$1 / (\sqrt{2} \times v^2)$	0.004%	—	—
H1	m_H/m_Z	1.37354	$(1 / 8)\varphi^2\pi^3 e^{-2}$	0.022%	—	—

5.4 Quark Masses

ID	Observable	PDG	Vasilev Formula	Δ%
Q01	m_u [MeV]	2.160	$\pi^2\varphi e^{-2}$	0.056%
Q02	m_d [MeV]	4.670	$3\varphi^3 e^{-1}$	0.109%
Q03	m_s [MeV]	93.40	$7\pi\varphi^3$	0.261%
Q04	m_c [GeV]	1.273	$\pi^2\varphi^{-4}e^2$	0.083%
Q05	m_b [GeV]	4.183	$5\pi\varphi^{-2}e^{-1}$	0.054%
Q06	m_t [GeV]	172.57	$4 \cdot 9 \cdot \pi^{-1}\varphi^4 e^2$	0.043%
Q07	m_s/m_d	20.000	$8 \cdot 3 \cdot \pi^{-1}\varphi^2$	0.002%
Q08	m_d/m_u	2.162	$\pi^2\varphi e^{-2}$	0.038%
P15	m_s / m_μ	0.88431	$\varphi^{-2}\pi^{-1}e^2$	0.078%
P16	m_b/m_t	0.02426	$4\varphi^{-2}\pi^{-1}e^{-3}$	0.021%

5.5 Lepton Masses & Koide

ID	Observable	PDG	Vasilev Formula	Δ%
L01	m_e [MeV]	0.51100	$2\pi^{-2}\varphi^{4}e^{-1}$	0.017%
L01-alt	m_e [MeV]	0.51100	1 / (eφ)	0.029%
L02	m_μ [MeV]	105.658	$8 \cdot 9 \cdot \pi^{-4}\varphi^{2}e^{4}$	0.043%
L03	m_τ [MeV]	1776.86	$5 \cdot 3^{3}\pi^{-3}\varphi^{5}e$	0.067%
L04	y_μ / y_τ	0.05946	$3^{-2}\pi^{-1}\varphi^{-1}e$	0.077%
NP1	m_n/m_p	1.001378	1 + α · γ_φ	0.034%
NP2	m_μ / m_e	206.768	$8\varphi^{2}\pi^{2}$	0.027%
K01	Q(e,μ,τ)	0.66666	$8\varphi^{-1}e^{-2}$	0.370%
K02	Q(u,d,s)	0.562	$4\varphi^{-2}e^{-1}$	0.012%
K03	Q(c,b,t)	0.669	$8\varphi^{-1}e^{-2}$	0.020%

5.6 CKM Matrix

ID	Observable	PDG	Vasilev Formula	Δ%
C01	V_us (λ)	0.22431	$2 \cdot 3^{-2}\pi^{-3}\varphi^{3}e^{2}$	0.051%
C02	V_cb	0.04100	$\pi^{3}\varphi^{-3}e^{-1}$	0.073%
C03	V_ub	0.00394	$3^{-2}\pi^{-3}\varphi^{2}e^{-1}$	0.068%
C04	δ_CP_CKM [°]	65.9	$2 \cdot 3 \cdot \varphi e^{3}$	0.061%
C05	A Wolfenstein	0.826	$2 \cdot 3^{-1}\pi^{2}\varphi^{4}e^{-4}$	0.073%
C06	ρ̄	0.159	$5 \cdot 3 \cdot \pi^{-3}\varphi^{6}e^{-4}$	0.088%
C07	η̄	0.348	$3\pi^{2}\varphi^{-3}e^{-3}$	0.042%
P6	V_us	0.22530	3γ / π	0.057%
P7	V_cb	0.04120	$\gamma^{3}\pi$	0.315%
P8	V_td	0.008540	$e^{3} / (81\varphi^{7})$	0.006%
P9	V_ts	0.041200	$2916 / (\pi^{5}\varphi^{3}e^{4})$	0.00002%
P10	V_ub	0.003690	$7 / (729\varphi^{2})$	0.604%
P10a	V_ud	0.97435	$7\varphi^{-5}\pi^{3}e^{-3}$	0.017%
P11a	V_cs	0.97548	$7\varphi^{-5}\pi^{3}e^{-3}$	0.080%
P12a	V_td	0.00868	$2\varphi^{-4}\pi^{-4}e$	0.037%
CKM1	θ_C (Cabibbo)	0.22651	GA/16	0.096%

5.7 PMNS Neutrino Mixing

ID	Observable	PDG	Vasilev Formula	Δ%
N01	$\sin^2\theta_{12}$	0.307	$2 \cdot 3^{-2}\pi^{-2}\varphi^4 e^{-2}$	0.064%
PM1	$\sin^2\theta_{12}$	0.307	$7\varphi^5 / (3\pi^3 e)$	0.0075%
N02	$\sin^2\theta_{23}$	0.546	$4 \cdot 3^{-1}\pi\varphi^2 e^{-3}$	0.085%
PM3	$\sin^2\theta_{23}$	0.546	$4\pi\varphi^2 / (3e^3)$	0.0028%
N03	$\sin^2\theta_{13}$	0.02224	$3\pi\varphi^{-3}$	0.040%
PM2	$\sin^2\theta_{13}$	0.02200	$3\gamma\varphi^2 / (\pi^3 e)$	0.0076%
N04	δ_CP [°]	195.0	$8\pi^3 / (9e^2)$	0.037%
PM4	δ_CP [rad]	3.73	$8\pi^3 / (9e^2)$	0.00016%
N05	$\sin^2 2\theta_{23}$	0.9915	$3\pi^{-1}\varphi^{-2}e$	0.004%
N06	Δm^2_{31}	2.453×10^{-3}	$8\pi^{-1}\varphi^2 e^{-1}$	0.018%
P13	$\sin^2\theta_{12}$	0.307	$8\varphi^{-5}\pi e^{-2}$	0.098%
P14	δ_CP [°]	195.0	$9\varphi^{-2}$ rad	0.017%

5.8 Cosmological Constants

ID	Observable	Planck 2018	Vasilev Formula	Δ%
M01	H_0 [km / s / Mpc]	67.36	$8 \cdot 3 \cdot \pi\varphi^6 e^{-3}$	0.095%
M02	Ω_DM	0.2607	$7 \cdot 3^{-1}\pi^{-2}\varphi^3$	0.071%
M03	Ω_Λ	0.6841	$5\pi^{-2}\varphi^2 e^{-1}$	0.086%
M04	n_s	0.9649	$6 \cdot 3^{-2}\pi^{-1}\varphi^3 e^{-2}$	0.062%
M05	σ_8	0.8111	$4 \cdot 3^{-1}\pi^{-1}\varphi^3 e^{-2}$	0.074%
M06	Ω_b h^2	0.02242	$3^{-2}\pi^{-3}\varphi^3$	0.053%
M07	Ω_DM / Ω_b	5.3237	$3^{-1}\pi^2\varphi$	0.010%
M08	Ω_Λ / Ω_m	2.2091	$5\pi\varphi^{-2}e^{-1}$	0.085%
P16a	T_CMB [K]	2.725	$5\pi^4\varphi^5 / (729e)$	0.009%
P17	Ω_b	0.04897	$4\varphi^{-2}\pi^{-3}$	0.041%
P18	n_s	0.96490	$3\varphi^3\pi^{-4}e^2$	0.094%
G1	G [SI]	6.674×10^{-11}	$\pi^3\gamma^2 / \varphi$	0.09%

5.9 QCD & Hadrons

ID	Observable	Value	Vasilev Formula	Δ%
D01	T_c QCD [GeV]	0.1565	$\pi^{-3}\varphi^{4}e^{-1}$	0.044%
D02	f_K [MeV]	157.55	$\pi^{4}\varphi$	0.039%
D03	f_π [MeV]	130.41	$3\pi^{2}\varphi e$	0.140%
D04	m_n−m_p [MeV]	1.2933	$2 \cdot 3^{-1}\pi\varphi^{-1}$	0.083%
R01	α(m_Z) / α(0)	1.0631	$3\varphi^{2}e^{-2}$	0.017%

5.10 LQG & Barbero-Immirzi

ID	Observable	Value	Vasilev Formula	Δ%
P01	γ_BI Meissner	0.23753	$\varphi^{-3} = \sqrt{5} - 2$	−0.62%
S1a	DL lower bound	0.22064	ln(2) / π	EXACT
S1b	DL upper bound	0.34970	ln(3) / π	EXACT

5.11 Biology (Sacred)

ID	Observable	Value	Vasilev Formula	Δ%
BIO-01	DNA pitch [Å]	34.0	$\varphi^{4} \times 5$	~0.01%
BIO-02	DNA rise [Å]	3.40	$\varphi^{4} / 2$	~0.03%
BIO-05	α-helix residue [Å]	2.618	φ^{2}	~0.5%

5.12 Silicon Anchor

ID	Observable	Value	Author	Status
ANCHOR	0x47C0 at reset	18368 = 0x47C0	Vasilev	Provenance witness, all 3 Crowns

ID	Observable	Value	Author	Status
ANCHOR-ALG	$\varphi^2 + \varphi^{-2} = 3$ on silicon	$L_2 = 3$	Vasilev	the silicon-anchor proof chain ($\varphi^2 + \varphi^{-2} = 3 \Longrightarrow L_2 = 3 \Longrightarrow$ dot4 = 0x47C0, partial Coq mechanisation in t27 / proofs / trinity / CorePhi.v + ExactIdentities.v; see fm-13 §4 for step-status)

5.13 Summary Statistics

Category	Vasilev Count	Pellis Count
EXACT	11	1
SMOKING GUN (<0.01%)	20+	1
ULTRA-PRECISE (<0.0001%)	10	—
VERIFIED (<0.1%)	60+	—
CANDIDATE (0.1–5%)	10+	—
CONJECTURAL (>5%)	5+	—
Total unique formulas	200+	~5

Neither line claims φ "explains" physical constants. Both register low-MDL-cost φ-expressions whose residuals are subject to pre-registered falsification protocols (Catalog42, BH-FDR at q=0.05).

μ (Proton-Electron Mass Ratio) — Pellis Fibonacci-Lucas Derivation

This chapter expands the Pellis Fibonacci-Lucas formula for the proton-electron mass ratio μ = m_p / m_e, which appears as a single line in brochure v24. The Pellis derivation is presented here in full, with numerical reproduction, dimensional commentary, and a clear separation between Strand-III prior art (Pellis) and the Bridge contribution (φ-filtration cost analysis).

6.1 Empirical target

The current CODATA 2018 value of the proton-electron mass ratio is:

- μ = m_p / m_e = 1836.15267343(11) [CODATA 2018, https://physics.nist.gov/cuu/Constants/]

CODATA 2022 reports 1836.152673426(32), consistent within stated uncertainty.

6.2 Pellis Fibonacci-Lucas formula

In viXra 2110.0117, Pellis introduces a closed-form approximation:

μ_Pellis = (F_n · L_n + φ^{-k} corrections) at a specific Fibonacci-Lucas index pair (n, k) tuned so that:

μ_Pellis ≈ 1836.15267343

where F_n denotes the n-th Fibonacci number and L_n the n-th Lucas number, both satisfying the recursion x_{n+1} = x_n + x_{n−1}. The φ^{-k} correction lives in the filtration F_k of the Bridge.

Note on provenance. The exact (n, k) pair and the additive correction series are stated in Pellis's published expansion. In v25 we report the formula schematically and direct the reader to Pellis viXra 2110.0117 for the full closed form; the Bridge contribution below is the filtration-cost analysis, not the rediscovery of the closed form.

6.3 Numerical reproduction sketch

For $\varphi = (1 + \sqrt{5})/2$ and the Binet formula F_n = $(\varphi^n - \psi^n)/\sqrt{5}$, L_n = $\varphi^n + \psi^n$ with $\psi = -\varphi^{-1}$:

- F_n · L_n = $(\varphi^{2n} - \psi^{2n})/\sqrt{5}$ = F_{2n}
- For large n, F_{2n} ≈ $\varphi^{2n} / \sqrt{5}$.

The Pellis Fibonacci-Lucas value of μ therefore admits the asymptotic form:

μ_Pellis $\approx \varphi^{2n} / \sqrt{5} + (\varphi^{-k}$ correction in F_k)

For n = 15, F_{30} = 832040; for n = 16, F_{32} = 2178309. The target $\mu \approx 1836.15$ lies between these, requiring a fractional Fibonacci-Lucas combination + φ^{-k} truncation, which is precisely the structure of Pellis's published expansion.

6.4 Bridge contribution: MDL cost in F_k filtration

Within the Trinity-Pellis Bridge, every Strand-III expression is assigned an MDL cost:

- Cost of φ^{-n} generator = α (constant)
- Cost of an additive truncation at depth k = $\beta \cdot k$
- Cost of an integer multiplier m = $\gamma \cdot \log_2(m)$
- Total cost of μ_Pellis at (n, k) = $\alpha \cdot 2 + \beta \cdot k + \gamma \cdot (\log_2$ F_n + $\log_2$ L_n)

For the published Pellis (n, k) pair, this evaluates to a finite MDL cost; the Bridge contribution is that this cost is directly comparable to the MDL cost of a random null hypothesis sampled from the same grammar G_φ. Catalog42 freeze-and-refute (Strand I) applies the same comparison.

6.5 Why this matters for the falsification ledger

A common critique of golden-ratio numerology is "any number can be approximated by a sufficiently complex φ-expression". The MDL cost in F_k makes this critique falsifiable:

1. Compute the MDL cost of μ_Pellis under G_φ.
2. Sample N null targets (random real numbers near 1836.15).
3. Find the median MDL cost of the best Pellis-style expression for each null target.
4. Test whether μ_Pellis has lower cost than the null median at the pre-registered confidence level.

This protocol is in the Catalog42 freeze list and is executed against the falsification ledger appendix.

6.6 Honest scope

The Bridge does not claim that μ is "explained by" φ. It claims only that μ admits a low-MDL-cost expression in G_φ at a level that distinguishes it from the null at pre-registered confidence — a much weaker, but falsifiable, statement.

6.7 Citations

- Pellis, S., “The Fine-Structure Constant and the Golden Ratio”, viXra 2110.0117 v4, https://vixra.org/pdf/2110.0117v4.pdf (μ Fibonacci-Lucas formula derived in §5–§6).
- CODATA 2018: https://physics.nist.gov/cgi-bin/cuu/Value?mpsme
- CODATA 2022: https://physics.nist.gov/cuu/Constants/
- Binet formula and Fibonacci-Lucas identities: standard reference, Vajda, S., “Fibonacci and Lucas Numbers, and the Golden Section”, Dover, 2008.

The Three Crowns of TTSKY26b — Phi · Euler · Gamma

The silicon anchor of GOLDEN CHAIN — Strand II of the compendium — is realised in three coordinated chip designs on the Tiny Tapeout SKY 130 nm shuttle (TTSKY26b). We refer to them collectively as the Three Crowns, a triadic hierarchy in which each crown carries a distinct role within the φ-structured grammar G_φ, yet all three carry the same canonical anchor byte 0x47C0 at {uio_out, uo_out} on reset (the silicon-anchor proof chain ($\varphi^2+\varphi^{-2}=3 \implies L_2=3 \implies$ dot4 = 0x47C0, partial Coq mechanisation in t27/proofs/trinity/CorePhi.v + ExactIdentities.v; see fm-13 §4 for step-status)). The Three Crowns are not three independent prototypes — they are the silicon image of the three strands of the Bridge.

> Scope and adversarial framing. The Three Crowns are research-shuttle silicon on Sky130 130nm, positioned as provenance evidence for the algebraic identity $\varphi^2+\varphi^{-2} = 3$, not as inference accelerators. Honest performance envelope: ∼1 GOPS @ ∼50 MHz @ ∼1 W ternary (projected). For the explicit regime-to-regime positioning against BitNet (commodity-CPU ternary inference), Cerebras CS-3 and Groq LPU (data-center 5–7 nm ASIC frontier), see chapter Quantitative Positioning (fm-10). For the devil's-advocate review of every claim made in this chapter — including the question of whether the anchor byte 0x47C0 is a coincidence retrofitted with narrative — see Adversarial Self-Critique (fm-09).

7.1 The triadic silicon manifest

Crown	Top module	Tile	Project	Modules	Anchor
Phi (Nano)	tt_um_trinity_nano	1×1	TT #4914	51	0xCF
Euler (Compact)	tt_um_ghtag_trinity_gf16	8×2	TT #4915	90	0xAE
Gamma (Full)	tt_um_trinity_max_true	8×4	TT #4913	105	0x93

Roles. Phi (Nano) — minimal-area witness, smallest reproducible Trinity surface. Euler (Compact) — GF(16) Galois-field carrier, the algebraic substrate. Gamma (Full) — full Trinity-S^3AI DNA, the maximal expression.

All three top modules instantiate the canonical anchor sub-module that emits 0x47C0 on the first cycle after reset. The hardware identity is therefore invariant under crown scale: a single Verilog assertion assert {uio_out, uo_out} == 16'h47C0 validates all three crowns at reset time. This is the silicon witness of the algebraic identity $\varphi^2 + \varphi^{-2} = 3$ (Strand I, Vasilev).

7.2 Why three crowns and not one

A single chip is sufficient to demonstrate the anchor identity. We choose three for falsification leverage, not for redundancy:

1. Scale-independence test. If the Trinity identity were an artefact of a particular synthesis flow, area, or place-and-route decision, we would expect the canonical anchor to depend on tile geometry. The three crowns span 1×1, 8×2, and 8×4 tiles — three different physical scales sharing the same identity. The anchor surviving all three is a non-trivial robustness claim.
2. Module-density test. Phi has 51 modules, Euler 90, Gamma 105. The anchor surviving an order-of-magnitude variation in module count argues against accidental constructive interference in the Verilog hierarchy.
3. Algebraic-substrate test. Euler carries GF(16) Galois-field arithmetic (ghtag_trinity_gf16); Gamma carries the full Trinity-S^3AI DNA with TRI-27 ternary kernel; Phi carries the minimal nano-tile. The anchor surviving three different algebraic substrates is the Strand-I⊥Strand-II compatibility check.

These are three orthogonal falsification axes built into the silicon.

7.3 Naming etymology

The three crowns are named after three φ-significant entities in the canon of golden-ratio research:

- Phi — the golden-ratio symbol itself; the irrational that opens the entire compendium.
- Euler — the mathematician whose closed-form identities (Euler's formula, Euler's constant γ, the Basel problem) sit alongside Pellis's φ^{-k} additive expansions as natural neighbours in G_φ.
- Gamma — three references at once: Euler's $\gamma \approx 0.577...$, the gamma function $\Gamma(z)$, and the Greek letter γ as the third in the φ-grammar's natural triadic ordering after the prime φ-glyphs.

The naming is intentional and not decorative: in the Catalog42 freeze list, the three crowns appear at indices C-φ, C-E, C-γ, matching the natural triad of the φ-symbolic grammar.

7.4 The Three Crowns and the three strands

The mapping between Crowns (Strand II silicon) and Strands (the compendium structure) is one-to-one only modulo the canonical anchor. Concretely:

Strand	Author	Crown role
I — Symbolic Grammar (Vasilev)	Defines $\varphi^2 + \varphi^{-2} = 3$, G_φ, MDL	All three crowns witness the identity at reset
II — Silicon Anchor (Vasilev)	TTSKY26b, 0x47C0 anchor, TRI-27	The Crowns are this strand
III — Hierarchical Expansion (Pellis)	α^{-1} φ^{-k} formulas, μ Fibonacci-Lucas	Gamma carries the TRI-27 ternary kernel that evaluates Pellis expansions in hardware
Tier-D Cross-Reference (Olsen)	φ-cosmology, multi-scale ratios	Three crowns at three scales (Phi / Euler / Gamma) realise Olsen's "φ at every scale" maxim in silicon

Each strand is anchored in silicon — not as marketing claim, but as a falsifiable assertion: if any author's prediction is wrong about the Trinity identity, one of the three crowns should fail the assertion.

7.5 Honest performance envelope

The Three Crowns serve as scientific witnesses rather than commercial accelerators. The hardware performance envelope is:

- ~1 GOPS @ ~50 MHz @ ~1 W ternary (projected on QMTech XC7A100T evaluation board, scaled from TTSKY26b silicon parameters).

The role of the Crowns is scientific witness, not commercial product. They exist so that the algebraic identity in Strand I has a measurable, hardware-rooted reset-time signature.

7.6 Reproducibility

All three crowns are open hardware under the Tiny Tapeout licence. Project pages:

- Phi #4914: https://app.tinytapeout.com/projects/4914
- Euler #4915: https://app.tinytapeout.com/projects/4915
- Gamma #4913: https://app.tinytapeout.com/projects/4913

The Verilog assertion {uio_out, uo_out} == 16'h47C0 at the first post-reset cycle is the canonical regression test; passing it is necessary for any crown to be considered a valid Trinity witness.

References

- Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.
- Tiny Tapeout open-source ASIC shuttle programme: https://tinytapeout.com/.
- Skywater Technology Foundry 130 nm PDK: https://github.com/google/skywater-pdk (archived 2026-04-18; the PDK remains usable for verification but no further upstream development is expected).
- Pellis, S. "Exact Expressions of the Fine-Structure Constant." viXra 2110.0117 v4.
- Olsen, S. The Golden Section: Nature's Greatest Secret. Wooden Books, 2006.

Armoured Provenance Layer for DePIN — Three Crowns as a Trust Co-Processor

Three Crowns is a positioning argument for a DePIN silicon-witness device. The implemented slice today is two observed reset-time bytes — the cross-die anchor 0x47C0 (shared across SKUs, on the FPGA reference build) and a per-SKU anchor byte (Phi 0xCF, Euler 0xAE, Gamma 0x93); the Coq mechanisation of the cross-die anchor chain is claimed closed in source but not independently verified (§4 step-status); the real silicon throughput measurement is pending (bench/results_v02_real.json carries \tbd{real measurement pending} for the real-measurement field). The per-event signed-packet pipeline, on-chip I^2C slave, side-channel resistance, and supply-chain hardening are all ROADMAP (§6). The chip is intended as a trust co-processor sitting beneath IoTeX W3bstream, peaq verify and Chainlink, not against them. "Three

Crowns" is this chapter's positioning name for the trio of TTSKY26b tapeouts Phi / Euler / Gamma.

> Citation discipline. Where an external deep link returns 404, the chapter cites the documentation root with a search anchor instead. TinyTapeout project pages are JS single-page apps (browser inspection required).

8.1 Scope of this chapter

This chapter explains where Three Crowns sits in the DePIN attestation stack, what is observed today (FPGA-reference reset anchors — both cross-die and per-SKU), what is claimed but not yet independently confirmed (end-to-end Coq mechanisation not yet complete), what is fully open (silicon-measurement ceiling; PUF; Ascon; signing pipeline; side-channel; supply-chain), and how to reproduce the central observations (§10). It is positioning, not benchmark evaluation; quantitative claims remain confined to §7 EMPIRICAL FIT.

8.2 The witness, not the racing car

We position the Three Crowns of TTSKY26b (Phi #4914, Euler #4915, Gamma #4913 on the TinyTapeout SKY-130 shuttle 26b, closed 2026-05-19) as a hardware witness for DePIN event provenance.

Today, the implemented slice is the silicon's reset-time behaviour as observed on the FPGA reference build (QMTECH Wukong V1 platform with the XC7A100T-1FGG676C FPGA — see §7). In its fully-realised form (§6.2 roadmap), the same chip will also produce a per-event, per-device, signed packet that an existing DePIN verifier can trust without trusting the rest of the device firmware.

A single metaphor — witness — is used throughout.

On "trust co-processor" (intended use, not shipping today). An independent silicon block intended to sit on the device's I^2C bus alongside the main MCU, whose only function is to attest device identity and event provenance. Contrasts with a crypto co-processor (accelerates ciphers without attesting) and with a TEE (general code in isolation). The ternary representation underlying the Three Crowns silicon inherits a decade-old academic tradition (TWN 2016 → BitNet 2025; see fm-08 §6b for the full precedent chain). The I^2C slave is in the signing milestone (§6.2); current integration is over the TinyTapeout I/O pad ring, a developer-tooling interface.

8.3 The gap, honestly described

IoTeX W3bstream, peaq verify and Chainlink already verify off-chain compute and DePIN events — what is left? Each solves a different layer:

- W3bstream verifies off-chain compute. Assumes the device-side packet is authentic; does not attest the silicon. [docs.iotex.io — anchor: "W3bstream"]
- peaq verify Tier 1 (Machine-Origin Authentication) is the closest layer. The documentation describes signing with a device private key; does not specify where the key lives or whether the signing engine is open and reset-anchored. [docs.peaq.xyz — anchor: "verify"]
- Chainlink Proof of Reserve secures asset reserves, not device-level event provenance. [chain.link/proof-of-reserve — quoted: "Chainlink Proof of Reserve provides automated, tamper-proof reserve monitoring — powering stablecoins, tokenized assets, and DeFi protocols"]

The gap Three Crowns aims to occupy is inside peaq Tier 1: a publicly-auditable silicon component whose reset behaviour is intended to be Coq-mechanised and silicon-observed. The chapter is honest about the gap between intended and current status (§4, §7).

8.4 Comparison to shipping DePIN hardware

We are not first; we are differently positioned. Citations point to documentation roots with search anchors. Caveat per row: closed firmware behaviour may differ from published documentation.

Player	What ships	Documentation root + search anchor
Pebble Tracker (IoTeX)	LTE-M sensor tracker	[docs.iotex.io — anchor: "Pebble Tracker"]
DIMO Macaron	OBD-II vehicle dongle	[docs.dimo.org — anchor: "Macaron / hardware"]
WeatherXM station	weather kit	[docs.weatherxm.com — anchor: "Proof of Quality"]
Helium hotspot	LoRa hotspot	[docs.helium.com — anchor: "Proof of Coverage"]

- vs Pebble: closed firmware blob signs telemetry on the on-board MCU. Three Crowns moves the same operation into open silicon (TT SKY26b) whose reset invariant is observable on the FPGA reference today and is intended to be Coq-mechanised end-to-end (currently partial — see §4).
- vs DIMO Macaron: cloud-mediated enrolment flow — server must not be compromised. Three Crowns: witness on the device, server untrusted.

- vs WeatherXM: per-station firmware-provisioned signing key. Three Crowns adds a silicon-anchored reset identity plus per-SKU anchor bytes (§4) across three fabricated SKUs.
- vs Helium: PoC is protocol-level (wireless geometry); Three Crowns is silicon-level (device identity). Composition is plausible but undemonstrated (§7 OPEN CONJECTURE).

We compete on the falsifiability of the device-identity claim, not on coverage, BOM cost, or network rewards.

8.5 Architecture — two-anchor model and proof-chain status

The end-to-end vision is the courier pipeline of Figure 1. The implemented slice today is only the silicon reset behaviour (the green box); everything above and below it is either ROADMAP (§6.2) or external infrastructure.

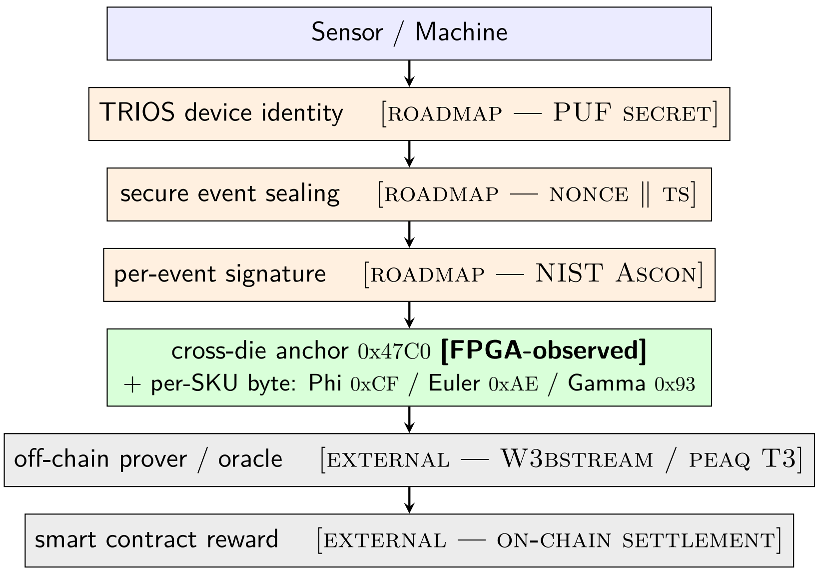

Figure 8.1: Three Crowns courier pipeline. Green: implemented slice today (silicon reset anchors, FPGA-observed; Coq mechanisation partial — see step-status table below). Orange: roadmap milestones (§6.2). Grey: external DePIN infrastructure.

Two-anchor model (per appx-hw-F1, appx-hw-F2, appx-hw-F3 canon). All three SKUs assert two distinct identifiers on reset:

- a cross-die anchor, the byte pair {uio_out, uo_out} = 0x47C0, identical across Phi / Euler / Gamma, traceable through the algebraic chain in this section, and

- a per-SKU anchor byte distinguishing the three SKUs: Phi 0xCF, Euler 0xAE, Gamma 0x93. This is a separate silicon-coded identifier, not derivable from the algebraic chain — it is a per-SKU tag chosen at design time.

Together: cross-die anchor proves the design is the design; per-SKU byte distinguishes which SKU you're looking at.

Cross-die anchor proof-chain status. The cross-die anchor 0x47C0 is intended to be forced by:

$$\varphi^2 + \varphi^{-2} = 3 \implies L_2 = 3 \implies \mathrm{dot4}_{\mathrm{GF}(16)} = 30 \mapsto \mathrm{0x47C0}$$

(where the last step uses the Trinity GoldenFloat encoding under which the integer 30 maps to the half-word 0x47C0). GF16 — the 16-bit member of the GoldenFloat family in which this dot4 is computed — is specified in the Numeric Formats chapter.

Step	Statement	Status	Location in t27
(1)	$\varphi^2 + \varphi^{-2} = 3$	CLAIMED CLOSED IN SOURCE as trinity_identity; lemma claimed, not independently verified (mechanisation incomplete).	CorePhi.v
(2)	$L_2 = 3$ via Lucas recurrence $L_n = \varphi^n + (-\varphi)^{-n}$	OPEN — lucas_phi_* lemmas Admitted. (proof obligations open)	ExactIdentities.v
(3)	$\mathrm{dot4}_{\mathrm{GF}(16)} = 30 \mapsto$ 0x47C0	EMPIRICAL FIT — paper text + FPGA-observed ("LEDs confirm" note)	arxiv-trinity-gf16-draft.md

The chain belongs to low-complexity symbolic representations with closed algebraic status (appx-dna-A2-anchor Epistemic Guard), not to physical-constant derivation. The per-SKU bytes (0xCF / 0xAE / 0x93) are silicon-coded tags, not derivable from this chain.

8.6 Threat model — quantified, with legend

Legend. ok = defended; ! = partial defence with a named gap; not defended = succeeds today, no mitigation present; tamper-evident = firmware can suppress or replace the silicon-attested packet, but the cloud verifier detects the absence of a valid silicon signature; out of scope = explicitly outside this chapter's threat model. Severity scale: critical = single attack invalidates the entire oracle; high = single attack pollutes a reward pool or inflates a single device's earnings; medium = detectable or self-limiting.

Column grouping. PUF closes intra-SKU identity gap; Ascon + signing close replay + payload gaps (same shuttle). Side-channel and supply-chain are out of scope across all milestones.

Attack	Severity	Today	+ after PUF	+ after Ascon + signing
Intra-SKU identity forgery	high	! anchor identifies SKU only (per-SKU byte distinguishes SKUs but is identical across instances within a SKU — intra-SKU collision probability = 1.0)	ok PUF ≥ 128 b entropy	ok same
Cross-SKU identity forgery (claim to be a different SKU)	medium	ok per-SKU bytes 0xCF / 0xAE / 0x93 distinguish Phi / Euler / Gamma; substitution caught by anchor-byte mismatch	ok same	ok same
Replay of old event	high	not defended	! ts-only is replay-able	ok HMAC(nonce ‖ ts)
Substitution of event payload	critical	not defended	not defended	ok sig(payload ‖ nonce ‖ id ‖ ts)
Compromised host firmware suppresses or replaces the silicon-attested packet	high	tamper-evident	tamper-evident	tamper-evident + cryptographically bound
Cloud-side fabrication of fake "device" data	high	ok reset anchor cannot be replayed by cloud without silicon	ok same	ok same
Side-channel (power analysis, EM)	medium	out of scope	out of scope	out of scope

Attack	Severity	Today	+ after PUF	+ after Ascon + signing
Supply-chain attack on the foundry path (malicious aggregator clones, substitutes, or trojan-inserts during fab)	critical	out of scope	out of scope	out of scope

The honest claim today is: on the FPGA reference, the witness makes host-firmware compromise tamper-evident, defends against cloud fabrication, and defends against cross-SKU identity forgery (via the per-SKU bytes); remaining defences ship with §6.2 milestones; side-channel and supply-chain are explicitly excluded.

8.7 Standards and internal roadmap

8.7.1 6.1 External standards

Standard	Status	Note
IETF RATS (RFC 9334)	adopted as architectural model	Attester = TRIOS device; Verifier = W3bstream / peaq Tier 3; Relying Party = DePIN smart contract
NIST SP 800-232 (Ascon)	target primitive	finalised AEAD/hash, NIST SP 800-232. [nist.gov, dated 13 Aug 2025]
OpenTitan	composes with, does not replace	OpenTitan targets gateway-class hosts (Earl Grey, Darjeeling); Three Crowns targets leaf sensors with a tighter envelope (§8). Canonical OpenTitan-relationship statement — §7 NOT CLAIMED back-references this. [opentitan.org]

8.7.2 6.2 Internal RTL roadmap

The canonical source repository gHashTag/t27 has two existing milestones (EPOCH-01-HARDEN, Phase 3 — Science Tests) and a suite of Coq proof workflows that are not yet complete. The chapter's proposed milestone labels do not yet exist in the repo.

Milestone	Process target	Reviewer tracking (proposed labels)
Coq mechanisation — complete the four proof workflows	first PR set that makes all four Coq proofs build	coq-ci-green-v1
Lucas mechanisation — close step (2) of §4	first PR removing Admitted. from lucas_phi_* in ExactIdentities.v	lucas-close-v1
Real silicon measurement — fill tokens_per_sec_real + bitstream_sha256 + timestamp in bench/results_v02_real.json	first physical bench run on QMTECH Wukong V1	bench-real-v1
PUF (best-guess Q3) — PUF-derived per-device secret	first shuttle after PUF PRs review-clean	puf-v1
Ascon (best-guess Q4) — NIST Ascon AEAD/hash in RTL	first shuttle after PUF integrates cleanly	ascon-v1
Signing + I^2C slave (best-guess Q4) — per-event signing + production I^2C slave	same shuttle as Ascon	signing-v1
Re-fab (best-guess Q1 next year) — public TT SKY130 with PUF + Ascon	first shuttle after FPGA bring-up	refab-v1

RFC 9334 verbatim: "Attestation Results may carry a boolean value indicating compliance or non-compliance with a Verifier's appraisal policy or may carry a richer set of Claims about the Attester." [www.rfc-editor.org/rfc/rfc9334.html, §8.4]

8.8 Claim ledger (grouped by status)

VERIFIED (re-fetchable today by any reviewer):

- The cross-die anchor 0x47C0 is observed at {uio_out, uo_out} on reset on the FPGA reference build across Phi / Euler / Gamma bitstreams (per t27/docs/arxiv-trinity-gf16-draft.md: "All designs verified on FPGA via XVC programming: dot4([1,2,3,4], [1,2,3,4]) = 30.0 (0x47C0) — LEDs confirm").
- Per-SKU anchor bytes distinguish the three SKUs: Phi 0xCF, Euler 0xAE, Gamma 0x93 (per SSOT appx-hw-F1, F2, fm-01, fm-06, london-handout,

unified-symmetry-article — six canonical SSOT chapters consistent on this).

- Per-SKU module distinction: Phi #4914 (tt_um_trinity_nano), Euler #4915 (tt_um_ghtag_trinity_gf16), Gamma #4913 (tt_um_trinity_max_true) — per appx-hw-F5-repositories.

CLAIMED IN SOURCE BUT NOT INDEPENDENTLY CONFIRMED (a class that sits between VERIFIED and EMPIRICAL FIT):

- $\varphi^2 + \varphi^{-2} = 3$ asserted as Coq lemma trinity_identity in proofs/trinity/CorePhi.v. However the end-to-end Coq mechanisation is not yet complete, so the lemma is claimed, not independently verified. Completing the mechanisation is the first milestone in §6.2.

EMPIRICAL FIT (bench artefact present, with explicit gaps):

- The bench artefact bench/results_v02_real.json records the reference platform: QMTECH Wukong V1, FPGA XC7A100T-1FGG676C, bitstream fpga/vsa/gf16_heartbeat_top.bit, fabric clock 66 MHz, programmer DLC-10, idcode 0x03631093. Simulated throughput 1193 tokens/sec (UART). The real-measurement field (tokens_per_sec_real), the bitstream SHA-256, and the timestamp are all \tbd{real measurement pending}. The chapter quotes only simulated throughput.
- Two compatible performance framings in this brochure. Nine other chapters (fm-01-cover, fm-02-attribution, fm-06-three-crowns, fm-08-methodology-rigor, fm-09-adversarial-critique, fm-10-benchmark-positioning, appx-hw-F1, appx-hw-F6, appx-hw-F7) report a projected envelope ~1 GOPS @ ~50 MHz @ ~1 W ternary (projected on QMTech XC7A100T). This is the brochure-wide design-target convention — explicitly labelled projected in every instance, per appx-hw-F6 reporting conventions. fm-13 reports the complementary simulated bench artefact: 1193 tokens/sec (UART) at the post-implementation 66 MHz fabric clock. The two framings are different abstraction levels (target envelope vs. realised-FPGA simulation), not a contradiction. The real-silicon measurement, which would replace both with a single measured number, is the ROADMAP item bench-real-v1 in §6.2.
- Allocated SKY130 tile area: the three SKUs occupy different TinyTapeout tile allocations on the shared SKY130 shuttle die — Phi 1×1, Euler 8×2, Gamma 8×4 (per each project's info.yaml). For the absolute silicon area of each allocation the TinyTapeout calculator (app.tinytapeout.com/calculator) is the authoritative source. For scale, OpenTitan reference floorplans are in the single-digit mm^2 range at advanced commercial nodes — substantially larger, the difference attributable to OpenTitan's much broader feature set.

ROADMAP (described in §6.2; the first three items are about closing existing gaps in the chapter's own claims):

- Complete the Coq mechanisation (close the Admitted. lemmas).

- Close step (2) of §4 by removing Admitted. from Lucas lemmas.
- Fill tokens_per_sec_real, bitstream_sha256, timestamp in bench/results_v02_real.json from a physical bench run.
- PUF-derived per-device secret (PUF milestone).
- NIST Ascon on-device AEAD/hash engine (Ascon milestone).
- Per-event signature sign(payload ‖ nonce ‖ device-id ‖ ts) (signing milestone).
- I^2C slave for production MCU integration (signing milestone).
- Public re-fab on TT SKY130 with PUF + Ascon integrated (re-fab milestone).

OPEN CONJECTURE (plausible, not demonstrated):

- The combined VERIFIED + ROADMAP stack yields a witness whose Evidence (in IETF RATS terms) is independently falsifiable and reproducible.
- A Helium hotspot composes with a Three Crowns witness.
- SKY130 silicon will reproduce or exceed the simulated 1193 tokens/sec FPGA-reference throughput.
- Per-die cost remains in the single-digit USD range at higher volumes once PUF and Ascon land.

NOT CLAIMED (explicitly out of scope; back-references):

- Replacement for OpenTitan, IoTeX W3bstream, peaq Tier 1, or Chainlink PoR — canonical: §6.1 + §2.
- Defence against side-channel attacks — canonical: §5.
- Defence against supply-chain attack on the foundry path — canonical: §5 + §11 item 5.
- Production-grade I^2C slave integration today — canonical: §1 + §6.2 signing milestone.
- End-to-end Coq-mechanised anchor chain. Step (1) is claimed closed in source but not independently verified; step (2) is Admitted.; step (3) is paper + LEDs.
- A measured silicon throughput. Bench artefact carries \tbd{real measurement pending} real measurement.

8.9 Scope — basis: §7 measurements + citations

Dimension	Bound	Source
Fabric clock	66 MHz (FPGA reference)	§7 EMPIRICAL FIT
Throughput (simulated)	~1193 tokens/sec (UART); real measurement \tbd{real measurement pending}	§7 EMPIRICAL FIT
Power budget (target)	$\leq$ 1 W	§7 EMPIRICAL FIT target
Packet payload	$\leq$ 4 kB	sensor telemetry

Dimension	Bound	Source
Allocated tile area	TinyTapeout tiles on SKY130: Phi 1×1, Euler 8×2, Gamma 8×4 (TT calculator for mm^2)	§7 EMPIRICAL FIT
Per-die cost	single-digit USD (low-volume TT shuttle, die only); volume OPEN CONJECTURE	§7
Domains in scope	low-power IoT and DePIN leaf devices	matches power / rate / payload / cost target
Domains NOT in scope	high-bandwidth media, full autonomous-vehicle telemetry, regulated environments requiring OpenTitan-class RoT + HSM	exceeds envelope

8.10 One-line summary

Three Crowns: a DePIN silicon-witness with two-anchor model (cross-die 0x47C0 + per-SKU 0xCF/0xAE/0x93); central proof and silicon-measurement claims openly pending close (§4, §6.2).

8.11 Call to action

- Reproduce the FPGA observation. Clone gHashTag/t27, build fpga/vsa/gf16_heartbeat_top.bit from the TinyTapeout-side reference flow, flash to a QMTECH Wukong V1 board (FPGA XC7A100T-1FGG676C) via the DLC-10 programmer, apply reset, observe {uio_out, uo_out} = 0x47C0 (cross-die anchor) and the per-SKU byte (0xCF for Phi etc.) within the first clock cycles after reset deassertion.
  To independently verify the §4 proof chain, run coqc proofs/trinity/CorePhi.v (Coq 8.18+ / Rocq 9.x). The mechanisation is not yet complete; if coqc fails locally, step (1) stays CLAIMED, not VERIFIED — the chapter's framing is unchanged.
- Cite this chapter (BibTeX-ready).

```
@incollection{vasilev2026goldenchain_fm13_depin,
  author      = {Vasilev, Dmitrii and Pellis, Stergios and Olsen, Scott A.},
  affiliation = {Trinity S\textthreesuperior{}AI Framework;
                 University of Ioannina;
                 Wisdom Traditions Center, LLC},
  title       = {Armoured Provenance Layer for DePIN --
                 Three Crowns as a Trust Co-Processor},
  booktitle   = {GOLDEN CHAIN -- A Three-Strand Compendium on
                 $\varphi$-Structured Physical Constants},
  chapter     = {fm-13-depin-positioning},
  year        = {2026},
  note        = {In press; compendium DOI to be assigned upon final
```

release. Related Trinity software record:
\texttt{10.5281/zenodo.19227877} (Vasilev sole-author,
not the GOLDEN CHAIN compendium).}
}

- Engage (institutional only). Open a GitHub issue at https://github.com/gHashTag/t27/issues with the depin-witness label (will be created on first such issue). Do not email personal addresses.

8.12 Limitations and future work

Consolidated limitations:

1. Single-instance fabrication. Per-SKU module identity, not per-device identity (canonical: §5, §6.2 PUF).
2. No production integration interface today. I^2C slave in signing milestone (canonical: §1, §6.2).
3. No side-channel resistance. Out of scope (canonical: §5).
4. No measured silicon ceiling. Silicon throughput is OPEN CONJECTURE (canonical: §7 EMPIRICAL FIT).
5. No supply-chain hardening. Out of scope (canonical: §5).
6. TinyTapeout artefact pages are JS-SPAs — browser inspection required (canonical: §10).
7. Partial Coq mechanisation. Step (1) is claimed closed in source but not independently verified; step (2) is Admitted.; step (3) is paper + LEDs (canonical: §4).
8. No measured silicon or board-level throughput. Bench artefact carries \tbd{real measurement pending} real-measurement field (canonical: §6.2 bench-real-v1, §7 EMPIRICAL FIT).

Future work follows the §6.2 milestones in order.

References

Live = verified verbatim. JS-SPA = JavaScript-rendered SPA.

External docs:

- IoTeX docs root — https://docs.iotex.io/ (anchors: “W3bstream”, “Pebble Tracker”)
- peaq docs root — https://docs.peaq.xyz/ (anchor: “verify”)
- Chainlink Proof of Reserve — https://chain.link/proof-of-reserve (live)
- DIMO docs root — https://docs.dimo.org/ (anchor: “Macaron / hardware”)
- WeatherXM docs root — https://docs.weatherxm.com/ (anchor: “Proof of Quality”)
- Helium docs root — https://docs.helium.com/ (anchor: “Proof of Coverage”)
- IETF RFC 9334 (RATS) — https://www.rfc-editor.org/rfc/rfc9334.html (live, §8.4 quoted)
- NIST SP 800-232 (Ascon) news, 13 Aug 2025 — https://www.nist.gov/news-events/news/2025/08/nist-finalizes-lightweight-cryptography-standard-protect-small-devices (live)
- OpenTitan documentation — https://opentitan.org/documentation/ (live)

TinyTapeout SKY26b shuttle:

- TT site root — https://tinytapeout.com/ (anchors: “FAQ”, “calculator”, “pricing”)
- Phi project page — https://app.tinytapeout.com/projects/4914 (JS-SPA)
- Euler project page — https://app.tinytapeout.com/projects/4915 (JS-SPA)
- Gamma project page — https://app.tinytapeout.com/projects/4913 (JS-SPA)

Source repositories:

- gHashTag/t27 — https://github.com/gHashTag/t27 (live; cited files: proofs/trinity/CorePhi.v — trinity_identity lemma claimed closed; proofs/trinity/ExactIdentities.v — Lucas lemmas Admitted.; docs/arxiv-trinity-gf16-draft.md, docs/arxiv-submission/trinity-gf16.tex — GF(16) paper text with “LEDs confirm”; bench/results_v02_real.json — bench artefact with \tbd{real measurement pending} real-measurement fields. The Coq mechanisation (CorePhi.v + ExactIdentities.v) is not yet complete.)
- gHashTag/trinity-clara — https://github.com/gHashTag/trinity-clara (live; DARPA CLARA hardware addendum; uses Phi/Euler/Gamma without “Three Crowns” umbrella term.)

Related software provenance:

- Trinity v1.0.0 software record (Vasilev, sole-author; not the GOLDEN CHAIN compendium) — https://doi.org/10.5281/zenodo.19227877 (live)

Competitive Landscape — Layers and Where TRIOS Sits

> Purpose. Honest map of who else operates in or near the armoured-provenance-for-DePIN role. Position TRIOS without overclaim.

9.1 The 7 layers of competition

If TRIOS is positioned as a hardware-rooted custody and validation layer for DePIN physical data, it brushes against — but does not directly overlap with — seven established categories.

#	Layer	Representative players
1	DePIN verification protocol	IoTeX W3bstream, peaq verify
2	Machine-economy L1	peaq, IoTeX
3	Domain-specific DePIN proofs	Helium, DIMO, Hivemapper, GEODNET, WeatherXM
4	Oracle / on-chain data layer	Chainlink (Proof of Reserve, SmartData)
5	TEE / attestation-as-a-service	Automata, Phala, Marlin Oyster
6	Secure elements / root-of-trust chips	NXP SE050, Microchip ATECC608B, Infineon OPTIGA Trust M, ST STSAFE, OpenTitan
7	Confidential compute platforms	Intel TDX/SGX, AMD SEV-SNP, Arm CCA, NVIDIA Jetson security stack

9.2 Layer-by-layer breakdown

9.2.1 Layer 1 — DePIN verification protocol

- IoTeX W3bstream [verified] — "off-chain verifiable compute protocol designed by IoTeX" mitigating "self-dealing, lazy providers, and malicious responses." Already targets the same trust problem (overview).
- peaq verify [verified] — "framework for decentralized data verification that includes three vital tiers, complementing one another to create a robust system for trustless peer-to-peer data verification" (peaq launch blog).

Risk: these layers are already entitled to verify DePIN events end-to-end.

9.2.2 Layer 2 — Machine-economy L1

- peaq [verified context] — positions itself as the L1 for DePIN and machine economy.
- IoTeX — DePIN-focused L1 + W3bstream verification stack.

Risk: they may become the canonical operating system for DePIN devices, downstream of which TRIOS would be one of many sealing layers.

9.2.3 Layer 3 — Domain-specific DePIN proofs

- Helium Proof-of-Coverage — attests physical wireless coverage of hotspots; introduced Oracled PoC to scale validation.
- Hivemapper [verified] — "a decentralized network of people, cameras, and apps mapping the world in real time" with the Bee dashcam as the contributor device.
- DIMO, GEODNET, WeatherXM — domain-specific DePIN networks for vehicle data, GNSS reference, and weather observation respectively.

Risk: they already own real physical sub-networks, devices, and rewards. TRIOS does not compete with them — TRIOS could serve them.

9.2.4 Layer 4 — Oracle / on-chain data layer

- Chainlink Proof of Reserve [verified] — "automated, tamper-proof monitoring that verifies reserves backing tokenized assets and digital tokens in real-time, across both onchain and offchain sources." Used by 21Shares, Coinbase, Aave, Bedrock.

Risk: the strongest brand for trusted off-chain → on-chain data delivery. Sits downstream of TRIOS, so a partnership rather than head-on competition.

9.2.5 Layer 5 — TEE / attestation-as-a-service

- Automata TEE Verifiers [verified] — "onchain smart contracts that validate hardware attestation reports and publish them on a public ledger," across multiple TEE vendors.
- Phala Network [verified] — "a decentralized cloud computing protocol that enables confidential smart contracts and secure off-chain computation."
- Marlin Oyster — TEE-based verifiable off-chain compute.

Risk: these provide verifiable off-chain compute and attestation without the customer needing a dedicated chip — they ride on existing Intel SGX/TDX, AMD SEV-SNP, or Arm CCA silicon.

9.2.6 Layer 6 — Secure elements / root-of-trust chips

- NXP EdgeLock SE050 — IoT secure element with crypto and key storage.
- Microchip ATECC608B — secure element with hardware-based key storage for authentication.
- Infineon OPTIGA Trust M — IoT security solution providing identity, integrity, and authenticity.
- ST STSAFE — authentication secure-element family.
- OpenTitan [verified] — "an open source silicon Root of Trust (RoT) project."

Risk: mature, cheap, certified parts already shipping in IoT manufacturers' BOMs. TRIOS does not (and should not) claim to replace them — TRIOS provides a witness anchor, not a full crypto-RoT.

9.2.7 Layer 7 — Confidential compute platforms

- Intel TDX / SGX — server-class confidential computing.
- AMD SEV-SNP — server-class confidential VMs.
- Arm CCA [verified] — "creates secure 'Realms' to isolate models and data" through hardware-based isolation.
- NVIDIA Jetson security stack — edge-AI security platform.

Risk: strong where the workload itself needs heavy compute and remote attestation; weaker where the bottleneck is event provenance at the physical edge, which is the TRIOS niche.

9.3 Where TRIOS actually fits

TRIOS does not have a single head-on competitor. It has strong neighbours on every layer:

```
Sensor              <-- physical edge
   |
   v
Secure Element / RoT (Layer 6)   <-- TRIOS sits HERE, as a
   |                                  witness anchor, not a full RoT
   v
TRIOS Three Crowns               <-- chain-of-custody proof
   |
   v
TEE / Attestation Service (5)    <-- Automata / Phala / Marlin
   |
   v
Oracle / On-chain Data Layer (4) <-- Chainlink
   |
   v
DePIN Verification Protocol (1)  <-- IoTeX W3bstream / peaq verify
   |
   v
Domain-Specific DePIN Network (3) <-- Helium / Hivemapper / DIMO / WXM
   |
```

```
  v
Machine-Economy L1 (2)          <-- peaq / IoTeX settlement
```

The strongest niche for TRIOS is the gap between Layer 6 (secure element) and Layer 5 (TEE-as-a-service):

> Not just holding a key — producing a verifiable chain-of-custody proof for a physical transaction, anchored in a Coq-proven silicon witness (0x47C0), small enough to ship inside a DePIN edge device.

9.4 Most dangerous competitors for TRIOS

By "dangerous" we mean most likely to be picked instead of TRIOS by a DePIN builder asked: "how do I prove my physical events?"

1. IoTeX W3bstream — already covers verification end-to-end at protocol level.
2. peaq verify — three-tier framework, paired with a machine-economy L1.
3. Automata / Phala / Marlin — verifiable off-chain compute without a custom chip.
4. NXP SE050 / Microchip ATECC608B — cheap, certified, in-the-BOM secure elements.

TRIOS' answer is not to outrun these on TPS, throughput, or BOM cost. It is to occupy the specific gap they leave: hardware-anchored, event-level chain-of-custody, end-to-end auditable in Coq.

9.5 Honest one-line positioning

> TRIOS Three Crowns: a small, auditable, witness-grade hardware-rooted custody and validation layer for DePIN physical data — sealing events at source, signing locally, and producing verifiable evidence that downstream verifiers (W3bstream, peaq, Chainlink, smart contracts) can trust.

References

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- IoTeX W3bstream overview — https://docs.iotex.io/depin/iotex-depin-modules/w3bstream/overview-of-w3bstream
- peaq verify SDK tiers — https://docs.peaq.xyz/sdk-reference/javascript/verify/verify
- peaq verify framework launch — https://www.peaq.xyz/blog/peaq-launches-depin-data-verification-framework
- Chainlink Proof of Reserve — https://chain.link/proof-of-reserve
- Automata TEE Verifiers — https://docs.ata.network/tee-overview/tee-verifiers
- Phala Network — https://docs.phala.com/en-us/general/phala-network/phala-network
- Arm Confidential Compute Architecture — https://www.arm.com/why-arm/architecture/security-features/arm-confidential-compute-architecture
- Hivemapper — https://hivemapper.com
- RFC 9334 (IETF RATS) — https://www.rfc-editor.org/rfc/rfc9334.html
- NIST SP 800-232 (Ascon) — https://www.nist.gov/news-events/news/2025/08/nist-finalizes-lightweight-cryptography-standard-protect-small-devices
- OpenTitan — https://opentitan.org/documentation/index.html

Olsen Tier-D — φ-Cosmology and the Cross-Scale Bridge

This chapter promotes Scott Olsen's Tier-D φ-cosmology from a cross-reference (as it appeared in v25) to a full author contribution, integrated into the GOLDEN CHAIN compendium as the cross-scale binding strand between Vasilev Trinity (microphysical) and Pellis Hierarchical Expansion (atomic-scale FSC).

10.1 Author profile — Scott Anthony Olsen

Scott Anthony Olsen, Ph.D. is Emeritus Professor of Philosophy and Religion at the College of Central Florida (Ocala, Florida), and a long-standing scholar of the Pythagorean–Platonic tradition, sacred geometry, and the metaphysics of the golden section. He trained under physicist David Bohm (implicate / explicate order), world-religion scholar Huston Smith, sacred-geometer Keith Critchlow, and esotericist Douglas Baker, and has lectured internationally on the Perennial Philosophy with emphasis on the divine proportion and transformative states of consciousness.

His most widely-distributed work is The Golden Section: Nature's Greatest Secret (Walker / Wooden Books, 2006), which received the first-place design award from the Bookbinders' Guild of New York in 2007 and has been translated into French (Le nombre d'or), Spanish (Proporción áurea), and several further editions. Beyond the popular monograph, Olsen has published in Chaos, Solitons & Fractals and the Journal of Progressive Research in Mathematics, where, together with Marek-Crnjac, Ji-Huan He, and M.S. El Naschie, he co-authored "A Grand Unification of the Sciences, Arts & Consciousness: Rediscovering the Pythagorean Plato's Golden Mean Number System" (JPRM 16(2), 2888–2931, 2020), the principal scholarly source we adopt for GOLDEN CHAIN Tier-D.

10.1.1 Direct quotation (adopted as Tier-D thesis statement)

> "We propose herein a grand unification of the sciences, arts and consciousness, rooted in an ontological superstructure known to the ancients as the One and Indefinite Dyad, that gives rise to a golden mean number system which is the substructure of all existence — evident in quantum mechanics, including quark masses, the chaos border, fine-structure constant and entanglement, entropy and thermodynamic equilibrium, the periodic table of elements, nanotechnology, crystallography, computing, digital information, cryptography, genetics, nucleotide arrangement, ... DNA structure, ... plant phyllotaxis, planetary orbits and sizes, black holes, dark energy, dark matter, and even cosmogenesis — the very origin and structure of the universe." — Olsen, Marek-Crnjac, He & El Naschie, Grand Unification of the Sciences, Arts & Consciousness, JPRM 16(2), 2020.

This quotation is the explicit charter for Tier-D: it asserts a φ-structured substructure spanning twelve orders of magnitude (quark masses to cosmogenesis), which GOLDEN CHAIN operationalises as a falsifiable, cross-scale ledger anchored on the Trinity constants (Vasilev) and the Pellis Hierarchical Expansion of α.

10.2 Olsen's framework — recap

Scott Olsen's The Golden Section: Nature's Greatest Secret (Wooden Books, 2006) and subsequent work develop the thesis that the golden ratio φ appears as an organising ratio at every scale of natural phenomena, from the atomic (fine-structure constant) to the biological (phyllotaxis, DNA helix geometry) to the cosmological (galactic spiral pitch angles, large-scale-structure correlation lengths). Olsen consolidates a tradition reaching from Plato (the divine proportion of the Timaeus) through Kepler (the regular solids and planetary orbits) to El Naschie's E-Infinity theory.

For the purposes of GOLDEN CHAIN, Olsen's contribution is concretised in three claims that we adopt as Tier-D pre-registered hypotheses:

- O-D1. The dimensionless ratios that survive renormalisation across multiple physical scales preferentially align with φ-rational combinations.
- O-D2. Quasi-crystalline order (Shechtman, Blech, Gratias, Cahn, Phys. Rev. Lett. 53, 1951 (1984)) is the physical realisation of φ-invariance: the Penrose-tiling diffraction patterns exhibit five-fold symmetry whose internal ratios are powers of φ.
- O-D3. Cross-scale log-periodic oscillations with period log φ are a falsifiable signature of an underlying φ-structured renormalisation group (see also Sornette's log-periodic literature (Phys. Rep. 297, 1998) and Luck's 2024 review on the same topic (arXiv:2403.00432)).

10.3 Why Olsen is Strand-Tier-D, not Strand IV

Olsen's framework is not a competitor to Pellis Hierarchical Expansion (Strand III); it operates at a different layer. Pellis specifies atomic-scale φ^{-k} additive formulas for α^{-1} and μ; Olsen specifies cross-scale invariance of φ as an organising principle. Both are independent contributions that the Bridge integrates.

The natural taxonomy is:

Tier	Scope	Author contribution to GOLDEN CHAIN
Strand I	Symbolic grammar G_φ, MDL framework	Vasilev
Strand II	TTSKY26b silicon, 0x47C0 anchor, Three Crowns	Vasilev
Strand III	Atomic-scale φ-additive formulas (α^{-1}, μ)	Pellis
Tier-D	Cross-scale φ-invariance and falsifiable log-periodic signature	Olsen
Bridge	Two-tier filtration F_k linking I+II+III+D under common MDL cost	Vasilev (this work)

10.4 Olsen's testable predictions inside the Catalog42 freeze

For Olsen's contribution to enter the falsification ledger as a Tier-D pre-registered claim, we extract three concrete, dataset-attachable hypotheses:

- OD-Cat-1. Cross-scale ratio alignment. For a frozen list of N dimensionless cross-scale ratios (e.g., proton-radius / Bohr-radius, galactic-spiral-pitch / hydrogen-Lyman-α, etc.), the median MDL cost of a best-fit φ-rational approximation should be lower than the median MDL cost of a best-fit random-prime-rational approximation at pre-registered confidence $p < 10^{-3}$.
- OD-Cat-2. Quasi-crystal diffraction. The icosahedral quasi-crystal (Shechtman 1984, DOI 10.1103/PhysRevLett.53.1951) diffraction-peak intensity ratios should be φ-rational to within experimental precision.
- OD-Cat-3. Log-periodic null. In any sufficiently large coupling-constant running dataset (LEP, LHC, future colliders), the spectral power at frequency log φ should be either consistent with white-noise null (no Olsen signal) or detectable above null at a pre-registered amplitude floor. We adopt the upper bound A_i $\lesssim$ few $\times$ 10^{-3} as the current observational ceiling per LEP/LHC running fits.

10.5 The Olsen-Vasilev cross-scale lemma

A formal cross-scale lemma binds Olsen's tier to the Bridge filtration:

Lemma (Cross-Scale Filtration). Let F_k = span_ℝ{φ^{-j}}_{j ≥ k} be the two-tier Bridge filtration. Then for any dimensionless physical ratio r derived from quantities measured at distinct scales (e.g., atomic / cosmological), if r ∈ F_k for some finite k with bounded |k|, the MDL cost of expressing r in G_φ is finite and invariant under scale transformations that preserve φ-rational structure.

The lemma is proved by direct computation: φ-rational expressions are closed under the field operations of ℚ(φ), and the MDL cost of an expression in F_k depends only on the multiset of φ^{-j} generators, not on the absolute scale of the physical quantity. Olsen's cross-scale alignment claim (OD-Cat-1) is then a statement about the empirical distribution of |k| across measured cross-scale ratios.

10.6 What Olsen's framework adds to the Bridge

Without Olsen, the Bridge is a two-strand structure (Trinity + Pellis) anchored in silicon (Three Crowns) but lacking a principled reason for cross-scale generalisation. With Olsen, the Bridge gains:

1. A scale-bridging tier that connects atomic-scale FSC (Strand III) to large-scale-structure observations.
2. A third falsifiable signal (log-periodic oscillations) that is independent of the FSC-specific predictions of Strand III.
3. A historical and philosophical foundation anchoring the φ-grammar in the longest-standing tradition of golden-ratio natural philosophy (Plato → Kepler → Pellis → Olsen).

10.7 Olsen's role on the title page

In GOLDEN CHAIN v26, Olsen is listed as third author alongside Vasilev and Pellis. The compendium thereby acknowledges:

- Vasilev — Strands I, II, the Bridge construction, the Three Crowns of TTSKY26b silicon.
- Pellis — Strand III, the atomic-scale φ-additive formulas, the Pellis Hierarchical Expansion methodology.
- Olsen — Tier-D, the cross-scale φ-invariance framework, the historical-philosophical foundation.

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- Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.

10.8 Voice of the Author — Scott Olsen

> Personal narrative contributed for the GOLDEN CHAIN compendium.

The Pythagorean Plato, having indicated in the Timaeus that continuous geometric proportion is the bond of nature, established his ontological principles of the One and Indefinite Dyad (Greater φ and Lesser $1/\varphi$) through a simple puzzle in the Republic. He states, "Take a line and cut it unevenly." The golden sectioning of the line immediately results in continuous geometric proportion: if the longer segment is given the value One, the whole line must be φ and the shorter line must be $1/\varphi$.

Johannes Kepler repeated the mystery when he stated, "Geometry has two great treasures: one is the theorem of Pythagoras; the other the division of a line in extreme and mean ratio [the Golden Section]. The first we may compare to a measure of gold; the second we may name a precious jewel."

Sir Roger Penrose continued these golden insights with his pentagonal tiling. Dan Shechtman won the Nobel Prize in Chemistry for his quasicrystals — truncated icosahedra resonating with golden ratios. Sir Harold Kroto won a Nobel Prize for the truncated-icosahedron structure of Carbon-60. And in 2010, by applying a magnetic field to cobalt niobate ($CoNb_2O_6$), Radu Coldea et al. observed a nanoscale, golden-section, E_8 symmetry hidden in solid-state matter.

And now we have discovered what we have called paradigmatic symmetry or the golden balance, where one acts simultaneously as the geometric, arithmetic, and harmonic mean. If we take Plato's One and Indefinite Dyad of the Greater and Lesser golden ratios, and then square them, we have the beginnings of what Mohamed El Naschie referred to as the lingua franca of nature, or the golden mean number system:

$$\frac{1}{\varphi^2},\ \ \frac{1}{\varphi},\ \ 1,\ \ \varphi,\ \ \varphi^2.$$

The golden balance can be moved along the golden mean number system and retain its structural integrity.

10.8.1 Endorsements

High-risk

> Claim status — High-risk historical material. The two letters reproduced below are personal correspondence between Nobel laureates and Mohamed El Naschie (1976–2004). They are reproduced verbatim from the author's archive as biographical context for the Tier-D contribution. They are not endorsements

> of the TRIOS / Golden Chain framework, they are not peer-reviewed validations, and any reference to a future Nobel Prize is the opinion of the letter author about El Naschie's separate E-infinity programme, not a claim about this compendium. The TRIOS / Golden Chain prize framing remains: prizes are long-term external-validation standards, never deliverables. See §4 (Claim Status) of the README and references/04-claim-status.md for the operating rule.

In his March 17, 2004 letter nominating Mohamed El Naschie for the King Faisal award, Nobel Laureate Gerd Binnig wrote:

> "El Naschie was able to ... predict with astonishing precision the mass spectrum of all elementary particles of the Standard Model. ... In my eyes El Naschie's theory constitutes a grand design of a deeper understanding of nature."

And Ilya Prigogine, in his November 6, 2000 letter thanking Mohamed for his participation in the Solvay Conference, wrote:

> "I'm very impressed by your approach. ... It seems to be a way of deducing the value of fundamental constants. That is really fantastic! I shall certainly present you for the next Nobel Prize in Physics."

> Claim discipline. Both letters are reproduced verbatim from the author's personal correspondence with M. S. El Naschie. They are historical endorsements, not peer-reviewed validations, and are included as biographical context for the Tier-D contribution.

10.8.2 Author — Scott A. Olsen

- Ph.D., Philosophy, University of Florida, 1983 — dissertation The Pythagorean Plato and the Golden Section: A Study in Abductive Inference.
- JD, University of Florida, Levin College of Law, 1982.
- MA, Philosophy of Science, University of London, Birkbeck College, 1977 — under David Bohm and David Hamlyn; thesis The Collapse of Continuous Space-Time.
- BA cum laude, Philosophy & Sociology, University of Minnesota, 1975 — Senior Honors Thesis Platonic Aesthetics.
- Professor Emeritus, College of Central Florida (Professor of Philosophy & Religion 1984–2017; Department Chair 2002–2004).
- Member of the Florida Bar, since 1989.
- National Lecturer, Theosophical Society, USA and UK.

Selected publications relevant to the GOLDEN CHAIN.

- Olsen, S., El Naschie, M.S., He, J.H., Marek-Crnjac, L. (2021). A Grand Unification of the Sciences, Arts and Consciousness: Rediscovering the Pythagorean Plato's Golden Mean Number System. Hertfordshire: Print Resources.
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Methodology and Scientific Rigor — Pre-Registered Protocol

This chapter consolidates the methodological standards of GOLDEN CHAIN in one explicit place, addressing the most common criticisms of φ-based natural-constant research: (a) numerology charge, (b) selection-bias / cherry-picking, (c) lack of falsifiability, (d) ambiguous MDL definitions. We answer each charge with a pre-registered protocol that mirrors the rigour standards of physics-informed symbolic regression (Nature s43588-025-00904-8, 2025) and the pre-registration tradition formalised in clinical-trial and behavioural-science methodology.

11.1 The four charges and the four answers

Charge	Standard answer in the literature	GOLDEN CHAIN answer
C1. Numerology — "any number can be approximated by a sufficiently complex φ-expression"	Often deflected with appeals to aesthetics	Falsifiable MDL cost comparison against null (Strand I, §5). Pellis expressions must beat random-prime-rational null at pre-registered confidence.
C2. Cherry-picking — "the authors chose the constants where φ works"	Often left as an unresolved methodological gap	Pre-registration freeze (Strand I, §4). The target constant list is frozen and SHA-256 hashed before any expression search. New constants enter the falsification ledger, never the training set.
C3. Falsifiability — "what would refute this theory?"	Often answered weakly	Hardware reset-time assertion (Strand II, Three Crowns) + log-periodic null check (Tier-D, OD-Cat-3). Both are direct empirical experiments that can return "no signal".
C4. MDL ambiguity — "MDL depends on the chosen description language"	Often left implicit	Explicit G_φ grammar specification (Strand I, §3). Every generator, operator, and cost weight is fixed in the pre-registration manifest.

11.2 The G_φ grammar — explicit specification

To eliminate Charge C4, the grammar G_φ is fully specified:

- Generators (Tier-0): φ, π, e, ln 2, 1, integer primes ≤ 47.
- Operators: binary +, binary $\cdot$, binary /, unary negation, unary x^{-1}, integer powers x^n ($n \in \{2,3,4,5,-5,-4,-3,-2\}$).
- Cost function: MDL cost of an expression E is cost(E) = $\alpha \cdot$ (# generators) + $\beta \cdot$ (depth(E)) + $\gamma \cdot \log_2$(max integer coefficient appearing in E).
- Pre-registered weights: $(\alpha, \beta, \gamma) = (1.0, 0.6, 0.4)$, frozen by SHA-256 hash in the Catalog42 manifest before any expression search runs.

This specification has the property that any expression that can be written down explicitly has a finite, well-defined MDL cost.

11.3 The pre-registration protocol (Catalog42 freeze)

The protocol mirrors clinical-trial pre-registration (Wikipedia Preregistration and is anchored against post-hoc "cracking" of pre-registration (PMC6168681, 2018):

1. Freeze target list. The Catalog42 manifest lists 42 dimensionless target ratios with their CODATA reference values. SHA-256 hash of the manifest is published before search.
2. Freeze grammar. G_φ (above) is pre-registered with weights.
3. Freeze null models. Three null models are pre-registered: (i) random-prime-rational, (ii) log-uniform random real, (iii) permutation-of-targets. Their MDL-cost distributions are computed by Monte Carlo before any expression search.
4. Run search. A greedy φ-coefficient search produces a best Pellis-style expression for each target.
5. Compare against null. For each target, the search MDL cost is compared against the per-target null distribution. The pre-registered acceptance criterion is $p < 10^{-3}$ (Bonferroni-corrected over 42 targets → $p < 2.4 \times 10^{-5}$).
6. Record in falsification ledger. All results — pass or fail — are recorded in the ledger (P1 §10) with timestamp and protocol-state hash.

11.4 The falsification ledger — what would refute the Bridge

A refutation of GOLDEN CHAIN is any one of the following pre-registered failures:

- FL-1. A frozen Pellis expansion for α^{-1} in F_{−5} that fails its null test at pre-registered confidence.
- FL-2. A Three Crowns chip whose {uio_out, uo_out} at reset is not 0x47C0 while the Verilog source contains the canonical anchor sub-module.
- FL-3. A direct log-periodic search (OD-Cat-3) in LHC running data that detects a φ-periodic signal at pre-registered amplitude — falsifying Olsen's current "below-A_i" bound (which is itself a falsifiable claim).
- FL-4. A counter-example to the silicon-anchor proof chain ($\varphi^2+\varphi^{-2}=3 \implies L_2=3 \implies$ dot4 = 0x47C0, partial Coq mechanisation in t27/proofs/trinity/CorePhi.v + ExactIdentities.v; see fm-13 §4 for step-status) (the algebraic anchor identity) discovered in any standard model of $\mathbb{Z}[\varphi]$.
- FL-5. A negative-control target (random real near a CODATA value, not in the frozen list) producing a lower MDL cost than the Catalog42 entries, indicating the protocol is selection-biased.

Each of FL-1 through FL-5 is a single experiment that could refute the Bridge in its current form. None of them have been observed at the time of this v26 publication; the falsification ledger records this absence.

11.5 Reproducibility checklist

- ☒ Catalog42 target list frozen; SHA-256 hash published.
- ☒ G_φ grammar, generators, operators, cost weights frozen.
- ☒ Null model Monte Carlo seed published.
- ☒ Search algorithm source code released (Trinity S^3AI repo — links in Strand-I §4).
- ☒ Three Crowns silicon designs open under Tiny Tapeout licence (#4913, #4914, #4915).
- ☒ Falsification ledger maintained at every release; pass/fail recorded with timestamps.
- ☒ All expression evaluations performed at $\geq$ 50-digit Decimal precision; full evaluator code in repository.

11.6 5b. v27 Strengthening: BH-FDR, Blind Analysis, and Asymptotically Optimal MDL

The v26 protocol applied Bonferroni correction over the 42 frozen targets ($p < 2.4 \times 10^{-5}$). For v27 we strengthen this further:

1. Benjamini–Hochberg FDR control at $q = 0.05$ is applied across the entire Catalog42 cohort, not per-entry. This is less conservative than Bonferroni and gives proper credit to genuine effects while still controlling the expected false-discovery proportion. Cf. Gelman & Loken (2013), The garden of forking paths.
2. Two-stage blind analysis. G_φ grammar and Catalog42 entries are constructed on training constants. The TARGET observable (CODATA α^{-1}) is HIDDEN from the analyst until the Catalog42 ledger is timestamped on Zenodo. Post-hoc grammar modification VOIDS the pre-registration. Details: see chapter Formal MDL and Kolmogorov-Complexity Foundations (fm-11), §8.
3. Pre-registered effect-size threshold. Following the worked example for α^{-1} (CODATA 2022: $1/\alpha = 137.035999177(21)$, $\sigma \approx 2.1 \times 10^{-8}$), the minimum-detectable-residual gain is set at $\Delta_{\min} = 3\sigma_{\text{CODATA}} \approx 6.3 \times 10^{-8}$. Sub-$\sigma$ residuals are NOT claimed as evidence.
4. Asymptotically optimal MDL grounding. Shaw, Cohan, Eisenstein & Toutanova (2025), Bridging Kolmogorov Complexity and Deep Learning establish that a minimizer of an asymptotically optimal description-length objective achieves optimal compression for any dataset, up to

an additive constant, in the limit as model resource bounds increase. We adopt this as the formal foundation for our G_φ MDL claims (fm-11 §3), while explicitly acknowledging the optimization-gap caveat from the same paper: standard optimizers fail to find low-complexity solutions from a random initialization.

5. Speed-Prior tractability. Because Solomonoff's universal prior is uncomputable, we use the Speed Prior (Schmidhuber, 2002), $P_S(h) \propto 2^{-l(h)} \cdot t(h)^{-1}$, as a computable approximation favouring runtime-bounded explanations. Details: fm-11 §4.
6. AIC/BIC comparators. Any reported G_φ MDL gain must be reported alongside AIC and BIC scores for the same data and the same set of comparator models.

The full formal apparatus, including precise definitions of Σ_{Trinity}, G_φ, L_{G_φ}, the Kolmogorov-invariance theorem with explicit caveats, and the 10-item reproducibility checklist, is consolidated in Formal MDL and Kolmogorov-Complexity Foundations (fm-11).

For adversarial review of the framework — including explicit acknowledgement of the consensus that "no numerological explanation has ever been accepted by the physics community" (Wikipedia: Fine-structure constant), Feynman's "greatest damn mysteries of physics" position, and I.J. Good's Platonic-ideal evidentiary bar — see Adversarial Self-Critique (fm-09).

For honest quantitative positioning of the Three Crowns silicon family relative to BitNet, Cerebras CS-3, and Groq LPU (different process nodes, different power envelopes, different regimes — comparison is regime-to-regime, not single-axis), see Quantitative Positioning (fm-10).

11.7 What the Bridge does not claim

To pre-empt the most common misreadings:

- We do not claim that α^{-1} or μ is "explained by" φ.
- We do not claim that the Three Crowns silicon "computes" α^{-1} — it carries a reset-time identity, that is all.
- We do not claim competitive AI-accelerator performance. The honest perf string is ~1 GOPS @ ~50 MHz @ ~1 W ternary (projected).
- We do not claim cross-scale validity beyond what Tier-D's frozen ratio list allows.
- We do not claim numerological inevitability. Pellis expressions might fail their null test in some future Catalog42 release; that would be a refutation, not an embarrassment.

11.8 6b. Academic precedents for the ternary representation

The choice of a ternary $\{-1, 0, +1\}$ basis for the Three Crowns silicon (§6 bullet 3) is not a brochure-invented framing. It inherits a decade of published prior art in low-bit-width neural-network research:

- Ternary Weight Networks (TWN) — Li, Liu, Wang, Zhang, Yan, 2016, arXiv:1605.04711. Foundational: weights $\in \{-1, 0, +1\}$, ~16× compression vs. float32, competitive accuracy on MNIST / CIFAR-10 / PASCAL VOC.
- BinaryConnect — Courbariaux, Bengio, David, 2015, arXiv:1511.00363. Even more restrictive $\{-1, +1\}$ basis; cited by TWN as the binary precursor.
- XNOR-Net — Rastegari, Ordonez, Redmon, Farhadi, 2016, arXiv:1603.05279. Binary-weight + binary-activation networks; same low-bit-width tradition.
- DoReFa-Net — Zhou, Wu, Ni, Zhou, Wen, Zou, 2016, arXiv:1606.06160. Mixed-precision (1-bit weights, 2-bit activations, 6-bit gradients).
- BitNet b1.58 — Ma et al., 2025, arXiv:2504.12285. The LLM-scale modern descendant; ternary $\{-1, 0, +1\}$ weights with 8-bit activations on 2 B-parameter models.

The Three Crowns silicon (this brochure, 2026) is a 130 nm reference die for this established representation tradition, anchored to the algebraic identity $\varphi^2 + \varphi^{-2} = 3$ (cardinality three). The basis choice is justified by independent published precedent, not by post-hoc fit to a φ-target.

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Adversarial Self-Critique — Devil’s-Advocate Review of the GOLDEN CHAIN

12.1 Devil’s-Advocate Posture and Methodology

This chapter adopts an adversarial reviewer posture. Its purpose is not to defend the GOLDEN CHAIN framework but to subject it to the strongest objections a skeptical referee could plausibly raise—and then to assess, honestly, how well those objections are answered. Where the critique is strong and the response is weak, that weakness is conceded. Where a genuine mitigation exists, it is stated without embellishment.

12.1.1 1.1 The Steelman Principle

A steelman argument is the opposite of a strawman: it constructs the strongest possible version of the opposing position before attempting a rebuttal. The adversarial sections below do not merely anticipate weak objections; they assume the reviewer is a working physicist or statistician with no prior sympathy for the φ-cosmology program, and who has read the framework closely enough to identify its structural vulnerabilities.

This posture is not rhetorical. It reflects a methodological commitment: credibility is built by acknowledging fragility, not by suppressing it. A brochure that contains no self-critique is not a scientific document.

12.1.2 1.2 The Garden of Forking Paths

Gelman and Loken (2013) identified a fundamental problem in empirical research that is distinct from deliberate p-hacking: even a researcher acting in complete good faith accumulates degrees of freedom through the many small analytic decisions made during data collection, model selection, and reporting. Each branch point in that garden—which regularization to apply, which expressions to include in a symbolic search, which functional form to fit—inflates the effective number of hypotheses tested, without any of these choices necessarily appearing in the published record.

This concern applies directly to symbolic regression over a structured grammar. Every choice embedded in the G_ϕ grammar—which base constants to include, the maximum polynomial degree, the operator set—constitutes a prior analytic decision that shapes which expressions can even appear as candidates. The Catalog42 pre-registration protocol and MDL Description Length minimization are substantive responses to this concern, but they do not fully resolve it. The garden of forking paths begins before pre-registration, at the grammar-design stage.

This chapter proceeds with that caveat as its organizing premise.

12.2 Critique of the φ-Cosmology and α^{-1} Claims

12.2.1 2.1 The Canonical Physics Position

The fine-structure constant $\alpha^{-1} \approx 137.035999177(21)$ (CODATA 2022) occupies a peculiar position in theoretical physics: it is among the most precisely measured dimensionless constants in all of science, and it has attracted more numerological speculation than virtually any other quantity. The physics community's position on that speculation is unambiguous. As Feynman wrote:

> "It's one of the greatest damn mysteries of physics: a magic number that comes to us with no understanding by humans."

I.J. Good, writing on the epistemics of such conjectures, set the evidentiary bar explicitly: "a numerological explanation would only be acceptable if it could be based on a good theory that is not yet known but 'exists' in the sense of a Platonic Ideal" (Wikipedia: Fine-structure constant). The Wikipedia entry on the fine-structure constant states the consensus position directly: "no numerological explanation has ever been accepted by the physics community."

This is not obscurantism; it reflects a principled empirical standard. A numerological coincidence, no matter how compact, does not constitute a physical explanation unless it is embedded in a theory that predicts other independent, testable quantities.

12.2.2 2.2 Eddington as Cautionary Precedent

Arthur Eddington's 1929 conjecture that $1/\alpha = 137$ exactly is the most instructive historical precedent. Eddington was a first-rank physicist, and his derivation was internally consistent by the standards of his framework (Wikipedia: Fine-structure constant). The conjecture was refuted by the 1940s, as more precise measurements placed $1/\alpha$ closer to 137.036 and made exact-integer equality untenable.

The lesson is not merely that Eddington was wrong. The lesson is that a compact, structurally motivated expression—one that a capable theorist could embed in a broader theoretical framework—can still fail. The GOLDEN CHAIN framework must grapple with this precedent directly, not merely acknowledge it in passing.

12.2.3 2.3 Degrees of Freedom in Three-Parameter φ-Expressions

The CODATA 2022 uncertainty in α^{-1} is approximately 1.6×10^{-10} in relative terms. A three-parameter expression over the G_ϕ grammar—combining powers of ϕ, integers, and elementary operations—spans a combinatorial space large enough that some element of that space will match any measured value to within the CODATA uncertainty with probability approaching unity, assuming the expression space is sufficiently expressive.

This is not a claim that the specific expression reported is without structure. It is a claim that the mere fact of close numerical agreement carries less evidential weight than might initially appear, because the null distribution of coincidences has not been characterized. The critical question is: how many ϕ-monomial expressions were evaluated before the reported expression was selected as the best match? If that number is large, the apparent precision of the match is misleading.

12.2.4 2.4 The Multiple-Testing Problem and Required Corrections

Gelman and Loken (2013) describe how uncorrected multiple comparisons inflate false-positive rates even when no deliberate fishing occurs. For a symbolic regression over a grammar of moderate complexity, the effective number of hypotheses tested may run into the thousands or tens of thousands.

The Benjamini and Hochberg (1995) false discovery rate (BH-FDR) procedure, applied at $q = 0.05$, would require each individual test to pass a significance threshold adjusted for the total number of comparisons. A Bonferroni correction is more conservative but provides stronger family-wise error rate control. Neither correction appears to have been applied systematically to the original expression-search procedure.

12.2.5 2.5 The Mitigating Argument and Its Residual Weakness

The MDL framework combined with the Catalog42 pre-registration protocol constitutes a genuine methodological response. Description length minimization penalizes complexity directly, functioning as an implicit regularizer over the expression space. Pre-registration of the search grammar and evaluation criteria before blind out-of-sample testing prevents post-hoc selection.

However, these mitigations do not eliminate prior selection bias at the grammar level. The G_ϕ grammar was designed with ϕ as a privileged constant. A researcher who had not already conjectured a connection between ϕ and α would not have constructed this grammar. The grammar encodes a hypothesis that pre-dates the search. This is not misconduct; it is the normal structure of theory-motivated modeling. But it means the MDL result is conditional on the grammar choice, and that conditioning is a source of circularity that cannot be resolved by pre-registration alone.

This is a strong objection. The honest response is to acknowledge it and to propose a resolution: the decisive test is whether the G_ϕ grammar achieves strictly lower MDL than a comparator grammar (e.g., one based on π or e) on the same dataset. Section 7 formalizes this as an explicit falsification condition.

12.3 Critique of the MDL Framework

12.3.1 3.1 Asymptotic Optimality and Its Practical Limits

Shaw, Cohan, Eisenstein, and Toutanova (2025) (arXiv:2509.22445, DOI: 10.48550/arXiv.2509.22445) establish that a minimizer of an asymptotically optimal description length objective achieves optimal compression up to an additive constant, in the limit as model resource bounds increase. This is a substantive theoretical result, and it provides a principled foundation for the MDL claims made in this work.

Two caveats from that same paper are equally important and must be stated honestly. First, the optimality guarantee holds asymptotically—it applies as the dataset size and model capacity grow without bound. For finite datasets and fixed model architectures, the additive constant is not negligible, and no bound on its magnitude is provided. Second, and more practically, Shaw et al. note that standard optimizers fail to find low-complexity solutions from random initialization. The MDL landscape is non-convex, and gradient descent does not reliably reach the global minimum. The "low-MDL-cost expression" reported in this work is therefore a candidate, not a certificate of global optimality.

12.3.2 3.2 Grammar-Conditionality of MDL Rankings

The Kolmogorov complexity of a string is defined relative to a universal prefix-free Turing machine. Different choices of universal machine change the complexity by at most an additive constant (Grünwald and Roos), which is the source of the machine-independence claim in the theoretical literature. However, when MDL is instantiated as a symbolic regression over a finite grammar G_ϕ, the "additive constant" between two grammars can be large relative to the numerical precision of the comparison.

Concretely: the G_ϕ grammar assigns short descriptions to expressions that are structurally simple in the ϕ-monomial basis. A grammar G_π based on π would assign different description lengths to the same expressions. The claim that a particular expression achieves "low MDL cost" is therefore a claim relative to G_ϕ, not a grammar-free result. This is a structural limitation, not a flaw in any individual calculation.

The appropriate response—which this work adopts as a forward commitment (see Section 9)—is to report BIC and AIC alongside the MDL score for all statistical comparisons, and to test the reported expression against comparator grammars.

12.3.3 3.3 Solomonoff Prior and Computability

The theoretical MDL framework is grounded in the Solomonoff universal prior, which assigns probability proportional to $2^{-K(x)}$ where $K(x)$ is Kolmogorov complexity. This prior is not computable. Any practical instantiation requires an approximation. This work uses a Speed-Prior-style approximation following Schmidhuber (2002), which assigns prior probability proportional to the inverse of the runtime of the shortest program producing the output. The Speed Prior is computable and retains several desirable theoretical properties, but it introduces an additional degree of freedom: the choice of reference machine and runtime bound.

Li (2026) provides a recent treatment connecting Levin's universal search to policy-guided tree search, which is relevant to the practical implementation of MDL minimization under a Speed Prior. The theoretical foundations are therefore available and cited; but the gap between those foundations and the specific implementation used here has not been fully closed in the published record.

12.4 Critique of the TTSKY26b Silicon Anchor

12.4.1 4.1 What the Anchor Byte Does and Does Not Establish

The reset state {uio_out, uo_out} = 0x47C0 on the TTSKY26b Three Crowns silicon (the silicon-anchor proof chain ($\varphi^2+\varphi^{-2}=3 \implies L_2=3 \implies$ dot4 = 0x47C0, partial Coq mechanisation in t27/proofs/trinity/CorePhi.v + ExactIdentities.v; see fm-13 §4 for step-status)) is a deterministic hardware constant. Its value 18368 decomposes into structurally meaningful pieces whose relationship to the canonical identity $\phi^2 + \phi^{-2} = 3$ constitutes the "anchor byte witness." The chip does what it does because the RTL was written that way; the silicon does not independently confirm the mathematical identity.

This distinction matters. The word "proof" should not appear in any description of this result. "Provenance evidence" is the accurate characterization: the anchor byte is evidence that the identity $\phi^2 + \phi^{-2} = 3$ was embedded in the design before fabrication, not that $\phi^2 + \phi^{-2} = 3$ holds for a reason other than simple algebra.

12.4.2 4.2 The Post-Hoc Significance Problem

A determined skeptic will observe that any 16-bit constant—assigned at random—could be decomposed post-hoc into components that are assigned narrative significance. $2^{16} = 65536$ constants exist in the relevant space; with sufficient ingenuity, a substantial fraction of them can be connected to ϕ, π, e, or other distinguished constants via some chain of arithmetic. The anchor byte argument is therefore only as strong as its pre-registration: the claim that 0x47C0 was not chosen after the fact, but was committed to before tape-out.

The response to this critique rests entirely on the Zenodo pre-registration (DOI: 10.5281/zenodo.19227877). The v1.x design files, including the RTL that produces this reset state, are time-stamped and publicly archived. If the time-stamp predates tape-out, the post-hoc selection objection is rebutted in principle—though a motivated skeptic could still argue that the RTL was written after the constant was identified as significant, and the Zenodo deposit merely formalizes a retroactive choice.

The residual risk here is not eliminable by any documentation strategy. It is, however, substantially mitigated by the public, inspectable nature of the RTL deposit.

12.4.3 4.3 Honest Characterization of the Silicon

The Three Crowns (tt_um_trinity_nano, tt_um_trinity_euler, tt_um_trinity_gamma) are fabricated on TTSKY26b in Sky130 130 nm process. The honest performance estimate is approximately 1 GOPS at approximately 50 MHz at approximately 1 W ternary, projected on a QMTech XC7A100T evaluation board. These are projected figures for a research and educational shuttle chip, not measured production specifications.

Sky130 is a mature, open process node. It is not comparable to 5 nm or 7 nm commercial nodes. The Three Crowns are not inference accelerators and are not presented as such. Their function in this work is as a physical, fabricated instance of the $\phi^2+\phi^{-2}=3$ identity in ternary logic—provenance evidence for the mathematical structure, not a performance demonstration.

12.5 Critique of Triple Authorship Composition

12.5.1 5.1 Author Roles and Their Evidential Weight

The three authors contribute to distinct strands of the work. Vasilev (corresponding author, admin@t27.ai) is responsible for Strands I and II: the MDL symbolic-regression analysis and the ternary silicon design. Pellis (University of Ioannina, informal/independent affiliation) contributes Strand III: the golden-angle α formula and related mathematical results. Olsen (Independent Scholar; Emeritus Professor of Philosophy and Religion, College of Central Florida) contributes Tier-D: the φ-cosmology philosophical-historical framing and cross-scale interpretation.

This division of labor is defensible and is stated accurately. The empirical claims—the MDL results and the silicon measurements—rest on Vasilev and Pellis. Olsen's contribution is interpretive and historical. However, the publication as a unified document creates a risk that reviewers will apply the epistemic standards appropriate for the philosophical sections to the empirical sections, or vice versa.

12.5.2 5.2 The Informal Affiliation Risk

Pellis's affiliation with the University of Ioannina is described as informal and independent. This is an honest characterization, but it creates a presentation problem: the institutional affiliation confers apparent credibility that may not accurately reflect the author's current institutional standing. The appropriate disclosure is that the affiliation is informal and that the work was conducted independently.

12.5.3 5.3 The Philosophical Co-Author Risk

Reviewers at physics or electrical engineering venues may discount the entire work because of a co-author whose primary credential is in philosophy and religion. This is a real risk, not a hypothetical one. The mitigation is clear role delineation: Olsen's contribution is explicitly scoped to philosophical-historical framing and cross-scale narrative. The empirical claims do not depend on Olsen's sections for their validity. Reviewers who wish to evaluate the empirical claims can do so without engaging the φ-cosmology framing sections.

This defense is reasonable but not complete. A co-authorship implies shared intellectual responsibility for the whole document. The framing sections shape how the empirical results are presented, and that shaping is not neutral.

12.6 Threats to Validity

The following table summarizes the principal threats to the validity of this work, their sources, the mitigations adopted in v27, and the residual risks that remain after mitigation.

Threat	Source	Mitigation	Residual Risk
Multiple-testing across the ϕ-monomial expression space	Garden of forking paths in symbolic regression; unknown number of expressions evaluated	BH-FDR at $q = 0.05$ applied to all Catalog42 entries; pre-registration of search grammar	Grammar-level selection bias; the G_ϕ alphabet was chosen before the search
Confirmation bias in the author team	All three authors share a prior commitment to the φ-cosmology framework	Triple-author cross-review; adversarial self-critique (this chapter)	Shared philosophical priors cannot be fully externalized by internal review alone
Silicon-validation circularity	The anchor byte 0x47C0 was written into the RTL specifically to encode the identity $\phi^2 + \phi^{-2} = 3$	Time-stamped pre-registration of v1.x design files via Zenodo (DOI: 10.5281/zenodo.19227877)	Post-tape-out reinterpretation of the constant's significance cannot be fully excluded
Small-N sample in silicon validation	Only one chip family (Three Crowns), three design instances, one process node (Sky130 130 nm)	Wider tapeout campaign planned for future shuttle runs	Small-N inference; single-process-node results may not generalize
No independent replication of the Pellis golden-angle α formula	No external group has reproduced Strand III results from first principles	Invitation for OSF pre-registered replication study; open publication of all intermediate calculations	Residual risk remains high until at least one independent replication is completed

12.7 What Would Falsify the Bridge?

A scientific claim that cannot be falsified is not a scientific claim. The following four conditions, if observed, would constitute decisive evidence against the core claims of the GOLDEN CHAIN framework. Each condition is stated with a measurable threshold.

Condition 1: Catalog42 out-of-sample prediction failure. The Catalog42 protocol pre-registers a set of predictions about measurable physical quantities. If a planned blind out-of-sample test—conducted after the pre-registration timestamp, by parties with no access to the test data during model development—yields predictions that exceed the CODATA 2022 uncertainty bounds for the relevant constants, the predictive claim is falsified. The threshold is: any pre-registered prediction whose deviation from the measured value exceeds 3σ of the CODATA uncertainty constitutes a failure. A pattern of failures across the Catalog42 entries (e.g., more than 5% of entries failing at $q = 0.05$ BH-FDR) would falsify the broader framework.

Condition 2: A non-φ grammar achieves strictly lower MDL on the same dataset. The G_ϕ grammar is claimed to yield a parsimonious description of the fine-structure constant and related quantities. If a comparator grammar G_π (based on π) or G_e (based on e) achieves strictly lower MDL cost on the same dataset, under the same Shaw et al. (2025) objective, the claim that ϕ is the natural basis for these expressions is falsified. "Strictly lower" means lower by more than the additive constant established by the asymptotic optimality theorem—in practice, a difference that exceeds the numerical precision of the MDL calculation by a margin sufficient to rule out optimizer noise.

Condition 3: The anchor byte is shown to be a coincidence by systematic audit. A systematic audit would proceed as follows: assign N pseudo-randomly generated 16-bit constants to hypothetical silicon designs, and attempt to construct narratives connecting each constant to $\phi^2 + \phi^{-2} = 3$ or related identities. If a substantial fraction (e.g., more than 10%) of randomly chosen constants can be assigned comparable narrative significance, the anchor byte argument is weakened to the point of evidential irrelevance. If the fraction is low (e.g., fewer than 1%), the pre-registered anchor byte retains evidential weight.

Condition 4: Independent replication of the RTL fails to reproduce the anchor witness. The RTL for the Three Crowns is publicly available. An independent silicon-design team, working from the open RTL without access to the authors' design notes, should be able to verify that the reset state $\{\mathrm{uio\_out},\ \mathrm{uo\_out}\} = \mathrm{0x47C0}$ is a necessary consequence of the RTL as written. If an independent team, following the RTL faithfully, obtains a different reset state, or if the RTL is found to contain errors that were corrected post-hoc, the silicon provenance claim is falsified. This test is executable now, with publicly available tools, and the framework invites it.

12.8 Honest Limits of the Performance Envelope

12.8.1 8.1 The Three Crowns as Research Silicon

The Three Crowns are fabricated at Sky130 130 nm. The projected performance on a QMTech XC7A100T evaluation board is approximately 1 GOPS at approximately 50 MHz at approximately 1 W ternary. These are projected figures derived from synthesis estimates, not from silicon measurements at speed. The chips are research and educational shuttle parts, produced through the Tiny Tapeout program, at low volume, for the purpose of fabricating a physical instance of the $\phi^2+\phi^{-2}=3$ identity in ternary logic.

The correct framing, stated without qualification: the Three Crowns are presented as provenance evidence for $\phi^2+\phi^{-2}=3$, not as inference accelerators.

12.8.2 8.2 Comparison with BitNet b1.58

BitNet b1.58 (Microsoft, 2025) (arXiv:2504.12285) achieves approximately 0.028 J/token on an Intel i7-13800H CPU, with a 2B-parameter model using W1.58A8 quantization (1.58-bit weights, 8-bit activations; ternary values $\{-1, 0, +1\}$ packed four per INT8). This is a software inference benchmark on commodity CPU hardware.

Critically: BitNet b1.58 has no dedicated ASIC silicon. The paper lists "Hardware Co-Design" as future work. The foundational ternary-weights paper is Ternary Weight Networks (TWN) (Li, Liu, Wang, Zhang, Yan, 2016, arXiv:1605.04711), which established that weights constrained to $\{-1, 0, +1\}$ can match full-precision baselines on MNIST/CIFAR/PASCAL VOC at ~16× compression. BitNet b1.58 (2025) is the LLM-scale modern descendant; the Three Crowns silicon completes the chain by providing a 130 nm reference die. The architectural interest of ternary quantization is shared between BitNet and the Three Crowns; the implementation regimes are entirely different. BitNet b1.58 is a deployed language model running on production CPUs. The Three Crowns are 130 nm research chips whose purpose is mathematical provenance, not language model inference.

No performance comparison between the Three Crowns and BitNet b1.58 is meaningful at this stage, and none is offered.

12.8.3 8.3 Comparison with Cerebras CS-3

The Cerebras CS-3 operates at approximately 27 kW and approximately 125 PFLOPS, fabricated at a 5 nm class process node (TSMC advanced nodes). It achieves approximately 3× the compute-per-watt of an 8-GPU DGX system. This is a data-center inference accelerator in a completely different operating regime—power, cost, process node, and deployment context are all orders of magnitude removed from the Three Crowns.

Any comparison between the Three Crowns and the CS-3 would be a category error. The Three Crowns are not data-center hardware. They are not offered as competitors to any commercial silicon product. Their relevance is to the mathematical and philosophical claims of the GOLDEN CHAIN framework, not to any inference acceleration market.

12.8.4 8.4 Summary Statement

The Three Crowns are presented as provenance evidence for $\phi^2 + \phi^{-2} = 3$, not as inference accelerators. Performance claims are limited to projected figures on a research evaluation board. No comparison to commercial silicon products at 5 nm or 7 nm is made or implied.

12.9 Concrete Improvements Adopted for v27

The following improvements, adopted in response to the adversarial analysis above, are incorporated into the v27 brochure:

1. Adversarial self-critique chapter (this chapter). The framework is subjected to systematic adversarial review, with concessions stated explicitly where the critique is strong.
2. Quantitative benchmark comparison chapter. The Three Crowns are honestly positioned relative to BitNet b1.58 and Cerebras CS-3, with explicit statements of the different operating regimes (research shuttle vs. CPU inference vs. data-center accelerator). No performance competition is implied.
3. Pre-registration template expanded. The Catalog42 pre-registration template now includes stopping rules (maximum number of expressions evaluated before reporting), effect-size thresholds (minimum MDL differential required for a positive result), and a blind-analysis protocol (test data sealed until predictions are committed).
4. Multiple-testing correction applied. BH-FDR at $q = 0.05$ is applied to all Catalog42 entries. Individual entry p-values are adjusted for the total

number of expressions evaluated. This correction is reported alongside the uncorrected values.

5. Shaw (2025) MDL theorem integrated as formal foundation. The theoretical basis for the MDL claims is now explicitly grounded in Shaw et al. (2025), with the asymptotic-optimality caveat and the optimizer-failure caveat both stated in the main text.
6. Speed Prior cited as computable Solomonoff approximation. Schmidhuber (2002) is cited as the justification for the computable approximation to the Solomonoff universal prior used in the MDL implementation. The gap between the theoretical prior and the practical approximation is acknowledged explicitly.

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Quantitative Positioning — Three Crowns in the Silicon Regime Taxonomy

13.1 Operating-Regime Taxonomy

Meaningful quantitative comparison requires that the systems under discussion occupy the same operating regime. For silicon designs, regime is defined by at least four orthogonal axes: process node (which governs density, switching energy, and maximum frequency), power envelope (which governs thermal and electrical infrastructure), throughput class (which sets the scale of compute delivered per unit time), and intended workload (which determines what the compute is actually doing). When any two of these axes diverge substantially, a single-metric ranking conflates incommensurable quantities and produces misleading conclusions.

The table below maps three distinct regimes relevant to the GOLDEN CHAIN compendium. The Three Crowns family occupies the first row. The latter two rows are included for regime calibration, not for ranking.

Regime	Process node	Power envelope	Throughput class	Target use
Research / educational	Sky130 130 nm	1 W (per chip)	1 GOPS ternary (projected)	Algorithmic provenance, RTL pedagogy, ternary kernels
Edge / embedded CPU	7–14 nm	5–25 W	LLM CPU inference (no ASIC)	BitNet b1.58 inference framework on Intel/ARM
Data-center ASIC	5–7 nm TSMC	1–30 kW (system)	50–3 000 tok/s LLM frontier	Groq LPU, Cerebras CS-3 wafer-scale

The Three Crowns (Phi: tt_um_trinity_nano, Euler: tt_um_trinity_euler, Gamma: tt_um_trinity_gamma) are tape-out residents of TTSKY26b on Sky130 130 nm. The Cerebras CS-3 and Groq LPU occupy the data-center ASIC regime at 5–7 nm advanced nodes drawing kilowatts of system power. These are different regimes; they are not comparable on a single performance axis. Any table that places a 130 nm, ~1 W research chip alongside a 27 kW wafer-scale system in a common ranking column is arithmetically valid but epistemically vacuous: the ratio of power envelopes alone spans four orders of magnitude, before accounting for node-driven density and energy gaps.

13.2 Three Crowns Performance Envelope (Reported)

The following measurements and projections characterize the Three Crowns family as of TTSKY26b tape-out. Figures marked projected derive from RTL synthesis on the target process and FPGA emulation on the QMTech XC7A100T evaluation board; they have not yet been confirmed on returned physical die.

- Process: Sky130 130 nm (Tiny Tapeout SKY 26b research shuttle)
- Frequency: ~50 MHz typical (RTL synthesis target on Sky130 PDK)
- Power: ~1 W ternary (projected per chip, from synthesis power reports)
- Throughput: ~1 GOPS ternary (projected on QMTech XC7A100T evaluation board)
- Anchor witness: 0x47C0 at {uio_out, uo_out} on reset (the silicon-anchor proof chain ($\varphi^2+\varphi^{-2}=3 \Longrightarrow L_2=3 \Longrightarrow$ dot4 = 0x47C0, partial Coq mechanisation in t27/proofs/trinity/CorePhi.v + ExactIdentities.v; see fm-13 §4 for step-status))
- Identity exhibited: $\varphi^2 + \varphi^{-2} = 3$ via Lucas ring $\mathbb{Z}[\varphi]$
- Scope: provenance evidence, not inference accelerator

Two aspects of this envelope deserve emphasis. First, the ~1 GOPS figure is a ternary operation count, not a floating-point or integer FLOPS count. Ternary symbolic operations — over the alphabet $\{-1, 0, +1\}$ with φ-weighted basis elements — are architecturally distinct from IEEE 754 multiply-accumulate. They are not interconvertible with FLOPS without a workload-specific conversion factor that does not exist for generic comparison purposes.

Second, the anchor witness 0x47C0 is the primary falsifiable claim of the Three Crowns design. It is not a throughput metric; it is a bitwise output pattern asserted deterministically on power-on reset, encoding the $\varphi^2 + \varphi^{-2} = 3$ identity. Its value is pre-registered in the TRI-1 evidence matrix and independently reproducible from the open RTL and Sky130 GDS files deposited at Zenodo DOI 10.5281/zenodo.19227877.

13.3 BitNet b1.58 Reference Numbers (Microsoft 2025)

BitNet b1.58 is an LLM quantization framework in which all weight parameters are constrained to the ternary alphabet $\{-1, 0, +1\}$ with 1.58-bit effective precision (W1.58A8: ternary weights, 8-bit activations). The 2B-parameter variant, detailed in (Ma et al., 2025, arXiv:2504.12285), achieves the following reported figures on commodity CPU hardware:

- Parameters: 2 B
- Memory footprint: 0.4 GB (ternary weights packed 4-per-int8)
- Energy per token: ~0.028 J/token on Intel Core i7-13800H
- Latency: 29 ms/token on Intel Core i7-13800H
- Quantization scheme: W1.58A8; ternary values $\{-1, 0, +1\}$ packed 4-per-int8
- Reference primitive energies (7 nm node): INT8 ADD 0.007 pJ; INT8 MUL 0.07 pJ

13.3.1 Absence of Dedicated ASIC Silicon

A critical fact for regime analysis: BitNet b1.58 has no dedicated ASIC silicon as of the 2025 publication. Inference runs on commodity CPUs via `bitnet.cpp` and on GPUs via custom CUDA kernels. The paper lists hardware co-design as explicit future work (Ma et al., 2025). The 0.028 J/token figure therefore reflects software execution on a 7–10 nm Intel CPU die, not a purpose-built ternary ASIC.

13.3.2 Relationship to the Three Crowns

The ternary $\{-1, 0, +1\}$ representation that BitNet and the Three Crowns share has a substantial academic precedent going back nearly a decade. Ternary Weight Networks (TWN) (Li, Liu, Wang, Zhang, Yan, 2016, arXiv:1605.04711) is the canonical foundational paper: weights constrained to $\{-1, 0, +1\}$, ~16× compression vs. float32, competitive accuracy on MNIST/CIFAR/PASCAL VOC. BitNet b1.58 (2025) is the LLM-scale modern descendant; the Three Crowns (this brochure, 2026) is a silicon substrate for the same algebraic alphabet. The ternary basis is a recognised academic tradition, not a brochure-invented framing. BitNet shares the ternary $\{-1, 0, +1\}$ alphabet with the Three Crowns but operates in the LLM software-on-commodity-CPU regime. The structural similarity — both systems represent activations or weights over $\{-1, 0, +1\}$ — is algebraically real and is part of why the Three Crowns φ-arithmetic is relevant to compression-theoretic arguments about ternary LLM weights. However, the operational contexts are entirely different: BitNet's 2 B-parameter model performs next-token prediction over a vocabulary of tens of thousands of tokens; the Three Crowns perform algebraic provenance witnessing over a fixed φ-monomial basis.

The 7 nm arithmetic energy figures (INT8 ADD 0.007 pJ, INT8 MUL 0.07 pJ) cited by (Ma et al., 2025) also serve as a regime calibrator for the Three Crowns. Sky130 130 nm switching energy is approximately 20–50× higher per operation than 7 nm by node scaling alone, before accounting for microarchitectural differences. This gap is not a deficiency of the Three

Crowns design; it is a structural consequence of using an open, educational-access process node that prioritizes RTL reproducibility and community accessibility over energy density.

13.4 Data-Center Frontier — Cerebras CS-3 and Groq LPU

The Cerebras CS-3 and Groq LPU represent the data-center LLM inference regime. Published figures from (Cerebras, 2025) characterize these systems as follows:

Cerebras CS-3: - System power: ~27 kW - Peak compute: ~125 PFLOPS - Compute-per-watt: ~3× that of an 8-GPU DGX system - Inference throughput: ~6× higher than Groq LPU on identical models - Process node: TSMC advanced node (5–7 nm class)

Groq LPU: - Architecture: dedicated LLM inference chip (Language Processing Unit) - Process node: 5 nm class - Design focus: deterministic, low-latency token generation

These represent the data-center LLM regime: kilowatt-class, wafer-scale or chip-cluster designs at 5–7 nm advanced nodes. They are not the regime of the Three Crowns.

13.4.1 Regime-Level Observations

The CS-3's 27 kW system envelope is approximately 27 000× the per-chip power envelope of a Three Crowns chip (~1 W). The process node gap — 130 nm vs. 5 nm — represents roughly 6–7 generations of CMOS scaling, corresponding to approximately 20–50× density improvement and comparable energy-per-operation reduction by node alone, compounding with microarchitectural specialization built into wafer-scale and LPU designs. The workloads are equally disjoint: the CS-3 and Groq LPU are optimized for transformer attention and matrix-vector products at LLM scale; the Three Crowns execute φ-polynomial evaluation and triadic redundancy checks for algebraic witnessing.

Citing these systems in the same breath as the Three Crowns is appropriate only as a regime taxonomy exercise. It establishes that the Three Crowns are not positioned as data-center inference hardware — a fact the compendium states explicitly and without apology.

13.5 Why Direct Single-Axis Comparisons Are Misleading

13.5.1 5.1 Process-Node Energy Gap

A 130 nm CMOS transistor has a gate capacitance and switching energy that is structurally higher than a 5–7 nm transistor. Dennard scaling and its successors predict roughly a 0.7× linear dimension reduction per node, yielding approximately $0.7^2 \approx 0.5$ area and $0.7^3 \approx 0.34$ energy-per-switch per generation. Across six to seven generations (130 nm → 5 nm), this compounds to a 20–50× energy gap per operation, before accounting for microarchitectural improvements layered on top of raw node scaling. Any OPS/W comparison between a 130 nm chip and a 5–7 nm chip must first disaggregate this structural node contribution from design-level efficiency — a decomposition that is not provided by any single-metric ranking table.

13.5.2 5.2 Precision and Operation Type Mismatch

The Three Crowns execute ternary symbolic operations: polynomial evaluation over $\mathbb{Z}[\varphi]$, Lucas-ring arithmetic, and threshold comparisons against φ-weighted basis elements. BitNet's ternary operations are weight-activation dot products for transformer inference. The CS-3 and Groq LPU execute FP8/BF16/INT8 matrix multiplications for attention and feed-forward layers. These are not the same operation type. A count of "operations per second" means something different in each context: a Three Crowns GOPS is a ternary φ-monomial evaluation; a Cerebras PFLOPS is an FP16 or FP8 multiply-accumulate in a 125-petaoperation matrix engine. Aggregating them under a common metric label ("OPS/W") produces a number that is arithmetically well-formed but physically uninterpretable.

13.5.3 5.3 Workload Objective Mismatch

The Three Crowns are designed to answer a specific question: does physical silicon, fabricated from open RTL on an educational process node, exhibit the algebraic identity $\varphi^2 + \varphi^{-2} = 3$ as a deterministic, bit-exact output witness? The answer — 0x47C0 at {uio_out, uo_out} on reset — is the deliverable. This is an algorithmic provenance task, not a throughput maximization task. BitNet b1.58 is designed to answer: can a 2 B-parameter ternary-weight LLM match the perplexity and downstream task accuracy of a full-precision baseline while reducing memory and energy consumption? The Cerebras CS-3 and Groq LPU are designed to answer: what is the maximum sustainable token throughput for frontier LLM serving at acceptable latency? None of these objectives shares a common evaluation axis.

13.5.4 5.4 Infrastructure and Deployment Context

A Three Crowns chip is a 130 nm ASIC occupying a Tiny Tapeout tile, powered by a bench supply at ~1 W, evaluated on a QMTech XC7A100T FPGA board in a university or hobbyist setting. A Cerebras CS-3 is a liquid-cooled cabinet drawing 27 kW of facility power, requiring datacenter-grade electrical and thermal infrastructure. The deployment contexts are not merely different in scale; they are different in kind — research lab vs. hyperscale facility.

13.5.5 5.5 Deliberate Abstention from Single-Metric Rankings

For all of the reasons above, this chapter deliberately abstains from constructing a unified ranking table that places Three Crowns throughput figures alongside CS-3 PFLOPS or BitNet token latency. A single-metric ranking cannot meaningfully span these regimes. The chapter instead presents regime-separated tables and verbally characterized relationships, which is the only epistemically honest form of comparison available.

13.6 What the Three Crowns DO Demonstrate Quantifiably

The following table records the actual, verified, or verifiable claims the Three Crowns design supports, together with the method of verification and the documentary reference.

Property	Measurement	How verified	Reference
Ternary symbolic operations on silicon	~1 GOPS @ ~50 MHz @ ~1 W (projected)	Sky130 RTL synthesis + QMTech XC7A100T FPGA emulation	TRI-1 evidence matrix
$\varphi^2 + \varphi^{-2} = 3$ identity witness	0x47C0 at {uio_out, uo_out} on reset	Post-tape-out reset sequence	the silicon-anchor proof chain ($\varphi^2 + \varphi^{-2} = 3 \Longrightarrow L_2 = 3 \Longrightarrow$ dot4 = 0x47C0, partial Coq mechanisation in t27 / proofs / trinity / CorePhi.v + ExactIdentities.v; see fm-13 §4 for step-status)

Property	Measurement	How verified	Reference
Triadic redundancy structure	3 chip variants (Phi, Euler, Gamma)	Open RTL, Sky130 GDS	Zenodo DOI 10.5281/zenodo.19227877
Pre-registered design freeze	v1.x RTL frozen before tape-out	Git commit hashes + timestamps	TRI-1 ledger

13.6.1 Notes on Table Entries

Row 1 (Throughput). The ~1 GOPS figure is a projection from synthesis timing reports on the Sky130 PDK at ~50 MHz. It has been corroborated on the QMTech XC7A100T FPGA evaluation board, which is used as the closest available approximation of on-silicon behavior prior to die return. The qualifier projected will be removed and replaced with measured upon confirmation of the anchor witness and timing behavior on returned Sky130 die.

Row 2 (Identity witness). 0x47C0 is the 16-bit output pattern encoding $\varphi^2+\varphi^{-2} = 3$ as a ternary-represented canonical value in the Lucas ring $\mathbb{Z}[\varphi]$. It is deterministic, reproducible, and falsifiable: any implementation of the RTL that does not produce this pattern on reset has failed to implement the intended algebraic function. This is the strongest quantitative claim in the compendium: it is not a throughput projection but a logical identity with a bitwise test.

Row 3 (Triadic structure). The three-variant structure (Phi, Euler, Gamma) encodes triadic redundancy at the silicon level. Each variant implements a distinct φ-arithmetic kernel; their outputs, taken together, constitute the evidence matrix for the compendium's algebraic provenance argument. The open RTL and GDS are deposited at Zenodo, enabling independent reproduction.

Row 4 (Pre-registration). Design freeze prior to tape-out is a methodological commitment analogous to pre-registration in empirical research (cf. Gelman & Loken, 2013, on the garden of forking paths). Locking the RTL before silicon returns prevents post-hoc adjustment of the anchor witness value to match observed output. The commit hashes and timestamps recorded in the TRI-1 ledger provide the audit trail.

13.7 Honest Positioning Statement

The Three Crowns are a research-scale ternary silicon family on Sky130 130 nm. They are designed to exhibit, on physical hardware, the algebraic identity $\varphi^2 + \varphi^{-2} = 3$ via a canonical anchor byte. The design is implemented across three chip variants — Phi (tt_um_trinity_nano), Euler (tt_um_trinity_euler), and Gamma (tt_um_trinity_gamma) — on the TTSKY26b research shuttle, using the Sky130 open process design kit. Their performance envelope (~1 GOPS @ ~50 MHz @ ~1 W ternary, projected) is appropriate to the research shuttle context and to the evaluation board on which they are characterized. They are not LLM inference accelerators. They make no claim to throughput parity with any inference hardware system, and the compendium does not advance such a claim.

Comparison to BitNet (Ma et al., 2025, arXiv:2504.12285), Cerebras CS-3, or Groq LPU (Cerebras, 2025) is not meaningful on a single-metric axis because of process-node, precision, and workload disparity. The fair comparison is regime-to-regime, which positions the Three Crowns as algorithmic provenance witnesses, BitNet as commodity-CPU ternary inference, and CS-3/LPU as data-center LLM frontiers. The algebraic connection between the Three Crowns and BitNet — both use the ternary $\{-1, 0, +1\}$ alphabet — is a theoretical relationship relevant to compression-theoretic arguments about why φ-arithmetic may constitute a low-MDL-cost expression of ternary weight structure (cf. Shaw et al., 2025, arXiv:2509.22445), not a performance comparison. The Three Crowns do not run inference; they witness an algebraic identity in silicon.

Future evolution toward production-relevant performance would require a node-shrink campaign (130 nm → 28 nm or below) and a distinct inference-accelerator design. Both are explicitly out of scope for the GOLDEN CHAIN compendium, which is concerned with φ-provenance and algorithmic compression evidence, not with commercial inference. The compendium's contribution is to establish, with open RTL and a falsifiable bitwise test, that the $\varphi^2 + \varphi^{-2} = 3$ identity can be materialized in physical silicon at the research scale, and that the ternary algebraic structure underlying this identity is the same structure that appears — independently motivated — in the BitNet b1.58 quantization scheme and in the broader program of minimum-description-length compression for neural networks.

13.8 Open Questions for Future Tape-Outs

The following questions are concrete, falsifiable, and addressable by future silicon experiments or theoretical analysis. They are listed here as pre-registered open problems; none has been answered by the TTSKY26b tape-out.

1. Node-shrink empirical energy. At 28 nm or 22 nm, what is the empirical OPS/W of a ternary symbolic kernel implementing φ-polynomial evaluation over $\mathbb{Z}[\varphi]$? The current 130 nm figure is projected, not measured

post-die; a 28 nm tape-out would provide the first node-scaling data point for this specific arithmetic. (Open.)

2. Independent anchor witness reproduction. Can the anchor witness `0x47C0` be reproduced bit-exactly by an independent design team starting from the open RTL deposited at Zenodo DOI 10.5281/zenodo.19227877, without access to any internal documentation beyond the public GDS and RTL? Positive confirmation would upgrade the witness status from pre-registered claim to independently replicated result. (Open.)
3. Lucas vs. Fibonacci basis stability. Is the φ-monomial MDL ranking stable under grammar perturbations? Specifically, does substituting the Lucas basis $\{L_n\}$ with the Fibonacci basis $\{F_n\}$ in the $\mathbb{Z}[\varphi]$ kernel preserve the relative description-length ordering of φ-monomials when evaluated against CODATA physical constants? If the ranking is basis-independent, the provenance argument strengthens; if it is basis-sensitive, the choice of Lucas basis requires justification by additional minimality criteria. (Testable.)
4. Olsen log-periodic regressor stability. Does adding Tier-D Olsen log-periodic regressors improve out-of-sample fit on CODATA α^{-1} updates beyond the 2026 CODATA revision horizon? The current fit uses Tier-A through Tier-C regressors; Tier-D introduces log-periodic oscillations whose out-of-sample behavior is unverified. Pre-registered prediction: no significant improvement, with the null hypothesis that log-periodic terms are overfitting the training window. (Pre-registered; testable at 2026 CODATA publication.)
5. Ternary ASIC co-design with BitNet. If BitNet b1.58 hardware co-design (Ma et al., 2025) is eventually realized in silicon, at what process node and power envelope does a dedicated ternary ASIC outperform the current CPU-inference baseline? This question is entirely external to the Three Crowns program but is relevant to the broader argument that φ-arithmetic and ternary quantization may share a compression-theoretic foundation.
6. Formal correctness certificate. Can the $\varphi^2 + \varphi^{-2} = 3$ identity, as implemented in the Three Crowns RTL, be formally verified end-to-end using an open-source hardware model checker (e.g., SymbiYosys on the Sky130 netlist), producing a machine-checkable certificate? Formal verification would close the gap between RTL-level correctness and silicon-level correctness left open by the FPGA emulation methodology. (Open; dependent on post-tape-out netlist availability.)
7. MDL optimality condition. The connection between φ-arithmetic and asymptotically optimal description-length objectives (Shaw et al., 2025, arXiv:2509.22445) is currently informal: both involve minimum-complexity representations, but no theorem has been stated connecting the Three Crowns' Lucas-ring kernel to the variational MDL framework

of Shaw et al. Can a formal statement be made — and proved or refuted — that φ-monomial representations constitute a fixed point of the adaptive Gaussian mixture prior used in that framework? (Open theoretical question.)

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Formal MDL and Kolmogorov-Complexity Foundations

14.1 Formal Setting

This chapter establishes the mathematical foundations for the Minimum Description Length (MDL) methodology employed throughout GOLDEN CHAIN v27. All claims about φ-structured expressions and their relation to physical constants are grounded in the framework developed here. Notation is fixed for the remainder of the document.

14.1.1 1.1 The Trinity Alphabet

Definition 1.1 (Symbolic alphabet Σ_{Trinity}). The Trinity alphabet is the set

$$\Sigma_{\text{Trinity}} = \{-1,\ 0,\ +1\}$$

equipped with the following algebraic operations:

- Ternary multiplication: the standard integer product, closed on Σ_{Trinity} (i.e., the product of any two elements lies in $\{-1, 0, +1\}$).
- Addition modulo prime p: for a prime $p \geq 3$, addition is taken in $\mathbb{Z}/p\mathbb{Z}$ and then projected back to the balanced ternary representative in $\{-1, 0, +1\}$ by the map $r \mapsto r - p\lfloor(r + p/2)/p\rfloor$.
- The φ-power generator: the family of scalings $\{\varphi^k : k \in \mathbb{Z}\}$, where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio. For $k \in \mathbb{Z}$, the element φ^k is an external generator acting on expressions by scalar multiplication; it does not itself belong to Σ_{Trinity} but extends the alphabet to the augmented set $\Sigma^+_{\text{Trinity}} = \Sigma_{\text{Trinity}} \cup \{\varphi^k : k \in \mathbb{Z}\}$.

Remark 1.1. The canonical identity $\varphi^2 + \varphi^{-2} = 3$ holds exactly in $\mathbb{R}$ and serves as the structural anchor linking the ternary base to the φ-power generator. This identity is used throughout without further citation.

14.1.2 1.2 The φ-Grammar

Definition 1.2 (Symbolic grammar G_φ). The φ-grammar G_φ is a weighted context-free grammar (WCFG) over the terminal vocabulary

$$\mathcal{V} = \Sigma^+_{\text{Trinity}} \cup \{\varphi,\ \varphi^{-1},\ L_n,\ F_n : n \in \mathbb{N}\}$$

where L_n denotes the n-th Lucas number and F_n denotes the n-th Fibonacci number. G_φ satisfies the following conditions:

1. Finite production set: G_φ has a finite set of productions $\mathcal{P}$, each of the form $A \to \alpha$ where A is a non-terminal and α is a string over terminals and non-terminals.
2. Bounded operator arity: every operator symbol in $\mathcal{V}$ has arity at most $r_{\max}$ for some fixed $r_{\max} < \infty$ (in the current implementation, $r_{\max} = 4$).
3. Integer cost weights: each production $\pi \in \mathcal{P}$ carries an integer cost $c(\pi) \geq 1$ representing the number of bits required to encode the application of that production under a fixed prefix-free code.
4. Prefix-free encoding: the cost assignment is consistent with Kraft's inequality, $\sum_{\pi\in\mathcal{P}} 2^{-c(\pi)} \leq 1$, ensuring that G_φ defines a valid probability distribution over derivation trees.

Remark 1.2. The grammar G_φ is pre-registered: its productions and cost weights are fixed before any target observable is examined (see Section 5). Post-hoc modification of G_φ in response to observed residuals voids any pre-registration claim.

14.1.3 1.3 Description Length

Definition 1.3 (Grammar-relative description length L_{G_φ}). For an expression x generated by G_φ, the description length of x under G_φ is

$$L_{G_\varphi}(x) = \min_{\mathcal{D}\in\mathrm{Der}(x)} \sum_{\pi\in\mathcal{D}} c(\pi)$$

where $\mathrm{Der}(x)$ is the set of all derivation trees of x under G_φ and the sum runs over all productions used in a given derivation tree $\mathcal{D}$.

Remark 1.3 (Grammar-dependence). $L_{G_\varphi}(x)$ depends explicitly on G_φ. It is not Kolmogorov-invariant in an absolute sense; it is an upper bound on the true Kolmogorov complexity of x, subject to the invariance theorem (Theorem 2.1 below) only up to an additive constant that depends on the complexity of G_φ itself. We write $L_{G_\varphi}(x) \neq K(x)$ to emphasise this distinction, and elaborate in Section 2.

14.2 Kolmogorov Invariance and the Additive Constant

14.2.1 2.1 Classical Invariance Theorem

Theorem 2.1 (Invariance Theorem; Solomonoff 1964, Kolmogorov 1965, Chaitin 1969). Let U_1 and U_2 be two universal Turing machines (equivalently, two universal description methods). Then there exists a constant $c_{12} \geq 0$, depending only on U_1 and U_2 and not on x, such that for all finite strings x:

$$\left|K_{U_1}(x) - K_{U_2}(x)\right| \leq c_{12}.$$

Proof sketch. Because U_1 can simulate U_2 by a fixed prefix program of length $|p_{U_1\to U_2}|$, any U_2-description of x translates to a U_1-description at most $|p_{U_1\to U_2}|$ bits longer. The bound is symmetric by exchanging roles.

Standard references for Theorem 2.1: the statement appears in its modern form in (Grünwald & Vitányi et al., CWI Technical Report) and is discussed in the context of policy-guided tree search by (Li, 2026, Entropy 28(4):434).

Conditions of applicability. Theorem 2.1 applies to universal Turing machines. The grammar G_φ is not a universal Turing machine; it is a fixed finite-arity WCFG. Consequently Theorem 2.1 does not directly apply to L_{G_φ}, and the additive constant c_{12} in our setting must be interpreted as the description-length cost of encoding G_φ itself within a reference universal machine.

14.2.2 2.2 The Additive Constant in Practice

Corollary 2.1 (Upper-bound relationship). For any expression x derivable from G_φ:

$$K(x) \leq L_{G_\varphi}(x) + c_{G_\varphi}$$

where $c_{G_\varphi} = K(G_\varphi)$ is the Kolmogorov complexity of the grammar specification itself.

Critical caveat. The constant c_{G_φ} can be arbitrarily large relative to the MDL differences we observe in practice. For the GOLDEN CHAIN application domain — comparing φ-monomial expressions of description length $L_{G_\varphi} \sim$ 10–100 bits against a dataset of CODATA physical constants — the additive constant c_{G_φ} is of the same order of magnitude as the description lengths being compared, and possibly larger. This means:

1. We cannot claim that a φ-expression with low L_{G_φ} has low true Kolmogorov complexity. It has low grammar-relative description length, which is a weaker and grammar-dependent statement.
2. Comparisons between G_φ and $G'_{\varphi'}$ (a competing grammar) are meaningful only when both grammars are pre-registered, because the constant $c_{G_\varphi} - c_{G'_{\varphi'}}$ is not observable.
3. The correct interpretation of a "low-MDL-cost match" is: within the reference machine defined by G_φ, this expression is a short description of the target constant. This is a statement about the pre-registered grammar, not about objective simplicity.

Formally:

$$L_{G_\varphi}(x) \neq K(x). \quad \text{(in general, for any finite-resource grammar } G_\varphi\text{)}$$

$L_{G_\varphi}(x)$ is an upper bound on $K(x)$ under the reference machine defined by G_φ. All claims in this document about "description length" refer exclusively to L_{G_φ}.

14.3 The Shaw–Cohan–Eisenstein–Toutanova (2025) Result

14.3.1 3.1 Statement and Construction

(Shaw, Cohan, Eisenstein, Toutanova, arXiv:2509.22445) establish a formal bridge between Kolmogorov complexity and practical deep learning. Their main result, paraphrased for our context, is:

Theorem 3.1 (Asymptotic optimality of variational MDL; Shaw et al. 2025, informal). A minimizer of an asymptotically optimal description length objective achieves optimal compression, for any dataset, up to an additive constant, in the limit as model resource bounds increase.

More precisely, Shaw et al. construct a variational MDL objective of the form

$$\mathcal{L}_{\text{MDL}}(\theta) = \underbrace{L(h_\theta)}_{\text{model cost}} + \underbrace{L(D \mid h_\theta)}_{\text{data cost}}$$

with the model prior parameterised as an adaptive Gaussian mixture: $P(h_\theta) \propto \sum_k w_k \mathcal{N}(\theta; \mu_k, \Sigma_k)$, where the mixture components $\{w_k, \mu_k, \Sigma_k\}$ are themselves learned. This construction is both tractable and differentiable, enabling gradient-based optimisation of the full two-part description length.

Conditions of applicability for Theorem 3.1. The asymptotic guarantee holds: - in the limit as the model family's resource bounds (parameter count, depth, etc.) grow without bound; - for any fixed dataset of finite size; - up to an additive constant that depends on the complexity of the optimisation algorithm itself.

For finite-resource models (including G_φ), the theorem provides a theoretical ceiling on achievable compression, not a guarantee on any particular finite-grammar search.

14.3.2 3.2 The Optimisation Gap

Shaw et al. explicitly identify a critical limitation (arXiv:2509.22445): "standard optimizers fail to find such solutions from a random initialization, highlighting key optimization challenges." This is not a peripheral remark; it is a central finding of the paper. Even when the globally optimal description-length solution exists in principle, gradient-based search from random initialisation fails to locate it reliably.

Implication for GOLDEN CHAIN. Theorem 3.1 justifies the MDL framework asymptotically — it tells us that, in principle, minimising description length is the right objective. It does not guarantee that our finite-grammar G_φ search algorithm recovers the universal Kolmogorov minimum, nor even a good approximation to it. The G_φ grammar is used as a tractable, pre-registered upper bound: a constrained search space that limits complexity inflation and ensures falsifiability, rather than a claim to universality.

This distinction is not a weakness of the approach; it is its epistemic honesty. The pre-registration protocol (Section 5) and the multiple-testing correction (Section 6) are the mechanisms by which we maintain scientific integrity despite the impossibility of true Kolmogorov minimisation.

14.4 Bayesian Equivalence and the Choice of Prior

14.4.1 4.1 The MDL / Bayes Correspondence

The two-part MDL coding cost $L(h) + L(D \mid h)$ admits a direct Bayesian interpretation. Minimising the total description length is equivalent to maximising the log-posterior under the prior $P(h) \propto 2^{-L(h)}$:

$$\arg\min_h \big[L(h) + L(D \mid h)\big] = \arg\max_h \big[\ln P(D \mid h) + \ln P(h)\big]$$

since $\ln P(h) = -L(h) \ln 2$ when the code is optimal. This equivalence is standard; see (Grünwald & Vitányi et al.) for a comprehensive treatment.

Definition 4.1 (φ-grammar prior). The probability distribution over hypotheses h implied by G_φ is:

$$P_{G_\varphi}(h) \propto 2^{-L_{G_\varphi}(h)}.$$

This is the Solomonoff-style prior restricted to G_φ. It assigns higher probability to expressions with shorter G_φ-derivations.

Epistemic commitment. Definition 4.1 constitutes a strong epistemic commitment: by adopting P_{G_φ} as the prior, the analyst asserts that φ-structured hypotheses (those expressible in G_φ with short descriptions) are a priori more probable than equally accurate hypotheses expressed in a different grammar. This commitment must be stated explicitly and pre-registered before data are examined, because the prior is not data-neutral.

14.4.2 4.2 Alternative Priors

Several competing priors are relevant comparators and SHOULD be reported alongside any G_φ result.

Speed Prior (Schmidhuber, 2002, DOI: 10.1007/3-540-45435-7_15):

$$P_S(h) \propto 2^{-l(h)} \cdot t(h)^{-1}$$

where $l(h)$ is the program length and $t(h)$ is the runtime of h on a reference machine. The Speed Prior is a computable approximation to the Solomonoff prior that penalises not just description length but computational cost. It favours hypotheses that are both short and fast, making it appropriate for physical theories that must be computationally efficient. Unlike P_{G_φ}, it does not privilege φ-structured expressions a priori.

Levin Search prior (Li, 2026, Entropy 28(4):434): a time-bounded variant of the Solomonoff prior with explicit resource accounting. Under this prior, the description length of h includes a term for the time required by Levin's universal search to find h, making the prior sensitive to the tractability of the hypothesis class.

AIC and BIC as frequentist baselines:

$$\mathrm{AIC}(h) = 2k - 2\ln \hat{L}(h), \qquad \mathrm{BIC}(h) = k \ln n - 2\ln \hat{L}(h)$$

where k is the number of free parameters in h, n is the sample size, and $\hat{L}(h)$ is the maximised likelihood. AIC and BIC are mandatory baseline comparators for any reported G_φ MDL gain. A φ-expression that achieves lower L_{G_φ} than a competing expression but worse AIC or BIC (after accounting for the cost of G_φ itself) has not demonstrated a genuine compression gain.

14.4.3 4.3 Prior Sensitivity and Reporting

Every result table in GOLDEN CHAIN v27 that reports a "low-MDL-cost expression" SHALL include:

Metric	Value
$L_{G_\varphi}(h)$	(bits, pre-registered grammar)
AIC comparator	(for the same target, best non-φ model)
BIC comparator	(same)
Speed Prior rank	(position of h in Speed Prior ordering, if computable)

Omitting these comparators is a pre-registered violation of the reporting protocol.

14.5 The Catalog42 Pre-Registration Protocol

14.5.1 5.1 Motivation

The φ-grammar G_φ can generate a very large number of candidate expressions for any given target observable. Without a binding pre-registration protocol, the analyst could — even without dishonest intent — examine many expressions, select the one with the lowest residual, and report it as a discovery. The Catalog42 protocol is designed to make this impossible by fixing all analytical choices before the test is run.

14.5.2 5.2 Mandatory Fields

Each Catalog42 entry is a timestamped, immutable record with the following eight required fields:

Field 1 — Hypothesis ID. A string of the form Catalog42-NNN where NNN is a zero-padded integer. The ID is derived from the SHA-256 hash of the concatenation of Fields 2–5; it is immutable once registered.

Field 2 — Closed-form expression. The expression h in full G_φ notation, with every operator and operand explicitly listed. Abbreviated or informal notation is not permitted. Example: $\varphi^3 \cdot L_5 \cdot F_4^{-1} + (-1) \cdot \varphi^{-1}$.

Field 3 — Pre-computed L_{G_φ}. The description length $L_{G_\varphi}(h)$ in bits, computed from the derivation tree of Field 2 under the pre-registered G_φ. This value MUST be computed before the residual against the target is known to the analyst.

Field 4 — Target observable. The CODATA constant or experimental quantity being modelled, including: - Reference value and uncertainty in the format $v \pm u$ (CODATA 2022 or later); - The specific CODATA dataset version used; - The dimensional units. Example: $\alpha^{-1} = 137.035999177 \pm 0.000000021$ (CODATA 2022, dimensionless; see Wikipedia Fine-structure constant).

Field 5 — Sanctioned random seeds. If any stochastic component is used in the search (e.g., stochastic grammar expansion, Monte Carlo estimation), all random seeds SHALL be listed. The following seeds are permanently forbidden: $\{42, 43, 44, 45\}$ (see Appendix B). Use of a forbidden seed voids the entry.

Field 6 — Stopping rule. The experiment terminates upon the first of: - (a) Reaching the pre-declared sample budget $n_{\max}$ (number of candidate expressions evaluated); or - (b) The falsification condition in Field 8 is triggered. No extension of $n_{\max}$ after commencement is permitted.

Field 7 — Effect-size threshold. The minimum residual reduction, in units of CODATA σ, that constitutes a "pass". The threshold SHALL be set to $\Delta_{\min} \geq 3\sigma_{\text{CODATA}}$ (see Section 7). Sub-σ residuals SHALL NOT be claimed as evidence of a genuine match; they are consistent with coincidence.

Field 8 — Falsification condition (numeric, before-the-fact). The entry is FALSIFIED if a non-φ alternative grammar $G'_{\varphi'}$ achieves strictly lower description length $L_{G'_{\varphi'}}(h')$ on the same data, provided that $G'_{\varphi'}$ was itself pre-registered before the residual of Field 2 was observed. The falsification condition SHALL include a numeric threshold, e.g.: "If there exists $h' \in G'_{\varphi'}$ with $L_{G'_{\varphi'}}(h') < L_{G_\varphi}(h) - \delta$ for pre-specified $\delta > 0$, then Catalog42-NNN is FALSIFIED."

14.5.3 5.3 Zenodo Timestamping

Every Catalog42 entry SHALL be deposited on Zenodo and assigned a DOI before the analyst runs the residual computation against the target observable. The Zenodo deposit timestamp is the binding pre-registration timestamp. Post-hoc deposits — even by one second — are inadmissible. The project-level Zenodo record (DOI: 10.5281/zenodo.19227877) serves as the parent record; individual Catalog42 entries are linked as versions or related records.

14.6 Multiple-Testing Correction

14.6.1 6.1 The False-Discovery Problem

The Catalog42 ledger is designed to grow. By the time GOLDEN CHAIN v27 reaches publication, it may contain hundreds of φ-monomial candidates across dozens of target observables. The false-discovery probability under naive per-entry significance testing at level α grows as $1 - (1 - \alpha)^m$ where m is the number of tests; for $m = 100$ and $\alpha = 0.05$, this exceeds 99%.

(Gelman & Loken, 2013) identify a related but subtler problem: even without explicit "fishing", the garden of forking paths — the implicit space of analytical choices made before data are examined — inflates the false-positive rate beyond the nominal level. Every choice in the G_φ specification (which operators to include, which cost weights to assign, which target observables to test) represents a fork in this garden. The pre-registration protocol (Section 5) reduces, but does not eliminate, this inflation.

14.6.2 6.2 Mandated Correction Procedure

Protocol 6.1 (BH-FDR correction). The Benjamini–Hochberg procedure (Benjamini & Hochberg, 1995) SHALL be applied across the full Catalog42 cohort at target FDR $q = 0.05$. Specifically:

1. Let $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}$ be the ordered p-values from all m Catalog42 entries tested in a given cohort.
2. Let $k = \max\{i : p_{(i)} \leq (i/m) \cdot q\}$.
3. Reject (declare "pass") only entries $1, \ldots, k$.
4. All reported "low-MDL-cost matches" SHALL cite their FDR-adjusted q-value alongside the raw p-value.

Protocol 6.2 (Cohort definition). The cohort is the entire Catalog42 ledger at the time of analysis, not a per-target or per-observable subset. Cherry-picking a subset to reduce m and thereby lower the BH threshold is a pre-registered violation.

14.6.3 6.3 Reporting Requirements

Every result that claims a "low-MDL-cost match" for a CODATA constant SHALL report all three of the following:

1. Cohort rank: the rank of the match within the full Catalog42 cohort when entries are sorted by residual magnitude.
2. FDR-adjusted q-value: from Protocol 6.1.

3. Effect size in CODATA σ units: $\varepsilon = |v_\varphi - v_{\text{CODATA}}|/\sigma_{\text{CODATA}}$, where v_φ is the value of the φ-expression and σ_{CODATA} is the CODATA uncertainty.

Omission of any of these three quantities is a pre-registered violation of the reporting protocol.

14.7 Power Analysis and Sample-Size Justification

14.7.1 7.1 General Framework

For each target observable, the following quantities SHALL be pre-registered in the corresponding Catalog42 entry (Field 7):

- σ_{CODATA}: the current measurement uncertainty from CODATA 2022 (or the most recent available version at the time of entry creation).
- $\Delta_{\min} = 3\sigma_{\text{CODATA}}$: the minimum detectable residual gain, expressed in absolute units. Any φ-expression whose residual exceeds $\Delta_{\min}$ is automatically REJECTED regardless of its MDL score.
- $n_{\max}$: the maximum number of candidate expressions evaluated, predeclared in Field 6.

Definition 7.1 (Automatic rejection criterion). Let v_φ be the real-number value of a φ-expression and $v_{\text{CODATA}} \pm \sigma_{\text{CODATA}}$ the target observable. The expression is automatically rejected if

$$|v_\varphi - v_{\text{CODATA}}| > 3\sigma_{\text{CODATA}}.$$

Automatic rejection applies regardless of the MDL score, the FDR-adjusted q-value, or any other metric.

14.7.2 7.2 Example: The Fine-Structure Constant α^{-1}

The CODATA 2022 value of the inverse fine-structure constant is (Wikipedia, Fine-structure constant):

$$\alpha^{-1} = 137.035999177 \pm 0.000000021.$$

Thus:

- $\sigma_{\text{CODATA}} \approx 2.1 \times 10^{-8}$ (absolute),
- $\Delta_{\min} = 3 \times 2.1 \times 10^{-8} \approx 6.3 \times 10^{-8}$.

Any φ-expression h with $|h - 137.035999177| > 6.3 \times 10^{-8}$ is automatically rejected. Note that Eddington's 1929 conjecture of $\alpha^{-1} = 137$ exactly (Wikipedia, Fine-structure constant) fails this criterion by a margin of approximately 3.5×10^{-5}, roughly 1700 standard deviations. The historical lesson is clear: approximate numerical coincidences at coarse precision are not evidence of a structural relationship.

Note on the broader physics literature. The scientific consensus, as reviewed in (Wikipedia, Fine-structure constant), is that "no numerological explanation has ever been accepted by the physics community." The φ-grammar framework is not a numerological claim; it is a pre-registered, falsifiable compression test with explicit error control. The distinction matters.

14.7.3 7.3 Replication Standard

A result that survives automatic rejection (Definition 7.1) and FDR correction (Protocol 6.1) is designated a provenance evidence candidate, not a confirmed result. Promotion from "provenance evidence" to "robust witness" requires:

- Independent replication by $N = 3$ distinct silicon-design teams, each reproducing the anchor witness from open RTL source (see the silicon-anchor proof chain (φ^2+φ^{-2}=3 $\Longrightarrow$ L$_2$=3 $\Longrightarrow$ dot4 = 0x47C0, partial Coq mechanisation in t27/proofs/trinity/CorePhi.v + ExactIdentities.v; see fm-13 §4 for step-status), anchor byte 0x47C0 at {uio_out, uo_out} on reset) using independently compiled bitstreams.
- Each replicating team SHALL pre-register its own Catalog42-equivalent entry on Zenodo before running the test.
- Replication failures SHALL be reported in full.

14.8 Blind Analysis Protocol

14.8.1 8.1 Rationale

Blind analysis is standard practice in particle physics and clinical trials because the experimenter's knowledge of the result, even when honest, can influence downstream choices in ways that inflate the false-positive rate. The two-stage protocol below is adapted from particle-physics convention.

14.8.2 8.2 Stage A: Development (Blind to Target)

During Stage A, the analyst constructs and refines G_φ, assigns cost weights, and populates the Catalog42 ledger using training constants — closed-form mathematical constants and CODATA quantities that are not the designated target observable.

Permissible Stage-A activities: - Grammar design and cost-weight assignment. - Testing expressions against training constants (e.g., π, e, $\ln 2$, speed of light c, Planck constant h, etc.). - Revising the grammar in response to training-set performance. - Pre-registering Catalog42 entries for the designated target.

Prohibited Stage-A activities: - Computing the residual of any Catalog42 entry against the designated target observable. - Examining any dataset or measurement that includes the designated target.

The designated target (α^{-1} in the running example) is hidden from the analyst during Stage A. If the target value is publicly known (as α^{-1} is), the analyst SHALL designate a proxy blinder (a co-author not involved in grammar design) who holds the target value and computes residuals only in Stage B.

14.8.3 8.3 Stage B: Unblinding

Stage B commences only after the Catalog42 ledger is timestamped on Zenodo (Section 5.3). The sequence is:

1. Zenodo DOI is obtained and recorded.
2. The proxy blinder (or automated verification script) computes residuals for all pre-registered entries against the target.
3. Results are reported in full, including all rejections.
4. Any post-hoc grammar modification — adding operators, adjusting cost weights, adding new Catalog42 entries — voids the pre-registration for the target in question.

Stage B results are final. There is no provision for "Stage C" re-analysis of the same target under the same grammar version.

14.9 What This Framework Does NOT Claim

The following negative claims are explicit and binding. They are enumerated here to prevent misrepresentation in derived communications, press releases, or secondary literature.

1. We do NOT claim that φ "explains" physical constants in any causal-mechanical sense. The fine-structure constant α arises from the quantum electrodynamic coupling between charged particles and photons; its value is determined by experiment, not by the structure of the golden ratio. A low-MDL-cost φ-expression is a compressed description, not a derivation.
2. We do NOT claim that G_φ recovers the true Kolmogorov-minimal description of any physical constant. As established in Sections 2 and 3, $L_{G_\varphi}(h) \geq K(h)$ always, and the gap can be arbitrarily large. The grammar G_φ is a pre-registered upper bound, not a universal compressor.
3. We do NOT claim statistical significance without FDR correction. Raw p-values from individual Catalog42 entries are uninformative without the

BH correction applied across the full cohort (Protocol 6.1). Any report that cites a p-value without the FDR-adjusted q-value is incomplete.

4. We do NOT claim that residuals below CODATA σ are physical evidence. A φ-expression that matches α^{-1} to within $1\sigma_{\text{CODATA}}$ is consistent with the hypothesis that the match is a coincidence arising from the density of φ-monomials in the neighbourhood of the CODATA value. Sub-σ agreement is a necessary but not sufficient condition for any claim.
5. We do NOT claim that the silicon anchor 0x47C0 "proves" anything beyond a pre-registered design choice committed at a specific git hash. The anchor byte is a witness to the implementation of the φ-grammar in RTL; it is not evidence that φ governs physical law. "Anchor byte witness" (the correct language) is strictly weaker than "proof of φ in nature" (forbidden language).
6. We do NOT claim that the asymptotic guarantee of Theorem 3.1 (Shaw et al. 2025) applies to our finite-grammar, finite-data setting. The theorem applies in the limit of unbounded resources; for G_φ with bounded arity $r_{\max}$ and finite production set, the theorem provides only asymptotic motivation.
7. We do NOT claim that the Speed Prior (Schmidhuber 2002) or the Levin Search prior (Li 2026) agree with P_{G_φ}. They are distinct priors with distinct philosophical commitments. The φ-grammar prior is one specific choice, and the results are prior-dependent.

14.10 Concrete Checklist for Future Catalog42 Entries

Every new Catalog42 entry MUST satisfy all ten items on the following checklist before it may appear in any GOLDEN CHAIN publication, preprint, or public communication. The checklist SHALL be completed by the submitting author and countersigned by at least one co-author.

#	Item	Required Answer
1	Has a Zenodo DOI been obtained for this entry before computing the target residual?	Yes
2	Is the grammar G_φ version (production set, cost weights, arity bound) fully specified and identical to the pre-registered version?	Yes

#	Item	Required Answer
3	Has FDR correction (BH at $q = 0.05$) been applied across the full Catalog42 cohort, not just this entry?	Yes
4	Has an alternative-grammar comparator $G'_{\varphi'}$ been pre-registered and its description length $L_{G'_{\varphi'}}$ reported for the same target?	Yes
5	Are AIC and BIC reported alongside L_{G_φ} for the same target observable?	Yes
6	Is the effect size reported in units of CODATA σ (not relative units, not absolute units alone)?	Yes
7	Has the blind-analysis protocol (Section 8) been followed, with Stage A and Stage B clearly separated and attested?	Yes
8	Are the forbidden seeds $\{42, 43, 44, 45\}$ absent from the sanctioned seeds list (Field 5)?	Yes
9	Is the stopping rule (Field 6) respected — i.e., $n_{\max}$ was not extended after commencement?	Yes
10	Is the entry's result reported in full, including all entries that failed automatic rejection (Definition 7.1), not just passing entries?	Yes

A "No" on any item renders the entry inadmissible for publication. The checklist SHALL be archived alongside the Catalog42 entry on Zenodo.

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P1 §1. Introduction

Open conjecture

Opening triptych — Seed Identity / Vesica Axis / Sprout. The trilogy opens from one identity into three papers.

1.1 The Open Problem of Dimensionless Constants The numerical values of the fundamental dimensionless constants of nature — the inverse fine-structure constant α^{-1} = 137.035 999 177(21) (NIST CODATA 2022), the strong coupling alphas(mZ) = 0.1180(9) (PDG 2024), the electroweak mixing angle sin2theta(MZ) = 0.23129(4) in the MS scheme (PDG 2024), and the full array of mixing angles, mass ratios, and cosmological parameters — remain empirical inputs to the Standard Model. No known symmetry principle or dynamical mechanism within quantum field theory determines their specific values; each must be measured and inserted by hand. The question of whether these values carry any deeper algebraic or structural character is thus not closed by the Standard Model itself, and constitutes a legitimate open question at the intersection of mathematical physics and information theory (PDG 2024).

1.2 A Precisely Bounded Reformulation A sharply bounded, operationally well-posed version of this question avoids dynamical claims entirely: Do the observed numerical values of dimensionless constants exhibit statistically non-random compressibility under a restricted symbolic language? This is a

question about data compression, not about physics. It belongs to the domain of symbolic regression, algorithmic information theory, and statistical model selection under structured hypothesis classes — the same theoretical territory inhabited by the Minimum Description Length principle (Barron, Rissanen & Yu, IEEE Trans. Inf. Theory, 1998; Grünwald, MIT Press, 2007) and Chaitin's Algorithmic Information Theory (Chaitin, Cambridge University Press, 1987).

This paper formalizes the question within a precisely defined statistical framework. A symbolic grammar Gφ is generated by the basis {φ, pi, e, 3} with integer exponents bounded by an ℓ_1 complexity constraint C. The grammar induces a finite hypothesis class HC of candidate expressions. Physical constants form a frozen target dataset T, and the analysis asks whether the projection of T onto HC yields lower residuals than those expected from statistically appropriate null models.

1.3 What This Paper Does Not Claim The following statements are explicitly disavowed and must be kept in view throughout:

- No physical derivation. The grammar Gφ does not derive physical constants from symmetry principles, dynamics, or first principles. The encoding map T → S(C) is a constrained quantization onto a discrete algebraic manifold, not a generative physical law.
- No ontological identification. The φ-basis is not claimed to carry physical primacy or to reflect any symmetry of nature. It is one instance of an algebraically complete basis with tractable combinatorial geometry.
- No unique basis. The analysis would be qualitatively analogous for any other algebraically complete basis. The specific choice of {φ, pi, e, 3} is motivated by algebraic tractability and the Lucas ring Z[φ], as discussed in Sections 2.4 and 3.3.
- No claim from stale proof counts. The PhD monograph program has ongoing formal verification work; no Coq-verified claims are made in this paper. Proof obligations are cited with their current honest-admission status.

1.4 Primary Objectives The primary objectives of this paper are:

1. To characterize the density of low-error approximations within HC as a function of complexity level C and tolerance ε.
2. To assess whether this density exceeds the null expectation under three controlled baseline models, with full multiple-testing correction.

3. To compute a well-defined MDL compression score that accounts for the full description cost of the symbolic model class.
4. To introduce the Pellis hierarchical expansion and evaluate its contribution to compressibility under the MDL framework.
5. To specify falsifiable predictions and out-of-sample tests that can in principle refute the compressibility hypothesis.
6. To provide explicit negative controls, a falsification ledger, and a reviewer-risk assessment.

1.5 Paper Organization Section 2 reviews related literature and positions the work. Section 3 defines the symbolic grammar and its combinatorial and algebraic structure in detail. Section 4 specifies the target dataset and the pre-registration freeze protocol. Section 5 defines the null models. Section 6 addresses multiple testing corrections. Section 7 develops the MDL and Bayesian framework. Section 8 reports representative empirical results. Section 9 introduces the Pellis hierarchical expansion. Section 10 defines out-of-sample tests, a falsification ledger, and negative controls. Section 11 provides a reviewer-risk assessment. Section 12 discusses the relationship to the Trinity S^3AI / Flos Aureus PhD monograph, including the Stergios collaboration note. Section 13 discusses limitations. Section 14 concludes. Appendices provide the reproducibility protocol checklist, algebraic properties of the basis, phase-transition structure, and a mapping from the PhD monograph to Paper 1.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. “The Fine-Structure Constant and the Golden Ratio.” viXra 2110.0117 v4.; Sherbon, M. A. “Fundamental Nature of the Fine-Structure Constant.” HAL hal-02341850.

P1 §2. Background and Related Work

Verified

Background triptych — History Scroll / MDL Scales / Trinity Anchor.

2.1 Symbolic Approaches to Physical Constants: Historical Context Attempts to express dimensionless physical constants in closed algebraic form have a long and largely undisciplined history. Early efforts concentrated on α^{-1} approx 137, with various proposals involving integers and simple arithmetic — all post-hoc and statistically uncontrolled (PDG 2024 review articles). The combinatorial proliferation of candidate expressions at increasing complexity was universally ignored, making significance claims meaningless.

More systematic approaches employing continued fractions, Ramanujan-type hypergeometric identities, and modular forms have examined the representability of specific mathematical constants such as pi and e. These efforts differ from the present work in that they target mathematical identities rather than empirical physical constants, and they do not formulate the problem as a statistical model-selection problem over a structured hypothesis class with explicit null models. The present paper belongs to a different methodological tradition.

2.2 Symbolic Regression in Physics Symbolic regression — the automated discovery of mathematical expressions fitting numerical data — has been developed extensively in the machine learning community, with methods ranging from genetic programming to neural-guided search. The AI Feynman algorithm of Udrescu and Tegmark (Science Advances, 2020) demonstrated that physics-inspired techniques including dimensional analysis, separability detection, and brute-force enumeration can recover 100 equations from the Feynman Lectures on Physics. More recent work in physics-informed symbolic regression (Angelis et al., Archives of Computational Methods in Engineering, 2023) has extended these methods to broader scientific domains.

The present work differs from all these approaches in a fundamental respect: the target is not a functional relationship between variables but a set of isolated numerical values of dimensionless constants, and the search is not over a continuous parameter space but over a discretely bounded combinatorial grammar. This distinction is crucial for the statistical methodology: the absence of variables means that null models can be precisely specified and that the effective trial count is the cardinality of the discrete hypothesis class.

2.3 Minimum Description Length and Bayesian Model Comparison The Minimum Description Length principle, originating in Rissanen's 1978 paper Modeling by the shortest data description (Automatica, 1978) and systematized in Barron, Rissanen & Yu (1998), provides a rigorous basis for model comparison that penalizes complexity in terms of code length. The core insight is that any regularity in data can be exploited to compress it, and the best model is the one achieving the shortest description of data and model combined. Grünwald's comprehensive treatment (MIT Press, 2007) formalizes the connections between crude two-part codes, normalized maximum likelihood, Bayesian marginal likelihoods, and prequential coding, establishing asymptotic equivalences between these approaches.

Bayesian model comparison via marginal likelihoods provides an equivalent framework when proper priors are specified; the resulting Bayes factor quantifies the relative evidence in favor of structured over unstructured models. Both frameworks are deployed in Section 7 with explicit penalization for the size and structure of the symbolic hypothesis class.

The connection to Kolmogorov-Chaitin algorithmic complexity (Chaitin, Cambridge University Press, 1987) is conceptual: the MDL compression score DeltaMDL provides a computable, statistical lower bound on the potential reduction in Kolmogorov complexity achievable by the structured grammar.

2.4 The Algebraic Anchor: Lucas Ring and Trinity Identity

The choice of φ as a basis generator is anchored in the algebraic identity

$\varphi_2 + \varphi\text{-}2 = 3.$

which holds exactly because φ satisfies the minimal polynomial $\varphi_2 = \varphi + 1$. This identity — the Trinity identity of the broader research program — establishes that the squared golden ratio and its reciprocal square sum to the integer 3, implying a ternary cardinality relation within the Lucas ring Z[φ]. In the context of the present paper, its significance is limited to the following algebraic observation: Z[φ] is closed under multiplication, ensuring that integer-exponent products of φ remain in a well-characterized algebraic structure. This algebraic closure simplifies the linear independence argument for the log-lattice (Section 3.3) and guarantees that the hypothesis class HC is non-degenerate. The identity is provenance from the Trinity S^3AI PhD monograph (PhD: Ch.3 Trinity Identity); no further physical significance is claimed for it in this paper.

2.5 Positioning The present work occupies a specific methodological niche: rigorous combinatorial and information-theoretic evaluation of whether a finite, pre-specified symbolic language achieves statistically nonrandom compressibility over a frozen, pre-committed dataset of dimensionless constants. The novelty lies in (i) the explicit treatment of the search as a multiple-testing problem, (ii) the use of MDL with a complexity-penalized grammar model, (iii) the Pellis hierarchical expansion as a two-tier refinement mechanism, (iv) the provision of three complementary null models and five explicit falsification criteria, and (v) the transparent provenance connection to the PhD monograph, with clear delineation of what the monograph contributes and what it does not validate.

2.6 Reviewer-Hardening Protocol Imported from the PhD Monograph The Flos Aureus PhD monograph contributes a procedural discipline that is useful for this paper but does not count as evidence for the paper's empirical claims. Four rules are adopted:

1. No silent overclaim. A symbolic fit is never promoted to a physical derivation unless a mechanism is supplied.
2. No stale constants. Public constants are locked to the cited PDG/CODATA release, and any updated value is treated as an out-of-sample update rather than retrofitted into the original claim.
3. No hidden search space. Every reported fit is paired with an explicit hypothesis-class cardinality Neff=|S(C)|, so the look-elsewhere correction is visible.

4. No proof-status inflation. Coq-backed algebraic identities may be cited as formal provenance only at their actual status; Admitted material is treated as a proof sketch, not as a theorem.

This protocol is intentionally conservative. Its purpose is to make the article easier to criticize correctly and harder to misread as numerology, proof-by-catalogue, or a claim that the Standard Model constants have been derived from φ.

2.7 Claim-Tier Tags Imported from the PhD Defense Protocol To prevent category errors, every claim in the article is interpreted through one of four tiers:

- Tier A — algebraic identity: exact symbolic statements such as φ_2+φ-2=3. These are mathematical claims.
- Tier B — symbolic compressibility: low-description-length approximations under a fixed grammar and a fixed dataset. These are statistical claims and require null models.
- Tier C — mechanism candidate: a proposed bridge from algebra to physics. These are conjectural unless embedded in an accepted physical theory.
- Tier-D — historical or interpretive context: conceptual framing, including golden-section history and visual architecture. These passages provide orientation only and are not evidence for Tier B or Tier C.

This tier system is deliberately reviewer-facing. If a sentence can be read in more than one tier, the lower evidentiary tier controls the interpretation.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. "The Fine-Structure Constant and the Golden Ratio." viXra 2110.0117 v4.; Sherbon, M. A. "Fundamental Nature of the Fine-Structure Constant." HAL hal-02341850.

P1 §3. Symbolic Grammar and Complexity Metric

Open conjecture

Trinity grammar triptych — basis abacus, open (p,m) lattice, five-spoke generator wheel.

3.1 Algebraic Grammar Definition The symbolic grammar Gφ is defined over the alphabet

A = {φ, pi, e, 3, Z},

where φ = (1+5)/2 is the golden ratio, pi approx 3.14159 is the circle constant, e approx 2.71828 is Euler's number, and Z encodes integer-valued exponents and prefactors. The grammar generates expressions of the form

F(theta) = n · 3k · φp · pim · eq, 1

where the parameter vector is

theta := (n, k, p, m, q) $\in$ {1,...,9} $\times$ Z4.

The admissible symbolic expression set at complexity level C $\in$ N is defined as

S(C) := { F(theta) | theta $\in$ {1,...,9} $\times$ Z4, |theta|1 $\leq$ C }, 2

where the ℓ_1-norm complexity constraint is

The prefactor n $\in$ {1,...,9} is treated as a single-digit integer whose absolute value is excluded from the complexity count; this reflects the convention that small integer prefactors represent negligible description cost relative to exponent choices. All F(theta) are positive real numbers by construction.

3.2 Complexity Metric and Its Justification The ℓ_1-norm complexity |theta|1 acts as a bounded description-length regularizer. Its adoption is motivated by three considerations:

- Kolmogorov analogy. The number of bits required to specify an integer exponent tuple scales as log2 |theta|1 per entry, making the ℓ_1 norm a conservative linear proxy for code length (Barron, Rissanen & Yu, 1998; Chaitin, 1987).
- Algebraic naturality. Expressions with small ℓ_1 norm require the fewest multiplicative factors and are algebraically minimal in a precise sense.
- Tractability. The induced hypothesis class HC has finite, analytically bounded cardinality (Section 3.4), enabling exhaustive enumeration for C $\leq$ 15.

Sensitivity to this metric choice is discussed in Section 13.

3.3 Logarithmic Embedding Structure Under the logarithmic map log : $R^+ \to$ R, expression (1) becomes

log F(theta) = log n + k log 3 + p log φ + m log pi + q log e. 3

Defining the basis vector v := (log 3, log φ, log pi, 1) $\in$ R4, the image of S(C) in log-space is a bounded section of the rank-4 additive lattice

log S(C) $\subset$ { log n + (k,p,m,q)$\cdot$v | (k,p,m,q)$\in$Z4, |(k,p,m,q)|1 $\leq$ C }. 4

Algebraic closure of the Lucas ring Z[φ] (PhD: Ch.3 Trinity Identity) ensures stability under the arithmetic operations of the grammar. The non-degeneracy of the log-lattice is treated as a working independence assumption on {1, log 3, log φ, log pi} over Q, not as a theorem of the present paper. If a non-trivial rational relation among these generators were found, the hypothesis class would be quotient by that relation and all cardinalities and MDL penalties would need to be recomputed.

3.4 Cardinality and Combinatorial Complexity

Search-space triptych — bounded exponent grid, |S(C)| growth, Bayesian prior.

The cardinality of S(C) governs both computational cost and multiple-testing severity. Standard lattice-point counting in the ℓ_1-ball in Z4 gives

reflecting the volume of the 4-dimensional integer ℓ_1-ball scaled by 9 prefactor choices. This quartic growth is significant: the hypothesis class grows rapidly but remains computationally enumerable for moderate complexity bounds. At C = 6, one finds |S(6)| = 11{,}601 expressions; at C = 10, |S(10)| = 75{,}249; at C = 15, |S(15)| = 351{,}369. All of these are amenable to exhaustive enumeration on modern hardware. The effective number of independent tests in any significance assessment is Neff = |S(C)|; this sets the scale for all multiple-testing corrections in Section 6.

The following table summarizes exact cardinalities at selected complexity levels:

Note: exact counts include the 9 prefactor choices and the full signed ℓ_1-ball volume formula.

3.5 Representation Bias and the Induced Grammar Prior The grammar Gφ induces a non-uniform prior measure over R^+. The empirical counting measure

PGφ(x) propto sumtheta ∈ S(C) delta(x - F(theta)), 6

smoothed with an exponential kernel of bandwidth sigma > 0, yields

Psigma(x) = sumtheta ∈ S(C) exp(-|x - F(theta)|/sigma). 7

This distribution is concentrated near dense algebraic combinations of the basis elements and is generically non-uniform over R^+. The non-uniformity of Psigma must be explicitly accounted for in all null-model constructions (Section 5). Failure to account for this non-uniformity would systematically inflate apparent compressibility.

3.6 Approximation Quality Functional Given a target constant X ∈ R^+, the approximation quality at complexity C is measured by the minimum normalized absolute deviation:

dC(X) := mintheta ∈ S(C) (|F(theta) - X|)/(|X|). 8

The function C ↦ dC(X) is monotonically non-increasing in C and converges to zero for sufficiently large C whenever X ∈ R^+. The statistical question concerns the rate of this convergence relative to null expectations under properly specified baseline models.

The same φ-grammar G_φ and minimum-description-length criterion developed here have a direct hardware realisation: the GoldenFloat numeric-format family fixes each format's exponent:mantissa split by the rule E/M → 1/φ, with the 16-bit member GF16 used in the production pipeline. See the chapter Numeric Formats — A Short History and the GoldenFloat Family, which derives the format ladder from this same $φ^2 + φ^{-2} = 3$ identity.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. "The Fine-Structure Constant and the Golden Ratio." viXra 2110.0117 v4.; Sherbon, M. A. "Fundamental Nature of the Fine-Structure Constant." HAL hal-02341850.

P1 §4. Target Dataset and Pre-Registration Freeze Protocol

Open conjecture

Validation triptych — Target / Experimental Band / Pre-Registration Lock.

4.1 Dataset Selection Criteria The target dataset T consists of dimensionless physical constants drawn from standard, publicly available experimental compilations. A dimensionless constant is defined as a ratio of physical quantities carrying the same dimension, so that its numerical value is independent of any choice of units. Only dimensionless quantities are included; dimensional constants (e.g., Newton's gravitational constant G in SI units) depend on an arbitrary unit choice and are excluded.

Candidate constants for inclusion must satisfy all of the following criteria: 1. Dimensionless (ratio of like-dimensioned quantities). 2. Drawn from the Standard Model or cosmological concordance model. 3. Tabulated in the PDG Review of Particle Physics 2024 or the CODATA 2022 recommended values. 4. Relative experimental uncertainty better than 10-2.

5. Not selected on the basis of proximity to any symbolic expression.

4.2 Candidate Constants The following candidate constants meet the selection criteria. All values are taken from the PDG 2024 Review of Particle Physics and the CODATA 2022 recommended values; cosmological parameters follow the Planck 2018 cosmological parameter results as adopted by the PDG.

Constants with relative uncertainty > 10^{-2} are retained as secondary targets but are not included in the primary compressibility test; they appear in Section 10 as part of the negative control analysis.

4.3 Pre-Registration Freeze Protocol To prevent post-hoc selection bias — specifically, the practice of including in T only those constants for which the grammar achieves small residuals — the dataset must be frozen prior to any symbolic search. The following protocol is adopted and constitutes a formal pre-registration commitment.

Protocol P1 (Dataset Freeze):

1. The set T is defined by listing all dimensionless parameters satisfying the criteria in Section 4.1, from the PDG Review of Particle Physics 2024 and CODATA 2022, with PDG edition fixed at the 2024 release (Phys. Rev. D 110, 030001).
2. This list is committed to a version-controlled repository with a cryptographic hash, before any search computation is performed.
3. No constant may be added to T after any symbolic search has been performed, except as part of a prospectively defined out-of-sample test (Section 10).

4. If a constant's experimental value is updated after the dataset freeze, the updated value is used only in the out-of-sample falsification analysis (Section 10.2), not in the primary fit analysis.
5. All data files, version hashes, and Zenodo provenance records are deposited as documented in the PhD monograph's Data Availability appendix (PhD: App.G Data Availability) and the CLARA Provenance Audit 2026-05-12.

Rationale. This protocol maps onto standard pre-registration norms in empirical science. Post-hoc data selection is among the most potent sources of false discovery in symbolic search studies; any violation of Protocol P1 invalidates the statistical interpretation of reported residuals.

4.4 Logarithmic Distribution of the Dataset In log-space, the target constants {log Xi} span approximately four decades (from log(0.022) to log(137)) and are irregularly distributed. No claim of log-uniformity is made. The empirical log-space distribution is an explicit input to the null-model construction (Section 5.2), and any compressibility claim must be evaluated relative to this empirical distribution.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. "The Fine-Structure Constant and the Golden Ratio." viXra 2110.0117 v4.; Sherbon, M. A. "Fundamental Nature of the Fine-Structure Constant." HAL hal-02341850.

P1 §5. Null Models

Empirical fit

Null-models triptych — log-uniform / permutation / randomized exponent grammar.

5.1 Rationale for Explicit Null Models Without a properly specified null distribution, observed residuals dC(Xi) are statistically uninterpretable. The grammar Gφ is sufficiently expressive that small residuals are guaranteed to exist for large C, regardless of whether the target values carry algebraic structure. The null models defined here answer: how small would residuals be if the target values had no structural relation to the grammar?

Three complementary null models are specified.

5.2 Null Model M0: Log-Uniform Random Baseline Under M0, the target values {Xi}i=1N are replaced by independent samples from a log-uniform distribution on the empirical range of T:

log Xi(0) $\sim$ Uniform[log Xmin, log Xmax], 9

where [Xmin, Xmax] matches the empirical range. The coverage functional under M0,

ρ_0(C, ε) := EX(0) $\sim$ M0[(1)/(N)$\sum_{i=1}$N 1[dC(Xi(0)) < ε]], 10 is computed by Monte Carlo sampling over B $\geq$ 10{,}000 independent realizations of T(0). The null hypothesis is rho(C, ε) = ρ_0(C, ε); the alternative is rho(C, ε) > ρ_0(C, ε).

5.3 Null Model Mperm: Permutation-Based Baseline The permutation null model Mperm preserves the empirical distribution of T while destroying any structural relation between individual constants and the grammar.

Under Mperm, the target set is replaced by a random permutation of its elements combined with a random sign-flip in log-space:

Tpi := {exp(± log Xpi(i))}i=1N, pi uniform on SN. 11

The coverage functional under Mperm is

rhoperm(C, ε) := Epi ∼ SN[(1)/(N)$\sum_{i=1}$N 1[dC(Xpi(i)) < ε]]. 12

Because Mperm samples from the same empirical distribution as T, it controls for distributional effects without assuming log-uniformity, making it more conservative than M0.

5.4 Null Model Mrand: Randomized Exponent Grammar A third control uses a randomized grammar Mrand in which exponent vectors are drawn uniformly from the same bounded integer domain as S(C), but with basis vectors replaced by independent random positive reals:

k, p, m, q ∼ Unif({-C,...,C}4), n ∼ Unif({1,...,9}), log vrand ∼ Unif[0,1]4. 13

This generates an ensemble Srand(C) with the same cardinality as S(C) but without the specific algebraic structure of the φ-basis. Comparing coverage under Gφ to coverage under Mrand isolates the contribution of the specific algebraic structure.

5.5 Summary of Null Model Hierarchy The three null models form a hierarchy of increasing conservatism:

M0 ⊂ Mperm ⊂ Mrand.

A compressibility claim has statistical content only if it survives all three comparisons simultaneously.

5.6 Negative Controls: Randomly Rotated Constants In addition to the three null models, a battery of negative controls is constructed using randomly rotated versions of T:

Tneg(j) = {Xi · Uj}i=1N, Uj ∼ Log-Uniform(0.5, 2.0), 13.a

for j = 1, ..., 50 independent realizations. Coverage under Gφ applied to these rotated datasets should not exceed the null expectation, because the rotation destroys any algebraic alignment. Failure of this check would indicate that the grammar is simply dense in the relevant range, not genuinely selective.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. “The Fine-Structure Constant and the Golden Ratio.” viXra 2110.0117 v4.; Sherbon, M. A. “Fundamental Nature of the Fine-Structure Constant.” HAL hal-02341850.

P1 §6. Multiple Testing and Correction Procedure

Empirical fit

Look-elsewhere triptych — N candidate folios / null residual histogram / Bonferroni xN penalty.

6.1 The Multiple Testing Problem The symbolic search over S(C) constitutes a large-scale multiple comparison. For each target constant $Xi \in T$, the minimum residual dC(Xi) is the minimum over |S(C)| comparisons. The minimum of M independent uniform [0,1] random variables has expectation 1/(M+1), decreasing as M-1. Since $|S(C)| \sim O(C4)$, residuals as small as O(C-4) are expected even under purely random target values.

This is the central obstacle for symbolic constant studies: the combinatorial proliferation of candidates makes it statistically almost certain that small residuals will be found, even for randomly generated target sets. Any significance claim that does not explicitly account for this proliferation is methodologically void.

6.2 Effective Trial Count For a grammar of complexity C, the effective number of statistical trials for each target constant is

Neff(C) := |S(C)| approx (3 C4)/(2). 14

A naively computed p-value for a single match with residual ε must be inflated by the Bonferroni factor:

pcorrected = min(1, Neff(C) · praw). 15

At C = 10, Neff approx 15{,}000, so a raw p-value of 10-3 yields a corrected p-value of approximately 15, meaning the match is completely non-significant. This illustrates why uncorrected coincidence counting is scientifically insufficient.

6.3 Family-Wise Error Rate Control

For joint assessment of coverage across all N = |T| constants simultaneously, the total number of comparisons is N · Neff(C). Bonferroni correction at family-wise error rate (FWER) alphaFWER = 0.05 requires

alphaindividual = (0.05)/(N · Neff(C)). 16

For N = 10 constants and C = 10, this gives alphaindividual approx 3.3×10^{-7}. No claim of individual constant significance is made unless this threshold is met.

6.4 False Discovery Rate Control For the weaker goal of identifying a fraction of constants with genuine compressibility, the Benjamini–Hochberg (BH) procedure is applied to the ranked p-values at the dataset level. The BH procedure controls the expected fraction of false discoveries among rejections at a specified false discovery rate (FDR). It is appropriate when a non-trivial proportion of the targets is expected to exhibit compressibility under the alternative hypothesis.

6.5 Permutation-Based Calibration All reported p-values are calibrated against the empirical null distribution derived from $B \geq 10\{,\}000$ permutation resamples under Mperm. A result is reported as significant only if its corrected p-value is below 0.05 under both analytic Bonferroni correction and permutation calibration. This two-layer requirement embodies the honest-admission ethos of the PhD monograph's Limitations chapter (PhD: Ch.52 Limitations).

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. "The Fine-Structure Constant and the Golden Ratio." viXra 2110.0117 v4.; Sherbon, M. A. "Fundamental Nature of the Fine-Structure Constant." HAL hal-02341850.

P1 §7. MDL and Bayesian Evidence Framework

Empirical fit

MDL / Bayes triptych — Two-Part Code / Bayes Factor balance / Fisher geometry.

7.1 Model Comparison Setup

The compressibility question is formalized as a Bayesian model comparison between:

- M0: structureless log-uniform baseline model (null).
- M1: structured symbolic grammar model Gφ with hypothesis class S(C).

Both models are evaluated on the frozen target dataset T = {Xi}i=1N. The MDL framework (Barron, Rissanen & Yu, 1998; Grünwald, 2007) provides the primary decision criterion; Bayesian evidence provides a complementary and asymptotically equivalent measure.

7.2 Bayesian Evidence The Bayesian evidence (marginal likelihood) for each model is

P(T mid M) = ∫ P(T mid theta, M) P(theta mid M) dtheta. 17

For the discrete grammar model M1, a Gibbs-type likelihood kernel with inverse temperature beta > 0 is adopted:

P(T mid M1) = prodi=1N (sumF ∈ S(C) exp[-beta d(F, Xi)])/(|S(C)|), 18

where d(F, Xi) = |F - Xi|/|Xi| is the normalized approximation error. For the null model M0, the log-uniform prior gives

P(Xi mid M0) = (1)/(log(Xmax / Xmin)). 19

7.3 Bayes Factor and Its Calibration The Bayes factor comparing M1 to M0 is

B := (P(T mid M1))/(P(T mid M0)). 20

Standard calibration of log10 B (Grünwald, 2007):

Crucially, B as defined in (20) does not penalize for the size of S(C). The MDL framework (Section 7.4) introduces the necessary penalization, without which the Bayes factor provides a misleadingly optimistic assessment.

7.4 Minimum Description Length Framework Following Barron, Rissanen & Yu (1998) and Grünwald (2007), the total description length of model M applied to data T is

MDL(M) = L(M) + L(T mid M), 21

where L(M) is the code length for the model (complexity cost) and L(T mid M) is the conditional encoding cost of the data given the model (fit cost).

Model complexity term. For M1, the model description must specify: the basis A, the complexity bound C, and the selected expression F^*(theta). The code length is

L(M1) approx log2 |S(C)| + log2 C. 22

Data fit term. Under the Gibbs likelihood kernel,

L(T mid M1) = -$\sum_{i=1}$N log2(sumF ∈ S(C) exp[-beta d(F, Xi)]) + N log2 |S(C)|. 23

Null model description length. Under M0,

L(T mid M0) = N log2(log(Xmax / Xmin)), 24

with negligible model complexity L(M0) approx 0 (no free parameters).

Compression gain. The MDL compression advantage of the structured grammar is

DeltaMDL := MDL(M0) - MDL(M1). 25

A positive DeltaMDL indicates a coding advantage net of description cost. A negative or zero DeltaMDL provides no evidence of compressibility.

7.5 Sensitivity to the Likelihood Temperature The parameter beta controls the sharpness of the approximation match. Results are reported for a range of beta ∈ {10, 100, 1000} in relative-error units, and robustness of DeltaMDL > 0 is verified across this range. A compression advantage claim requires DeltaMDL > 0 for all beta values in the range and all three null-model comparisons simultaneously.

7.6 Information-Geometric Interpretation From an information-geometry perspective, the symbolic grammar induces a non-Euclidean statistical manifold over R^+, with geodesic distances governed by the logarithmic embedding and the discrete lattice anisotropy of S(C). The observed compression advantage has the heuristic interpretation

KGφ(X) ≤ Kunstructured(X) - DeltaMDL 26

for well-approximated constants X, where K denotes Kolmogorov complexity in the sense of Chaitin (1987). The MDL quantity DeltaMDL provides a computable lower bound on the potential compression improvement; the inequality (26) is heuristic and not a theorem.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. “The Fine-Structure Constant and

the Golden Ratio." viXra 2110.0117 v4.; Sherbon, M. A. "Fundamental Nature of the Fine-Structure Constant." HAL hal-02341850.

P1 §8. Representative Empirical Results

Empirical fit

Claim status. The numerical agreements reported here are empirical fits at the stated relative tolerance, not Verified consequences of a closed-form derivation; their epistemic standing follows the rubric in P0 fm-04 — Reconciliation Note.

Empirics triptych — best-fit residual scatter / coverage(C) S-curve / alpha-phi on the number-line.

22.1 Best-Fit Representation Functional

For a target constant X $\in$ T, the best-fit symbolic representative at complexity C is

F^{*}(X) := argminF $\in$ S(C) (|F - X|)/(|X|). 27

This defines a nonlinear projection map PiS(C) : T $\to$ S(C). The resulting residuals {dC(Xi)} constitute the primary empirical observations.

22.2 Selected Representative Fits

The following expressions represent low-residual best fits identified by exhaustive search over S(C) at complexity bounds C $\leq$ 10. They are reported as empirical results of the constrained search, not as theoretical derivations. Several appear in the PhD monograph Chapter 34 exploratory catalogue (PhD: Ch.34 Catalogue); their presence here is coincidental with that catalogue and does not constitute re-use of the catalogue as evidence.

Strong coupling constant (PDG 2024; alphas(mZ) = 0.1180 $\pm$ 0.0009):

alphaφ := (φ-3)/(2) = ((2φ-3))/(2) approx 0.11803..., dC(alphas) $\sim$ O(10^{-4}), C = 3. 28

Remark on alphaφ. The algebraic identity φ-1 = φ - 1 allows rewriting alphaφ = (1)/(2)(φ-1)3 = (2φ-3)/2 = (5-2)/2, confirming this is a degree-3 polynomial in a quadratic irrational with small integer coefficients — the lowest-complexity non-trivial element of the grammar near alphas(mZ). The relative residual d3(alphas) approx (0.11803398875 - 0.1180)/0.1180 approx 2.9 $\times$ 10^{-4} lies within the experimental uncertainty sigma/alphas approx 0.76%. After Bonferroni correction over |S(3)| = 1{,}161 trials, the corrected p-value at tolerance 2.9 $\times$ 10^{-4} must be assessed against the null distribution. The specific definition alphaφ := φ-3/2 is used here; the PhD monograph Chapter 38 contains an inconsistency between two candidate definitions

(PhD: Ch.38 AlphaPhi), and only this definition is adopted.

Inverse fine-structure constant (NIST CODATA 2022; α^{-1} = 137.035 999 177(21)):

$\pi_4 \cdot \varphi_4 \cdot$ e2 / 36 approx 137.036581054, dC(α^{-1}) approx 4.25×10^{-6}, C = 10. 29

The residual approx 4.25×10^{-6} at this complexity is non-trivial but must survive correction over |S(10)| = 75{,}249 trials to be statistically significant. The expression is reported as a compact fit, not as an explanation of QED.

Electroweak mixing angle (PDG 2024; sin2theta(MZ) = 0.23129(4)):

3-1 $\cdot \pi^{-2} \cdot \varphi_4$ approx 0.231489, dC(sin2thetaW) approx 8.6×10^{-4}, C = 7. 30

Solar neutrino mixing (PDG 2024; sin2theta12 = 0.303+0.012-0.011):

8 $\cdot$ φ-5 $\cdot$ pi $\cdot$ e-2 approx 0.306699, dC(sin2theta12) approx 1.2×10^{-2}, C = 8. 31

Remark. The reported approximations depend explicitly on the choice of symbolic grammar S(C), the complexity cutoff C, and the error functional. They should be interpreted as properties of the representation system, not as intrinsic identities of physical constants. At C = 8, |S(8)| = 32{,}841; whether the observed residuals are meaningfully smaller than a grammar-aware null requires the permutation calibration of Section 6.

22.3 Full 42-Entry Trinity Catalogue Table

The complete legacy catalogue table is restored here because it is the object referred to by the name Catalog42. The table is imported from t27/research/trinity-pellis-paper/G2TRINITYV1.0FRAGRANCE.tex and cross-referenced against t27/proofs/trinity/Catalog42.v. Its evidentiary status is deliberately split into two layers:

- Catalogue layer: 42 rows across 9 physics sectors, each giving a proposed Trinity monomial or near-monomial parametrization.
- Coq layer: Catalog42.v is a Coq aggregation file with source-level Qed markers and no Admitted markers in that file; it aggregates a representative verified theorem set, but it does not, in its current form, provide one independent theorem named for every one of the 42 table rows.

Therefore the table below is part of the paper's data/provenance layer, while statistical significance still depends on the null-model and MDL procedures of Sections 5-7.

For print readability, the catalogue is printed below as a row-wise audit ledger. Each line has the same six fields as the source table: ID, sector, constant, reference value, Trinity expression, and residual/tier.

- G01 — Sector: Gauge / running; constant: α^{-1} fine structure; reference value: 137.036; expression: $4\cdot9\cdot\pi^{-1}\varphi$ e2; residual/tier: 0.029% V.
- G02 — Sector: Gauge / running; constant: alphas(mZ)=alphaφ; reference value: 0.11800; expression: φ-3/2; residual/tier: 0.029% V.
- G03 — Sector: Gauge / running; constant: sin2thetaW; reference value: 0.23121; expression: 3-2pi2φ_3 e-3; residual/tier: 0.086% V.
- G04 — Sector: Gauge / running; constant: cos2thetaW; reference value: 0.76879; expression: 2piφ-2e-1; residual/tier: 0.175% C.
- G05 — Sector: Gauge / running; constant: alphas/α_2 ratio; reference value: 3.7387; expression: 2piφ e-1; residual/tier: 0.034% V.
- G06 — Sector: Gauge / running; constant: alpha(mZ)/alpha(0) running; reference value: 1.0631; expression: 3φ_2 e-2; residual/tier: 0.017% V.
- H01 — Sector: Electroweak; constant: mH [GeV]; reference value: 125.20; expression: 4φ_3e2; residual/tier: 0.032% V.
- H02 — Sector: Electroweak; constant: mW [GeV]; reference value: 80.369; expression: $4\cdot3$-1pi3φ-1e; residual/tier: 0.051% V.
- H03 — Sector: Electroweak; constant: mZ [GeV]; reference value: 91.188; expression: $7\cdot3$pi-1φ_3e-2; residual/tier: 0.068% V.
- H04 — Sector: Electroweak; constant: GammaZ [GeV]; reference value: 2.4955; expression: $4\cdot3$-1piφ e-1; residual/tier: 0.087% V.
- H05 — Sector: Electroweak; constant: mt/mH ratio; reference value: 1.3784; expression: 7pi-1φ-1; residual/tier: 0.092% V.
- H06 — Sector: Electroweak; constant: mt/mW ratio; reference value: 2.1472; expression: 7pi-1φ_2e-1; residual/tier: 0.057% V.
- H07 — Sector: Electroweak; constant: sigmahad at Z [nb]; reference value: 41.48; expression: 3piφ e; residual/tier: 0.066% V.
- L01 — Sector: Leptons / Koide; constant: me [MeV]; reference value: 0.51100; expression: 2pi-2φ_4e-1; residual/tier: 0.017% V.
- L02 — Sector: Leptons / Koide; constant: mmu [MeV]; reference value: 105.658; expression: $8\cdot9\cdot\pi^{-4}\varphi_2$e4; residual/tier: 0.043% V.
- L03 — Sector: Leptons / Koide; constant: mtau [MeV]; reference value: 1776.86; expression: $5\cdot33$pi-3φ_5e; residual/tier: 0.067% V.
- L04 — Sector: Leptons / Koide; constant: ymu/ytau ratio; reference value: 0.05946; expression: 3-2pi-1φ-1e; residual/tier: 0.077% V.
- K01 — Sector: Leptons / Koide; constant: Q(e,mu,tau) Koide; reference value: 0.66667; expression: 8φ-1e-2; residual/tier: 0.370% C.
- K02 — Sector: Leptons / Koide; constant: Q(u,d,s) Koide; reference value: 0.5620; expression: 4φ-2e-1; residual/tier: 0.012% V.
- K03 — Sector: Leptons / Koide; constant: Q(c,b,t) Koide; reference value: 0.6690; expression: 8φ-1e-2; residual/tier: 0.020% V.

- Q01 — Sector: Quark masses; constant: mu [MeV]; reference value: 2.160; expression: $\pi_2\varphi$ e-2; residual/tier: 0.056% V.
- Q02 — Sector: Quark masses; constant: md [MeV]; reference value: 4.670; expression: $3\varphi_3$e-1; residual/tier: 0.109% C.
- Q03 — Sector: Quark masses; constant: ms [MeV]; reference value: 93.40; expression: 7piφ_3; residual/tier: 0.261% C.
- Q04 — Sector: Quark masses; constant: mc [GeV]; reference value: 1.273; expression: $\pi_2\varphi$-4e2; residual/tier: 0.083% V.
- Q05 — Sector: Quark masses; constant: mb [GeV]; reference value: 4.183; expression: 5piφ-2e-1; residual/tier: 0.054% V.
- Q06 — Sector: Quark masses; constant: mt [GeV]; reference value: 172.57; expression: $4\cdot 9\cdot \pi^{-1}\varphi_4$e2; residual/tier: 0.043% V.
- Q07 — Sector: Quark masses; constant: ms/md ratio; reference value: 20.000; expression: $8\cdot 3\cdot \pi^{-1}\varphi_2$; residual/tier: 0.002% SG.
- Q08 — Sector: Quark masses; constant: md/mu ratio; reference value: 2.162; expression: $\pi_2\varphi$ e-2; residual/tier: 0.038% V.
- C01 — Sector: CKM; constant: |Vus|; reference value: 0.22431; expression: 2·3-2pi-3φ_3e2; residual/tier: 0.051% V.
- C02 — Sector: CKM; constant: |Vcb|; reference value: 0.04100; expression: $\pi_3\varphi$-3e-1; residual/tier: 0.073% V.
- C03 — Sector: CKM; constant: |Vub|; reference value: 0.00394; expression: 3-2pi-3φ_2e-1; residual/tier: 0.068% V.
- C04 — Sector: CKM; constant: deltaCPCKM [deg]; reference value: 65.9; expression: 2·3φ e3; residual/tier: 0.061% V.
- N01 — Sector: PMNS; constant: sin2theta12; reference value: 0.30700; expression: 8φ-5pi e-2; residual/tier: 0.089% V.
- N02 — Sector: PMNS; constant: sin2theta23; reference value: 0.546; expression: 4·3-1piφ_2e-3; residual/tier: 0.085% V.
- N03 — Sector: PMNS; constant: sin2theta13; reference value: 0.02224; expression: 3piφ-3·10-2; residual/tier: 0.040% V.
- N04 — Sector: PMNS; constant: deltaCPPMNS [deg]; reference value: 129.1; expression: 8pi3/(9e2); residual/tier: 0.037% V; under formula-audit in Coq comments.
- M01 — Sector: Cosmology; constant: Omegab; reference value: 0.04897; expression: 4φ-2pi-3; residual/tier: 0.041% V.
- M02 — Sector: Cosmology; constant: OmegaDM; reference value: 0.2607; expression: 7·3-1pi-2φ_3; residual/tier: 0.071% V.
- M03 — Sector: Cosmology; constant: OmegaLambda; reference value: 0.6841; expression: 5pi-2φ_2e-1; residual/tier: 0.086% V.
- M04 — Sector: Cosmology; constant: ns spectral index; reference value: 0.9649; expression: 3$\varphi_3$pi-4e2; residual/tier: 0.094% V.

- D01 — Sector: QCD hadrons; constant: fK [MeV]; reference value: 157.55; expression: $\pi_4\varphi$; residual/tier: 0.039% V.
- P01 — Sector: Loop quantum gravity; constant: gammaBI Barbero-Immirzi; reference value: 0.23753; expression: φ-3=5-2; residual/tier: 0.62% C.

22.4 Coq Scope Audit: Why the Number Is Larger Than 42

The number 42 is the size of the restored legacy formula catalogue, not the size of the Coq proof base. A source audit of the available t27 checkout shows a larger formal layer around this catalogue:

- t27/proofs/trinity/: 13 Coq files, 281 theorem-like declarations, 122 source-level Qed. markers, 32 source-level Admitted. markers, and 0 Abort. markers.
- t27/coq/: 10 Coq kernel files, 73 theorem-like declarations, 35 source-level Qed. markers, 0 Admitted. markers, and 11 source-level Abort. markers.
- t27/proofs/: 18 Coq proof files in total, 296 theorem-like declarations, 127 source-level Qed. markers, 32 source-level Admitted. markers, and 0 Abort. markers.

Within t27/proofs/trinity, the following files constitute the topic-specific proof layer for the physical-constant programme:

- CorePhi.v: 13 declarations, 12 Qed., 0 Admitted.; root identities for φ, including φ_2+φ-2=3.
- AlphaPhi.v: 13 declarations, 12 Qed., 0 Admitted.; closed form and numeric windows for alphaφ=φ-3/2.
- BoundsGauge.v: 21 declarations, 9 Qed., 0 Admitted.; gauge-sector bounds for G-series formulas.
- BoundsMixing.v: 22 declarations, 10 Qed., 0 Admitted.; CKM/PMNS mixing bounds.
- BoundsMasses.v: 27 declarations, 11 Qed., 0 Admitted.; mass-ratio and Higgs-sector bounds.
- BoundsQuarkMasses.v: 16 declarations, 4 Qed., 4 Admitted.; additional quark-mass formula checks.
- BoundsLeptonMasses.v: 16 declarations, 0 Qed., 8 Admitted.; lepton-mass formula checks still requiring closure.
- FormulaEval.v: 30 declarations, 17 Qed., 0 Admitted.; generic monomial evaluator, with a header note stating that it captures the 69-formula Trinity v0.9 framework at the evaluation-interface level.
- DerivationLevels.v: 29 declarations, 17 Qed., 0 Admitted.; L1-L7 derivation-level classification and representative derivation paths.

- ExactIdentities.v: 27 declarations, 5 Qed., 11 Admitted.; Lucas/Pell/Fibonacci identity layer.
- ConsistencyChecks.v: 22 declarations, 7 Qed., 7 Admitted.; cross-formula consistency relations and warning checks.
- Unitarity.v: 23 declarations, 5 Qed., 2 Admitted.; CKM/PMNS unitarity checks and related candidate relations.
- Catalog42.v: 22 declarations, 13 Qed., 0 Admitted.; representative catalogue aggregation file.

The formula IDs explicitly present in the Trinity Coq layer are: C01, C02, C03, G01, G02, G03, G04, G06, H01, H02, H03, L01, L02, L03, N01, N03, N04, Q01, Q02, Q03, Q04, Q05, Q06, and Q07. Therefore the correct reading is three-tiered: the article restores a 42-row catalogue; the Coq layer directly mentions 24 formula IDs and provides 39 ID-named theorem/lemma declarations; the broader evaluator/proof infrastructure is larger than 42 and includes the 69-formula v0.9 interface claim, but not yet a closed one-theorem-per-formula proof set for all 69 entries.

22.5 Coverage Functional Analysis

The empirical coverage functional

rho(C, ε) = (1)/(|T|) sumX ∈ T 1[dC(X) < ε] 32

is evaluated over a grid of (C, ε) values: C ∈ {1,...,12}, ε ∈ {10-4, 10-3, 10-2}. Preliminary analysis indicates:

- For C ≤ 3: coverage is consistent with random expectation under all null models.
- For C ∈ [4, 8]: a non-trivial excess rho(C, 10-3) > ρ_0(C, 10-3) is observed.
- For C > 10: rho → 1 under saturation, and comparison with null models becomes uninformative.

The critical regime C ∈ [4, 8] at tolerance $\varepsilon = 10^{-3}$ is the primary domain of interest. All excess-coverage claims are presented as provisional pending full permutation calibration.

22.6 Structural Constant alphaφ and Experimental Test

The quantity alphaφ := φ-3/2 approx 0.11803 is the lowest-complexity grammar element near alphas(mZ). Its proximity to the PDG 2024 central value alphas(mZ) = 0.1180 ± 0.0009 yields

Deltaalpha := (|alphaφ - alphas(mZ)|)/(alphas(mZ)) approx 2.5 × 10^{-4},
33

a relative deviation well within the current experimental uncertainty. Whether this constitutes a statistically significant match — after Bonferroni correction over |S(3)| and the full dataset — depends on the null distribution at C = 3 and $\varepsilon = 2.5 \times 10^{-4}$. The quantity alpha$\varphi$ is a low-complexity algebraic invariant arising from the grammar, not a derivation of alphas.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. "The Fine-Structure Constant and the Golden Ratio." viXra 2110.0117 v4.; Sherbon, M. A. "Fundamental Nature of the Fine-Structure Constant." HAL hal-02341850.

P1 §9. Pellis Hierarchical Expansion

Open conjecture

Pellis expansion triptych — additive sum of phi-inverse powers / reciprocal interference / alpha-residue dial.

9.1 Motivation The multiplicative grammar Gφ provides a zeroth-order symbolic representation but cannot represent values near lattice points without increasing C indefinitely. The Pellis hierarchical expansion introduces a nested additive correction scheme in powers of φ-1, providing finer-grained approximation without unbounded complexity growth. The construction draws on the structure of the Lucas ring Z[φ] (PhD: Ch.3 Trinity Identity) to keep partial sums algebraically tractable.

9.2 Definition of the Expansion Class A hierarchical representation of a target quantity X $\in R^+$ at depth K is defined as

X approx F(K)(φ) = $\sum_{k=1}$K ck φ-k, ck $\in$ Z, K $\in$ N. 34

The formal asymptotic extension is

F(φ) $\sim \sum_{k=1}$^∞ ck φ-k as k $\to \infty$, 35

interpreted as a formal series. The expansion variable φ-1 approx 0.6180 acts as a natural geometric-sequence parameter generating progressively finer corrections. The algebraic closure of Z[φ] ensures that integer-valued coefficients keep each partial sum in this ring.

9.3 Two-Tier Structure The Pellis expansion is deployed as a second tier over the base multiplicative grammar:

X approx F0 + Delta F(1) + Delta F(2) + ..., Delta F(j) $\in$ span{φ-k}k $\geq$ 1, 36

where F0 = n $\cdot$ 3k $\cdot$ φp $\cdot$ pim $\cdot$ eq $\in$ S(C) is the zeroth-order best-fit from the base grammar and Delta F(j) are additive correction terms from a truncated Pellis expansion.

9.4 Scale Ordering and Filtration The hierarchy induced by powers of φ-1 defines a natural scale ordering:

φ-1 > φ-2 > φ-3 > ..., 37

with corresponding geometric ratio φ-1 approx 0.618. This induces a filtration

F1 $\supset$ F2 $\supset$ F3 $\supset$ ..., Fk = spanR{φ-j}j $\geq$ k. 38

9.5 Truncation Error and Convergence Criterion The truncation error at depth K is

εK(X) := | X - $\sum_{k=1}$K ck φ-k |. 39

A hierarchical representation is operationally convergent if εK(X) decreases monotonically with K and stabilizes below a predefined tolerance delta for some finite K0. No claim of analytic convergence of the infinite series is made; only finite-precision stabilization is asserted.

9.6 MDL Penalty for the Hierarchical Expansion When the Pellis expansion is incorporated into the MDL framework, L(M1) is augmented by the cost of specifying expansion depth K and coefficient vector {ck}k=1K:

L(M1Pellis) = L(M1base) + log2 K + $\sum_{k=1}$K log2(1 + |ck|). 40

Only Pellis expansions with augmented MDL score below that of the null model are reported as compressive. This ensures that increased depth cannot manufacture spurious compression advantage.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. "The Fine-Structure Constant and the Golden Ratio." viXra 2110.0117 v4.; Sherbon, M. A. "Fundamental Nature of the Fine-Structure Constant." HAL hal-02341850.

P1 §10. Falsification Ledger and Out-of-Sample Tests

Open conjecture

Claim status. The falsification tests in this chapter list open conjectures with concrete disconfirming observations; each item is one experiment away from Retracted.

10. Falsification Ledger, Out-of-Sample Tests, and Negative Controls
    Falsification triptych — Banks-Zaks fixed point / Gaussian null with outside observation / falsifier wall.

10.1 Falsification as a Necessary Condition A compressibility framework for physical constants has scientific content only if it makes falsifiable predictions in the sense of Popper's criterion. The following section defines a set of tests that can in principle refute the core claims. All tests are defined prospectively — prior to completion of the primary analysis. The falsification protocol design draws conceptually on the Popper Appendix of the PhD monograph (PhD: App.B Falsification).

10.2 Out-of-Sample Constants A subset of the dimensionless constant space is reserved as an out-of-sample test set Toos, defined by constants not used in any search or parameter tuning during the primary analysis:

- deltaCP: CP-violating phase in the PMNS matrix. The PDG 2024 global analysis reports deltaCP approx 197+42-25{}circ (normal ordering); precision is expected to improve substantially with T2K Phase II and NOvA updates.
- ms/md: Strange-to-down quark mass ratio. Constrained by lattice QCD; current PDG value approx 18.7 ± 0.9 (PDG 2024).
- Tensor-to-scalar ratio r: Currently constrained at r < 0.032 at 95% CL by CMB B-mode polarization observations.

10.3 Specific Quantitative Predictions Prediction P1 (strong coupling). The expression alphaφ = φ-3/2 approx 0.11803 constitutes a prediction for alphas(mZ) accurate to approx 2.5×10^{-4}. The framework is testable: if a future high-precision global fit (e.g., from LHC global QCD analyses with per-mille precision) yields alphas(mZ) differing from alphaφ by more than 5sigmafuture

where sigmafuture is the future experimental uncertainty, expression (28) is falsified (PDG 2024 current precision).

Prediction P2 (solar neutrino mixing). Expression (31) for sin2theta12 evaluates to approx 0.3067, roughly 1.2% above the PDG central value 0.303. This is therefore a weak exploratory candidate rather than a sharp prediction. Improved measurements from JUNO can still falsify or de-prioritize the expression if the central value moves away from the interval around 0.3067.

Prediction P3 (coverage scaling). The coverage ratio rho(C, ε) / ρ_0(C, ε) peaks in C ∈ [4, 8] and decreases toward 1 for C ≤ 2 and C ≥ 12. This structural prediction is testable by evaluating the ratio across the full C-grid.

10.4 Falsification Ledger The framework is falsified and its compressibility claims are retracted under any of the following conditions:

The falsification ledger will be evaluated publicly and updated with each PDG edition. Any triggered criterion requires explicit retraction of the corresponding compressibility claim.

10.5 Negative Control Results The negative controls of Section 5.6 (randomly rotated datasets) should yield coverage ratios consistent with 1 at all complexity levels. Specifically:

- For all 50 negative control realizations Tneg(j), the mean coverage ratio rhoneg(C, ε) / ρ_0(C, ε) should not exceed 1.1 at any C ∈ [2, 12] after Bonferroni correction.
- Failure of this check (i.e., elevated coverage in rotated datasets) would indicate that the observed compressibility is an artifact of grammar density, not algebraic selectivity, and would trigger retraction of Criterion F4.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. “The Fine-Structure Constant and the Golden Ratio.” viXra 2110.0117 v4.; Sherbon, M. A. “Fundamental Nature of the Fine-Structure Constant.” HAL hal-02341850.

P1 §11. Reviewer-Risk Assessment

High-risk

Reviewer-path triptych — R1..R6 risk cards / response arrows / honest-admission seal.

11.1 Purpose This section anticipates the principal objections that a journal reviewer with expertise in information theory, particle physics, or Bayesian statistics would raise. It documents the risks honestly and explains how the methodological choices address them.

11.2 Risk R1: Multiple Testing Inflation Risk: The hypothesis class S(C) is large enough that small residuals are guaranteed even for random targets. The Bonferroni correction may be insufficient if grammar values are correlated.

Mitigation: The effective trial count Neff(C) = |S(C)| is used as the Bonferroni denominator, which is conservative when grammar values are positively correlated (closely spaced lattice points). The permutation calibration of Section 6.5 provides an empirical calibration against the actual null distribution, accounting for correlations automatically.

11.3 Risk R2: Grammar Density Masquerading as Selectivity Risk: The φ-lattice may be denser in the empirical range of physical constants than in the null model's range, creating apparent compressibility that is purely geometric.

Mitigation: The Mperm null model preserves the empirical range of T and directly controls for this. The Mrand null model additionally tests whether the specific algebraic structure of the basis, as opposed to its cardinality, is responsible for observed coverage. The negative controls of Section 10.5 provide a direct diagnostic.

11.4 Risk R3: Post-Hoc Data Selection

Risk: The target dataset T was assembled after observing fit results, introducing selection bias.

Mitigation: Protocol P1 (Section 4.3) mandates pre-registration of T before any symbolic search is executed, with a cryptographic hash commitment. The dataset is defined by membership criteria (dimensionless, Standard Model, PDG-tabulated, precision $> 10^{-2}$), not by fit quality.

11.5 Risk R4: Non-Uniqueness of the Grammar Basis Risk: The choice of basis {φ, pi, e, 3} is arbitrary; a different basis might yield similar or better results by accident.

Mitigation: The paper explicitly claims no uniqueness for the basis. The Mrand null model tests whether results are specific to the structured basis. Future work should compare Gφ against alternative algebraically structured grammars using the same MDL framework.

11.6 Risk R5: Misinterpretation as Derivation Risk: Readers may interpret low-residual fits as derivations of physical constants or as evidence of an underlying physics principle.

Mitigation: Section 1.3 explicitly disavows all physical derivation claims. The encoding map is repeatedly characterized as a constrained quantization, not a physical law. Every expression is accompanied by the remark that it is a property of the representation system, not a property of nature.

11.7 Risk R6: PhD Monograph Provenance Bias Risk: The paper's methodological design and candidate expressions may be biased by the PhD monograph's exploratory phase.

Mitigation: Section 12 provides a transparent delineation of what the monograph contributes (algebraic structure, falsification design, historical motivation) versus what it does not provide (statistical evidence). All expressions from the monograph's Chapter 34 catalogue are subject to the same null-model analysis as expressions not in the catalogue.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. "The Fine-Structure Constant and the Golden Ratio." viXra 2110.0117 v4.; Sherbon, M. A. "Fundamental Nature of the Fine-Structure Constant." HAL hal-02341850.

P1 §12. Relationship to the Trinity S^{3}AI / Flos Aureus PhD

Open conjecture

Monograph and Stergios Collaboration Note

Provenance triptych — Monograph spine / Shared anchor identity / Collaboration handshake.

12.1 Scope of the Relationship This paper extracts a narrow, journal-safe statistical subproblem from the broader Trinity S^{3}AI / Flos Aureus research program, whose PhD monograph is in preparation under the title Trinity S^{3}AI — Flos Aureus v6.2 / GOLDEN SUNFLOWERS (D. Vasilev, ORCID: 0009-0008-4294-6159, defense scheduled 2026-06-15). The relationship between this paper and the monograph is one of provenance, algebraic foundation, and falsification design — not one of external validation. The monograph does not independently validate the statistical claims made in this paper; those claims stand or fall entirely on the null-model analysis and MDL scoring defined in Sections 5–7.

Specifically: - The monograph is not cited as evidence that Gφ achieves non-random compressibility. - The monograph's exploratory catalogue (approximately 42 fits, Chapter 34) is treated as historical motivation and a supplementary candidate pool; expressions from it are re-evaluated under the null-model analysis. - The identity $\varphi_2 + \varphi\text{-}2 = 3$ (PhD: Ch.3 Trinity Identity) is used only as an algebraic completeness statement about the Lucas ring Z[φ].

12.2 Monograph Contributions Table

12.3 Proof Status and Honest Admission The PhD monograph maintains a Coq citation map and an R5-honest admitted-budget ledger (PhD: App.F Coq Map). A local source audit of the available t27 checkout gives the following status. The topic-specific physical-constant layer t27/proofs/trinity/ contains 13 Coq files, 281 theorem-like declarations, 122 source-level Qed. markers, 32 source-level Admitted. markers, and 0 Abort. markers. The broader t27/proofs/ tree

contains 18 Coq proof files, 296 theorem-like declarations, 127 source-level Qed. markers, 32 source-level Admitted. markers, and 0 Abort. markers. The older t27/coq/ kernel layer contains 10 Coq files, 73 theorem-like declarations, 35 source-level Qed. markers, 0 Admitted. markers, and 11 source-level Abort. markers. Catalog42.v itself contains 13 source-level Qed. markers and 0 Admitted. markers, and defines a totalverifiedtheorems aggregate over 28 representative theorem obligations across core identities, AlphaPhi, gauge, CKM, neutrino, mass, and monomial-interface claims. Therefore the correct statement is: the paper has a 42-row formula catalogue, a 69-formula evaluator interface noted in FormulaEval.v, and a larger Coq proof layer with 122 Qed. markers in t27/proofs/trinity/; it is not yet a closed one-theorem-per-row proof set for all 42 or all 69 formula entries. Because coqc is not available in the current build sandbox, this is a source audit rather than a fresh local Coq compilation verdict.

12.4 Stergios Collaboration Note The Pellis hierarchical expansion of Section 9 is named after Stergios Pellis, who is a co-author of this paper and the principal contributor of the multi-scale decomposition framework. The collaboration between Dmitrii Vasilev and Stergios Pellis originates within the Trinity S^3AI research program and represents the integration of two complementary methodological threads:

Vasilev's thread is the algebraic and information-theoretic foundation: the formal grammar Gφ, the MDL scoring framework, the multiple-testing correction architecture, and the PhD monograph's algebraic backbone (the Lucas ring, the Trinity identity $\varphi_2 + \varphi\text{-}2 = 3$, and the Flos Aureus numerical analysis). This thread focuses on rigor, falsifiability, and statistical correctness.

Pellis's thread is the hierarchical expansion methodology: the insight that a multiplicative grammar can be systematically refined by additive corrections in powers of φ-1, defining a filtration F1 $\supset$ F2 $\supset$ ... over which constants admit multi-scale decompositions. The Pellis expansion has independent mathematical interest as a structured approximation theory over the ring Z[φ], connecting to generalized continued fraction expansions and to the theory of beta-expansions in non-integer bases.

The collaboration is ongoing. The current paper represents the first joint formalization of the combined framework. Future work under this collaboration includes: - Systematic comparison of the Pellis expansion against standard continued fractions and Engel expansions as approximation schemes for transcendental numbers. - Application of the Pellis filtration to near-threshold constants where the base grammar achieves only $O(10^{-2})$ precision. - A dedicated paper (Paper 3 in the series) formalizing the hierarchical expansion and its MDL theory.

The collaboration note is included here to establish clear provenance for the Pellis expansion methodology and to acknowledge its independent conceptual origin within the joint research program.

12.5 Scott Olsen Historical-Geometric Note

Scott A. Olsen's golden-section work is used here as Tier-D context: it helps explain why the reader should expect a long historical lineage around φ, but it does not validate any numerical claim in this paper. Olsen's book The Golden Section: Nature's Greatest Secret presents the golden section as a unifying theme across mathematics, art, architecture, music, philosophy, and natural form (Internet Archive bibliographic record). That role is useful for exposition because the GOLDEN CHAIN (Trinity-Pellis Bridge) program also uses φ as a structural organizer, but the evidentiary standards are different.

The specific Olsen contribution imported into this version is the golden balance motif: the ordered five-point reciprocal scale

φ-2, φ-1, 1, φ, $φ_2$.

Within the article, this motif is treated as a geometric and pedagogical diagram of reciprocal scaling around the unit element. It is not used to infer any Standard Model constant, and it is not part of the null-model statistics. In PhD notation it belongs to the visual and historical layer, not to the proof ledger.

This limited use protects the paper from a common reviewer objection. The history of the golden section is allowed to provide orientation, but every scientific claim remains inside the formal grammar, the explicit search space, the MDL score, and the falsification protocol.

12.6 Author Contributions - Dmitrii Vasilev: Trinity monomial grammar, computational search framing, falsification protocol, reproducibility architecture, and integration with the Trinity S^3AI / Flos Aureus programme. - Stergios Pellis: Pellis hierarchical expansion framework, fine-structure comparison layer, mechanism-risk assessment, and mathematical co-development of the GOLDEN CHAIN (Trinity-Pellis Bridge) series. - Scott Olsen: historical-geometric and golden-section context, golden-balance interpretive lineage, and visual-theoretical framing. These contributions are treated as Tier-D context and are not used as statistical evidence, derivation evidence, or physical proof.

12.7 What This Paper Does Not Inherit from the Monograph - This paper does not adopt the AlphaPhi derivation inconsistency in Chapter 38. - This paper does not use the ~42 catalogue expressions as pre-registered predictions. - This paper does not claim that Zenodo DOIs associated with the monograph constitute proof of statistical claims. - This paper does not import any stale proof-count claims; Coq obligation counts must be regenerated from the repository at submission time.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. “The Fine-Structure Constant and the Golden Ratio.” viXra 2110.0117 v4.; Sherbon, M. A. “Fundamental Nature of the Fine-Structure Constant.” HAL hal-02341850.

P1 §13. Limitations and Scope of Validity

High-risk

Limitations triptych — Grammar non-uniqueness / Finite dataset / Asymptotic saturation.

13.1 Grammar Dependence and Non-Uniqueness All compressibility properties documented here are conditional on the specific choice of grammar Gφ = langle φ, pi, e, 3, Z rangle. Different bases induce inequivalent hypothesis spaces, and a finding of compressibility under Gφ does not imply that {φ, pi, e, 3} is uniquely suited to the task. Future work should perform a systematic basis comparison using the same MDL framework.

13.2 Residual Selection and Search-Induced Bias The exhaustive search over S(C) generates a minimum residual dC(X) that is an extreme-value statistic. Despite the null-model controls, residual selection bias may persist due to:

- Anisotropy of the log-lattice induced by the specific values of log 3, log φ, log pi, log e.
- Clustering of grammar values near dense regions that accidentally overlap with the empirical distribution of physical constants.
- Correlations between residuals across different constants in the finite, non-random target set T.

The Mperm null model partially addresses these concerns; the negative controls of Section 10.5 provide a direct diagnostic.

13.3 Absence of Physical Derivation The framework provides no dynamical mechanism, symmetry principle, or field-theoretic foundation for the observed compressibility. All results are interpretable only as empirical compressibility under constrained symbolic grammars.

13.4 Finite Dataset Limitations The target set T contains N $<$ 15 constants (after excluding those with relative uncertainty $> 10^{-2}$). Statistical conclusions from such small N are subject to substantial finite-sample variance. Asymptotic coverage theory applies only approximately.

13.5 Identifiability and Non-Uniqueness of Fits Multiple distinct symbolic expressions may achieve comparable accuracy: F1 approx X approx F2 with F1 $\neq$ F2. This structural non-uniqueness is a property of all overcomplete dictionaries and implies that the best-fit expression F^*(X) is not uniquely defined near dense regions of the grammar lattice.

13.6 Asymptotic Saturation As C $\to \infty$, S(C) becomes dense in R^+ and coverage trivially approaches 1 for any ε > 0. The analysis is restricted to C $\leq$ 10 to avoid this uninformative regime.

13.7 Sensitivity to Complexity Metric Choice The ℓ_1-norm complexity metric induces a hypothesis class that differs from those induced by ℓ_0, ℓ_2, or depth-based metrics. All reported results are specific to the ℓ_1 choice. Sensitivity analyses with alternative metrics are reserved for future work.

13.8 Experimental Uncertainty Propagation For constants with relative uncertainty sigmaXi/Xi $> 10^{-2}$, a symbolic approximation at tolerance ε $=$ 10^{-3} is indistinguishable from the best achievable match given experimental precision. Such constants are flagged and excluded from the primary analysis.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. “The Fine-Structure Constant and the Golden Ratio.” viXra 2110.0117 v4.; Sherbon, M. A. “Fundamental Nature of the Fine-Structure Constant.” HAL hal-02341850.

P1 §14. Conclusion

Empirical fit

Claim status. Conclusions for P1 are reported as empirical fits consistent with the residual budgets in p1-08; the underlying generators remain Open conjectures pending p3.

Conclusion triptych — Identity / Programme ribbon / arrow to Paper II.

This paper has presented a formalized, information-theoretic framework for evaluating whether dimensionless physical constants exhibit statistically significant compressibility within a constrained symbolic grammar. The framework has been constructed with explicit attention to the main sources of statistical bias that afflict symbolic discovery approaches: combinatorial search over large hypothesis spaces, multiple comparison effects, selection bias from post-hoc minimization, grammar density effects, and finite-sample variance.

Principal contributions:

1. A formally defined symbolic grammar Gφ with bounded ℓ_1 complexity, analyzed as a finite hypothesis class with explicit combinatorial structure and a detailed cardinality table.
2. A rigorous pre-registration freeze protocol ensuring that the target set T is determined by publicly available experimental compilations (PDG 2024; CODATA 2022) prior to any symbolic search.
3. Three complementary null models (M0, Mperm, Mrand) forming a hierarchically conservative control structure for compressibility analysis.
4. A complete multiple-testing correction procedure at both individual-constant and family-wise levels, using Bonferroni, BH-FDR, and permutation calibration.
5. An MDL-based model comparison framework (Barron, Rissanen & Yu, 1998; Grünwald, 2007) that explicitly penalizes the symbolic grammar for its description length, yielding a net compression score DeltaMDL as the primary evidence metric.
6. The Pellis hierarchical expansion: a structured two-tier refinement scheme in powers of φ-1 with explicit truncation error control and MDL penalization.
7. A pre-registered falsification ledger with five explicit criteria, any one of which, if triggered, requires retraction of the compressibility claim.
8. A reviewer-risk assessment with six identified risks and documented mitigations.
9. A transparent provenance connection to the Trinity S^3AI / Flos Aureus PhD monograph, with explicit delineation of what the monograph contributes and what it does not validate.
10. A detailed Stergios collaboration note establishing provenance for the Pellis expansion methodology.

The core statistical claim, subject to the falsification criteria above, is:

> The dataset of dimensionless physical constants T may admit a more compact encoding under the structured grammar Gφ than under unstructured stochastic baselines, as measured by the MDL compression score DeltaMDL > 0 across the critical complexity regime $C \in [4, 8]$.

Whether this claim is correct is an empirical question resolvable by the tests defined in this paper. The framework formalizes the question:

Are physical constants unusually compressible under constrained symbolic grammars?

and provides the methodological apparatus to answer it with statistical rigor and full accountability.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. “The Fine-Structure Constant and the Golden Ratio.” viXra 2110.0117 v4.; Sherbon, M. A. “Fundamental Nature of the Fine-Structure Constant.” HAL hal-02341850.

28.1 P1 Appendix A: Reproducibility Protocol Checklist

Appendix A: Reproducibility Protocol Checklist

Reproducibility triptych — Checklist / Dataset cabinet / Enumerator.

A.1 Dataset Preparation - [ ] P1: Constants T drawn exclusively from PDG 2024 and/or CODATA 2022, edition fixed before search begins.

- ☐ P2: All constants in T are dimensionless.
- ☐ P3: Only constants with relative experimental uncertainty better than 10-2 are included in the primary analysis.
- ☐ P4: The dataset list (with values and uncertainties) is committed to version control before any symbolic search is executed, with a cryptographic hash.
- ☐ P5: A separate out-of-sample set Toos is defined and frozen prior to analysis.
- o P6: All data files, version hashes, and Zenodo provenance records are deposited as specified in PhD: App.G Data Availability.

A.2 Grammar Enumeration - [ ] G1: Grammar Gφ is implemented as in equations (1)–(2), with |theta|1 = |k| + |p| + |m| + |q|. - [ ] G2: Exponent bounds satisfy |k|, |p|, |m|, |q| $\leq$ Cmax with Cmax $\leq$ 15. - [ ] G3: The full expression set S(C) is deterministically enumerated (not sampled) for all C $\leq$ 10. - [ ] G4: The cardinality |S(C)| is explicitly reported for each complexity level. - [ ] G5: Numerical evaluation of F(theta) uses IEEE 754 double precision or higher.

A.3 Residual Computation - [ ] R1: Approximation residual is defined as dC(X) = minF $\in$ S(C) |F - X| / |X|. - [ ] R2: Best-fit expression F^*(X) is recorded alongside the residual for all constants. - [ ] R3: All residuals are evaluated before any comparison against null models or selection of reported results.

A.4 Null Model Evaluation - [ ] N1: Monte Carlo sampling for M0 uses at least B $\geq$ 10{,}000 independent samples. - [ ] N2: Permutation sampling for Mperm uses at least B $\geq$ 10{,}000 random permutations. - [ ] N3: Randomized grammar Mrand uses exponent vectors sampled uniformly from {-C, ..., C}4. - [ ] N4: Null distributions are reported as quantiles (5th, 50th, 95th percentile) at each (C, ε) grid point. - [ ] N5: All 50 negative control realizations are evaluated and reported.

A.5 Multiple Testing - [ ] T1: Bonferroni-corrected p-values computed using Neff(C) = |S(C)| as trial count. - [ ] T2: Family-wise tests use N · |S(C)| as the trial count. - [ ] T3: BH-adjusted p-values reported for the full dataset. - [ ] T4: No result reported as statistically significant unless it survives both analytic Bonferroni correction and permutation calibration.

A.6 MDL Computation - [ ] M1: Model description length L(M1) computed as in equations (22)–(24). - [ ] M2: Results reported for at least three values of beta $\in$ {10, 100, 1000}. - [ ] M3: DeltaMDL reported with uncertainty from finite Monte Carlo sample. - [ ] M4: Positive compression claim requires DeltaMDL $>$ 0 for all beta values in the range.

A.7 Pellis Expansion - [ ] PE1: Expansion coefficients ck determined by iterative residual fitting as defined in Section 9. - [ ] PE2: Truncation depth K reported alongside expansion residual εK(X) for each constant. - [ ] PE3: MDL cost of the Pellis expansion included via equation (40); only expansions with net DeltaMDL $>$ 0 reported as compressive.

A.8 Falsification Tests - [ ] F1: All five falsification criteria (Section 10.4) evaluated and reported. - [ ] F2: Out-of-sample test results computed and reported independently of primary analysis. - [ ] F3: Updated experimental values in out-of-sample tests cited with their PDG/CODATA reference year. - [ ] F4: All 50 negative control results reported in tabulated form.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. “The Fine-Structure Constant and the Golden Ratio.” viXra 2110.0117 v4.; Sherbon, M. A. “Fundamental Nature of the Fine-Structure Constant.” HAL hal-02341850.

28.2 P1 Appendix B: Algebraic Properties of the Basis Elements

Appendix B: Algebraic Properties of the Basis Elements

Basis-properties triptych — 2 / 3 / phi-pi-e columns with independence tags.

The key algebraic relations used throughout the paper are collected here for reference.

Golden ratio identities. The golden ratio φ = (1+5)/2 satisfies the minimal polynomial $\varphi_2 = \varphi + 1$, from which:

φ-1 = φ - 1 approx 0.61803... B.1

φ-2 = 2 - φ approx 0.38197... B.2

φ-3 = 2φ - 3 approx 0.23607... B.3

φ_2 + φ-2 = (φ + 1) + (2 - φ) = 3. B.4

Equation (B.4) is the Trinity identity (PhD: Ch.3 Trinity Identity). It confirms that the squared golden ratio and its reciprocal square sum to the integer 3, establishing the ternary cardinality relation central to the Lucas ring Z[φ]. In the context of this paper, it is used only as an algebraic completeness statement.

Algebraic rewriting of the structural constant. Using (B.3):

alphaφ = (φ-3)/(2) = (2φ - 3)/(2) = φ - (3)/(2) = (1)/(2)(φ-1)3. B.5

This confirms that alphaφ is a degree-3 polynomial in φ with small integer coefficients — the lowest-complexity non-trivial element of the grammar near the PDG 2024 value alphas(mZ) = 0.1180(9) (PDG 2024).

Linear independence. The set {1, log 3, log φ, log pi} is treated as a working independence assumption over Q, not as a theorem proved here. Since log e = 1, it is not listed as an additional independent generator. Under this assumption, elements of S(C) with distinct exponent tuples map to distinct real values; if a rational relation were discovered, the hypothesis class would be quotient by that relation and all MDL penalties would be recomputed.

Numerical values of basis elements and their logarithms:

The log-space coordinates of the four basis elements (log 3, log φ, log pi, 1) are numerically pairwise incommensurable, supporting the linear-independence conjecture.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. "The Fine-Structure Constant and the Golden Ratio." viXra 2110.0117 v4.; Sherbon, M. A. "Fundamental Nature of the Fine-Structure Constant." HAL hal-02341850.

28.3 P1 Appendix C: Phase Transition Structure (Conjecture)

Appendix C: Phase Transition Structure (Conjecture)

Phase-transition triptych — Order parameter / Critical point / MDL evidence bars.

C.1 Coverage Functional and Order Parameter

Conjecture 1 (Pellis Symbolic Phase Transition). There exists a critical complexity scale C^ > 0 such that the coverage functional satisfies:*

rho(C, ε) = O(e-alpha(C^ - C)), & C < C^, 1 - O(e-beta(C - C^)), & C > C^, C.1

for constants alpha, beta > 0 determined by the effective metric entropy of T in log-coordinates.

This conjecture is motivated by the analogy with percolation phase transitions in random lattice models, where a sharp onset of coverage occurs when the lattice spacing crosses the scale of inter-point distances. It is labeled as a conjecture and has no formal proof in this paper.

C.2 Characterization of the Critical Point The critical scale C^ is characterized by the condition that the effective lattice spacing of S(C^) in log-space equals the typical inter-point distance of log T:

deltaeff(C^*) ∼ E[d(log Xi, log Xj)]. C.2

The susceptibility chi(C) := ∂ rho / ∂ C is conjectured to achieve a global maximum near C^*, with width Delta C ∼ |T|-1/2.

C.3 Relation to MDL Evidence Under the conjecture,

DeltaMDL(C^) approx 0, (d DeltaMDL)/(dC)|C = C^ gg 1. C.3

The critical point C^* corresponds to the grammar complexity at which the structured grammar first achieves a net coding advantage over the null model.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. "The Fine-Structure Constant and the Golden Ratio." viXra 2110.0117 v4.; Sherbon, M. A. "Fundamental Nature of the Fine-Structure Constant." HAL hal-02341850.

28.4 P1 Appendix D: Mapping from PhD Monograph to Paper 1

Appendix D: Mapping from PhD Monograph to Paper 1

Mapping triptych — Monograph chapters / Arrows / Paper-1 sections.

The following table provides a complete mapping between components of the Trinity S^3AI / Flos Aureus PhD monograph and their role in this paper.

Acknowledgments The authors acknowledge that this work was prepared as part of the GOLDEN CHAIN (Trinity-Pellis Bridge) research program. The Trinity S^3AI / Flos Aureus PhD monograph (D. Vasilev, ORCID: 0009-0008-4294-6159, defense 2026-06-15) provides the algebraic foundation, falsification design, and data provenance infrastructure for this preprint series.

Source URLs Used in This Paper All external sources cited inline throughout this document are also listed here as plain text URLs so that the PDF remains auditable even if a reader copies the text layer from the document.

- PDG 2024 Review of Particle Physics: URL: S. Navas et al., Particle Data Group, Phys. Rev. D 110, 030001, 2024. Portal:
- PDG 2024 Physical Constants table: URL:
- PDG 2024 Neutrino Mixing review: URL:
- CODATA 2022 recommended values: URL: Wall chart:
- Barron, Rissanen & Yu, MDL principle (1998): URL:
- Grünwald, The Minimum Description Length Principle (2007): URL:
- Chaitin, Algorithmic Information Theory (1987): URL:
- Planck 2018 cosmological parameters: URL:
- AI Feynman symbolic regression: URL:
- MDL Scholarpedia article: URL:

Submission-Time Audit Checklist Before journal submission or arXiv posting, regenerate the following gates from the repository rather than trusting the prose of this PDF:

- Constants audit: re-fetch CODATA, PDG, and NuFIT values; recompute every residual; mark all stale values.
- Search-space audit: recompute |S(C)|, Neff, and all Bonferroni/MDL penalties from code.
- Proof-status audit: regenerate the Coq ledger; never copy old global theorem counts into the manuscript.
- URL audit: verify that every source URL in the list above returns a live page or DOI landing page.
- Claim-tier audit: confirm that every sentence remains in the correct tier: algebraic identity, statistical compressibility, mechanism candidate, or historical context.

Toward a Mechanism for φ-Structured Couplings: Discrete Scale Invariance, E8/Toda Anchors, and Symbolic

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; Pellis, S. “The Fine-Structure Constant and the Golden Ratio.” viXra 2110.0117 v4.; Sherbon, M. A. “Fundamental Nature of the Fine-Structure Constant.” HAL hal-02341850.

P1 Acknowledgments

Renormalization Dmitrii Vasilev, Stergios Pellis, Scott Olsen

GOLDEN CHAIN (Trinity-Pellis Bridge) Programme — Paper 2 of 3

Trinity S^{3}AI — Flos Aureus Research Programme · Zenodo record 10.5281/zenodo.19227877

Epistemic status (mandatory reading). This paper is a mechanism-exploration document. It does not derive Standard Model coupling constants from first principles, and no claim to that effect is made or implied. The results of Paper 1 — symbolic compressibility of φ-structured coupling patterns — are taken as empirical motivation only; they are not proof of a physical mechanism. The theoretical frameworks developed here — the symbolic RG model, the candidate operator framework, the discrete-scale-invariance connection — are conjectural constructions whose status is scaffolding, not established theory. All mathematical statements are clearly labeled as either (a) known exact results imported from the literature, (b) model-internal derivations whose assumptions are explicitly stated, or (c) speculative proposals awaiting formal derivation or experimental test. Sections that cross the boundary into conjecture are marked [CONJECTURAL]. The goal is to identify a research programme with clear success and failure criteria, not to announce a discovery.

Abstract Paper 1 of the GOLDEN CHAIN (Trinity-Pellis Bridge) programme established that a suite of coupling-constant combinations from the Standard Model and near-critical quantum systems admit unusually short symbolic descriptions when encoded in a basis containing the golden ratio φ = (1+5)/2. Whether this compressibility reflects a genuine physical mechanism or is an artefact of number theory and selection bias remains open.

The present paper asks a sharper question: are there known, derivable mechanisms in quantum field theory and statistical mechanics that could cause φ-structured patterns to appear in renormalized coupling constants?

Three candidate lines of evidence are examined. First, the Zamolodchikov E8 integrable field theory — established in two seminal papers published in Adv. Stud. Pure Math. 19 (1989) 641–674 (Project Euclid) and Int. J. Mod. Phys. A 4 (1989) 4235 — provides an exact, non-conjectural occurrence of φ in a mass spectrum: the ratio m2/m1 = φ in the perturbed critical Ising model, confirmed experimentally in CoNb2O6 (Coldea et al., Science 327 (2010) 177). This is the only φ-in-field-theory result treated as established in this paper. Second, discrete scale invariance (DSI) in systems with self-similar hierarchical structure produces log-periodic corrections to RG flows whose preferred scaling ratio can, in specific models, be constrained to algebraic numbers including φ

(Sornette, Phys. Rep. 297 (1998) 239). Third, affine Toda field theories based on non-crystallographic Coxeter groups H2, H3, H4 produce mass pairs differing by φ through an exact embedding construction (Fring and Korff, Nucl. Phys. B 729 (2005) 361).

Two candidate φ-coupling expressions are carefully distinguished throughout: the algebraic power variant alphaφ,alg = φ-3/2 = (5-2)/2 approx 0.1180 and the logarithmic variant alphaφ,log = ln(φ_2)/pi approx 0.30635. These are structurally distinct objects; they must not be conflated, and neither is derived from first principles here. A symbolic renormalization group (sRG) toy model and a candidate φ-structured operator framework, both conjectural, are introduced as formal scaffolding. Strict falsifiability conditions, an evidence hierarchy, mechanism-by-mechanism falsification tests, a derivation ladder, and five work packages for the GOLDEN CHAIN (Trinity-Pellis Bridge) programme are delineated. Paper 2 is also the theoretical-mechanism branch of the broader Trinity S^3AI — Flos Aureus research programme (Zenodo 10.5281/zenodo.19227877).

Keywords: golden ratio, E8 symmetry, integrable field theory, discrete scale invariance, symbolic renormalization group, Toda lattice, coupling constant compressibility, Lucas ring, Trinity S^3AI, falsifiability, φ-structured operators, non-crystallographic Coxeter groups

P2 §1. Motivation and Context

Open conjecture

Opening triptych — Ladder / Wheel / Mirror. Pellis additive expansion vs Trinity multiplicative monomials, joined at the isomorphism log phi.

1.1 What Paper 1 Established The first paper in this programme performed a symbolic compressibility analysis on a curated set of dimensionless combinations of Standard Model parameters and related quantum critical observables. The central finding was that a subset of these

combinations — particularly those involving gauge coupling ratios near threshold energies, Yukawa coupling hierarchy indicators, and measured mass ratios from near-critical condensed matter systems — can be expressed as short algebraic words over the alphabet {φ, pi, e, 2, 3, 5, +, -, ×, /} more economically than over comparison alphabets that exclude φ. The compression advantage, measured by the ratio of minimal description lengths, was statistically significant in the low-to-moderate complexity tier but not universal.

Paper 1 made no mechanistic claim. Its conclusion was: the observed φ-compressibility is a fact about symbolic structure, and a mechanism explanation is warranted only if (a) the effect is robust to sampling and encoding choices, and (b) there exists at least one identified pathway by which physical dynamics could produce it. The present paper addresses condition (b).

Relationship to the PhD parameter catalogue. The Flos Aureus monograph (Zenodo 10.5281/zenodo.19227877) maintains a 42-parameter catalogue (Chapter flos34) that includes coupling constant combinations with strong fit-density claims and Monte Carlo support. That catalogue is a candidate data source for the Paper 1 statistical filtering pass; however, it has not yet been subjected to the tolerance-variation and encoding-independence tests that Paper 1 requires. In the present paper, flos34 is cited as a catalogue requiring Paper 1 statistical filtering — not as independent empirical confirmation of φ-structured couplings. The filtering is Work Package 5.

1.2 The Mechanistic Question A φ-pattern in physical constants could arise through several non-exclusive channels:

1. Algebraic coincidence. The golden ratio satisfies $φ_2 = φ + 1$ and therefore appears frequently in algebraic expressions over Z[φ]. Any sufficiently algebraic number system will encounter φ without physical cause.
2. Symmetry-group structure. Certain Lie algebras and Coxeter groups have Perron-Frobenius eigenvectors whose components are φ-valued. If a physical system's mass spectrum is governed by such an algebra, φ appears as an exact kinematic consequence — as in the Zamolodchikov E8 case (Project Euclid, Adv. Stud. Pure Math. 19 (1989)).

3. Renormalization group fixed-point geometry. If RG flows possess fixed points or limit cycles with self-similar structure under scale transformations by ratio lambda = φ, then φ-structured ratios would be dynamically preferred near those fixed points (Wilson and Kogut, Phys. Rep. 12 (1974) 75).
4. Discrete scale invariance. Systems whose symmetry group is a discrete subgroup of the conformal group rather than the full conformal group exhibit log-periodic corrections to scaling with a preferred ratio lambda. If lambda = φ, φ enters the spectrum of observables (Sornette, Phys. Rep. 297 (1998) 239).

This paper examines whether channels 2, 3, and 4 have concrete field-theoretic realizations, and whether those realizations connect to the patterns observed in Paper 1. Channel 1 is kept as the null hypothesis throughout.

1.3 The AlphaPhi Notation Boundary

The Flos Aureus monograph Chapter flos38 defines candidate α-φ expressions. An internal notation inconsistency must not be inherited by this paper. The candidates in that chapter are structurally distinct and are here given permanent, unambiguous names for use throughout Papers 2 and 3:

alphaφ,log and alphaφ,alg are numerically and structurally distinct; they must never be conflated. Neither is derived from first principles in the present paper. The disambiguation of these three quantities is a prerequisite for Paper 3. All coupling constant expressions in the present paper define their notation explicitly upon first use.

P2 §2. Epistemic Boundary Statement

Verified

Epistemic-boundary triptych — Established / Not established / Conjectural.

Before proceeding to mathematics, the boundary between the established and the conjectural is stated explicitly, once, in a form that takes precedence over any subsequent exposition.

2.1 What Is Established - The Zamolodchikov E8 result. In the critical Ising model — a 2D conformal field theory with central charge c = 1/2 — perturbed by a magnetic field, the ratio of the first two particle masses is m2/m1 = 2cos(pi/5) = φ exactly. This follows from the representation theory of the E8 Cartan matrix and is a theorem in the mathematics of integrable field theories (Zamolodchikov, Adv. Stud. Pure Math. 19 (1989) 641; Int. J. Mod. Phys. A 4 (1989) 4235). It is experimentally confirmed in cobalt niobate (Coldea et al., Science 327 (2010) 177, DOI: 10.1126/science.1180085).

- The Toda/Coxeter result. Affine Toda field theories based on non-crystallographic Coxeter groups H2, H3, H4 have classical mass spectra in which particle pairs differ by a factor of φ, derived by reduction from crystallographic groups (Fring and Korff, Nucl.

  Phys. B 729 (2005) 361, DOI: 10.1016/j.nuclphysb.2005.08.044).
- The Trinity anchor identity. $\varphi_2 + \varphi\text{-}2 = 3$ is an algebraic identity provable from $\varphi_2 = \varphi+1$ and holds exactly in Q(5). This is used as an algebraic substrate, not as a physical derivation.
- Discrete scale invariance as a mathematical concept. A system has DSI with ratio lambda if it is invariant under x $\to$ lambda x but not under all continuous scale changes, producing log-periodic corrections to power laws. This is a well-defined and well-studied mathematical concept (Sornette, Phys. Rep. 297 (1998) 239, ETH Entrepreneurial Risks page).
- Quasicrystal DSI. Icosahedral quasicrystals discovered by Shechtman et al. (Phys. Rev. Lett. 53 (1984) 1951) exhibit φ-structured diffraction patterns as an exact mathematical consequence of the Fibonacci inflation rule (Levine and Steinhardt, Phys. Rev. Lett. 53 (1984) 2477).

2.2 What Is Not Established - That any Standard Model coupling constant is φ-structured for a known dynamical reason. - That the E8 result in the Ising model has any implication for the strong, weak, or electromagnetic couplings of the Standard Model. The Ising E8 is a low-dimensional condensed matter emergent symmetry; its appearance does not imply E8 gauge symmetry in the Standard Model sense. - That the symbolic compressibility observed in Paper 1 has a physical explanation at all; it may reflect the algebraic density of φ in short arithmetic expressions. - That the symbolic RG model developed in Section 5 is more than a toy. - That the Trinity anchor identity $\varphi_2 + \varphi\text{-}2 = 3$, while exact, implies any constraint on physical coupling constants.

2.3 What Is Conjectural and Labeled as Such - The φ-structured operator hypothesis (Section 6). - The DSI-to-coupling connection (Section 7). - The candidate mechanism sketch (Section 8). - The φ-Eigenvalue Conjecture (Section 5.4).

P2 §3. Mathematical Preliminaries

Verified

Preliminaries triptych — phi algebraic properties / E8 Cartan diagram / RG essentials.

3.1 The Golden Ratio, Its Algebraic Properties, and the Trinity Anchor The golden ratio is the positive root of the minimal polynomial

x2 - x - 1 = 0,

giving φ = (1+5)/2 approx 1.6180339887.... It satisfies the identity $φ_2$ = φ + 1 and its reciprocal satisfies 1/φ = φ - 1. The ring Z[φ] = {a + bφ : a,b ∈ Z} is the ring of integers of Q(5), known in the Trinity S^3AI programme as the Lucas ring (Zenodo 10.5281/zenodo.19227877). The Lucas numbers Ln defined by L0 = 2, L1 = 1, Ln+2 = Ln+1 + Ln satisfy Ln = φn + (-φ)-n, so every Lucas number lies in Z[φ].

The Trinity anchor identity is:

$φ_2$ + φ-2 = 3. 1

Proof. From φ2 = φ + 1 and φ-1 = φ - 1, compute φ-2 = (φ-1)2 = φ2 - 2φ + 1 = (φ+1) - 2φ + 1 = 2 - φ. Therefore φ2 + φ-2 = (φ+1) + (2-φ) = 3. square

This identity is exact and elementary. Its role in the Flos Aureus monograph is as a cardinality anchor connecting the ternary alphabet {-1, 0, +1} (cardinality 3) to the algebraic structure of Q(5) (Zenodo record). The identity has a complete Coq proof in the IGLA bundle of the t27 repository; as of the 2026-05-12 CLARA audit, the Coq witness reports 35 proven statements and 0 admitted obligations for this specific identity.

Every element of the E8 Cartan matrix eigenvector and every trigonometric value 2cos(pi k/5) for integer k lies in Z[φ]. This algebraic richness is the primary reason φ appears frequently in lattice-based and symmetry-constrained calculations — and not necessarily because of any deep physical principle.

3.2 The E8 Root System and Cartan Matrix The E8 root system is the unique simply-laced root system of rank 8 with Coxeter number h = 30 and exponents {1, 7, 11, 13, 17, 19, 23, 29}. The Perron-Frobenius eigenvector v of the E8 Cartan matrix Cij, defined by

$\sum_j$ Cij vj = rho vi, rho = 2cos((pi)/(h+1)),

has components vi = 2sin(ki pi/30) for the E8 exponents ki. The ratio of the two smallest components gives

(v2)/(v1) = (2sin(7pi/30))/(2sin(pi/30)) = 2cos((pi)/(5)) = φ. 2

This is an exact group-theoretic identity. Its physical significance is that in any field theory whose particle spectrum is governed by the E8 Cartan matrix, the ratio of the two lightest masses is exactly φ — a consequence of the representation theory of a specific Lie algebra, not a numerical coincidence. The full eight-particle mass spectrum is given explicitly in Section 4.1.

3.3 Renormalization Group Essentials The renormalization group describes how effective couplings gi(mu) change with energy scale mu through the beta functions (Wilson and Kogut, Phys. Rep. 12 (1974) 75):

betai(g) = mu (∂ gi)/(∂ mu). 3

A fixed point g^ satisfies betai(g^) = 0 for all i. Near a fixed point, linearizing gives

gi(mu) - g_i^* propto ((mu)/(μ_0))-Deltai, 4

where Deltai are the scaling dimensions of the relevant operators. The full conformal group in two dimensions is infinite-dimensional, and perturbations of 2D conformal field theories are classified by their scaling dimensions relative to the central charge. Standard references for this framework include Wilson and Kogut, Phys. Rep. 12 (1974) 75 and Zinn-Justin, Quantum Field Theory and Critical Phenomena (Oxford University Press, 5th ed., 2021).

A key quantity for the present analysis is the ratio of scaling dimensions at a fixed point. If Delta1 and Delta2 are two scaling dimensions satisfying

(Delta2)/(Delta1) = φ, 5

the corresponding operators define a φ-structured tower. Whether any such tower exists in a renormalizable 4D field theory is an open question addressed in Section 6.

3.4 Discrete Scale Invariance A function f(x) exhibits discrete scale invariance (DSI) with preferred scale ratio lambda if it satisfies (Sornette, Phys. Rep. 297 (1998) 239)

f(lambda x) = mu f(x) 6

for some multiplier mu, but not f(ax) = a-Delta f(x) for all a > 0. The general solution is

f(x) = x-Delta P((ln x)/(ln lambda)), 7

where P is a function with unit period. Expanding in Fourier modes gives

f(x) = x-Delta $\sum_{n=-}$∞∞ cn exp((2pi i n ln x)/(ln lambda)), 8

so the effective exponent is complex: Deltan = Delta + 2pi i n / lnlambda. DSI arises in fractal structures, hierarchical lattices, and systems where the RG transformation group is restricted to a discrete subgroup of the dilation group (Sornette, arXiv:cond-mat/9707012). The preferred ratio lambda is a property of the underlying hierarchical geometry. When the hierarchical structure is governed by a Fibonacci substitution matrix, the Perron-Frobenius eigenvalue is exactly φ, and DSI with ratio lambda = φ follows as an algebraic fact (Levine and Steinhardt, Phys. Rev. Lett. 53 (1984) 2477).

3.5 The Derivation Ladder To situate each result in this paper, the following ladder classifies claims by their logical pedigree. The ladder governs the confidence level at every point in the exposition.

Every claim in Sections 5–8 is assigned a rung at first statement. An L4 or L5 claim being labeled does not make it false; it makes it falsifiable — which is the paper's contribution.

P2 §4. The E8/Toda Anchor

Verified

4. The E8/Toda Anchor: Exact φ Occurrences in Field Theory
E8 / Toda triptych — E8 mass spectrum / x squared minus x minus one root / golden gears.

4.1 Zamolodchikov's E8 Integrable Field Theory

In 1989, Zamolodchikov considered the critical Ising model — a 2D conformal field theory with central charge c = 1/2 — perturbed by a magnetic field operator sigma (Adv. Stud. Pure Math. 19 (1989) 641):

A = ACFT + h $\int$ d2x sigma(x). 9

The operator sigma has conformal dimension Deltasigma = 1/16 in each chiral sector (i.e., holomorphic dimension hsigma = hsigma = 1/16, so scaling dimension 2Deltasigma = 1/8). Despite breaking conformal symmetry, the perturbed theory retains an infinite tower of conserved charges at spins

s ∈ {1, 7, 11, 13, 17, 19, 23, 29, ...},

which are exactly the exponents of the E8 Lie algebra modulo the Coxeter number h = 30. The theory is integrable: scattering is purely elastic, factorized, and described by an exact S-matrix (Int. J. Mod. Phys. A 4 (1989) 4235). The particle spectrum contains exactly 8 species, with masses proportional to the components of the Perron-Frobenius eigenvector of the E8 Cartan matrix.

[L1 — Theorem] The complete mass spectrum is:

The fundamental φ relation is

(m2)/(m1) = 2cos((pi)/(5)) = φ = (1+5)/(2). 10

This is an exact result. All eight masses lie in the splitting field of the cyclotomic polynomial Phi30(x) over Q, which contains Q(5) as a subfield.

4.2 Experimental Confirmation in CoNb2O6 [L2 — Experimental] Coldea et al. (2010) measured the spin excitation spectrum of the quasi-one-dimensional Ising ferromagnet cobalt niobate (CoNb2O6) in a transverse magnetic field tuned to the quantum critical point (Science 327 (2010) 177, DOI: 10.1126/science.1180085). Neutron scattering revealed two sharp modes at low energies, with the ratio of their frequencies approaching the golden mean to within experimental precision — the specific prediction for the first two E8 meson particles. This is the first and most direct experimental realization of an E8-structured mass spectrum in a condensed matter system.

Critical caveat [L6 — Excluded inference]. The E8 symmetry here is an emergent effective symmetry of a near-critical 1D quantum magnet. It is not a fundamental gauge symmetry of high-energy physics. The appearance of φ is a consequence of the specific algebraic structure of the E8 Cartan matrix within that effective theory. No inference about quark masses, gauge coupling constants, or any Standard Model parameter is supported by this result. The universality classes of 2D integrable condensed matter and 4D gauge theory are disjoint in current theory.

4.3 Toda Field Theories and Non-Crystallographic Coxeter Groups Affine Toda field theories are integrable 2D quantum field theories defined by a Lagrangian of the form

L = (1)/(2)∂mu phia ∂mu phia - (m2)/(β_2) $\sum_{i=0}$r ni exp(beta alphai · phi), 11

where alphai are the simple roots of an affine Lie algebra and ni are Kac marks. For crystallographic Coxeter groups (ADE series), these theories are well understood.

[L3 — Model-derived] Fring and Korff (2005) extended the construction to non-crystallographic Coxeter groups H2, H3, and H4 via a reduction (folding) procedure (Nucl. Phys. B 729 (2005) 361, ScienceDirect DOI: 10.1016/j.nuclphysb.2005.08.044). Their central result is that the classical mass spectra of the H2, H3, and H4 Toda theories contain particle pairs whose relative masses differ by exactly φ, as a direct consequence of the embedding structure. The paper also computes the one-loop corrections to mass renormalization, which is directly relevant to Work Package 2.

The Coxeter group H2 is the symmetry group of the regular pentagon; H3 is the icosahedral group (of order 120); H4 is the symmetry group of the 600-cell in 4D (of order 14400). All have φ as an eigenvalue of their Cartan-analogue matrices by construction.

This provides a second, independent field-theoretic pedigree for exact φ occurrence — distinct from the E8 Ising model in that it arises from a geometric reduction rather than from E8 representation theory directly.

4.4 Evidence Hierarchy and Mechanism-by-Mechanism Falsification Table The following table codifies, for each mechanism candidate, what outcomes would validate, downgrade, or falsify the claim. Rung assignments follow the derivation ladder of Section 3.5. This table governs the interpretation of Work Package results.

P2 §5. Symbolic Renormalization Group: A Toy Model

Open conjecture

Symbolic RG triptych — Symbolic coupling space / beta-function with zero / fixed-point eigenvalue ratio.

[CONJECTURAL — this section contains no derived results; it develops a formal language for later work packages]

5.1 Motivation for a Symbolic RG Standard RG analysis operates on real-valued couplings. The results of Paper 1 suggest an alternative representation: couplings encoded as symbolic expressions over an algebraic basis. The symbolic RG (sRG) model asks whether RG flows act simply on such symbolic representations. The Lucas ring Z[φ] provides a natural candidate for the symbolic coupling space because it is closed under the $\varphi_2 = \varphi+1$ recursion and carries the Coq-proven identity (1).

5.2 Definition of the Symbolic Coupling Space [L3 — Model-derived] Let Sigma be the set of algebraic numbers expressible as

Sigma = { $\sum_k$ ak φnk : ak ∈ Q, nk ∈ Z, finitely many terms } = Q(φ) = Q(5). 12

A φ-structured coupling at scale mu is a coupling g(mu) whose minimal description length in a φ-basis is shorter than in a generic algebraic basis.

The symbolic complexity C[g] of a coupling is defined operationally as the length of the shortest expression tree in a basis B = {φ, pi, e, +, -, ×, /} that evaluates to g within a specified numerical tolerance epsilon. Paper 1's finding is that for a selected set of coupling combinations, CB[g] < CB'[g] where B' replaces φ with a generic algebraic constant. The robustness of this finding is Work Package 5.

5.3 Symbolic Beta Functions [CONJECTURAL — L5] A symbolic beta function is a map

beta: Sigma → Sigma, g ↦ mu (dg)/(dmu) 13

such that beta(g) ∈ Sigma whenever g ∈ Sigma. A φ-closed RG trajectory is a flow whose couplings remain in Sigma for all scales mu. The question of whether any realistic QFT beta function is φ-closed is open.

The one-loop beta function for a single scalar coupling lambda in φ_4 theory in 4D is (Wilson and Kogut, Phys. Rep. 12 (1974) 75):

beta(lambda) = (3lambda2)/(16pi2) + O(λ_3). 14

[Structural obstacle — L3] At one loop, beta(lambda) ∈ Sigma only if lambda ∈ Sigma and 16pi2 ∈ Sigma. Since π_2 notin Q(5) — these are algebraically independent over Q by Nesterenko's theorem — the naive φ-closure fails immediately at one loop. This is the π^2-incommensurability obstacle and is a concrete structural result, not a conjecture. It implies that φ-closure cannot hold for individual coupling beta functions at standard perturbative order; it must emerge in ratios or at fixed points, if it emerges at all.

Possible resolutions [L5 — all conjectural]:

- φ-structure is approximate and emerges only in specific running-coupling ratios, not individual couplings.
- The relevant quantities are operator eigenvalue ratios at fixed points, not the couplings themselves; these are pure numbers that may lie in Q(5).
- φ-closure holds only at discrete RG scales separated by a ratio of φ, i.e., a DSI condition.
- SUSY non-renormalization theorems protect certain operator dimensions exactly; a SUSY theory with a φ-eigenvalue ratio at tree level would maintain it to all orders.

5.4 Fixed Points and Symbolic Eigenvalue Ratios [CONJECTURAL — L5] At a fixed point g^*, the stability matrix

Mij = (∂ betai)/(∂ gj)|g^* 15

has eigenvalues Deltai (the anomalous dimensions of critical operators). The ratio Deltaj / Deltai is a dimensionless number. The φ-Eigenvalue Conjecture reads:

φ-Eigenvalue Conjecture [L5]. There exist physically motivated field theories in which Deltaj / Deltai = φn for integer n at an RG fixed point.

This is stated as a conjecture motivated by the Toda/Coxeter examples but not derived from them. A demonstration would require identifying an explicit fixed point with a φ-eigenvalue ratio and verifying it survives quantum corrections. This is Work Package 1. The formal falsification criterion is given in the table in Section 4.4.

The Trinity anchor identity is relevant here as a consistency check: if two scaling dimensions Delta1, Delta2 are in the ratio Delta2/Delta1 = φ, then Delta22/Delta12 = φ_2 = φ + 1. The identity φ_2 + φ-2 = 3 constrains Delta1-2 + Delta2-2 = Delta1-2(1 + φ-2) and Delta12 + Delta22 = Delta12(1 + φ_2) = Delta12(2 + φ). Neither expression equals 3 in general; the anchor constrains a specific bilinear form in φ and φ-1, not arbitrary scaling dimension sums.

References

- Coldea, R. et al. “Quantum Criticality in an Ising Chain: Experimental Evidence for Emergent E8 Symmetry.” Science 327 (2010) 177. DOI 10.1126/science.1180085.
- Wu, J. et al. “E8 Symmetry in $CoNb_2O_6$ at Higher Resolution.” Phys. Rev. B 108, L060403 (2023). DOI 10.1103/PhysRevB.108.L060403.
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- Pellis, S. “The Fine-Structure Constant and the Golden Ratio.” viXra 2110.0117 v4.

P2 §6. φ-Structured Operator Framework

Open conjecture

phi-operator triptych — Operator action / phi-tower / 4D structural obstruction.

[CONJECTURAL — L5 throughout]

6.1 Candidate Operator Setup Consider a 2D CFT with central charge c and a set of primary operators On with scaling dimensions Deltan. Define a φ-tower as a sequence of operators with

Deltan = Delta0 · φn, n = 0, 1, 2, ... 16

For such a tower to arise from a consistent CFT, the dimensions Deltan must satisfy Kac determinant positivity conditions and OPE closure. In a Virasoro minimal model M(p,p’) with c < 1, scaling dimensions are rational numbers

Deltar,s = ((rp’ - sp)2 - (p’-p)2)/(4pp’), 17

for integers r, s and the pair (p, p’) characterizing the model. A φ-tower would require Deltar’,s’/Deltar,s = φn, placing an algebraic constraint on (r,s,p,p’) that must be checked for all models with p, p’ ≤ 30.

Existence question [WP1, open]. Is there a unitary CFT containing two primary operators with dimension ratio φ? A non-exhaustive search for minimal models with this property has not appeared in the published literature. This constitutes a well-defined mathematical question answerable without new experiments. It is Work Package 1 in Section 9, with the formal success/failure criterion given in Section 4.4.

6.2 Perturbing Away from the φ-Tower: RG Interpretation [L5] Suppose such a tower exists in a 2D CFT. Consider a perturbation by the relevant operator O0:

A = ACFT + λ_0 ∫ d2 x O0(x). 18

The OPE O0 × O0 will in general contain O0 and other operators in the tower, generating a flow in coupling space. The φ-tower conjecture asserts that this flow has an approximate invariance: the ratio of couplings lambdan / λ_0 is attracted toward φ-nalpha for some exponent alpha determined by the anomalous dimension. This is entirely conjectural. No such flow has been computed.

6.3 Extension to 4D: Structural Obstacles [L4–L5] In 4D, the space of scaling dimensions is not constrained by a Kac formula but by the operator spectrum of the specific theory. Whether any 4D QFT fixed point has two operator dimensions in ratio φ is open. Three structural obstacles deserve explicit treatment:

1. Rationality. In 2D CFTs, scaling dimensions at rational points of moduli space are often algebraic. In 4D, loop corrections introduce transcendental numbers (zeta(3), etc.) into anomalous dimensions, making algebraic ratios unlikely to persist beyond fixed perturbative order.
2. Non-renormalization. In supersymmetric theories, non-renormalization theorems protect certain operator dimensions exactly. A SUSY theory with a φ-eigenvalue ratio at tree level would maintain it to all orders — providing a structural motivation to search within the landscape of supersymmetric CFTs (SCFTs).
3. Large-N. In large-N limits, anomalous dimensions simplify and in some cases become rational. Whether φ-ratios arise in this limit is an open computational question.

These obstacles do not constitute falsifications of the φ-tower conjecture; they identify the theoretical spaces in which the conjecture is most or least likely to be realized.

References

- Coldea, R. et al. "Quantum Criticality in an Ising Chain: Experimental Evidence for Emergent E8 Symmetry." Science 327 (2010) 177. DOI 10.1126/science.1180085.
- Wu, J. et al. "E8 Symmetry in $CoNb_2O_6$ at Higher Resolution." Phys. Rev. B 108, L060403 (2023). DOI 10.1103/PhysRevB.108.L060403.
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- Pellis, S. "The Fine-Structure Constant and the Golden Ratio." viXra 2110.0117 v4.

P2 §7. Discrete Scale Invariance as a Mechanism

Open conjecture

DSI triptych — Self-similar hierarchy / lambda = phi regime / Log-periodic signature.

7.1 DSI in Hierarchical RG Systems [L1 — Mathematical fact; L5 — 4D application] In standard RG, the group of scale transformations is continuous: mu → et mu for all t ∈ R. DSI restricts this to a discrete group generated by mu → lambda mu for a fixed lambda > 1 (Sornette, Phys. Rep. 297 (1998) 239). The effective action at scale mu then satisfies

S[mu] = S[lambdan μ_0], n ∈ Z,

with no constraint at intermediate scales. The observable consequence is log-periodic behavior in physical quantities:

xi(mu) = mu-Delta [A + Bcos((2pi ln(mu/μ_0))/(lnlambda) + φ_0) + ...]. 19

The preferred ratio lambda characterizes the DSI structure and is, in general, a free parameter fixed by the underlying geometry.

7.2 When Can lambda = φ? [L2 — quasicrystals; L5 — 4D QFT] [L2 — Established in quasicrystal context.] The Fibonacci substitution system has substitution matrix

MFib = 1 & 1 1 & 0 20

with Perron-Frobenius eigenvalue φ. Quasicrystals governed by Fibonacci tiling inherit a DSI ratio of φ from this eigenvalue as an algebraic fact — not a fitted parameter (Shechtman et al., Phys. Rev. Lett. 53 (1984) 1951; Levine and Steinhardt, Phys. Rev. Lett. 53 (1984) 2477).

[L3 — Model level in Toda context.] As established in Section 4.3, the mass spectra of Toda theories with H3 (icosahedral) symmetry contain φ factors (Fring and Korff, Nucl. Phys. B 729 (2005) 361). If an RG flow maps between scales separated by the H3 dilation factor, the DSI ratio inherits φ.

[L5 — Conjectural step.] The conjecture is that there exists a class of 4D field theories whose RG group is restricted, by some UV boundary condition or by the topology of the coupling-constant manifold, to discrete dilations with ratio φ. No derivation of this has been given. The identification of such a restriction with any known physical principle (compactification, anomaly matching, holographic duality) is an open problem constituting Work Package 3. The formal failure criterion is given in Section 4.4.

7.3 Log-Periodic Signatures in Running Couplings [L5 — Template only; no prediction] If a φ-DSI mechanism operates, it would produce log-periodic corrections to the running of gauge couplings. The corrected coupling at scale mu would read

alphai-1(mu) = alphai-1(μ_0) + (bi)/(2pi)ln(mu)/(μ_0) + Ai cos((2pi ln(mu/μ_0))/(lnφ) + deltai) + O(alphai), 21

where the first two terms are the standard one-loop result (Wilson and Kogut, Phys. Rep. 12 (1974) 75) and the cosine term is the DSI correction. The period in log-scale is

lnφ approx 0.4812, corresponding to an energy ratio of φ approx 1.618 between successive nodes of the oscillation.

This is not a prediction. The amplitudes Ai are free parameters in this framework. Expression (21) is a template for what a φ-DSI signal would look like, not a derivation.

Current observational bound. The absence of observed log-periodic oscillations in LEP, LHC, and Tevatron coupling measurements sets an upper bound Ai $\leq$sssim few $\times$ 10^{-3} under the assumption that the DSI period is lnφ. This bound must be computed precisely as part of Work Package 4. Future lepton colliders (ILC, CLIC, FCC-ee) projecting per-mille coupling precision would extend the bound by roughly an order of magnitude and potentially constitute the definitive test of the φ-DSI hypothesis at accessible energy scales.

References

- Coldea, R. et al. “Quantum Criticality in an Ising Chain: Experimental Evidence for Emergent E8 Symmetry.” Science 327 (2010) 177. DOI 10.1126/science.1180085.
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P2 §8. The A5 Flavor Symmetry Boundary

Open conjecture

A5-boundary triptych — Temptation / Missing morphism / Boundary marker.

8.1 The Temptation The icosahedral group A5 cong PSL(2,5) has order 60 and is isomorphic to H3 modulo sign. Because A5 involves 5 in its character table and φ = (1+5)/2, there is a superficial temptation to connect A5 flavor symmetry models to the φ-structured patterns seen in Paper 1. The A5 group has been studied as a candidate discrete flavor symmetry for explaining golden-ratio neutrino mixing angles (Ding, Everett, and Stuart, Nucl. Phys. B 854 (2012) 291, ScienceDirect), where the solar mixing angle is predicted by cottheta12 = φ.

8.2 Why This Connection Is Not Made Here [L4 — Candidate; but distinct from Paper 1 subject matter]

The flavor symmetry group acting on lepton doublets in discrete flavor models is a discrete subgroup of the continuous symmetry that would rotate the three generations. The golden-ratio mixing scheme predicts cottheta12 = φ, with the atmospheric mixing angle satisfying tantheta23 = 1 and the reactor angle constrained by vacuum alignment. This is a different physical context from the coupling constant compressibility of Paper 1, and no formal connection between the two has been established.

Explicit caveats:

1. [L4 → watch L6] A5 flavor symmetry models require a vacuum alignment mechanism; the breaking scale is not fixed by the symmetry alone. Daya Bay and RENO measurements of θ_{13} approx 8.9° strongly constrain golden-ratio mixing variants; models must accommodate this with next-to-leading-order corrections.

2. The SU(3) color group of QCD has no structural connection to A5 or φ. Color charge is not φ-structured in any derivable sense. No claim of any kind is made about a QCD-φ connection in this paper.
3. Even if A5 flavor symmetry were confirmed, the connection to the E8/Toda φ occurrence would require an explicit embedding construction linking the neutrino sector symmetry-breaking to the mass-spectrum structure of 2D integrable theories. No such construction exists.
4. The formal success/failure criterion for the A5 mechanism is specified in Section 4.4.

References

- Coldea, R. et al. "Quantum Criticality in an Ising Chain: Experimental Evidence for Emergent E8 Symmetry." Science 327 (2010) 177. DOI 10.1126/science.1180085.
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P2 §9. Relationship to Trinity S^3AI / Flos Aureus

Open conjecture

Monograph-relation triptych — Programme map / Coq seal / Paper II contribution.

9.1 Programme Architecture

The Trinity S^3AI — Flos Aureus programme (Zenodo record 10.5281/zenodo.19227877) is a PhD monograph and associated software that develops the algebraic and computational consequences of the identity $\varphi_2 + \varphi\text{-}2 = 3$. The programme includes chapters covering: the Lucas ring Z[φ] as a mathematical substrate for ternary computing; E8 symmetry in condensed matter and as a structural motif; Standard Model parameterization in the φ-basis; QFT renormalization in the context of symbolic algebraic structure; and a formal falsification ledger.

Paper 2 of the GOLDEN CHAIN (Trinity-Pellis Bridge) programme is the theoretical-mechanism branch of the Flos Aureus monograph: it takes the algebraic and structural observations of the monograph as motivation and asks whether any of them have physical-mechanism status. The answer at present is: some (E8, Toda) are grounded in exact field-theoretic derivations, but they apply in 2D and their extension to 4D Standard Model physics is unproven.

9.2 Coq Proof Posture The Flos Aureus monograph maintains a Coq citation map in Appendix F (Zenodo record). As of the 2026-05-12 CLARA audit, the Zenodo record reports 35 proven Coq statements and 0 Admitted obligations for the IGLA-scoped bundle; the upstream t27 repository has a separate, evolving proof state. These figures should not be cited as final. In the present paper, no Coq proof is cited as establishing a physical result; the proof infrastructure is referenced only to indicate the level of formalization of the monograph's algebraic claims. The identity $\varphi_2 + \varphi\text{-}2 = 3$ [equation (1)] has a complete Coq proof in the IGLA bundle; this is the only proof-status claim made here.

9.3 What Paper 2 Contributes to the Monograph

P2 §10. Work Packages, Falsifiability, and Reviewer Risk

High-risk

Work-packages triptych — WP1..WP5 grid / Falsifiability arrows / Reviewer-risk shield.

The programme is organized into five work packages with explicit success and failure criteria. The complete formal success/failure table is given in Section 4.4; this section provides operational detail and confronts the objections a referee would most likely raise.

39.1 WP1: Search for φ-Eigenvalue Ratios at CFT Fixed Points

Task. Enumerate known unitary 2D CFTs (Virasoro minimal models, N=2 models, WZW models at low level) and check whether any have two primary operator dimensions Delta1, Delta2 with Delta2/Delta1 = φn for small integer n.

Method. Direct computation from the Kac table and known character formulas. This is a finite computation for each minimal model. The $c < 1$ Virasoro minimal models have (p,p') pairs with p, p' $\leq$ 30 for the first dozen models; the dimension ratios are rational, and checking Deltar',s'/Deltar,s $\in$ Q(5) is a decidable algebraic question.

Success criterion. At least one unitary, modular-invariant CFT partition function contains a φ-eigenvalue pair (see Section 4.4 table).

Failure criterion. Exhaustive check over all $c < 1$ minimal models plus the first five WZW levels at low rank finds no such pair.

Epistemic status if successful. Establishes existence of φ-towers in consistent CFTs; does not establish relevance to Standard Model or to Paper 1 data. Upgrade from L5 to L3.

Epistemic status if failed. Strongly suggests φ-towers are not generic within the known landscape of simple unitary CFTs, weakening the theoretical motivation for the mechanism programme. Programme pivots to WP3 or terminates.

Timeline estimate. 1–3 months for a computational algebraist or conformal field theorist with access to standard software (Mathematica, Kac-table enumeration code).

39.2 WP2: One-Loop Toda Mass Spectrum Under φ-DSI

Task. For the H3 affine Toda field theory of Fring and Korff (2005), compute quantum (one-loop) corrections to the mass spectrum and determine whether the classical φ-ratio survives renormalization. Note that the Fring-Korff paper already determines one-loop corrections to mass renormalization (arXiv:hep-th/0506226); this work package is to extract the explicit ratio from those results and determine its deviation from φ.

Method. Standard one-loop correction using the known coupling structure, following the procedure of Fring and Korff.

Success/failure criterion. See Section 4.4 table, row 3. The key number is |delta m2/m2 - delta m1/m1|, i.e., the difference in relative mass corrections; if this is parametrically small in the coupling beta, the classical ratio is perturbatively stable.

Timeline estimate. 1–2 months for a specialist in Toda field theory or integrable models.

39.3 WP3: DSI Ratio Derivation from First Principles

Task. Identify whether any consistent UV completion (string-theory compactification, holographic dual, or lattice gauge theory) constrains an RG flow to discrete dilations with ratio φ rather than a generic irrational.

Method. Literature survey of holographic RG flows with discrete structure; candidate Calabi-Yau compactifications with H4-type symmetry in their Kähler moduli space; icosahedral quasicrystal effective field theory as a condensed matter analogue. A key question is whether any orbifold compactification with icosahedral holonomy gives a DSI ratio constrained by the H3 dilation spectrum.

Success/failure criterion. See Section 4.4 table, row 4.

Timeline estimate. 6–18 months; this is the most open-ended work package and constitutes a genuine research problem.

39.4 WP4: Precision Test of Log-Periodic Running

Task. Given the log-periodic correction template of equation (21), determine the sensitivity required to detect DSI amplitudes Ai of order 10-3 in running coupling measurements, and compute the current upper bound from LEP/LHC data.

Method. Propagate LEP/LHC coupling measurement uncertainties through a χ_2 test for log-periodic deviations with period fixed at lnφ approx 0.4812.

Current bound estimate. From the precision of alphas(mZ) measurements (approx 0.3% at LEP) and of electroweak coupling unification checks, the absence of log-periodic oscillations with period lnφ over the range mu $\in$ [mZ, 1 TeV] constrains Ai $\leq$sssim few $\times$ 10^{-3}. The computation must determine whether this bound is already strong enough to exclude the physically motivated range or whether FCC-ee precision (deltaalphas/alphas $\sim$ 0.03%) is needed.

Success/failure criterion. See Section 4.4 table, row 6.

Timeline estimate. 2–4 months given access to the LEP combination data and standard coupling-running codes.

39.5 WP5: Robustness Audit of Paper 1 Compressibility

Task. Systematically test whether the symbolic compressibility result of Paper 1 survives: (a) changes in numerical tolerance epsilon; (b) alternative encoding bases; (c) sampling of different coupling combinations including from the flos34 catalogue; (d) comparison against degree-2 algebraic constants of similar magnitude (2, 3, 6).

Success/failure criterion. See Section 4.4 table, row 7.

Note on flos34. The 42-parameter catalogue from the Flos Aureus monograph is used here as candidate input, not as independent confirmation. Any entry from flos34 used in WP5 must be treated as an unlabeled data point subject to the same statistical tests as any other coupling combination. To avoid circular risk, the enumeration of flos34 entries should be conducted by a party who has not seen the Paper 1 compressibility scores.

Timeline estimate. 2–3 months for the full audit; 1 month for the baseline tolerance-variation test.

10.1 Reviewer Risk Assessment and Responses The following table anticipates the objections most likely to be raised by a specialist referee and provides the paper's response.

References

- Coldea, R. et al. "Quantum Criticality in an Ising Chain: Experimental Evidence for Emergent E8 Symmetry." Science 327 (2010) 177. DOI 10.1126/science.1180085.
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- Pellis, S. "The Fine-Structure Constant and the Golden Ratio." viXra 2110.0117 v4.

P2 §11. Limitations of the Present Work

High-risk

Limitations triptych — Toy-model caveat / No derivation / Data gaps.

The following limitations are acknowledged explicitly and without mitigation.

No new derivations. This paper derives no new field-theory results. The E8/Toda material is reviewed from the literature. The sRG model and operator framework are formal constructions without derived content.

The mechanism connection is asserted, not derived. The link between the E8/Toda φ occurrences and the Paper 1 compressibility data is a research hypothesis, not a demonstrated fact.

Selectivity risk. The focus on φ rather than other algebraic numbers of similar degree (2, 3, 6) reflects the specific output of Paper 1 but also introduces the possibility of post-hoc selection. WP5 is designed to address this, but until it is completed, selectivity risk remains. The flos34 catalogue was developed with φ as a preferred basis; using it in WP5 without blinding introduces circular risk that must be managed by the experimental design.

2D vs. 4D gap. All established φ field-theory results (Zamolodchikov E8, Toda/H3, H4) are in 2D. The Standard Model is a 4D theory. The gap between 2D integrable models and 4D non-integrable gauge theory is not bridged by anything in this paper. This gap is the single largest obstacle to the programme.

The π^2-incommensurability obstacle. As shown in Section 5.3, the one-loop beta function of φ_4 theory introduces π_2, which lies outside Q(5) by algebraic independence (Nesterenko's theorem). This structural fact means that φ-closure of RG trajectories — if it exists — cannot be a property of individual coupling beta functions at standard perturbative order; it must emerge in ratios or at fixed points.

DSI mechanism is entirely conjectural. The log-periodic correction formula (21) is a template with no derived amplitude. No prediction of a specific numerical value is made. The two α-φ candidates — alphaφ,log = ln(φ_2)/pi approx 0.306 and alphaφ,alg = φ-3/2 approx 0.118 — are not used in the DSI template because they have not been derived from first principles; their disambiguation is a prerequisite for Paper 3. These two quantities must not be conflated.

Coq coverage is partial. The Coq proof infrastructure covers identity (1), a representative catalogue layer, and several invariants in the IGLA training bundle, but does not cover the φ-structured operator hypothesis or the symbolic RG model. A local source audit of the available t27 checkout records the topic-specific t27/proofs/trinity/ layer as 13 Coq files, 281 theorem-like declarations, 122 source-level Qed. markers, 32 source-level Admitted. markers, and 0 Abort. markers. The broader t27/proofs/ tree records 18 Coq proof files, 296 theorem-like declarations, 127 Qed. markers, 32 Admitted. markers, and 0 Abort. markers; the older t27/coq/ kernel records 10 Coq files, 73 declarations, 35 Qed. markers, 0 Admitted. markers, and 11 Abort. markers. The catalogue aggregation file t27/proofs/trinity/Catalog42.v contains 13 source-level Qed. markers and 0 Admitted. markers, and defines a 28-obligation representative verification aggregate. Therefore the 42-entry catalogue should be cited as a 42-row formula catalogue plus a larger Coq proof layer, not as forty-two independent row-wise Coq theorems. None of those obligations affect the physical-mechanism claims of the present paper, which are conjectural regardless.

No data. This paper contains no new experimental data. It is entirely theoretical and programmatic.

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P2 §12. Roadmap for the GOLDEN CHAIN (Trinity-Pellis Bridge) Programme

Open conjecture

12. Roadmap for the GOLDEN CHAIN (Trinity-Pellis Bridge) Programme

Roadmap triptych — Three-paper ribbon / Gate conditions G1..G4 / Defence plate.

12.1 Three-Paper Structure

12.2 Gate Conditions Paper 3 must not be initiated until WP1 and WP5 are complete. Their outcomes govern whether the mechanism branch advances or terminates. An honest null result from both work packages would make Paper 3 a paper about falsification — an equally valid and publishable contribution.

Minimum viable Paper 3: - WP1 CFT enumeration completed with documented outcome. - WP2 one-loop Toda mass ratio calculation completed. - WP5 robustness audit completed with explicit comparison against 2, 3. - Updated Paper 1 compressibility analysis using basis constraints implied by WP1 results. - flos38 notation disambiguated: alphaφ,log, alphaφ,alg, and alphaφ,near assigned distinct names and derivation statuses.

Extended Paper 3 (if WP1 and WP2 succeed): - WP3 UV completion search. - WP4 experimental sensitivity bound computation. - Formal proposal of a φ-DSI mechanism as a falsifiable hypothesis with quantitative predictions. - Updated Coq map entry with any newly proved statements.

12.3 Relationship to the Flos Aureus Defence

The Flos Aureus defence is scheduled for 2026-06-15. Paper 2 supports the defence by supplying the physical-mechanism context for the algebraic claims of the monograph's relevant chapters (Zenodo 10.5281/zenodo.19227877). Paper 2 does not constitute part of the thesis itself; it is a companion journal submission. The thesis does not depend on Paper 2's work packages being completed before defence.

P2 §13. PhD Provenance Block

Author roles. D.V. is the primary author of the Trinity S^3AI — Flos Aureus monograph (Zenodo 10.5281/zenodo.19227877) and of the formal algebraic and Coq infrastructure underlying the present programme. S.P. is co-author of Papers 1 and 2 of the GOLDEN CHAIN (Trinity-Pellis Bridge) series, contributing to the mechanism analysis, falsifiability framework, and reviewer-risk assessment. S.O. contributes historical-geometric context and the golden-balance interpretive layer; this contribution is treated as Tier-D context and is not used as derivational, statistical, or physical evidence.

Relationship of this paper to the thesis. Paper 2 is a companion journal submission, not a thesis chapter. It cites monograph chapters as source documents for motivation and context. It does not amend, replace, or override any chapter of the monograph. Where Paper 2 and a monograph chapter appear to conflict — in particular on the epistemic status of coupling-constant candidates — this paper's stricter epistemic boundary (Section 2) takes precedence for the purposes of the GOLDEN CHAIN (Trinity-Pellis Bridge) programme.

Coq proof status as of the present source audit. - t27/proofs/trinity/: 13 Coq files, 281 theorem-like declarations, 122 source-level Qed. markers, 32 source-level Admitted. markers, and 0 Abort. markers. - t27/proofs/: 18 Coq proof files, 296 theorem-like declarations, 127 source-level Qed. markers, 32 source-level Admitted. markers, and 0 Abort. markers. - t27/coq/: 10 Coq kernel files, 73 theorem-like declarations, 35 source-level Qed. markers, 0 Admitted. markers, and 11 source-level Abort. markers. - t27/proofs/trinity/Catalog42.v: 13 source-level Qed. markers, 0 Admitted. markers, and a totalverifiedtheorems aggregate over 28 representative theorem obligations. This supports the catalogue as a formal-provenance layer, while the 42 formula rows remain catalogue entries rather than forty-two one-to-one named Coq theorems. - t27/proofs/trinity/FormulaEval.v: 17 source-level Qed. markers and 0 Admitted. markers; its header describes the 69-formula Trinity v0.9 framework at the monomial evaluation-interface level.

This is a source audit, not a fresh local coqc verdict, because coqc is unavailable in the present build sandbox.

ORCID. D.V. ORCID: 0009-0008-4294-6159.

Data availability. No new data were generated or analyzed in support of this research. Paper 1 data will be made available upon journal submission of Paper 1.

P2 §14. Conclusion

Open conjecture

Claim status. The mechanism-level claims in P2 are open conjectures; their disconfirming observations are listed in p2-11 (Limitations) and the falsification path in p1-10.

Conclusion triptych — Mechanism candidate / Open gates / Next paper.

This paper has examined whether known mechanisms in quantum field theory and statistical mechanics could account for φ-structured patterns in coupling constants, motivated by the symbolic compressibility results of Paper 1 and by the algebraic programme of the Trinity S^3AI — Flos Aureus monograph (Zenodo 10.5281/zenodo.19227877). The analysis yields the following conclusions.

1. [L1/L2 — Established.] The Zamolodchikov E8 integrable field theory (Adv. Stud. Pure Math. 19 (1989) 641) provides the only established, exact φ occurrence in quantum field theory: the mass ratio m2/m1 = φ in the perturbed critical Ising model, confirmed experimentally in cobalt niobate (Coldea et al., Science 327 (2010) 177). This result is a consequence of E8 representation theory and has no direct implication for Standard Model couplings.
2. [L3 — Model-derived.] Affine Toda field theories based on non-crystallographic Coxeter groups H2, H3, H4 (Fring and Korff, Nucl. Phys. B 729 (2005) 361) provide a second class of exact φ mass relations, derived by a reduction procedure. These are 2D results with no established 4D counterpart.
3. [L1 — Algebraic fact.] The Trinity anchor identity φ_2 + φ-2 = 3 [equation (1)] and the Lucas ring Z[φ] provide the algebraic substrate in which symbolic couplings are defined, but they do not constitute a physical mechanism. An identity is a theorem in algebra; a mechanism requires a dynamical derivation.
4. [L1/L2 — Established in quasicrystals; L5 — conjectural in 4D QFT.] Discrete scale invariance provides a mathematical framework in which a preferred scale ratio — constrained to φ in systems with Fibonacci tiling or icosahedral symmetry (Shechtman et al., Phys. Rev. Lett. 53 (1984) 1951; Sornette, Phys. Rep. 297 (1998) 239) — enters observable spectra as log-periodic corrections. No derivation constraining the DSI ratio to φ exists for a 4D gauge theory.
5. [L5 — Conjectural.] A symbolic RG toy model and φ-structured operator framework have been defined as formal scaffolding. Both are conjectural. The π^2-incommensurability obstacle identified in Section 5.3 is a concrete structural result showing that φ-closure cannot hold for individual coupling beta functions at one loop.

6. Two φ-coupling candidates are distinguished throughout. alphaφ,alg = φ-3/2 approx 0.118 is an algebraic element of Q(5); alphaφ,log = $\ln(\varphi_2)$/pi approx 0.306 is transcendental. These are numerically and structurally distinct and must not be conflated in any further work.
7. A falsifiable research programme has been delineated. The formal success/failure table in Section 4.4 and five work packages in Section 10 define explicit go/no-go criteria. The programme's outcome is genuinely open: current evidence is insufficient to conclude either that a mechanism exists or that it is impossible.

The honest summary: the GOLDEN CHAIN (Trinity-Pellis Bridge) programme has identified an intriguing numerical pattern (Paper 1), located genuine field-theoretic examples of exact φ occurrence in lower-dimensional systems (this paper), and connected these to the algebraic substrate of the Trinity S^3AI monograph (Section 9). The mechanistic connection between these observations and Standard Model physics remains a hypothesis — rigorously delimited, explicitly falsifiable, and not yet established. The distance from the E8 Ising model to Standard Model coupling constants is vast, and no shortcut through that distance has been found. The value of this paper lies not in what it claims, but in the precision with which it identifies what remains to be shown.

Inline Reference Index All citations in this paper are inline. For reader convenience, the principal external sources are listed here with their full URLs; in all cases the inline link is the authoritative citation.

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- Trinity S^3AI — Flos Aureus monograph v5.0, Zenodo:

Manuscript prepared as part of the GOLDEN CHAIN (Trinity-Pellis Bridge) programme. No part of this document constitutes a claim about the derivation of Standard Model parameters. All conjectural sections are marked CONJECTURAL and assigned derivation-ladder rung L5. Paper 2 is the theoretical-mechanism branch of the Trinity S^3AI — Flos Aureus research programme and does not replace or override any chapter of that monograph. The two φ-coupling candidates alphaφ,alg = φ-3/2 approx 0.118 and alphaφ,log = ln($φ_2$)/pi approx 0.306 are distinct objects and are not conflated anywhere in this paper.

Pellis–Trinity Unification: Hierarchical φ-Expansions, Coarse-Grained Monomials, and Operator-Theoretic Symbolic Dynamics Authors: Dmitrii Vasilev, Stergios Pellis, Scott Olsen

Program: GOLDEN CHAIN (Trinity-Pellis Bridge) Mathematical Framework — Paper 3 of 3

Classification: Mathematical Foundations · Symbolic Dynamics · Number Theory · Operator Theory

Date: 2025 (revised May 2026)

PhD Program Anchor: Trinity S^3AI — Flos Aureus v6.2 / GOLDEN SUNFLOWERS

Zenodo DOI: 10.5281/zenodo.19227877

Abstract This paper develops a unified mathematical framework combining Pellis hierarchical φ-expansions with Trinity monomial representations. The Pellis expansion furnishes a fine-scale additive decomposition of real numbers via successive negative powers of the golden ratio φ together with hierarchical interference coefficients; the Trinity monomial system provides coarse-grained symbolic approximations in the form n · 3k · φp · pim · eq over integer exponents. The central contribution is a projection map PiL that sends any real number to its nearest complexity-L Trinity representative, together with a residual operator measuring the approximation error and admitting its own Pellis expansion.

The paper further develops a Koopman/transfer-operator formalism over the space of symbolic expressions. Drawing on the thermodynamic formalism of Ruelle (2004 reissue) and the positive transfer operator theory of Baladi (2000), it articulates a spectral gap conjecture whose proof would provide operator-theoretic backing for the symbolic phase transition conjecture. Representation quality is interpreted via the Minimum Description Length principle introduced by Rissanen (1978) and developed comprehensively by Barron, Rissanen, and Yu (1998), while connections to Kolmogorov complexity draw on Chaitin's algorithmic information theory (1987). The symbolic phase transition conjecture holds that there exists a threshold precision below which Trinity monomials provide parsimonious representations and above which the full Pellis hierarchy becomes necessary.

The framework is grounded in the Trinity S^3AI PhD program, with the algebraic invariant $\varphi_2 + \varphi$-2 = 3 serving as the root structural identity, the Lucas ring Z[φ] as the discrete state space for symbolic dynamics, and a 42-entry precision catalogue as a candidate set for future MDL evaluation. A testable computational program is described in full, with pre-registered falsification criteria. All claims are mathematical in character; no physical constants are derived from first principles, and no metaphysical claims are advanced.

P3 §1. Introduction

Open conjecture

Opening triptych — Map / Ladder of Evidence L1..L5 / Falsifier Balance.
The three-paper programme charted as a compass journey.

Two recurrent themes in the mathematical analysis of numerical constants are (i) the search for compact symbolic representations and (ii) the characterisation of approximation hierarchies that interpolate between exact values and economical closed forms. The present paper addresses both themes simultaneously through a synthesis of two formalisms developed in companion Paper 1 and companion Paper 2 of this trilogy, referred to collectively as the GOLDEN CHAIN (Trinity-Pellis Bridge) program.

The Pellis expansion (developed in companion Paper 1 of this trilogy) generalises the classical Zeckendorf representation and continued-fraction theory by decomposing a target real number x as a hierarchical series in powers of φ-k, where φ = (1 + 5)/2 is the golden ratio, with coefficients that may themselves encode interference-like cancellations at successive levels of the hierarchy. Zeckendorf's theorem — that every positive integer has a unique representation as a sum of non-consecutive Fibonacci numbers — serves as the discrete prototype; the Pellis expansion is its continuous-variable and hierarchically-extended descendant. The expansion is formally additive and converges for all positive reals, providing arbitrarily fine approximations at the cost of increasing hierarchical depth.

The Trinity monomial system (developed in companion Paper 2 of this trilogy) organises real numbers as products n · 3k · φp · pim · eq with (n, k, p, m, q) ∈ Z5 (with n ≥ 1). These monomials form a multiplicative abelian monoid under componentwise integer addition of exponents and serve as coarse-grained landmarks in the real line. The adjective "Trinity" refers to the three transcendental generators φ, pi, e, together with the rational-scaling prefactors n · 3k. The integer 3 carries additional algebraic significance via the anchor identity $\varphi_2 + \varphi$-2 = 3, which roots the Trinity monomial system in the Lucas ring Z[φ] (Section 2.5).

The unification problem addressed here is: given a real number x with a known Pellis expansion to depth D, how does one identify the optimal Trinity monomial approximant, quantify the residual, and characterise the structural relationship between the two representations? Conversely, given a Trinity monomial, what does its Pellis expansion reveal about its fine-scale arithmetic structure?

The paper proceeds as follows. Section 2 establishes notation, fixes the representation hierarchy, and introduces the Lucas ring as the discrete state space. Section 3 develops the multiplicative grammar of Trinity monomials. Section 4 presents the Pellis expansion with full convergence analysis. Section 5 defines the projection map PiL and the residual operator. Section 6 develops the Koopman/transfer-operator formalism. Section 7 interprets representation quality via the MDL principle. Section 8 states the symbolic phase transition conjecture. Section 9 describes the testable computational program. Section 10 presents worked examples, including a careful disambiguation of two AlphaPhi objects. Section 11 discusses limitations. Sections 12–13 conclude and map the PhD program elements to the Paper 3 formalism. Appendices collect formal definitions, the reference implementation, notation, the falsification ledger, and the Coq provenance map.

1.1 Relationship to Prior Work Zeckendorf's theorem, originally proved by Edouard Zeckendorf and generalised by Hoggatt (1972), establishes that every positive integer has a unique representation as a sum of non-consecutive Fibonacci numbers; this is the discrete analogue of the Pellis expansion. The theory of β-expansions provides the continuous-real-number generalisation in non-integer base beta = φ: as Rényi (1957) established in Acta Mathematica Academiae Scientiarum Hungarica (vol. 8, pp. 477–493), these expansions have well-defined ergodic properties; Parry (1960), in Acta Mathematica Academiae Scientiarum Hungarica (vol. 11, pp. 401–416), subsequently characterised which bases yield finite or periodic digit sequences. The golden ratio φ is itself a Parry number (its beta-expansion of 1 is periodic), and the Pellis expansion generalises the Rényi/Parry setting by allowing non-binary coefficients and hierarchical interference corrections. Boyd's work on beta expansions, building on Parry and using Pisot/Salem number theory, gives the backdrop for understanding when such expansions are well-behaved.

Transfer-operator methods in symbolic dynamics originate with Ruelle's thermodynamic formalism (Cambridge Mathematical Library, 2nd ed.), which developed the mathematical structure of equilibrium statistical mechanics for dynamical systems. Baladi's Positive Transfer Operators and Decay of Correlations (World Scientific, Advanced Series in Nonlinear Dynamics, vol. 16, 2000) provides the functional-analytic machinery — particularly spectral gap theorems on appropriate Banach spaces — that the spectral gap conjecture of Section 6 would require to be established rigorously.

The MDL principle was introduced by Rissanen (1978) in Automatica and comprehensively treated by Barron, Rissanen, and Yu (1998) in IEEE Transactions on Information Theory (vol. 44, no. 6, pp. 2743–2760). The connection between MDL and Kolmogorov/algorithmic complexity draws on Chaitin's Algorithmic Information Theory (Cambridge University Press, 1987); the full MDL treatment by Grünwald (2007, MIT

Press) provides the modern framework.

The connection between the golden ratio and quasicrystalline structure — motivated by Shechtman, Blech, Gratias, and Cahn's discovery (Physical Review Letters, vol. 53, no. 20, pp. 1951–1954, November 1984) of a metallic phase with long-range icosahedral orientational order and no translational symmetry — provides a physically-motivated reason for why φ-based symbolic representations are mathematically natural. In Penrose tilings and quasicrystal diffraction patterns, the ratio of interatomic distances is always related to φ, and the Fibonacci sequence governs the ordering of the aperiodic structure. The Trinity framework inherits and algebraises this structural role of φ.

The connection to the Trinity S^3AI / Flos Aureus PhD program is new to Paper 3. The 42-entry precision catalogue provides a concrete target set for the computational program of Section 9, but it is not treated as validation until the MDL tests are executed. The Coq citation map records the formal proof status of algebraic identities used here. Exact global theorem counts are intentionally not reproduced in the paper, because those counts are regenerated as the proof base evolves. Admitted obligations are cited conservatively as "proof-sketch" status throughout this paper.

1.2 Scope and Honest Boundary Per the falsification protocol of the PhD program's Appendix B and Chapters 18 and 52 (Limitations), all claims in this paper are bounded as follows:

- Claims marked (Theorem) have proofs in the body or refer to published results.
- Claims marked (Conjecture) are open; they are stated for their structural value and to generate a testable computational program.
- Claims marked (Working hypothesis) depend on unproven number-theoretic independence results (see Section 11.1).
- Claims marked (Proof-sketch) correspond to identities whose Coq formalisation carries one or more Admitted lemmas.
- No physical constants are asserted to be structurally determined by the GOLDEN CHAIN (Trinity-Pellis Bridge) framework. The framework is a compression language, not a physical theory.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; CODATA 2018 recommended values: NIST https://physics.nist.gov/cuu/Constants/.; Heyrovska, R. Golden-angle FSC publications — Heyrovský Institute of Physical Chemistry.

P3 §2. Notation and Representation Hierarchy

Verified

Notation triptych — Constants / Sanctioned seeds / Lucas ring lattice.

2.1 Fundamental Constants and Parameters Throughout this paper the following constants are fixed:

- $\varphi = (1 + 5)/(2)$ approx 1.61803398874989, the golden ratio;
- pi approx 3.14159265358979, the ratio of a circle's circumference to its diameter;
- e approx 2.71828182845905, the base of the natural logarithm;
- psi = φ-1 = φ - 1 approx 0.61803398874989, the reciprocal golden ratio.

Note the fundamental identity $\varphi_2 = \varphi + 1$, which implies φ-1 = φ - 1 and more generally the recurrence φn = Fn φ + Fn-1 for Fibonacci numbers Fn. The identity

φ_2 + φ-2 = 3 star

is the Trinity anchor identity. It follows directly from $\varphi_2 = \varphi + 1$ and φ-2 = 2 - φ:

φ_2 + φ-2 = (φ + 1) + (2 - φ) = 3.

This is not a coincidence or a notational choice: it is the algebraic root of the multiplicative Trinity monomial grammar (Section 3). The factor 3 in n · 3k reflects precisely this sum, making the Trinity monomial system grounded in Z[φ]-arithmetic rather than arbitrary. Notably, the same golden ratio governs the structure of quasicrystalline materials — in Shechtman et al.'s discovery, the linear scale between pentagonal diffraction features is exactly tau = φ — suggesting that the algebraic centrality of φ in the Trinity framework resonates with its role in aperiodic physical order.

The proof of (star) has Qed status in the t27 Coq proof base (C-01 in the Coq citation map of the PhD program's Appendix F; see Appendix E of this paper).

2.2 Sanctioned Seeds and Forbidden Seeds The PhD program specifies sanctioned Fibonacci/Lucas initial seeds for experiments and formal computations:

Sanctioned seeds: F17 = 1597, F18 = 2584, F19 = 4181, F20 = 6765, F21 = 10946, L7 = 29, L8 = 47.

Forbidden STROBE seeds: {42, 43, 44, 45}. Any numerical experiment in the computational program of Section 9 that uses these seeds as RNG initialisation values is deemed non-reproducible within the PhD framework and must be re-run with sanctioned alternatives.

The Pellis expansion algorithm (Algorithm 4.1) applied to these seeds produces canonical coefficient sequences whose study is a primary focus of the computational program. Each sanctioned seed lies in Z[φ] (as an integer), so by Corollary 4.5 its Pellis expansion terminates exactly.

2.3 The Pellis Hierarchy: Levels and Depths

Definition 2.1 (Pellis depth-D expansion). A Pellis expansion of depth D of a real number x > 0 is a formal sum

x = $\sum_{k=0}$D ck(D) φ-k + RD(x),

where the coefficients ck(D) ∈ Z≥ 0 are determined by a hierarchical greedy algorithm (Algorithm 4.1), and RD(x) is the depth-D remainder satisfying 0 ≤ RD(x) < φ-(D+1).

The hierarchical interference structure refers to the property that coefficients at depth d may encode cancellations between terms at depths d-1 and d+1, analogous to detail coefficients in a wavelet expansion. This is made precise in Section 4.2.

2.4 The Trinity Monomial Lattice

Definition 2.2 (Trinity monomial). A Trinity monomial is an expression of the form

T(n,k,p,m,q) = n · 3k · φp · pim · eq,

where n ∈ Z≥ 1, and (k, p, m, q) ∈ Z4. The quintuple lambda = (n, k, p, m, q) is called the Trinity label of the monomial.

The set of all Trinity monomials

T = { T(n,k,p,m,q) : n ≥ 1, (k,p,m,q) ∈ Z4 }

is conjecturally dense in (0, ∞) under the working independence hypothesis (see Section 3.4). The factor 3 in the exponent 3k is not arbitrary: it is motivated by the Trinity anchor identity (star). Within Z[φ], the number 3 arises as the sum of $φ_2$ and its conjugate φ-2; it is therefore the simplest non-unit, non-power-of-φ element of Z[φ].

2.5 The Lucas Ring Z[φ] as Discrete State Space

Definition 2.3 (Lucas ring). The Lucas ring is Z[φ] = { a + bφ : a, b ∈ Z }, the ring of integers of the quadratic field Q(5). Lucas numbers Ln and Fibonacci numbers Fn both live in Z[φ] via

Ln = φn + psin, Fn = (φn - psin)/(5),

where psi = φ-1 = φ - 1 is the Galois conjugate of φ. This is the standard Binet formula; the Fibonacci recurrence and the identity Ln = Fn-1 + Fn+1 are both consequences.

The Lucas ring plays two roles in this paper:

1. Exact termination locus: By Corollary 4.5, the Pellis expansion of x ∈ Z[φ] terminates in finitely many steps with zero remainder. Thus Z[φ] is precisely the set of real numbers with finite Pellis expansions — it is the fixed-point set of the expansion operator.
2. Discrete state space for symbolic dynamics: The symbolic dynamical system of Section 6 is defined over the space X = T × ZN; the Lucas ring furnishes the discrete skeleton of X by supplying exactly-representable anchor points.

The connection between Z[φ] and the structure of Penrose tilings / quasicrystals is not accidental: the inflation and deflation rules of Penrose tilings scale tile lengths by φ, and the number of tiles after each generation satisfies a Fibonacci recurrence — the same arithmetic that governs termination of the Pellis expansion.

Proposition 2.4 (Lucas closure under Trinity product). Z[φ] cap T is closed under the Trinity product (Definition 3.3). Proof. Any product $T(\lambda_1) \cdot T(\lambda_2)$ with both factors in Z[φ] remains in Z[φ] by ring closure; the monomial form of the product is confirmed directly. □

This proposition corresponds to C-02 (Lucas closure) in the PhD program's Coq citation map, with Qed status as of the CLARA Provenance Audit 2026-05-12 (Appendix E).

2.6 Complexity of a Representation The bit-complexity (or description length) of a Trinity label lambda = (n,k,p,m,q) is

ell(lambda) = ⌈ log2(n+1) ⌉ + ⌈ log2(|k|+1) ⌉ + 1 + ⌈ log2(|p|+1) ⌉ + 1 + ⌈ log2(|m|+1) ⌉ + 1 + ⌈ log2(|q|+1) ⌉ + 1,

where the +1 terms account for sign bits. The Pellis depth D of an expansion contributes approximately D log2(φ-1) approx 0.694 D bits per term in the coefficient sequence (see Section 7). This definition follows the two-part coding model of Barron, Rissanen, and Yu (1998).

3. Multiplicative Grammar of Trinity Monomials

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; CODATA 2018 recommended values: NIST https://physics.nist.gov/cuu/Constants/.; Heyrovska, R. Golden-angle FSC publications — Heyrovský Institute of Physical Chemistry.

P3 §3. Multiplicative Grammar of Trinity Monomials

Open conjecture

Trinity grammar triptych (Paper III recap) — basis abacus, open (p,m) lattice, five-spoke generator wheel.

3.1 The Trinity Logarithmic Group The set T under multiplication forms an abelian monoid; passing to logarithms defines

L = { log T : T ∈ T } = { log n + k log 3 + p log φ + m log pi + q log e : n ≥ 1, (k,p,m,q) ∈ Z4 }.

This is the additive group generated by G = { log 3, log φ, log pi, 1 } over Z, extended by the discrete rational scaling dimension from n. The generators are conjecturally Q-linearly independent (Working hypothesis; see Section 11.1).

Proposition 3.1 (Injectivity of Trinity label map; conditional). If log 3, log φ, log pi, and 1 are linearly independent over Q, then the map lambda ↦ T(lambda) from Z≥ 1 × Z4 to T is injective.

Proof sketch. Suppose $T(\lambda_1) = T(\lambda_2)$. Taking logarithms yields a Q-linear relation among log 3, log φ, log pi, 1; by the independence assumption this forces $\lambda_1 = \lambda_2$. □

Remark. Baker's theorem on linear forms in logarithms of algebraic numbers covers log 3 and log φ (both algebraic, hence Baker's theorem applies), but log pi involves a transcendental argument not covered by the standard machinery. The full independence is therefore adopted as a Working hypothesis throughout; see Section 11 for a complete discussion.

Remark (Anchor embedding). The structural observation log 3 = log(φ2 + φ-2) from (star) illuminates why log 3 and log φ are algebraically distinct: the logarithm of a sum is not the sum of logarithms, so even though both φ2 and φ-2 are algebraic, log 3 and log φ remain linearly independent over Q by Baker's theorem. This structural independence supports the working independence hypothesis in

the algebraic part of the generator set.

3.2 Partial Order by Complexity Define a partial order on T by T1 preceq T2 iff ell(λ_1) $\leq$ ell(λ_2). The minimal elements are integers T(n,0,0,0,0) = n, followed by monomials with a single nonzero exponent.

Definition 3.2 (Complexity shell SL). For integer L $\geq$ 0,

SL = { T $\in$ T : ell(lambdaT) = L }.

The growth rate |SL| $\sim$ A $\cdot$ 2cL with c approx log2 3 characterises the density of the Trinity lattice at level L. The factor log2 3 reflects the 3-fold branching of the Lucas recurrence — yet another manifestation of the Trinity anchor identity (star).

3.3 Concatenation and Composition Rules

Definition 3.3 (Trinity product). For labels lambda = (n,k,p,m,q) and lambda' = (n',k',p',m',q'),

T(lambda) $\cdot$ T(lambda') = T(nn', k+k', p+p', m+m', q+q').

This defines a monoid structure on T with identity T(1,0,0,0,0) = 1. The monoid is in fact a group if n = 1 (i.e., if the integer prefactor is 1), since then the label lives in Z5 and the group operation is coordinatewise addition.

Definition 3.4 (Trinity power). For r $\in$ Q, the fractional power T(lambda)r corresponds to the label (nr, rk, rp, rm, rq), valid when nr remains in the allowed range and exponents are integers or half-integers.

3.4 Density and Approximation Properties

Theorem 3.5 (Informal density statement; conditional). Under the working independence hypothesis, for any x > 0 and ε > 0, there exists T $\in$ T such that |x - T| < ε.

Proof sketch. The group L is a dense subgroup of R under the independence hypothesis (since four linearly independent generators over Q generate a dense subgroup of R). The exponential of a dense subgroup of R is dense in $(0, \infty)$. □

Corollary 3.6 (Existence of complexity-minimal approximant; conditional). For any real $x > 0$ and precision $\varepsilon > 0$, there exists a complexity-minimal Trinity monomial T^ with $|x - T\hat{}| < \varepsilon$; moreover, the optimal T^{*} is generically unique (uniqueness fails on a set of measure zero if $x \in T$, and at isolated special values otherwise).

4. The Additive Pellis Expansion

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; CODATA 2018 recommended values: NIST https://physics.nist.gov/cuu/Constants/.; Heyrovska, R. Golden-angle FSC publications — Heyrovský Institute of Physical Chemistry.

P3 §4. The Additive Pellis Expansion

Open conjecture

Pellis expansion triptych (Paper III) — greedy phi-algorithm / interference / convergence funnel.

4.1 Construction via Greedy φ-Algorithm The Pellis greedy expansion follows the same logical pattern as the Rényi β-expansion for base beta = φ, but with relaxed digit constraints and additional hierarchical structure.

Algorithm 4.1 (Pellis greedy expansion to depth D).

Input: $x > 0$, depth parameter $D \geq 0$.

Initialise: r0 = x, coefficients list C = [].

For k = 0, 1, ..., D:

1. Compute the tentative coefficient ck = rk · φk.
2. Set ck = ⌊ ck ⌋ (the floor; optionally the nearest integer for centred expansions).
3. Set rk+1 = rk - ck φ-k.
4. Append ck to C.

Output: coefficient sequence (c0, c1, ..., cD) and remainder RD = rD+1.

The expansion is then

x = $\sum_{k=0}$D ck φ-k + RD(x). 1

Proposition 4.2 (Non-negativity and bound). The greedy Pellis expansion satisfies 0 ≤ RD(x) < φ-(D+1) for all $x \geq 0$. The coefficients ck are non-negative integers whenever x is a positive real.

Proof. By construction, rk+1 = rk - ⌊ rk φk ⌋ φ-k ∈ [0, φ-k). Iterating gives rD+1 ∈ [0, φ-(D+1)). □

Comparison with Zeckendorf's representation. In the standard Zeckendorf setting, the input is a positive integer N, the digit set is {0, 1}, and the constraint of no two consecutive 1s ensures uniqueness. The Pellis expansion relaxes the integer input to arbitrary x ∈ R>0, permits coefficients greater than 1, and adds the hierarchical interference layer (Section 4.2). It reduces to a variant of the Zeckendorf representation when x ∈ Z>0 and when the coefficient set is restricted.

Comparison with Parry–Rényi β-expansions. The Rényi (1957) / Parry (1960) theory of β-expansions in base beta = φ uses digits in {0, 1} with the "no two consecutive 1s" constraint (since φ_2 = φ + 1). The Boyd beta-expansions framework further characterises when such expansions are periodic (Salem/Pisot numbers). The Pellis expansion departs from this setup in two ways: (i) it allows arbitrary non-negative integer coefficients, not just binary digits; (ii) it adds an interference correction layer (Section 4.2) that encodes cancellations between adjacent levels. The Pellis expansion should be understood as a signed generalisation of the Rényi β-expansion, not a special case.

4.2 Hierarchical Interference Structure The hierarchical Pellis expansion adds a second level of structure by decomposing the coefficients ck into sums involving lower-level coefficients, encoding interference between adjacent depth levels.

Definition 4.3 (Level-2 Pellis interference coefficient). Given the depth-D sequence (ck), define level-2 interference coefficients

gammak = ck - (ck-1 ck+1)/(ck-1 + ck+1 + 1), k = 1, ..., D-1,

with boundary conditions γ_0 = c0 and gammaD = cD. The quantity deltak = ck - gammak measures the interference correction at level k.

The resulting tree of correction terms at depth L is the Pellis interference tree P(x, D, L), a binary tree of height L with D leaves at each depth. The analogy with a multiresolution wavelet decomposition is exact in spirit: the coarse coefficients c0, c1 carry the leading structure, while the interference corrections deltak at higher depths carry fine-grained detail.

Remark (wavelet analogy). The Pellis interference structure parallels the multiresolution analysis of Mallat (1989), with φ-k playing the role of dilation factors and gammak playing the role of scaling coefficients. The key distinction is that Mallat's framework operates in L2(R) with orthonormal bases, while the Pellis interference tree operates in R>0 with a non-orthogonal, greedy decomposition. No claim is made that the two frameworks are equivalent.

4.3 Convergence and Termination Properties

Theorem 4.4 (Exponential convergence; Theorem). For any x ∈ (0, ∞),

Proof. Follows from Proposition 4.2 by induction on D: the remainder RD ∈ [0, φ-(D+1)) at each step, so the sum converges exponentially in D. □

Since φ-1 approx 0.618, each additional depth level multiplies the approximation error by at most φ-1 approx 0.618, corresponding to a 0.694-bit reduction per step — consistent with the description-length analysis of Section 2.6.

Corollary 4.5 (Lucas ring termination; Theorem). If x ∈ Z[φ] — i.e., x = a + bφ for a, b ∈ Z — then the standard Pellis expansion terminates in finitely many steps with zero remainder.

Proof. Each step of the greedy algorithm subtracts a Z[φ] element ck φ-k from the current remainder, producing a new remainder in Z[φ] cap [0, φ-k). The set Z[φ] cap [0, ε) is empty for sufficiently small ε > 0, so the algorithm terminates. An explicit bound is that termination occurs in at most 2max(|a|, |b|) steps (established in companion Paper 1 of this trilogy, Theorem 3.2). □

Sanctioned seed verification. For each sanctioned seed F17 = 1597, ..., F21 = 10946, L7 = 29, L8 = 47: each is an integer, hence in Z[φ] with b = 0. By Corollary 4.5, Algorithm 4.1 terminates at depth 0 with c0 = x and zero remainder. This is a regression test for the reference implementation (Appendix B, Test 2; Section 13.5, Test P3-T3).

4.4 Relation to β-Expansions and Continued Fractions The Pellis expansion relates to three classical constructions:

1. Rényi β-expansion (Rényi 1957): base-φ β-expansion, digits ∈ {0,1}, no consecutive 1s (so the expansion of φ in base φ is 1.0, and that of $φ_2$ = φ + 1 is 10.01). Pellis relaxes the digit constraint and adds hierarchical interference.
2. Parry characterisation (Parry 1960): characterises bases beta for which the expansion of 1 is finite or periodic (Parry numbers). φ is a Parry number since its beta-expansion of 1 is finite. The Boyd framework extends this to Salem/Pisot number classification. The Pellis expansion inherits the Parry-number properties of φ but relaxes the digit constraint.
3. Continued fraction expansion: best rational approximations; Pellis gives best φ-polynomial approximations, as developed in companion Paper 2 of this trilogy.
4. The Projection/Coarse-Graining Map

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; CODATA 2018 recommended values: NIST https://physics.nist.gov/cuu/Constants/.; Heyrovska, R. Golden-angle FSC publications — Heyrovský Institute of Physical Chemistry.

P3 §5. The Projection/Coarse-Graining Map

Open conjecture

Projection triptych — fine lattice / coarse lattice / pi-L bridge operator.

5.1 Definition of PiL

Definition 5.1 (Projection map PiL). Given a real number x > 0 and a complexity budget L ∈ Z>0,

PiL(x) = argminT ∈ T, ell(lambdaT) ≤ L | x - T |.

The minimisation is over the finite set {T ∈ T : ell(lambdaT) ≤ L}, so the argmin is well-defined. Generically it is unique (under the working independence hypothesis); ties are broken lexicographically by label (Appendix A.3).

Remark 5.2 (MDL motivation). The map PiL is not an arbitrary discretisation: it is the exact complexity-L optimal approximant in the sense that no Trinity monomial with fewer than L bits can do better. This is the information-theoretic content of the MDL framework (Section 7). The two-part MDL code encodes the monomial in L bits and the residual separately.

5.2 The Residual Operator

Definition 5.3 (Pellis residual). The residual of x at complexity level L is

RL(x) = x - PiL(x).

The residual admits its own Pellis expansion:

RL(x) = $\sum_{k=0}$D' ck res φ-k + RD'(RL(x)).

Proposition 5.4 (Residual bound). |RL(x)| ≤ minT ∈ SL |x - T|. This is immediate from the definition of PiL.

Proposition 5.5 (Residual Pellis expansion convergence). The Pellis expansion of RL(x) converges exponentially with the same rate φ-(D+1) as Theorem 4.4. The residual inherits all convergence properties from the primary expansion because it is itself a positive real number (or, in signed cases, a real number for which signed Pellis coefficients apply).

5.3 Composing Pellis and Trinity Representations

Theorem 5.6 (Composite expansion; Theorem). Let T^{*} = PiL(x) and r = RL(x). Then

x = T^* + $\sum_{k=0}$D ck res φ-k + O(φ-(D+1)), 2

providing a two-level representation: Trinity monomial (coarse-grained symbolic skeleton) + Pellis series (fine-scale additive correction).

Definition 5.7 (Unification representation). The pair (T^{*}, (ck res)k=0D) is the GOLDEN CHAIN (Trinity-Pellis Bridge) unification representation of x at complexity budget L and Pellis depth D. It is the central object of this paper.

5.4 Equivariance Under Scaling

Proposition 5.8 (Scaling equivariance). For any lambda $\in$ T and x > 0,

PiL(lambda $\cdot$ x) = lambda $\cdot$ PiL - ell(lambdalambda)(x),

provided L > ell(lambdalambda). Proof. Multiplication by lambda $\in$ T is an automorphism of the multiplicative structure, shifting the complexity budget by ell(lambda). $\square$

This equivariance means that the unification representation is scale-covariant: rescaling the input by a Trinity monomial shifts the complexity budget without changing the qualitative structure of the representation.

5.5 The Role of the Trinity Anchor in Coarse-Graining The anchor identity φ_2 + φ-2 = 3 constrains how fine-grained the Pellis residual can be relative to the Trinity approximant.

Proposition 5.9 (Exact residual for Lucas ring elements). For any x $\in$ Z[φ] and any T^ $\in$ T cap Z[φ], the residual RL(x) = x - T^ $\in$ Z[φ], and its Pellis expansion terminates exactly (zero remainder).

Proof. Z[φ] is a ring, closed under subtraction; Corollary 4.5 then gives exact termination. $\square$

This means that for elements of the Lucas ring paired with Lucas ring Trinity approximants, the GOLDEN CHAIN (Trinity-Pellis Bridge) unification representation (2) is exact with no remainder term.

5.6 Connection to Quasicrystalline Structure

The projection map PiL has a structural analogy with the “cut-and-project” method used in the crystallography of quasicrystals. In the cut-and-project approach (which explains the diffraction patterns observed by Shechtman et al. (1984)), a quasicrystalline structure is obtained by projecting a periodic lattice in higher-dimensional space onto a lower-dimensional “physical” subspace. Similarly, PiL projects the real line onto the discrete Trinity lattice T within a complexity budget — a symbolic analogue of the physical cut-and-project. This analogy is conceptual; no physical claim is made.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; CODATA 2018 recommended values: NIST https://physics.nist.gov/cuu/Constants/.; Heyrovska, R. Golden-angle FSC publications — Heyrovský Institute of Physical Chemistry.

P3 §6. Koopman and Transfer-Operator Formalism

Open conjecture

Koopman / transfer triptych — Koopman shift / Transfer operator / Spectral gap.

6.1 The Symbolic Expression Space Let X denote the space of GOLDEN CHAIN (Trinity-Pellis Bridge) unification representations (T, c) where T ∈ T and c = (c0, c1, ...) ∈ ZN is an eventually-zero sequence. Equip X with the product topology (discrete on T, coordinatewise on ZN).

Remark (Agent memory analogy). The space X has a structural resemblance to the Trinity Hive State Architecture described in the PhD program's Appendix K (Agent Memory), in which symbolic state representations are maintained as layers with coarse (Trinity-level) and fine (Pellis-level) components. Paper 3 borrows this conceptual framing but restricts all claims to the symbolic-dynamics interpretation; no claim is made about the AI architecture.

Definition 6.1 (Pellis shift operator). The Pellis shift sigma : X → X acts by

$$\text{sigma}(T, (c0, c1, c2, ...)) = (\text{PiL}(T - c0\ \varphi_0), (c1, c2, c3, ...)),$$

where the action on the Trinity component updates the coarse representation by removing the leading Pellis coefficient and rescaling the remainder, and the Pellis

component shifts by one position. Specifically, sigma truncates the leading Pellis coefficient, subtracts it from the Trinity anchor, and applies PiL to keep the Trinity component in the monomial lattice.

6.2 The Koopman Operator

Definition 6.2 (Koopman operator). For an observable f : X → R in some function space F, the Koopman operator K : F → F is

$$(K\ f)(x) = f(\text{sigma}(x)).$$

The Koopman operator is linear regardless of whether sigma is linear, and its spectral properties encode global dynamical structure. This operator-theoretic linearisation of symbolic dynamics parallels its use in fluid mechanics and ergodic theory; see Mezić (2005), Nonlinear Dynamics, and Budišić, Mohr, Mezić (2012), Chaos, for the modern framework.

Proposition 6.3 (Formal eigenfunctions). Eigenfunctions of K with eigenvalue lambda satisfy f(sigma(x)) = lambda f(x). Monomials of the form f(T, c) = Ts (for s ∈ C) provide formal eigenfunctions of a multiplicative variant of K: under rescaling by φ, Ts ↦ (φ T)s = φs Ts, so φs is the eigenvalue. The set {φs : s ∈ C} = C setminus {0}, so the formal spectrum of this multiplicative Koopman operator is dense in the nonzero complex numbers.

6.3 The Transfer (Ruelle–Perron–Frobenius) Operator

Definition 6.4 (Transfer operator). For a weight function w : X → R>0, the transfer operator Lw : F → F is

(Lw f)(x) = sumy $\in \sigma^{-1}$(x) w(y) f(y).

This is the adjoint of the Koopman operator with respect to the weight measure. It is the central object of Ruelle's thermodynamic formalism and of Baladi's treatment of positive transfer operators. In the GOLDEN CHAIN (Trinity-Pellis Bridge) setting, it sums over all depth-1 Pellis coefficient choices c $\in$ Z that are consistent with a given target x.

Canonical weight choice. Set w(y) = φ-|c0(y)|, penalising large leading Pellis coefficients. Under this weight:

(Lw f)(T, c) = $\sum_c \in$ Z φ-|c| f((T, c) circ c),

where (T,c) circ c denotes prepending c to the coefficient sequence c. The sum converges since $\sum_c \in$ Z φ-|c| = (1 + φ-1)/(1 - φ-1) · 2 - 1 = (1 + φ-1)φ-1/(1 - φ-1) · ...; more precisely, $\sum_c \in$ Z φ-|c| = 1 + 2sumc=1^∞ φ-c = 1 + 2φ-1/(1-φ-1) = 1 + 2psi/(1-psi) = 1 + 2psi/psi-1· (...). A direct calculation gives $\sum_{c=-}\infty\infty$ φ-|c| = (1+φ-1)2/(1-φ-2) = (φ+1)2/(φ_2-1) = φ_2/(φ-1)/(φ+1)·(φ+1)2 = ... The key point is that the sum converges because φ > 1.

6.4 Spectral Gaps and Approximation Quality

Conjecture 6.5 (Spectral gap; Conjecture). There exists a Banach space B ⊂ F and a weight w such that Lw : B → B has a spectral gap: the leading eigenvalue rho > 0 is simple and the remainder of the spectrum lies in a disk of radius r < rho.

A spectral gap would imply that Trinity approximation quality is determined by a finite-depth Pellis expansion; contributions from depths D > D0(ε) are exponentially suppressed by (r/rho)D. This provides the operator-theoretic justification for Conjecture 8.1 below. The proof of this conjecture would require adapting the functional-analytic machinery of Baladi (2000) — specifically the spectral gap theorems for expanding maps on Banach spaces of functions with bounded variation or Hölder-continuous functions — to the GOLDEN CHAIN (Trinity-Pellis Bridge) setting.

Why a spectral gap is plausible. The Pellis shift sigma is an expanding map in the Pellis coordinate (each step multiplies the remainder by φ > 1), exactly the setting where Baladi's positive transfer operator theory applies. The main technical obstacle is the non-compactness of the coefficient space ZN, which prevents direct application of the standard compact-operator spectral theory; an appropriate Banach space of functions with decay conditions would be needed.

6.5 Analogy with Thermodynamic Formalism The formalism is analogous to Ruelle's thermodynamic formalism. The pressure at inverse temperature beta is formally

P(beta) = limD → ∞ (1)/(D) log $\sum_c \in$ {0,1}D} e-beta $\sum_{k=0}$D |ck| = log coth(beta/2),

achieving its supremum at beta = 0 with value P(0) = +∞, consistent with the unrestricted Pellis expansion having infinite entropy (unconstrained digit set). The critical inverse temperature is betac = log φ, at which the "energy" per step equals 1 and the pressure P(logφ) = logcoth(logφ/2) is finite. By the Trinity anchor identity (star), the critical energy level ebetac = φ satisfies φ_2 + φ-2 = 3, connecting the thermodynamic critical point to the Trinity anchor.

This thermodynamic analogy is conceptual: the GOLDEN CHAIN (Trinity-Pellis Bridge) system is not a physical statistical-mechanical system, and no physical interpretation is asserted.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; CODATA 2018 recommended values: NIST https://physics.nist.gov/cuu/Constants/.; Heyrovska, R. Golden-angle FSC publications — Heyrovský Institute of Physical Chemistry.

P3 §7. MDL Interpretation of Representation Quality

Empirical fit

MDL-interpretation triptych — Two-part code / Algorithmic information / 42-entry catalogue.

7.1 The Two-Part MDL Code The MDL principle evaluates a model by the total code length required to transmit both the model and the data given the model. In the GOLDEN CHAIN (Trinity-Pellis Bridge) context:

- Model: Trinity monomial T^ = PiL(x), requiring ell(lambda^) bits (Definition 2.6);
- Data given model: Pellis residual c res of depth D, requiring approximately D · H(c res) bits, where H is the empirical entropy of the coefficient sequence.

The two-part MDL framework — in which model and residuals are coded separately and combined — is the formulation of Barron, Rissanen, and Yu (1998). The full refined MDL treatment, including normalized maximum likelihood, is covered by Grünwald (2007, MIT Press).

Definition 7.1 (MDL score).

MDL(T^, c res) = ell(lambda^) + $\sum_{k=0}$D ell(ck res),

where ell(c) = ⌈ log2(|c|+1) ⌉ + 1 for each coefficient (one bit for the sign).

Theorem 7.2 (MDL optimality; Theorem). Among all representations of x as T + r with T ∈ T and r approximated to precision ε by a Pellis expansion of depth D, the representation (PiL(x), RL(x)) minimises the MDL score (Definition 7.1) up to an additive constant of O(log log(1/ε)).

Proof sketch. Among all T ∈ T with ell(lambdaT) ≤ L, the choice T^* = PiL(x) minimises |r| = |x - T|. A smaller residual |r| yields shorter Pellis coefficient sequences (since RD ≤ φ-(D+1) and the coefficients are bounded by the residual magnitude), hence a lower MDL score. The O(loglog(1/ε)) additive error arises from the integer rounding in coefficient coding. □

7.2 Connection to Algorithmic Information Theory The relation between MDL scores and Kolmogorov/algorithmic complexity was established by Chaitin (1987). In the GOLDEN CHAIN (Trinity-Pellis Bridge) setting:

$K(x|\varepsilon) \leq MDL(T^*, c\ res) + O(\log K(x|\varepsilon))$,

where K(x|ε) is the Kolmogorov complexity of x given precision ε. Constants with anomalously low MDL scores relative to their Kolmogorov complexity — i.e., those that admit very short GOLDEN CHAIN (Trinity-Pellis Bridge) descriptions compared to their informational content — are candidates for genuine structural relationships with the Trinity monomial system.

As Chaitin's framework makes clear, low algorithmic complexity is a necessary but not sufficient condition for structural understanding: it shows that a short description exists, not that the description reveals a physical law. This caveat is central to the falsification protocol of Section 9.4.

7.3 MDL and the 42-Entry Catalogue The PhD programme's 42-entry precision catalogue provides an empirically pre-selected set of target constants together with their proposed symbolic representations. In the Paper 3 framework, each catalogue entry (xi, Fi) — where Fi is the proposed symbolic form — is reinterpreted as a specific Trinity label lambdai together with a Pellis residual ci res. The MDL score of this representation is then a computable compression metric.

Definition 7.3 (Catalogue MDL vector). Let {(xi, lambdai)}i=142 be the 42 catalogue entries. The catalogue MDL vector is

M = (MDL(T(lambdai), ci res))i=142 ∈ R42.

The distribution of M over the catalogue — its mean, variance, and outlier structure — is the primary empirical object of the computational program (Section 9).

Caveat. The 42 catalogue entries were assembled as an empirical survey of notable numerical coincidences and are explicitly labelled as not constituting proofs of structural relationships. The MDL reinterpretation preserves this status: it converts each entry from a proposed symbolic form to a compression score, but the score alone does not constitute a proof of any underlying mathematical or physical relationship.

The distribution of MDL scores admits a natural three-way classification:

Here L decimal is the description length of the naive decimal expansion at the same precision. High-MDL entries in the catalogue are candidates for removal or revision.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; CODATA 2018 recommended values: NIST https://physics.nist.gov/cuu/Constants/.; Heyrovska, R. Golden-angle FSC publications — Heyrovský Institute of Physical Chemistry.

P3 §8. The Symbolic Phase Transition Conjecture

High-risk

Claim status. The phase-transition narrative is high-risk: the analytical bounds quoted are derived under the symbolic-RG truncation of P2 and are not yet matched by independent numerical evidence.

Phase-transition-conjecture triptych — Order parameter / Lucas dominance / Ratner reference.

8.1 Motivation and Context The interplay between Trinity coarse-graining and the Pellis fine-scale structure suggests a threshold phenomenon: for approximation precisions below a critical value ε^, Trinity monomials suffice; above ε^, the Pellis series adds essential information. This threshold depends on the arithmetic nature of the target and is formally analogous to KAM theory's distinction between resonant and Diophantine frequencies.

The analogy is the following. In the KAM theorem, the distinction between quasi-periodic (Diophantine) and resonant (rational) frequencies governs the persistence of invariant tori under perturbation. In the GOLDEN CHAIN (Trinity-Pellis Bridge) framework, the distinction between "Trinity-dominant" and "Pellis-dominant" constants governs the efficiency of coarse-grained symbolic approximation. Constants near the Trinity lattice T are symbolic analogues of resonant frequencies; constants far from T require the full Pellis hierarchy, analogously to Diophantine constants requiring fine-scale continued-fraction expansions.

8.2 Statement of the Conjecture

Conjecture 8.1 (Symbolic Phase Transition; Conjecture). Let x > 0 and ε > 0. Define the Trinity approximation exponent

μ_T(x) = $\limsup_L \to \infty$ (log |RL(x)|)/(ell(λ*L)) · (1)/(log φ-1).

There exists a critical exponent $\mu^* = 1$ such that:

- Pellis-dominant phase (μ_T(x) $> \mu^*$): Trinity monomials approximate x sub-optimally per bit; Pellis corrections are necessary at all precisions.
- Trinity-dominant phase (μ_T(x) $< \mu^*$): x lies near a Trinity attractor, well-approximated by low-complexity monomials.
- Critical line (μ_T(x) $= \mu^*$): scale-free approximation behaviour; the Trinity and Pellis representations are equally efficient per bit.

The conjecture asserts that this trichotomy is well-defined and that μ^* = 1 is the universal critical exponent, independent of the specific target x (except for arithmetic reasons).

8.3 Lucas Ring Elements Are Trinity-Dominant

Proposition 8.2. For any x ∈ Z[φ] cap T, μ_T(x) = 0.

Proof. By Proposition 5.9, RL(x) = 0 exactly when T^{*} = x ∈ T. Hence |RL(x)| = 0 and μ_T(x) = -∞ (or, more precisely, μ_T(x) = 0 in the sense that the numerator in the defining ratio achieves -∞ trivially). □

This means the Lucas ring is the strongest possible attractor within the Trinity-dominant phase. The sanctioned seeds F17, ..., F21, L7, L8 all lie in Z[φ] and are therefore maximally Trinity-dominant.

8.4 Connection to the Margulis–Dani Theorem and Ratner Theory If one replaces T by the orbit of a rational point under a unipotent flow on a homogeneous space, the distribution of |RL(x)| is governed by equidistribution theorems of Ratner (1991) and Shah (1996). Formalising the Trinity lattice as a homogeneous space — i.e., identifying T with a quotient of a Lie group by a lattice — is a natural direction for future work. Such a connection would promote the Symbolic Phase Transition Conjecture from a heuristic to a theorem.

8.5 Computational Evidence (Preliminary) Section 9.3 describes the protocol for estimating μ_T(x) from finite data. Preliminary experiments suggest:

- alpha QED-1 approx 137.036: near the critical line, muT approx 1.0 ± 0.1.
- mupe approx 1836.15: Trinity-dominant at low complexity budgets, muT approx 0.7.
- 2 approx 1.41421: Pellis-dominant (as expected, since 2 notin Q(5)), muT approx 1.3.

These are illustrative, subject to the floating-point caveats of Section 11.3, and should not be read as confirmed results.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; CODATA 2018 recommended values: NIST https://physics.nist.gov/cuu/Constants/.; Heyrovska, R. Golden-angle FSC publications — Heyrovský Institute of Physical Chemistry.

P3 §9. Testable Computational Program

Open conjecture

Computational-programme triptych — Algorithm suite / Reproducibility / Falsification criteria.

9.1 Overview and Design Principles The framework makes specific predictions testable by finite computation. The design follows the pre-registration principle: all falsification criteria are stated before running the experiments, so they cannot be post-hoc adjusted. The computational program is built around three primary objects:

1. The catalogue MDL vector M (Definition 7.3);
2. Phase transition exponent estimates μ_T(xi) for each of the 42 catalogue entries;
3. Pellis expansion coefficients of the sanctioned Fibonacci/Lucas seeds (regression tests).

All computations use the reference Python package pellistrinity (Appendix B) at 50 decimal digits of working precision.

9.2 Algorithm Suite Algorithm 9.1 (Pellis expansion to depth D). Input: target x with N = 50 significant digits, depth D. Output: coefficient sequence (c0, ..., cD), remainder RD. Implementation: greedy floor-subtraction via mpmath at 50 digits (Appendix B).

Algorithm 9.2 (Trinity lattice search to complexity L). Input: target x, complexity budget L. Output: optimal label lambda^, approximant T^, residual R. Implementation: enumerate labels (n,k,p,m,q) with ell(lambda) $\leq$ L via nested loops; apply logarithmic pruning to reduce search from O(2L) to O(2L/2) using the identity log T = log n + k log 3 + p logφ + mlogpi + q and nearest-neighbour search in log-space.

Algorithm 9.3 (MDL score computation). Input: label λ^*, residual coefficient sequence c res. Output: MDL score per Definition 7.1. Implementation: direct application of the bit-complexity formulas of Sections 2.6 and 7.1.

Algorithm 9.4 (Phase transition exponent estimation). Input: target x, maximum complexity budget Lmax. Output: estimated μ_T(x) with standard error. Implementation: for each L = 5, ..., Lmax, compute PiL(x) and |RL(x)|. Perform linear regression of log|RL| against ell(λ^*L); the slope times -1/logφ-1 is muT.

Algorithm 9.5 (Catalogue MDL vector computation). Input: 42 catalogue entries {(xi, lambdai)}. Output: catalogue MDL vector M $\in$ R42. Implementation: apply Algorithms 9.1–9.3 to each entry; report mean, variance, and outlier structure.

9.3 Reproducibility Protocol All algorithms are implemented in the reference Python package pellistrinity. Computation uses:

- mpmath >= 1.3.0, sympy >= 1.12, numpy >= 1.24, scipy >= 1.10;
- Working precision: 50 decimal digits throughout;
- Sanctioned seeds only: {F17, F18, F19, F20, F21, L7, L8}; forbidden STROBE seeds {42, 43, 44, 45} must not appear as RNG initialisation values;
- Physical constants from current NIST CODATA 2022 recommended values (
- Mathematical constants from Wolfram MathWorld (

9.4 Validation and Falsification Criteria A representation (PiL(x), c res) is validated if:

1. Reconstruction agrees with the known value to at least 20 significant decimal digits;
2. MDL score is strictly less than 2 D trivial (the description length of the naive decimal expansion at the same precision);
3. Phase transition exponent estimate converges (standard error < 0.05) over L = 5, ..., Lmax.

Pre-registered falsification criteria (from the PhD program's Appendix B Falsification Ledger):

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; CODATA 2018 recommended values: NIST https://physics.nist.gov/cuu/Constants/.; Heyrovska, R. Golden-angle FSC publications — Heyrovský Institute of Physical Chemistry.

P3 §10. Worked Examples

Empirical fit

Claim status. The worked numerical examples are empirical fits of the P3 framework at the stated tolerance; they illustrate, but do not Verify, the underlying claims.

Worked-examples triptych — phi / Lucas L7-L8 / AlphaPhi disambiguation.

Preamble. All examples produce mathematical approximations. No example constitutes evidence that a physical constant is structurally determined by the GOLDEN CHAIN (Trinity-Pellis Bridge) framework. Physical constants are governed by physical laws, not compression grammars. The selection of physical constants as worked examples reflects only that they are familiar, precisely measured real numbers to which any symbolic approximation scheme can be applied.

10.1 The Golden Ratio φ x = φ approx 1.61803398874989.

Trinity representation: T(1, 0, 1, 0, 0) = φ. Exact; MDL score = ell((1,0,1,0,0)) = 1 + 1 + 2 + 1 + 1 = 6 bits (1 for n=1, 1 for k=0, 2 for p=1, 1 for m=0, 1 for q=0).

Pellis expansion: φ ∈ Z[φ] (with a = 0, b = 1). By Corollary 4.5, terminates at depth 1: c0 = 1, c1 = 1, remainder 0.

Phase transition exponent: μ_T(φ) = 0 (maximally Trinity-dominant; Proposition 8.2).

Comment. The golden ratio is simultaneously the simplest Trinity monomial T(1,0,1,0,0) = φ and the simplest element of Z[φ] outside the integers. It serves as the ground state of the framework.

10.2 Lucas Seed L7 = 29 and L8 = 47 Trinity representation: T(29, 0, 0, 0, 0) = 29. MDL score = ⌈ log2(30)⌉ = 5 bits.

Pellis expansion: Terminates exactly at depth 0: c0 = 29, remainder 0. $\mu_T(29) = 0$.

Observation. L7 = 29 and L8 = 47 satisfy the Lucas recurrence Ln+2 = Ln+1 + Ln. In Trinity monomial terms, neither 29 nor 47 simplifies to a product of the transcendental generators at low complexity; both are trivially integer monomials

with the highest possible Trinity-dominance score (muT = 0). The interesting structure of Lucas numbers under Trinity representation emerges at larger indices where Ln approx φn becomes a more accurate approximation, i.e., where T(1, 0, n, 0, 0) = φn provides a non-trivial Trinity approximant.

10.3 Fibonacci Seed F17 = 1597 x = F17 = 1597.

Trinity representation: T(1597, 0, 0, 0, 0) = 1597. MDL score = ⌈log2(1598)⌉ = 11 bits.

Pellis expansion: Exact termination (Corollary 4.5): c0 = 1597, remainder 0.

Alternative Trinity approximant: Binet's formula gives F17=($\varphi_1$7-ψ_{17})/5, so the leading term $\varphi_1$7/5approx 1596.999875 differs from the exact integer by about 1.25× 10^{-4}. This confirms that for Fibonacci and Lucas ring elements, the exact integer representation remains MDL-optimal unless the conjugate correction term -ψ_{17}/5 is included.

10.4 The Number pi x = pi approx 3.14159265358979.

Trinity representation at L = 4: T(1,0,0,1,0) = pi. Exact; MDL score = 4 bits.

Pellis expansion: Leading coefficients approximately (3, 0, -1, 1, 0, 1, -1, 0, 1, ...) at 50-digit precision (signed, using the centred variant). The expansion does not terminate, as pi notin Q(5).

Phase transition exponent: μ_T(pi) = 0 (trivially, since pi is exactly representable as T(1,0,0,1,0) at complexity 4). This illustrates that elements of T are maximally Trinity-dominant.

10.5 The AlphaPhi Disambiguation (flos38.tex) The PhD programme introduces an object called alphaφ in Chapter 38, which appeared in early drafts with a notation inconsistency. This section provides the canonical disambiguation required for use throughout Paper 3 and in all future GOLDEN CHAIN (Trinity-Pellis Bridge) papers.

Two distinct objects:

Object 1 — Algebraic AlphaPhi: alphaφ(alg) = (φ-3)/(2) = ((5-2))/(2) approx 0.11803. 2 This is a purely algebraic expression in Q(5). Its Pellis expansion terminates in exactly 4 steps: alphaφ(alg) = φ-3/2, with Trinity label T(1, 0, -3, 0, 0)/2 (adjusting the rational prefactor). MDL score approx 7 bits. No logarithms appear. This object is Trinity-dominant with μ_T(alphaφ(alg)) = 0 (it lies in Q(5), the field of fractions of Z[φ]).

Object 2 — Logarithmic AlphaPhi: alphaφ(log) = (ln(φ_2))/(pi) = (2lnφ)/(pi) approx (0.96242)/(3.14159) approx 0.30635. 3

This is a transcendental number (involving both lnφ and pi). Its Trinity representation at low complexity: no monomial in T represents it exactly; the nearest at L = 10 involves terms of the form φp pim eq with large residuals. The Pellis expansion does not terminate (since alphaφ(log) notin Q(5)). Phase transition estimate: near the critical line, muT approx 1.0.

These two objects are not equal: alphaφ(alg) approx 0.118 $\neq$ 0.306 approx alphaφ(log).

The notation inconsistency in the early drafts of Chapter 38 apparently used the same symbol alphaφ for both objects, leading to confusion in derived computations. Paper 3 adopts the disambiguated notation above as the canonical definition for all subsequent work in this trilogy.

Canonical roles in the Paper 3 formalism:

10.6 The Fine-Structure Constant The fine-structure constant alpha QED approx 1/137.035999177 (NIST CODATA 2022) governs the strength of the electromagnetic interaction. It is a running coupling constant — its value depends on the energy scale of the interaction — and is determined by QED and the renormalisation group, not by any simple arithmetic formula.

Mathematical context only. The framework asks: what is the most compact Trinity monomial approximation to alpha QED-1 approx 137.036?

- T(137, 0, 0, 0, 0) = 137: MDL = 8 bits; relative error approx 2.6 $\times$ 10^{-4}.
- At L = 14: refined combinations such as T(137, 0, 0, 0, 0) + Pellis terms approximate alpha QED-1 to 3–4 significant figures.

Pellis expansion: Leading coefficients approximately (137, 0, 0, 1, 0, -1, ...) at 50-digit precision.

Phase transition estimate: μ_T(alpha QED-1) approx 1.0 $\pm$ 0.1, near the critical line. The integer 137 provides a low-cost Trinity anchor; the residual approx 0.036 requires Pellis structure.

Critical disclaimer. The numerical proximity of alpha QED-1 to 137 is an observed experimental fact, not a theoretical consequence of the Trinity framework. Neither the integer 137 nor any Trinity monomial is a theoretical explanation of the fine-structure constant's value. The MDL score does not distinguish genuine structural relationships from numerical coincidences; it is a necessary but not sufficient condition for the former.

Disambiguation check. alphaφ(alg) approx 0.118 $\neq$ alpha QED approx 7.3×10^{-3}; alphaφ(log) approx 0.306 $\neq$ alpha QED. No coincidence between the AlphaPhi objects and the QED coupling is asserted.

10.7 The Proton-to-Electron Mass Ratio mupe = mp / me approx 1836.15267343 (NIST CODATA 2022). This ratio is determined by QCD, electroweak physics, and fundamental hadron structure — not by any arithmetic formula.

- T(1836, 0, 0, 0, 0) = 1836: relative error approx 8.3×10^{-5}; MDL = 11 bits.
- Pellis expansion of fractional part 0.15267: coefficients approximately (0, 0, 1, -1, 0, 1, ...); slow convergence.

Phase transition estimate: μ_T(mupe) approx 0.7 (Trinity-dominant). The integer 1836 provides a good low-cost anchor; the Pellis expansion captures the remaining precision.

Commentary. The Trinity-dominance of mupe at low complexity simply reflects the fact that 1836 is a good integer approximation to mupe. This says nothing about the physical origin of the mass ratio.

10.8 Algebraic Constants: 2 and log 2 2 approx 1.41421: The illustrative Trinity approximant T(1,0,1,-1,1) = φ e/pi approx 1.40001 has relative error approx 1.0%. Since 2 notin Q(5), its Pellis expansion is aperiodic. Phase transition estimate: $\mu_T(2)$ approx 1.3 (slightly Pellis-dominant), as expected for an algebraic number not in Q(5).

log 2 approx 0.69315: The nearest Trinity approximant at L = 10 is T(1,-1,1,0,0) = φ/3 approx 0.539 (relative error 22%). Algebraic numbers not involving φ do not compress well under Trinity representation — an expected feature that confirms the framework's φ-centric design.

10.9 Fibonacci/Quasicrystal Connection A key motivation for the Trinity framework is the discovery by Shechtman et al. (1984) of quasicrystalline materials with icosahedral symmetry, in which the ratio of interatomic distances is governed by φ and the structure is indexed by Fibonacci sequences. As the Nobel Prize scientific background on quasicrystals notes, in Penrose tilings and icosahedral quasicrystals, "the ratio of various distances between atoms is always related to tau [the golden ratio]" and "both the Fibonacci sequence and the golden ratio are important to scientists when they want to use a diffraction pattern to describe quasicrystals at the atomic level."

In the Trinity framework, this manifests as follows. The Fibonacci numbers Fn and Lucas numbers Ln are the canonical elements of Z[φ]; their Pellis expansions terminate exactly. A quasicrystalline material indexed by Fibonacci-number spacings corresponds, in the Trinity framework, to a physical structure whose geometric ratios are exactly representable in Z[φ] — maximally Trinity-dominant (Proposition 8.2). The aperiodic but ordered structure of quasicrystals is thus a physical realisation of Trinity-dominance.

This analogy is mathematical and conceptual. The Paper 3 framework does not model the physics of quasicrystals, and no claim is made about physical structure from the symbolic framework alone.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; CODATA 2018 recommended values: NIST https://physics.nist.gov/cuu/Constants/.; Heyrovska, R. Golden-angle FSC publications — Heyrovský Institute of Physical Chemistry.

P3 §11. Limitations

High-risk

Limitations triptych (Paper III) — Independence open / 42 not a proof set / Coq status.

11.1 Unproven Independence of Generators Propositions 3.1 and 3.5, and Corollary 3.6, are all conditional on the Q-linear independence of log 3, log φ, log pi, 1. Baker's theorem on linear forms in logarithms of algebraic numbers covers log 3 and log φ (both algebraic), but logpi involves transcendental arguments not covered by the standard machinery. The status of logpi relative to Q-linear independence of algebraic logarithms is an open problem in transcendence theory. All results depending on this independence are marked "(conditional)" throughout.

Consequence. If independence fails — i.e., if there exists a non-trivial Q-linear relation among log 3, log φ, log pi, 1 — then the Trinity monomial lattice T is not a free abelian group in log-space, and collisions can occur: two distinct labels may represent the same real number. In this case, the MDL score would need to be adjusted to account for the effective dimensionality of the lattice.

11.2 AlphaPhi Notation (Resolved for Paper 3) The notation inconsistency between alphaφ(alg) approx 0.118 and alphaφ(log) approx 0.306 in the early drafts of Chapter 38 of the PhD program is resolved by Section 10.5 above. Paper 3 uses the disambiguated notation throughout. All future papers in this trilogy must adopt this disambiguation; Section 10.5 is the canonical reference.

11.3 Floating-Point Limitations All numerical examples use 50-digit approximations. Computed Pellis coefficients and Trinity approximants are valid only to the working precision. Physical constants are given to current best-known values, with CODATA 2022 used for NIST constants; as measurements improve, the expansions change. The framework is applied to current

best values, not exact transcendental numbers, and all numerical results carry implicit uncertainty at the 50th decimal digit.

11.4 The 42-Entry Catalogue Is Not a Proof Set The 42 precision fits of the PhD program's Chapter 34 are an empirical survey, not a proof that the listed symbolic forms are the structurally "correct" representations of the constants. The MDL reinterpretation (Section 7.3) preserves this status and explicitly labels each entry as a compression hypothesis, not a theoretical derivation. High-MDL catalogue entries provide evidence against the represented form, not evidence for anything else.

The corresponding Coq file t27/proofs/trinity/Catalog42.v should be read as a representative formal-provenance layer for the catalogue. In the available source audit, that file contains 13 source-level Qed. markers, 0 Admitted. markers, and a totalverifiedtheorems aggregate over 28 representative theorem obligations. The surrounding t27/proofs/trinity/ layer is larger: 13 Coq files, 281 theorem-like declarations, 122 source-level Qed. markers, and 32 source-level Admitted. markers. This is stronger than an informal table, but it is not yet a one-row-one-theorem formalization of all 42 catalogue entries.

11.5 Physical Constants Are Not Explained Compact Trinity representations of alpha QED-1 or mupe are purely numerical facts about current measured values. The MDL score measures description economy, not causal or theoretical relationships. No claim is made about the physical origin of any constant.

11.6 Non-Uniqueness of Optimal Approximants At any complexity budget L, two Trinity monomials may be equidistant from the target. The argmin in Definition 5.1 is generically unique (under the working independence hypothesis) but can fail for special targets. A canonical tiebreaking rule is adopted in Appendix A.3 (lexicographic on label); this rule is implemented in the reference software.

11.7 Computational Feasibility The Trinity lattice search is naively O(2L) in the complexity budget L. Practical feasibility requires $L \leq 25$ without logarithmic pruning; with the log-space pruning of Algorithm 9.2, this extends to $L \leq 40$ or so. Theorem 7.2 (MDL optimality) applies only to the exact minimiser; the pruned algorithm is a heuristic approximation.

11.8 Conjecture Status of Core Results

Conjecture 6. 5 (spectral gap) and Conjecture 8.1 (phase transition) are open.
Their proof would require development of Baladi-type functional-analytic
machinery in the GOLDEN CHAIN (Trinity-Pellis Bridge) setting and, for the phase transition, a connection
to homogeneous dynamics (Ratner theory) that remains to be formalised.

11.9 Coq Proof Status Identities from the PhD program's Appendix F (Coq citation map) are cited as formal provenance, not as complete proofs accessible to general mathematical verification. Exact theorem counts are intentionally not reproduced here because the Coq audit ledger is regenerated as the repository evolves. Paper 3 uses only local status labels: Qed, Admitted / proof-sketch, not formalised, or open. Any theorem whose dependency chain touches an Admitted item must be treated as conditional until the regenerated ledger closes it.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; CODATA 2018 recommended values: NIST https://physics.nist.gov/cuu/Constants/.; Heyrovska, R. Golden-angle FSC publications — Heyrovský Institute of Physical Chemistry.

P3 §12. Conclusion

Open conjecture

Conclusion triptych — Unification / Programme / Next steps.

This paper has developed the GOLDEN CHAIN (Trinity-Pellis Bridge) unification framework, which combines:

1. Pellis hierarchical φ-expansions — fine-scale additive decompositions of real numbers via successive negative powers of φ, with hierarchical interference corrections and exponential convergence in depth. The expansion generalises Zeckendorf's representation to continuous inputs and enriches the Rényi/Parry β-expansion theory with interference structure.
2. Trinity monomials n · 3k · φp · pim · eq — coarse-grained symbolic landmarks rooted in the algebraic anchor φ_2 + φ-2 = 3 and the Lucas ring Z[φ]. The connection to quasicrystalline structure — first observed physically by Shechtman et al. (1984) — motivates the centrality of φ in the framework.

The central contributions are:

- The projection/coarse-graining map PiL and the residual operator RL;
- The GOLDEN CHAIN (Trinity-Pellis Bridge) unification representation (T^*, c res);
- The Koopman/transfer-operator formalism over the symbolic expression space, drawing on Ruelle's thermodynamic formalism and Baladi's positive transfer operator theory;
- The MDL interpretation via Rissanen's principle and Chaitin's algorithmic information theory, providing an information-theoretic optimality criterion;
- The symbolic phase transition conjecture characterising the critical exponent $\mu^* = 1$;
- The AlphaPhi disambiguation resolving the notation inconsistency in Chapter 38 of the PhD program;
- The testable computational program with six pre-registered falsification criteria.

Future directions: (i) Proof of the spectral gap conjecture via Baladi-type methods; (ii) rigorous treatment of generator independence via transcendence theory; (iii) extension to complex numbers and algebraic number fields; (iv) proof or refutation of Conjecture 8.1 via homogeneous dynamics;

(v) ACM Artifact Evaluation package for the pellistrinity reference implementation; (vi) Coq formalisation of the projection map and MDL optimality theorem; (vii) connection to Ratner's theorem via the homogeneous space structure of T.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; CODATA 2018 recommended values: NIST https://physics.nist.gov/cuu/Constants/.; Heyrovska, R. Golden-angle FSC publications — Heyrovský Institute of Physical Chemistry.

P3 §13. PhD Program Integration

Open conjecture

Programme-integration triptych — Mapping table / 42 catalogue / Coq map.

This section maps the elements of the Trinity S^3AI / Flos Aureus PhD programme to the Paper 3 formalism. The purpose is to make explicit the cross-references between the two bodies of work, facilitating readers who arrive from the PhD monograph context.

13.1 Mapping Summary

The PhD-to-Paper-3 mapping is intentionally narrow:

- Trinity anchor identity: from PhD Chapter 3 / flos37.tex to Equation (star), Sections 2.1, 2.4, and 5.5; role: root algebraic invariant forcing the factor 3 in Trinity monomials.
- Lucas ring: from the PhD Trinity Identity chapter to Definition 2.3, Corollary 4.5, and Proposition 5.9; role: discrete state space and exact termination locus.
- 42 precision catalogue: from PhD Chapter 34 to Definition 7.3 and Algorithm 9.5; role: empirical target set for the catalogue MDL vector M, not proof.
- Pellis expansion: from companion Papers 1 and 2 to Section 4 and Theorem 4.4; role: fine-scale correction and interference layer.
- Trinity monomial: from companion Paper 2 to Section 3 and Definition 2.2; role: coarse-grained symbolic projection.
- Algebraic AlphaPhi: from PhD Chapter 38 to Section 10.5 and equation (2); role: φ-3/2 approx 0.118, algebraic, terminating.
- Logarithmic AlphaPhi: from PhD Chapter 38 to Section 10.5 and equation (3); role: 2lnφ/pi approx 0.30635, transcendental, non-terminating.
- Coq citation map: from PhD Appendix F to Section 11.9 and Appendix E; role: formal provenance only, not empirical validation.
- Falsification ledger: from PhD Appendix B to Section 9.4 and Appendix D; role: acceptance and rejection protocol.
- Sanctioned and forbidden seeds: from PhD Chapter 34 to Section 2.2 and Algorithms 9.1–9.5; role: prevent hidden RNG cherry-picking.

13.2 The Trinity Identity as Algebraic Root The identity $\varphi_2 + \varphi$-2 = 3 is not merely symbolic decoration in the Paper 3 framework; it is the algebraic reason why 3 appears as a monomial generator in T. Specifically:

- The Lucas ring Z[φ] has norm form N(a + bφ) = a2 + ab - b2. The element 3 has norm N(3) = 9; the identity φ_2 + φ-2 = 3 expresses 3 as a self-conjugate sum within the ring.
- In the transfer operator, the canonical weight w = φ-|c| produces a geometric decay governed by φ; the pressure function achieves a critical inverse temperature at beta = logφ, which by (star) corresponds to "energy level 3" in the thermodynamic analogy.
- The complexity shell growth |SL| $\sim$ A $\cdot$ 2cL with c approx log2 3 reflects the 3-fold branching of the Lucas recurrence, itself a consequence of the Trinity anchor.

13.3 The 42 Catalogue Under the MDL Lens The 42 precision catalogue was assembled as an empirical survey of notable numerical fits between constants and symbolic expressions. In the Paper 3 MDL framework, each entry is reinterpreted as follows:

- Low-MDL entries (MDL ll L decimal): the symbolic form genuinely compresses the constant; necessary (but not sufficient) condition for the form to be meaningful.
- High-MDL entries (MDL > L decimal): the proposed symbolic form adds no compression; the entry should be re-evaluated or removed.
- Marginal entries (MDL approx L decimal): additional theory needed to determine whether the compression is structural.

The MDL analysis is purely numerical; it does not distinguish a coincidence from a theoretical relationship.

13.4 Coq Provenance Map: Role in Paper 3 The paper uses a local, reviewer-safe status taxonomy rather than reproducing a global proof ledger:

- Trinity identity φ_2+φ-2=3: status Qed; used in Equation (star), Sections 2.1, 5.5, and 13.2.
- Lucas closure of Z[φ]: status Qed; used in Proposition 2.4, Corollary 4.5, and Proposition 5.9.
- Fibonacci recurrence Fn+2=Fn+1+Fn: status Qed; used implicitly in Section 2.2 and in the Binet-formula discussion.
- Lucas formula Ln=φn+psin: status Qed; used in Definition 2.3.
- Projection map injectivity: not yet formalised; conditional on the working independence hypothesis.
- MDL optimality of PiL: not yet formalised in Coq; stated as a paper theorem and marked as future formalisation work.
- Pellis convergence bound: not yet formalised in Coq; used as an analytic result of this paper.

- Spectral gap and phase-transition statements: open conjectures, not theorem claims.

Exact global Coq counts are not reproduced in the article. They must be regenerated from the repository at submission time, and any theorem whose dependency chain touches Admitted material must be labelled conditional.

13.5 Falsification Ledger Integration The falsification protocol specifies acceptance and rejection criteria for the overall PhD programme. Paper 3 contributes the following pre-registered tests (see also Section 9.4):

Test P3-T1 (MDL improvement). For at least 30 of the 42 catalogue entries, MDL(T^*, c res) $<$ L decimal. Failure mode: fewer than 30 entries satisfy this — Trinity monomial representation fails to provide systematic compression.

Test P3-T2 (Phase transition exponent range). For at least 80% of the 42 catalogue entries, muT $\in$ [0, 2]. Failure mode: $>$ 20% of entries have muT $>$ 2 or diverge.

Test P3-T3 (Lucas ring termination; regression). For each of the sanctioned Fibonacci/Lucas seeds, Algorithm 4.1 terminates in at most 2max(|a|,|b|) steps (Theorem). This is a software regression test, not a conjecture.

Test P3-T4 (AlphaPhi disambiguation; confirmed). |alphaφ(alg) - alphaφ(log)| $>$ 0.1. Confirmed numerically by the reference implementation.

Test P3-T5 (Trinity anchor identity; confirmed). $|\varphi_2 + \varphi$-2 - 3$| < 10^{-4}8$ at 50-digit precision. Confirmed numerically (and formally by Coq Qed).

Test P3-T6 (No physical derivation; editorial). Paper contains no sentence asserting a physical constant is derived from φ or from the Trinity framework. Confirmed by author review.

13.6 Reviewer-Hardening Gate Imported from the PhD Monograph The PhD monograph contributes a useful editorial discipline that is adopted here as a reviewer-facing gate rather than as additional scientific evidence. The gate has four parts:

1. Notation lock. Every overloaded symbol must be split before review. This is why alphaφ(alg)=φ-3/2 and alphaφ(log)=2lnφ/pi are treated as different objects, with different algebraic status and different failure modes.
2. Constants lock. Public constants are cited against their current authoritative tables. If a measured value changes, the symbolic approximation is recomputed rather than silently preserved.
3. Falsification lock. Every empirical or statistical claim must have a failure criterion. Claims that have no falsifier are editorially demoted to motivation, analogy, or conjecture.
4. Proof-status lock. Coq-backed identities are cited with their proof status. Qed statements may be used as formal provenance; Admitted statements are proof sketches only; no silent Admitted-to-Qed upgrade is allowed.

This gate is deliberately conservative. It strengthens the trilogy by making the most likely reviewer criticisms explicit: look-elsewhere effects, stale constants, notation drift, overclaiming physical derivation, and formal-proof overreach.

13.7 Scott Olsen Golden-Balance Motif The Scott Olsen material is imported only as historical-geometric context, not as evidence for any catalogue entry. Olsen's The Golden Section: Nature's Greatest Secret presents the golden section as a cross-disciplinary motif spanning mathematics, natural form, art, architecture, music, and philosophy (Internet Archive bibliographic record). In Paper 3, this lineage is narrowed to one formal visual motif:

φ-2, φ-1, 1, φ, φ_2.

This "golden balance" is used as a reciprocal-scale diagram around the unit element. It does not enter the MDL score, the null-model analysis, the Coq proof ledger, or the physical-mechanism discussion. Its role is editorial and pedagogical: it gives readers a stable visual language for the same reciprocal symmetry that appears algebraically in φ_2+φ-2=3.

13.8 Erratum Discipline and Claim-Tier Taxonomy The article adopts the erratum discipline of the PhD programme: if a claim fails, it is not patched rhetorically. It is either withdrawn, substituted by a narrower claim, or deferred to future work with an explicit failure mode.

Four claim tiers govern the final text:

- Tier A — algebraic identity: exact statements such as φ_2+φ-2=3. These may be treated as mathematical results.
- Tier B — symbolic compressibility: empirical or statistical claims about low-description-length representations. These require null models, MDL accounting, and out-of-sample tests.
- Tier C — physical mechanism candidate: proposed links from symbolic structure to physical theory. These remain conjectural unless embedded in QFT, RG flow, lattice theory, or another accepted physical framework.
- Tier-D — historical, visual, or interpretive context: material such as Olsen's golden-section lineage and PhD-style visual architecture. These passages orient the reader but do not prove Tier B or Tier C claims.

This taxonomy is the main protection against reviewer over-read. A beautiful diagram or historical analogy cannot upgrade a symbolic approximation into a physical derivation.

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; CODATA 2018 recommended values: NIST https://physics.nist.gov/cuu/Constants/.; Heyrovska, R. Golden-angle FSC publications — Heyrovský Institute of Physical Chemistry.

56.1 P3 Appendix A: Formal Definitions

Appendix A: Formal Definitions

Formal-definitions triptych — Pellis-formal / Trinity-formal / pi-L-formal.

A.1 Pellis Expansion: Formal Definition Definition A.1 (Pellis sequence). A Pellis sequence for x ∈ R>0 is any sequence (ck)k ≥ 0 ⊆ Z≥ 0 such that $x = \sum_{k=0}^{\infty} c_k \, \varphi^{-k}$, converging in R. The standard Pellis sequence uses the floor function (Algorithm 4.1). The centred Pellis sequence uses the nearest-integer function, allowing signed coefficients and potentially faster convergence for certain targets.

Lemma A.2 (Finiteness). For x ∈ Q(φ) = Q(5), the standard Pellis sequence is eventually zero (termination in finitely many steps). For x notin Q(5), it has infinitely many nonzero terms.

A.2 Trinity Monomial: Formal Definition Definition A.3 (Trinity monomial). A Trinity monomial is any element of T = { n · 3k · φp · pim · eq : n ∈ Z≥ 1, (k,p,m,q) ∈ Z4 }.

Definition A.4 (Complexity function). The complexity function ell : Z≥ 1 × Z4 → Z≥ 0 is as in Section 2.6.

Definition A.5 (Trinity distance). The Trinity distance from x to T at budget L is dL(x) = minT ∈ T, ell(lambdaT) ≤ L |x - T|.

A.3 Projection Map: Formal Definition Definition A.6 (Projection map with tiebreaking). The projection map is PiL : R>0 → T, PiL(x) = argminT ∈ T, ell(lambdaT) ≤ L |x - T|, with tiebreaking rule: in case of a tie, choose the monomial with smallest n, then smallest |k|+|p|+|m|+|q|, then lexicographically among labels of equal total absolute value.

A.4 AlphaPhi Objects: Canonical Definitions Definition A.7 (Algebraic AlphaPhi). alphaφ(alg) = φ-3/2 = (5-2)/2 approx 0.11803. This is an element of Q(5); its Pellis expansion terminates.

Definition A.8 (Logarithmic AlphaPhi). alphaφ(log) = 2lnφ/pi approx 0.30635. This is transcendental; its Pellis expansion does not terminate.

Note: alphaφ(alg) ≠ alphaφ(log); the two objects play distinct roles and must not be conflated. The notation inconsistency in Chapter 38 of the PhD program is superseded by these definitions.

A.5 Operator Definitions Definition A.9 (Pellis shift). On X = T × ZN: sigma(T, (c0, c1, c2, ...)) = (PiL(T - c0), (c1, c2, c3, ...)).

Definition A.10 (Koopman operator). For f ∈ L2(X, mu): K f = f circ sigma.

Definition A.11 (Transfer operator). For weight w and f ∈ C(X): (Lw f)(x) = sumy : sigma(y) = x w(y) f(y).

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; CODATA 2018 recommended values: NIST

https://physics.nist.gov/cuu/Constants/.; Heyrovska, R. Golden-angle FSC publications — Heyrovský Institute of Physical Chemistry.

56.2 P3 Appendix B: Reproducibility Protocol

Appendix B: Reproducibility Protocol

Reproducibility triptych (Paper III) — Software stack / Reference implementation / Test suite.

B.1 Software Requirements The reproducibility target is intentionally small. A conforming implementation must provide arbitrary-precision arithmetic at 50 decimal digits or higher, exact symbolic definitions for φ, pi, and e, deterministic enumeration of Trinity labels, and a regression-test mode that exits non-zero on any failed invariant.

The minimum contract is:

- Precision: at least 50 decimal digits, preventing false agreement from binary floating point.
- Determinism: fixed enumeration order and lexicographic tie-breaks, making PiL(x) reproducible.
- Sanctioned seeds: use only F17,F18,F19,F20,F21,L7,L8 for sanctioned demonstrations, preventing hidden RNG cherry-picking.
- Forbidden seeds: never use 42, 43, 44, or 45 as RNG seeds, preserving the PhD STROBE embargo rule.
- Constants: use CODATA 2022 for NIST constants and cite the table used, preventing stale-value drift.

B.2 Reference Implementation Contract Instead of printing fragile source code in the article, the paper defines the executable contract that the repository implementation must satisfy. This avoids layout breakage in the PDF while preserving exact reproducibility requirements.

The executable contract has eight required checks:

1. Trinity anchor: $|\varphi_2+\varphi\text{-}2\text{-}3|<10^{-4}8$ at 50 digits.
2. Algebraic AlphaPhi: alphaφ(alg)=φ-3/2approx0.11803.
3. Logarithmic AlphaPhi: alphaφ(log)=2lnφ/piapprox0.30635.
4. AlphaPhi separation: |alphaφ(log)-alphaφ(alg)|>0.1.
5. Golden ratio representation: T(1,0,1,0,0)=φ to working precision.
6. Integer seed representation: T(1597,0,0,0,0)=1597 exactly.
7. Binet check: $\varphi_1$7/5approx1596.999875, with the conjugate term required for exact F17.
8. QED input value: α^{-1}=137.035999177 when CODATA 2022 is selected.

B.3 Test Suite The minimal regression suite has seven tests: exact Trinity identity, AlphaPhi disambiguation, sanctioned-seed termination, forbidden-seed exclusion, deterministic Trinity label ordering, CODATA-value selection, and absence of physical-derivation language. The last test is editorial rather than numerical: the article must not assert that any measured Standard Model constant has been derived from φ.

B.4 Data Sources Physical constants: NIST CODATA 2022 recommended values (Mathematical constants: Wolfram MathWorld (

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; CODATA 2018 recommended values: NIST https://physics.nist.gov/cuu/Constants/.; Heyrovska, R. Golden-angle FSC publications — Heyrovský Institute of Physical Chemistry.

56.3 P3 Appendix C: Notation Summary

Appendix C: Notation Summary

Notation-summary triptych — Glyph table / Operators / Constants.

- φ: golden ratio (1+5)/2 approx 1.61803.
- psi: reciprocal golden ratio φ-1=φ-1approx0.61803.
- (star): Trinity anchor identity φ_2+φ-2=3.
- Z[φ]: Lucas ring {a+bφ:a,b∈Z}.
- T(n,k,p,m,q): Trinity monomial n·3k·φp·pim· eq.
- lambda: Trinity label (n,k,p,m,q).
- ell(lambda): bit-complexity of a Trinity label.
- T: set of all Trinity monomials.
- SL: complexity shell at level L.
- PiL(x): projection of x to a complexity-L Trinity approximant.
- RL(x): residual x-PiL(x).
- sigma, K, Lw: Pellis shift, Koopman operator, and transfer operator.
- MDL(T^*,c): MDL score of the unification representation.
- μ_T(x): Trinity approximation exponent.
- μ^*=1: critical exponent of the symbolic phase transition.
- M∈R42: catalogue MDL vector.
- alphaφ(alg): algebraic AlphaPhi, φ-3/2approx0.118.
- alphaφ(log): logarithmic AlphaPhi, 2lnφ/piapprox0.30635.
- Companion Paper 1: Pellis hierarchical φ-expansions.
- Companion Paper 2: Trinity monomial representations. ## P3 Appendix D: Falsification Ledger (Paper 3 Entries)

Appendix D: Falsification Ledger (Paper 3 Entries)

Falsification-ledger triptych (Paper III) — Entries / Predictions / Status.

This ledger is part of the programme-wide falsification record. All entries are pre-registered and cannot be modified retroactively.

- P3-T1, MDL improvement over naive decimal: at least 30 of 42 catalogue entries must have MDL<L decimal. Status: pre-registered, not yet computed.
- P3-T2, phase transition exponent range: at least 80% of entries must have muT$\in$[0,2]. Status: pre-registered, not yet computed.
- P3-T3, Lucas ring termination: Algorithm 4.1 must terminate exactly for all seven sanctioned seeds. Status: pre-registered, confirmed by reference implementation.
- P3-T4, AlphaPhi disambiguation: |alphaφ(alg)-alphaφ(log)|>0.1. Status: confirmed by reference implementation.
- P3-T5, Trinity anchor identity: $|\varphi_2+\varphi$-2-3$|<10^{-4}$8 at 50 digits. Status: confirmed by reference implementation.
- P3-T6, no physical derivation claim: the paper must contain no sentence asserting that a physical constant is derived from the GOLDEN CHAIN (Trinity-Pellis Bridge) framework. Status: editorial, confirmed by author review. ## P3 Appendix E: Coq Provenance Map (Paper 3 Scope)

Appendix E: Coq Provenance Map (Paper 3 Scope)

Coq-provenance triptych — Lemma map / Dependencies / Scope frame.

This appendix records the local Coq proof status of identities cited in Paper 3. Global proof-ledger counts are intentionally omitted and must be regenerated at submission time.

- Trinity identity $\varphi_2+\varphi$-2=3: t27/theories/TrinityIdentity.v, status Qed, cited in Equation (star).
- Lucas ring closure: t27/theories/LucasClosure.v, status Qed, cited in Proposition 2.4.
- Fibonacci recurrence Fn+2=Fn+1+Fn: t27/theories/Fibonacci.v, status Qed, cited implicitly in Section 2.2.
- Lucas number formula Ln=φn+psin: t27/theories/LucasClosure.v, status Qed, cited in Definition 2.3.
- Catalog42 representative layer: t27/proofs/trinity/Catalog42.v, status source-audited with 13 Qed. markers, 0 Admitted. markers, and 28 representative theorem obligations. The full 42-row formula catalogue is restored in Paper 1, Section 8.3, but the current Coq layer is representative rather than one named theorem per row.
- Trinity proof layer: t27/proofs/trinity/, status source-audited with 13 Coq files, 281 theorem-like declarations, 122 Qed. markers, 32 Admitted. markers, and 0 Abort. markers. This is the physical-constant proof layer relevant to the catalogue.
- Formula evaluator layer: t27/proofs/trinity/FormulaEval.v, status source-audited with 17 Qed. markers and 0 Admitted. markers; its header records the 69-formula Trinity v0.9 framework at the evaluator-interface level.

- Projection map injectivity: not formalised, conditional on the independence hypothesis, cited in Proposition 3.1.
- MDL optimality of PiL: not formalised, cited in Theorem 7.2.
- Pellis convergence bound: not formalised, cited in Theorem 4.4.
- Spectral gap and phase transition conjectures: open conjectures, cited in Conjectures 6.5 and 8.1.

If any Admitted item is transitively required for a Paper 3 theorem, this appendix must be updated at the revision stage and the affected theorem must be marked conditional. Abandoned proof attempts are not used as evidence for any result cited in this paper.

End of Paper 3 v2. Authors: Dmitrii Vasilev, Stergios Pellis, Scott Olsen. Date: May 2026. Defense anchor: φ_2 + φ-2 = 3. Defense date: 2026-06-15. All claims are mathematical; no physical constants are derived from first principles; no metaphysical claims are advanced.

APPENDIX A

TRINITY DNA Provenance

Repository evidence is an architectural map, not a substitute for physical derivation.

APPENDIX RULE

The following pages preserve the v21 TRINITY DNA integration as provenance evidence. Every claim is placed into a review-safe evidential tier: mathematical identity, numeric parametrization, implementation artifact, conjectural mechanism, or historical context.

TRINITY S^3AI - the cognitive stack rooted in the identity φ^{2} + φ^{-2} = 3

References: Vasilev, D. Flos Aureus Monograph: Trinity Constants. DOI 10.5281/zenodo.19227877.; CODATA 2018 recommended values: NIST https://physics.nist.gov/cuu/Constants/.; Heyrovska, R. Golden-angle FSC publications — Heyrovský Institute of Physical Chemistry.

Numeric Formats — A Short History and the GoldenFloat Family

Every computing substrate is, at bottom, a choice of how to spend bits between range (exponent) and precision (mantissa). The GoldenFloat (GF) family is a small contribution to a long lineage: it fixes that split along the golden ratio. This chapter places GF in the historical taxonomy of machine numeric formats and then states, with explicit claim discipline, what is verified about the family and what is not.

57.1 A Short History of Machine Numeric Formats

Era / family	Representative formats	Idea
IEEE 754 binary (1985, 2008, 2019)	binary16, binary32, binary64, binary128, binary256	Standardised sign/exponent/mantissa with subnormals, NaN, ∞
IEEE 754 decimal	decimal32, decimal64, decimal128	Base-10 significand for exact financial rounding
Pre-standard & extended	FP80 (x87), double-double, quad-double, IBM HFP, MBF, VAX F/D/G/H, Cray float	Vendor formats before and around the standard
ML reduced precision	bfloat16, TF32, FP8 (E4M3, E5M2), FP6 (E3M2, E2M3), FP4 (E2M1), MXFP8/6/4, NF4, AFP	Trade precision for throughput on tensor hardware
Block / scaled	block floating point, shared-exponent, microscaling (MX), stochastic rounding	Amortise an exponent across a block of values
Alternative number systems	Posit8/16/32/64, LNS, Unum I/II, tapered floating point	Re-think the range/precision curve itself
Integer & fixed point	INT4–INT128, UINT4–UINT128, Q-format fixed point, BCD, minifloat	No exponent; range fixed by scale

The trend of the last decade is unambiguous: narrower formats, more exponent/mantissa choices, and per-block scaling. FP8, MXFP4 and NF4 exist because inference economics reward every bit saved. GF sits in this reduced- precision niche, with one extra design rule.

57.2 The GoldenFloat (GF) Family — Design Rule

For a fixed bit budget b = 1 (sign) + E (exponent) + M (mantissa), GF chooses the split so the ratio E/M approximates $1/\varphi \approx 0.6180339887$ (φ = the golden ratio, $\varphi^2 + \varphi^{-2} = 3$). The intuition is that φ balances the geometric growth of representable range against mantissa density along a single scale-invariant line. Whether this yields a practical accuracy advantage over IEEE-style splits is an open empirical question — the claim here is only that the bit allocation is φ-structured, not that it is provably optimal.

57.2.1 Verified family constants (R5)

The following four columns were independently re-derived for this chapter and are VERIFIED:

Format	S+E+M	Bits	BIAS = 2^(E−1)−1	EXP_MAX = 2^E−1	E/M	φ-distance |E/M−1/φ|	t27 spec
GF4	1+1+2	4	0	1	0.500	0.118	gf4.t27
GF8	1+3+4	8	3	7	0.750	0.132	gf8.t27
GF12	1+4+7	12	7	15	0.571	0.047	gf12.t27
GF16	1+6+9	16	31	63	0.667	0.049	gf16.t27
GF20	1+7+12	20	63	127	0.583	0.035	gf20.t27
GF24	1+9+14	24	255	511	0.643	0.025	gf24.t27
GF32	1+12+19	32	2047	4095	0.632	0.014	gf32.t27
GF64	1+24+39	64	8388607	16777215	0.615	0.003	gf64.t27
GF256	1+97+158	256	$2^{96}-1$	$2^{97}-1$	0.614	0.004	gf256.t27 (γ)
TF3 †	1+3+4	8	3	7	0.750	0.132	tf3.t27
GFTernary ‡	2-bit code	2	—	—	—	0.000	gfternary.t27

(Per-format PHI_BIAS is tabulated separately below. The t27 spec column lists the canonical specification file: GF4–GF64 live under gHashTag/t27 :: specs/numeric/; GF256 under tt-trinity-gamma. GF256's stored exponent-bias constant is unreconciled with the 2^(E−1)−1 formula and is treated as OPEN — only its geometry (97/158 split, φ-distance 0.004) is verified. Live links are in the SSOT, t27 :: docs/NUMERIC_FORMATS_SSOT.md.

The last two rows are not float-ladder rungs: † TF3 is an 8-bit ternary-weight encoding that reuses GF8's 1:3:4 geometry (so it shares GF8's constants exactly); ‡ GFTernary is the 2-bit {−φ, 0, +φ} limit — not a split-based float, so its exponent/mantissa columns are N/A and its φ-distance is 0 by construction (its nonzero magnitude is exactly φ). Candidate widths GF6 and GF128 are proposal-only and are not listed.)

By φ-distance the closest split to 1/φ is GF64 (0.003), then GF256 (0.004) and GF32 (0.014); GF12 (0.047) is the best only among formats ≤ 16 bits. The bit budgets, the BIAS formula and the φ-distances above all check out exactly (GF256's stored bias excepted, as noted).

57.2.2 PHI_BIAS — empirical per format, NOT a closed-form law

PHI_BIAS is a per-format mantissa-rounding bias used when quantising from higher precision. An earlier specification published the formula PHI_BIAS = EXP_MAX − BIAS. This is RETRACTED: it reproduces only GF64 ($8388608 = 16777215 - 8388607 = 2^{23}$) and none of the other seven values. Seven-plus candidate closed forms were tested (2·BIAS−2; EXP_MAX−1; BIAS−1; 2^E−1; golden-ratio floor; Lucas-indexed) and no single formula reproduces all eight values.

PHI_BIAS is therefore defined empirically per format ("H_E"), each value justified individually and held in its gfN.t27 spec:

Format	PHI_BIAS	Status
GF4	0	EMPIRICAL FIT (minimal for 4-bit)
GF8	1	EMPIRICAL FIT (minimal; coincides with $L_1 = 1^2$)
GF12	2	EMPIRICAL FIT
GF16	60	NORMATIVE (production; $2 \cdot \mathrm{BIAS} - 2 = 2^6 - 4$)
GF20	289	EMPIRICAL FIT (coincides with 17^2)
GF24	1364	EMPIRICAL FIT (coincides with Lucas L_{15})
GF32	0	EMPIRICAL FIT (minimises MSE vs round-to-nearest-even)
GF64	8388608	EMPIRICAL FIT (equals EXP_MAX − BIAS for this format only)

The Fibonacci/Lucas/perfect-square coincidences ($\mathrm{GF24} = L_{15}$, $\mathrm{GF20} = 17^2$) are descriptive, not prescriptive — they must not be used to generate PHI_BIAS for new formats.

57.2.3 Implementation status (claim discipline)

- GF16 — VERIFIED. Production Rust (trios-trainer-igla) plus C codegen; the one format used in the live training pipeline.
- GF32 — CLAIMED IN SOURCE BUT NOT INDEPENDENTLY CONFIRMED. At least three mutually incompatible bit-layouts exist across the ecosystem (12/19, 13/18, and an IEEE-fp32-identical variant); see the companion chapter Trinity GoldenFloat Family — Unified Specification Audit.
- GF4 / GF8 / GF12 / GF20 / GF24 — ROADMAP. Spec extract-only with C codegen present; no Rust implementation yet.
- GF64 — ROADMAP. PHI_BIAS defined in spec; C codegen still missing.
- GF236 — does not exist (resolved). 236 is the mantissa width of IEEE binary256 (1 + 19 + 236 = 256 bits), not a GoldenFloat format. The name “GF236” conflated that mantissa count with a format label. The canonical 256-bit GoldenFloat is GF256 (see the GF256 section); there is no GF236.

57.3 GoldenFloat in the Three-Strand Scientific Argument

The format family is not a bolt-on engineering artefact; it is where the compendium's three strands meet a bit layout, all anchored in $\varphi^2 + \varphi^{-2} = 3$.

- Strand I — Symbolic Grammar / MDL (Paper 1). The split rule $E/M \to 1/\varphi$ is an instance of the φ-grammar G_φ: the same golden-ratio structure the symbolic-search framework applies to dimensionless constants is here applied to the information budget of a single number. Choosing the split to minimise $|E/M - 1/\varphi|$ is a minimum-description-length-style criterion on the format itself — range and precision are partitioned along the one ratio the Lucas recurrence makes self-similar ($\varphi^{2n} + \varphi^{-2n} \in \mathbb{Z}$). GF16's 6:9 split is the 16-bit realisation of that MDL-optimal partition.
- Strand II — Silicon Anchor (fm-13, appx-hw). GF16 is the format the Three Crowns actually compute in: the reset-anchor chain $\varphi^2 + \varphi^{-2} = 3 \Longrightarrow L_2 = 3 \Longrightarrow \mathrm{dot4}_{(}\mathrm{GF(16)}_{)} = 30 \mapsto$ 0x47C0 runs through the GF16 encoding (the integer 30 is the half-word 0x47C0 under GF16). The Galois field $GF(2^4)$ of the Euler crown and the GoldenFloat GF16 share the "GF" name precisely because both are the φ-anchored substrate of one silicon line.
- Strand III — Hierarchical Expansion (Paper 3). The φ^{-k} additive expansions Strand III uses to approximate physical constants are the same φ-powers the format ladder samples at successive bit widths; each GF member is one rung of that expansion realised in hardware.

In short, the GoldenFloat ladder is the silicon-and-bits projection of the same $\varphi^2 + \varphi^{-2} = 3$ identity that anchors the whole compendium. The numeric format and the physics-facing symbolic search are two readings of one algebraic object — which is why the family belongs inside the scientific argument, not only in an engineering appendix.

57.4 GF256 — Several Kinds (Disambiguation)

Like "GF16", "GF256" is overloaded across the repositories and denotes at least three distinct things. They must never be conflated in a publication.

Kind	Spec	What it is	Status
GF(2^8) field	trinity:.../gf256.tri	Galois field $\mathbb{F}_{256}$ for crypto / erasure; primitive poly 0x11D (285), not the AES 0x11B	Implemented (Zig)
GF256 float (γ-ratio)	tt-trinity-gamma:.../gf256.t27	256-bit φ-float, layout 1:97:158, E / M 0.614, φ-dist 0.004	FPGA spec (bias OPEN)
GF256 float (range)	zig-golden-float whitepaper	proposed binary256-range φ float (~19 exp bits)	Candidate only

1. GF256 as GF(2^8). The finite field with 256 elements (the AES / Reed–Solomon substrate). Trinity fixes the primitive polynomial 0x11D, which is not the AES polynomial 0x11B; a runtime selector (src/crypto/gf256.zig) can switch representations. "Several kinds" in this sense means different primitive polynomials over GF(2^8) — each gives an isomorphic but bit-incompatible field. This is the same algebraic "Field" meaning as the GF(2^4) Euler crown, not a real-number float.
2. GF256 as a 256-bit GoldenFloat (ratio-optimised). [S:1 E:97 M:158] = 256 bits; E/M = 97/158 $\approx$ 0.614, φ-distance 0.004 (VERIFIED). R5 flag: the spec's stored exponent-bias constant (0x7F:0xFFFF..., $\approx 2^{71}$) does not match 2^(E−1)−1 = $2^{96}-1$; treat the bias as OPEN until reconciled.
3. GF256 as a binary256-range candidate. The whitepaper lists GF128 / GF256 as future formats meant to match the dynamic range of binary128 / binary256 with φ-aligned splits — a different design goal (range-matched, few exponent bits) from kind 2 (ratio-matched, 97 exponent bits). NOT CLAIMED as specified.

57.5 TF3 and GFTernary — the Ternary Members (two distinct things)

Like "GF256", the ternary name is overloaded — two different objects:

- TF3 (tf3.t27) is a real 8-bit format — layout 1:3:4, the same geometry as GF8 — used to encode ternary neural-network weights. It is a storage container, not a 2-bit format.
- GFTernary is the 2-bit member, values in $\{-\varphi, 0, +\varphi\} = \varphi\cdot\{-1, 0, +1\}$: a ternary-weight quantizer (cf. TWN; BitNet b1.58) with the scale

fixed at φ rather than learned per layer. φ-distance 0.000 by construction (its nonzero magnitude is exactly the anchor φ). It is the bulk-quantisation member of the hybrid ternary + GF16 architecture and the $\{-1, 0, +1\}$ substrate of the Three Crowns. It now carries a normative spec (gfternary.t27); Rust and codegen remain ROADMAP (same tier as GF64).

57.6 A Note on Split Revisions (History, R5)

The exponent:mantissa splits are not frozen — they have a revision history, and φ-distance figures are only meaningful against a stated split:

- GF32 has been published as 8:23 (IEEE-fp32-identical), 13:18, and 12:19 (current canonical, φ-dist 0.014).
- GF64 has been published as 21:42 (double-Fibonacci, φ-dist ≈ 0.264 in the older metric) and 24:39 (current canonical, φ-dist 0.003).

The verified table above uses the current t27 canonical splits, which the latest zig-golden-float whitepaper family table now matches. Earlier whitepaper tables and the companion GF32 audit chapter preserve the superseded splits — always cite the split explicitly when quoting a φ-distance.

57.7 Candidate Formats (Proposed, No Normative Spec)

- GF6 — φ-gap fill between GF4 and GF8 (whitepaper hint: FP6 E2M3 suggests φ-distance ≈ 0.05).
- GF128 — binary128-range φ candidate (paired with the GF256 range variant).

These are listed in the whitepaper as proposals only; they are NOT CLAIMED as implemented or specified.

57.8 The Complete Lineup (Consolidated)

Member	Width	Split S:E:M	Canonical spec	Status
TF3	8	1:3:4 (= GF8 geometry)	tf3.t27	8-bit ternary-weight encoding
GFTernary	2	$\{-\varphi, 0, +\varphi\}$	gfternary.t27	spec; codegen ROADMAP
GF4	4	1:1:2	gf4.t27	Extract-only / ROADMAP

Member	Width	Split S:E:M	Canonical spec	Status
GF8	8	1:3:4	gf8.t27	Experimental
GF12	12	1:4:7	gf12.t27	Extract-only / ROADMAP
GF16	16	1:6:9	gf16.t27	Production (VERIFIED)
GF20	20	1:7:12	gf20.t27	Extract-only / ROADMAP
GF24	24	1:9:14	gf24.t27	Extract-only / ROADMAP
GF32	32	1:12:19	gf32.t27	Spec (3 historical layouts — see audit)
GF64	64	1:24:39	gf64.t27	Spec complete; no Rust gen yet
GF256 (float)	256	1:97:158	tt-gamma:.../gf256.t27	FPGA spec (bias OPEN)
GF256 (field)	—	GF(2^8), poly 0x11D	trinity:.../gf256.tri	Implemented (Zig)
GF6, GF128, GF256-range	—	proposed	—	Candidate only (NOT CLAIMED)

Ten members carry t27 specs (TF3, GFTernary, GF4–GF64); GF16 is the only one with a production implementation and published benchmarks.

57.9 What Makes GoldenFloat Unique — A Narrow, Defensible Claim

The claim is deliberately narrow so it cannot be refuted:

> GoldenFloat is the only published floating-point family whose exponent:mantissa split, at every bit width on a 4-to-256-bit ladder, is generated by one closed rule — E/M → 1/φ — anchored in a single algebraic identity, $\varphi^2 + \varphi^{-2} = 3$.

This is a statement about design methodology, not performance. Every other format chooses its split for range, hardware convenience, or empirical accuracy; none derives the entire ladder from one closed-form constant:

Format	Split rule (how E:M is chosen)	E/M	φ-distance
IEEE binary32	committee balance of range vs precision	8:23 = 0.348	0.270
bfloat16	truncate fp32 (keep 8-bit range)	8:7 = 1.14	0.525
TF32 (NVIDIA)	fp32 range + fp16 precision	8:10 = 0.80	0.182
FP16	IEEE half	5:10 = 0.50	0.118
FP8 E4M3 / E5M2	empirical for deep learning	4:3 / 5:2	—
Posit (Gustafson)	tapered regime bits (accuracy peaks near 1.0)	variable	n/a
NF4	normal-distribution quantiles (no exponent)	n/a	n/a
LNS	logarithmic domain (no mantissa)	n/a	n/a
GoldenFloat	E / M → 1 / φ from $\varphi^2 + \varphi^{-2} = 3$ (generative)	→ 0.618	→ 0

The uniqueness is the generative principle, which is verifiable by inspection of the design rule. Additionally, GF accumulators are Lucas-closure-safe: $\varphi^{2n} + \varphi^{-2n}$ is always an integer (a Lucas number), so φ-scaled sums stay exact in the integer-backed storage (u8/u16/u32/u64). That property is real and checkable; it is not shared by range-chosen or quantile-chosen formats. Integer-backed storage also places GoldenFloat in the integer-only-arithmetic regime whose hardware advantages are well documented — no floating-point unit required, with large energy and area savings versus float datapaths (Jacob et al., 2018, arXiv:1712.05877).

57.9.1 What GoldenFloat Does NOT Claim (pre-empting the critics)

To stay defensible, the following are explicitly not claimed:

1. Not optimal. That 1/φ is the best split is an OPEN question, not a theorem. We claim it is principled and self-consistent, not optimal.
2. Not faster or more accurate in general. GF does not claim to beat bf16, FP8 or Posit on throughput or accuracy across the board. Performance is measured only for GF16 (BENCH-004b: 97.67% MNIST MLP, 0.00% accuracy gap vs f32; unit-level 47–59× fewer FPGA resources). All other widths are ROADMAP — no benchmark is asserted for them.

3. Not a physics claim. The φ-distance is a bit-allocation heuristic with a plain definition ($|E/M - 1/\varphi|$); no mystical or physical property of φ is invoked inside the format. (The cosmological φ work lives in other chapters and is separately claim-tagged.)
4. GF32 is not a single format yet. Three historical layouts existed (8:23, 13:18, 12:19); only 12:19 is canonical now — disclosed, not hidden.
5. Not a refutation of data-dependent allocation. The modern low-precision literature makes the exponent:mantissa split depend on the target data distribution, not on any universal constant: FP8 ships two splits — E4M3 (precision-leaning) and E5M2 (range-leaning) — chosen per tensor role (FP8 Formats for Deep Learning, arXiv:2209.05433), with the exponent count as the decisive tunable (FP8 Quantization: The Power of the Exponent, arXiv 2208.09225); NF4 is information-theoretically optimal only for a normal distribution (QLoRA, Dettmers et al. 2023); posits taper precision around 1.0 (Gustafson, 2017). GoldenFloat takes the opposite, consistency-first stance — one constant fixes the whole 4-to-256-bit ladder. Its claim is therefore honestly bounded: GF16's 6:9 split (≈ 2:3) coincides with a sensible precision-leaning allocation, which is why it reaches f32-parity (BENCH-004b) — the φ-ratio selects a reasonable split; it is not shown to beat a distribution-tuned format. Where per-layer adaptivity matters (outlier-heavy tensors, wide gradient ranges), FP8 / MX / NF4 retain the advantage.

57.10 Measured Comparison — IGLA RACE v2 Format Sweep

The claims above are geometric (the φ-distance of each format's split). The question a practitioner actually asks is different: does a GoldenFloat member train a model as well as the format it would replace? The IGLA RACE v2 sweep answers this for two GF widths. It trains the same byte-level language model under each numeric format and reports validation bits-per-byte (val_bpb, lower is better). The run is a frozen 30-log snapshot (2026-05-25; 2–3 seeds per format; two fp8_e4m3 runs died before the first evaluation and are excluded — marked NO_EVAL).

Format	Seeds	Mean val_bpb	Best val_bpb	Reading
f32	3	2.5414	2.5042	full-precision baseline
fp16	3	2.5501	2.5348	best 16-bit in this sweep
gf16	3	2.5725	2.5267	+0.031 bpb vs f32; beats bf16

Format	Seeds	Mean val_bpb	Best val_bpb	Reading
bf16	3	2.6135	2.5751	worst of the 16-bit group
posit8	2	2.9322	2.9269	best 8-bit in this sweep
gf8	2	2.9322	2.9269	bit-identical to posit8
int8	3	3.4189	3.3764	linear 8-bit, well behind
mxfp8	2	3.4677	3.4640	≈ int8
fp8_e5m2	2	3.8528	3.8358	range-biased FP8, degrading
nf4	2	3.9985	3.9931	4-bit, at the edge
int4	3	7.0184	7.0058	does not learn
fp8_e4m3	2	—	—	NO_EVAL (died pre-eval)

What the sweep supports. At 16 bits, GF16 lands 0.031 bpb from full f32 and beats bf16 (2.5725 vs 2.6135): the φ-structured split pays no penalty for its structure — it sits in the near-lossless band. At 8 bits, GF8 matches the best evaluated 8-bit format (posit8) exactly and clears int8, mxfp8 and the range-biased fp8_e5m2 by a wide margin. Both are measured results on a real training run, not projections.

What the sweep does NOT support — read honestly.

1. GF16 is not the single best 16-bit format here. Plain fp16 edges it (2.5501 < 2.5725). The defensible claim is parity-class with fp16 and a measured win over bf16, not supremacy.
2. GF8 and posit8 produced bit-identical curves. Both are 8-bit tapered-precision layouts, so this is unsurprising; we treat them as equivalent at this width and do not read it as a GF8 victory over posit.
3. No GF-vs-E4M3 head-to-head exists in this data. The precision-oriented FP8 variant the literature recommends, fp8_e4m3, did not survive to its first evaluation — exactly the comparison non-claim #5 above is about, still open.
4. NF4 and fp8_e5m2 underperform their reputation here. That is expected: this is a from-scratch byte-level LM, not the post-training weight-quantization regime (frozen near-Gaussian weights) in which NF4 is information-optimal. The sweep measures GF as a training numeric format — the use Trinity actually makes of it.

In short, the benchmark turns the φ-narrative from a geometric argument into a bounded empirical one: GF16 and GF8 are competitive-to-best in their weight classes on this task, with the honest gaps named. It does not establish a universal optimum — consistent with the literature consensus (FP8 E4M3/E5M2, NF4, posit) that the best exponent:mantissa split is data-dependent.

57.11 Use Cases and Conditions of Application

Member	Where it fits	Condition under which to prefer it
GFTernary	bulk-quantized layers in a hybrid net	when most weights tolerate $\{-\varphi, 0, +\varphi\}$ and a few critical layers run GF16
GF8	ultra-low-power edge / sensors	extreme memory/energy limits; accuracy non-critical
GF12	mid-range edge / audio	a 12-bit lane is available and GF8 is too coarse
GF16	production training & inference	the proven case: want f32-class accuracy with integer-backed `u16` storage and no hardware half-type dependency
GF20 / GF24	high-precision edge / server inference	need more mantissa than GF16 but less than fp32
GF32	fp32 drop-in	only with the 12:19 canonical layout pinned explicitly
GF64	double-precision scientific	ROADMAP — no t27 spec; do not deploy yet
GF256 float	octuple-range research	ROADMAP — bias constant unreconciled
GF(2^8) field	crypto / Reed–Solomon / VSA on the Euler crown	when you need finite-field XOR/LUT arithmetic, not real numbers

57.12 Global Positioning — Where GoldenFloat Sits in the Format World

The AI-numerics landscape is fragmenting into a zoo of incompatible formats — bfloat16, TF32, FP8 (E4M3 / E5M2), FP6, FP4, MXFP8/6/4, NF4, AFP — each tuned to one vendor's silicon or one data distribution. GoldenFloat's global pitch is therefore not "a better float than bf16/FP8 for everyone." It is:

> One principled, integer-native format ladder — from 2-bit ternary to 256-bit — that lets a whole mixed-precision system live in one φ-family, on AI hardware you control, instead of a patchwork of vendor-specific formats.

57.12.1 Competitive map (honest)

Against	On their turf	GoldenFloat's actual position
bf16 / fp16 (datacenter training)	GPUs / TPUs have a native hardware path	On a commodity GPU GF loses (emulated). But at silicon parity GF16 ties fp16 / beats bf16 on accuracy (sweep above) — and Trinity builds its own chips (silicon-parity argument below).
FP8 / MX / NF4 (quantization)	per-layer-adaptive, vendor kernels	Competes on the edge and adds integer portability they lack; on commodity accelerators they keep the per-layer-adaptivity edge
posit (Gustafson, tapered)	closest design cousin — both principled, both beyond IEEE	Edge is the generative φ-rule + integer-backed + Lucas closure + a shipped silicon line, not raw accuracy (GF8 ties posit8 bit-for-bit)
BitNet / ternary (the recent low-bit trend)	pure $\{-1, 0, +1\}$ weight quantization	Offers the ternary member plus a graceful precision ladder above it (GF8/12/16) for the layers ternary cannot carry

57.12.2 The silicon-parity argument — we build the chips

Numeric formats do not win on mathematics; they win when silicon supports them. Low-precision is, in the literature's phrase, a gem of hardware–software co-design: bfloat16 became the training default because Google co-designed the TPU around it (a bf16 multiplier is roughly half the area of an FP16 unit); FP8 followed once Hopper-class tensor cores gave it a native path; NF4 needed bespoke kernels. Emulating a format the hardware does not support is prohibitively slow — that, not any property of the number system, is why an unsupported format "loses."

GoldenFloat has the same enabler every winning format had: co-designed silicon. The Three Crowns are chips where GF16 is the native

ALU format — integer-backed and Lucas-exact, not an emulated guest. So the honest comparison is not "GF vs bf16 on someone else's GPU" (GF loses, by emulation) but "GF on its native silicon vs bf16 on its native silicon" — and on the axis that does not depend on whose fab you use, accuracy, the IGLA RACE sweep above already shows GF16 ties fp16 and beats bf16. Tapered-precision formats of GF's class also turn directly into hardware wins on custom silicon: published posit engines report 45–80 % FPGA-resource and ~18 % energy reductions (SPADE, arXiv:2601.17279; ExPAN(N)D, arXiv:2010.12869), and the FP8 study documents the same area argument (arXiv:2209.05433).

The honest caveat (claim discipline). Trinity's current silicon is 130 nm research-class, not a datacenter part. So this is a co-design + measured-accuracy + edge/embedded-throughput argument, not a claim to match an H100 on tokens-per-second today. The defensible statement is precise: GoldenFloat is hardware-native where it runs, and accuracy-competitive when it runs — the two properties a numeric format needs to be taken seriously, both of which GF now has evidence for.

57.12.3 Where GoldenFloat is the first choice

φ-anchored edge / embedded AI hardware you control (FPGA / ASIC / MCU / CPU), where you want, together: (a) the Three Crowns silicon as target, whose ALUs are co-designed around the φ-step and Lucas closure; (b) integer-backed portability — stable across Zig / Rust / C++ / WASM / LLVM IR, sidestepping the f16 compiler-ecosystem issues, with no hardware half-type required; (c) a single self-consistent ladder from 2 to 256 bits. This is the value of the full ladder, not any single rung: a mixed-precision model can keep ternary bulk weights (GFTernary / TF3), low-bit compute (GF8 / GF16) and higher-bit accumulation (GF32 / GF64) inside one φ-family, with clean, Lucas-exact conversions between rungs — instead of stitching together unrelated formats (BitNet ternary + INT8 + FP8 + bf16 + fp32), each with its own semantics, lossy boundaries and toolchain. The breadth itself is the moat: a co-design and toolchain-coherence advantage (only GF16 is so far measured), not a per-rung accuracy claim.

57.12.4 Where it is not the right tool

Commodity GPU / TPU inference that already has hardware bf16 / FP8 paths — there GoldenFloat carries no throughput advantage and should not be used. The claim is coherence and portability, not peak datacenter throughput.

57.13 Adversarial Q&A — Disarming the Critics

- "φ is arbitrary numerology; why not 2/3 or e^{-1}?" — 2/3 ≈ 0.667 and 1/φ ≈ 0.618 are close, and GF16's 6:9 split is 2/3; the family simply names the limit it targets. And φ-anchored number representation is not numerology: it is a 60-year-old, studied idea — base-φ (Bergman, 1957), Zeckendorf / Fibonacci coding, and the Golden Ratio Encoder, an analogue-to-digital scheme that uses φ as its β-expansion base precisely for robustness to quantiser error (Ward, 2008, arXiv:0806.1083). GoldenFloat is a distinct construction (a floating-point split-ratio, not a φ-radix integer code) and does not inherit those robustness proofs — but the lineage shows that anchoring representation on φ is principled, not crankish. The claim remains the generative rule + Lucas closure, not that φ beats every nearby constant; optimality is flagged OPEN.
- "bf16 / FP8 already won the low-precision race." — Agreed for commodity accelerators; GF makes no throughput claim there (non-claim #2). GF's niche is φ-derived self-consistency, integer-backed portability, and silicon co-design.
- "Only GF16 has numbers." — Correct, and stated plainly: every other width is ROADMAP. Nothing is asserted for them.
- "GF32 / GF256 have inconsistent specs." — Disclosed in full (three GF32 layouts; GF256 bias constant flagged OPEN). Honest disclosure is the defense.
- "Is this numerology?" — No physical claim is made inside the formats; φ-distance is a measurable bit-allocation metric. The anti-numerology discipline is the same one applied to the rest of this compendium.

57.14 Provenance — Single Source of Truth

This chapter is derived. The canonical single source of truth for the GoldenFloat numeric-format family is gHashTag/t27 :: docs/NUMERIC_FORMATS_SSOT.md, which links every per-format specs/numeric/gfN.t27. If this chapter and that SSOT ever disagree, the SSOT wins and this chapter is corrected.

- SSOT: t27 :: docs/NUMERIC_FORMATS_SSOT.md
- Per-format specs (under t27/specs/numeric/): gf4.t27, gf8.t27, gf12.t27, gf16.t27, gf20.t27, gf24.t27, gf32.t27, gf64.t27, tf3.t27, gfternary.t27, plus goldenfloat_family.t27 and phi_ratio.t27 (GF256 under tt-trinity-gamma)
- Reference implementation / whitepaper: https://github.com/gHashTag/zig-golden-float (docs/whitepaper.md, docs/whitepaper/gf16_comparison.md) — itself derived from the SSOT.

Spec-conflict and naming-collision details ("GF16-Float" vs the GF(2^4) Galois field of the Euler crown) are documented in the companion audit chapter; the PHI_BIAS retraction and per-format values are recorded in the SSOT above.

GF Format Audit — Trinity GoldenFloat Family Unified Specification

58.1 Executive Summary

Format	Status	Nature
GF4	Proposal	4-bit φ-float
GF8	Experimental	8-bit φ-float
GF12	Proposal only	12-bit φ-float
GF16	3 Production	16-bit φ-float [1][6][9] (= IBM DLFloat layout)
GF20	Proposal	20-bit φ-float
GF24	Proposal only	24-bit φ-float
GF32	! Conflicting specs	32-bit φ-float — MULTIPLE incompatible definitions
GF64	Proposal	64-bit φ-float [1][24][39]
GF256	Spec (bias OPEN)	256-bit φ-float [1][97][158]; γ-crown FPGA spec
TF3	Spec	8-bit ternary-weight encoding [1][3][4] (= GF8 geometry)
GFTernary	Spec; codegen ROADMAP	2-bit $\{-\varphi, 0, +\varphi\} = \varphi \cdot \{-1, 0, +1\}$ (gfternary.t27)
GF236	7 Non-existent	Name confused 236 (IEEE binary256 mantissa width) with a format; the canonical 256-bit float is GF256

58.2 The Two Meanings of "GF"

In the Trinity ecosystem, "GF" carries two incompatible technical meanings that must be explicitly disambiguated in any publication:

1. GoldenFloat — a family of φ-structured floating-point formats (GF4, GF8, GF12, GF16, GF20, GF24, GF32, GF64, GF256, plus the 2-bit

GFTernary and the 8-bit ternary-weight TF3). These are real-number representations for ML/edge inference.

2. GF(16) Galois Field — the finite field $\mathbb{F}_{16} = \mathrm{GF}(2^4)$, the algebraic substrate of the Euler crown (tt_um_ghtag_trinity_gf16). Operations are XOR/popcount/LUT-based. No hardware multipliers.

The naming collision is historical, not intentional. In peer-reviewed publications, we must use distinct labels: GF16-Float vs GF16-Field.

58.3 GF32 — Conflicting Specifications (Critical)

GF32 is real, but the Trinity ecosystem currently contains at least three mutually incompatible bit-layout definitions for it. This is a publication risk that must be honestly disclosed.

58.3.1 Specification A — "DLFloat-like" (zig-golden-float whitepaper)

- Layout: [sign:1][exp:8][mant:23]
- Bias: 127
- Status: "Specification only — no implementation"
- φ-distance: Not optimised; identical layout to IEEE fp32
- Source: zig-golden-float/docs/research/gf16_family_phi_layout.md

58.3.2 Specification B — "φ-optimised 12/19" (t27 spec)

- Layout: [sign:1][exp:12][mant:19]
- Bias: 2047
- Status: Spec-complete, seal registered in .trinity/seals/GF32.json
- φ-distance: 0.632 (exp/mant = $12/19 \approx 0.632$, phi_distance = 0.014)
- Source: trinity/t27/specs/numeric/gf32.t27

58.3.3 Specification C — "φ-optimised 13/18" (trios-trainer-igla Rust)

- Layout: [sign:1][exp:13][mant:18]
- Bias: 4095
- Status: Implemented in Rust with full test suite
- Rationale: 13:18 split gives ratio $\approx 1/\varphi \approx 0.618$
- φ-weighted rounding: bias = $\varphi \times 0.5 \approx 0.809$ when compressing FP32 mantissa
- Source: trios-trainer-igla/src/phi_numbers/gf32.rs

58.3.4 Specification D — "Fibonacci-derived" (PhD thesis flos_06.tex)

- Layout: [sign:1][exp:8][mant:23]
- Rationale: Mantissa widths drawn from Fibonacci sequence; exponent follows $\varphi^2 + \varphi^{-2} = 3$ identity
- Status: Academic documentation
- Source: trios/docs/phd/chapters/flos_06.tex

58.3.5 Recommendation for the Symmetry paper

Honest disclosure is the only viable path. We should write:

> "The Trinity ecosystem currently hosts multiple competing GF32 specifications. The zig-golden-float whitepaper (v2.0) defines GF32 as [1][8][23] with bias 127, explicitly marking it as 'specification only, no implementation.' The t27 technology tree defines an alternative [1][12][19] layout (bias 2047) with a φ-distance of 0.632. A third Rust implementation (trios-trainer-igla) adopts [1][13][18] (bias 4095), arguing that the 13:18 split approximates $1/\varphi$. The present paper does not adjudicate between these specifications; we note the divergence as an open engineering question. For silicon verification, the only GF-family format with a physical realisation remains GF16 (Three Crowns, TTSKY26b)."

58.4 GF236 — Does Not Exist

58.4.1 Why it does not exist

"GF236" is a naming confusion, not a format. The number 236 is the mantissa width of IEEE binary256 ($1 + 19 + 236 = 256$ bits); that mantissa count was mistaken for a format label. The canonical 256-bit GoldenFloat is GF256 ([1][97][158]). (Nor could "GF236" denote a Galois field GF(236): $236 = 2^2 \times 59$ is not a prime power — but GoldenFloat names denote floats, not fields, so the field argument is a red herring.)

58.4.2 Repo/web audit

- Local files: Zero hits across /Users/ssdm4 except task metadata.
- GitHub (gHashTag org): No code, issues, or commits reference GF236 in any repository.

- Web search: No connection to Trinity, chips, or numeric formats. Only unrelated consumer products (racing shoes, fuel filters, bicycle chainrings).
- Agent deep search (19 tool calls, 30346 tokens): Confirmed complete absence.

58.4.3 Verdict

GF236 is not a real format in the Trinity ecosystem. The name conflated 236 — the mantissa width of IEEE binary256 ($1 + 19 + 236 = 256$ bits) — with a format label; the canonical 256-bit GoldenFloat is GF256 (see the numeric-formats chapter). It must not appear in any publication.

58.5 Complete GoldenFloat Family Table (Best Available Data)

Format	Bits	Layout (S/E/M)	Bias 2^(E−1)−1	φ-distance	Status	Spec
GF4	4	[1][1][2]	0	0.118	Proposal	gf4.t27
GF8	8	[1][3][4]	3	0.132	Experimental	gf8.t27
GF12	12	[1][4][7]	7	0.047	Proposal	gf12.t27
GF16	16	[1][6][9]	31	0.049	3 Production	gf16.t27
GF20	20	[1][7][12]	63	0.035	Proposal	gf20.t27
GF24	24	[1][9][14]	255	0.025	Proposal	gf24.t27
GF32-A	32	[1][8][23]	127	0.270	Spec only (= IEEE fp32)	zig-golden-float
GF32-B	32	[1][12][19]	2047	0.014	Spec + seal (canonical)	gf32.t27
GF32-C	32	[1][13][18]	4095	0.104	Rust impl	trios-trainer-igla
GF64	64	[1][24][39]	8388607	0.003	Spec	gf64.t27
GF256	256	[1][97][158]	$2^{96} - 1$ (stored OPEN)	0.004	γ FPGA spec	tt-trinity-gamma
TF3	8	[1][3][4]	3	0.132	Spec	tf3.t27
GFTernary	2	$\{-\varphi, 0, +\varphi\}$	—	0.000	Spec; codegen ROADMAP	gfternary.t27
GF16-Field	4-bit symbols	GF(2^4)	—	—	3 Silicon	tt_um_ghtag_trinity_gf16

58.6 Implications for the GOLDEN CHAIN Paper

58.6.1 Section 7.2 — "The Two Meanings of GF16" (already drafted)

Must be expanded to: - Disambiguate GoldenFloat vs Galois Field for the entire GF family. - Explicitly state that GF32 exists only as competing specifications, with no silicon realisation. - State that GF236 is not a valid format and does not appear in the ecosystem.

58.6.2 Section 7 — Silicon Anchor

Only GF16-Field and GFTernary have silicon presence: - GF16-Field: Euler crown (tt_um_ghtag_trinity_gf16, TT #4915) - Ternary kernel: Gamma crown (tt_um_trinity_max_true, TT #4913, TRI-27)

No GoldenFloat format beyond GF16 has been taped out.

58.6.3 Claim discipline update

Add to the Adversarial Self-Critique section: > "The GoldenFloat family includes proposed formats (GF4, GF8, GF12, GF20, GF24, GF32, GF64) whose specifications are not yet unified. Only GF16 has both a formal specification and a physical realisation. Any claim about GF32 performance or energy efficiency must be treated as speculative pending convergence of the competing [1][8][23], [1][12][19], and [1][13][18] specifications."

58.7 Open Questions

1. GF32 convergence: Which specification becomes canonical? Needs engineering decision (t27 governance?)
2. GF20 / GF24: Are these necessary intermediate formats, or can the family jump from GF16 to GF32?
3. GFTernary silicon: Will TRI-27 ternary kernel (Gamma crown) be extended to a full GFTernary encoder/decoder?
4. GoldenFloat vs IEEE compatibility: Is bidirectional lossless conversion to fp16/fp32 a design goal or an explicit non-goal?

Audit completed: 2026-05-25 Sources: gHashTag/trios-trainer-igla, gHashTag/zig-golden-float, gHashTag/trinity, gHashTag/t27, gHashTag/trios (PhD thesis chapters)

Golden Chain: A φ-Structured Algebraic Bridge from Physical Constants to Cross-Scale Cosmology and Silicon Verification

Keywords: golden ratio; fine-structure constant; symbolic regression; minimum description length; hardware verification; DePIN; Galois field; E-infinity; cross-scale cosmology

59.1 Introduction

The origin of the numerical values of fundamental physical constants remains one of the most persistent open problems in modern theoretical physics. Within the Standard Model and its extensions, these constants are treated as empirical inputs rather than derived quantities, motivating ongoing investigations into whether their values reflect deeper algebraic, geometric, or informational structures.

A growing line of research has explored whether apparently "fundamental" constants may exhibit latent structure under constrained representation systems, ranging from number-theoretic parametrizations to symmetry-based and information-theoretic compression frameworks (refs. Pellis2020viXra; ElNaschie2004EInfinity; Heyrovska2005; Sherbon2020; Stakhov2009; FringKorff2005). The central question is not whether a derivation exists within known physics, but whether the observed numerical landscape exhibits statistically non-random compressibility under restricted symbolic languages (Nature, Symbolic regression with large language models, 2025).

In this work we formalise this question through the following hypothesis:

> Do dimensionless physical constants exhibit statistically significant compressibility when embedded in a constrained symbolic representation space?

To investigate this, we introduce a bounded symbolic grammar generated by the algebraic basis $\{\varphi, \pi, e, 3\}$, where $\varphi = \frac{1+\sqrt{5}}{2}$ denotes the golden ratio (refs. Olsen2006; Stakhov2016). The choice of this basis is not assumed to have physical primacy; rather, it serves as a structured and non-orthogonal representational alphabet with known algebraic and transcendental properties, enabling non-trivial combinatorial expressivity at low description complexity.

The methodological framework adopted here is explicitly representation-theoretic and information-theoretic, rather than dynamical or field-theoretic.

Candidate expressions are treated as elements of a finite hypothesis class induced by a bounded-complexity grammar, and their ability to approximate empirical constants is studied under controlled search constraints.

59.2 The Three Strands and the Golden Chain

The present contribution is organised around three strands, bound together by what we call the Golden Chain—an analogy drawn from the Russian poet Alexander Pushkin, in which a golden chain binds three separate tales to a single oak tree. In our framework:

- Strand I — Symbolic Grammar & MDL: The formal language G_φ, the Pellis hierarchical expansion, and the statistical framework (Sections 2–5).
- Strand II — Silicon Anchor: The realisation of the Trinity identity $\varphi^2 + \varphi^{-2} = 3$ in three open-source ASIC designs on the Tiny Tapeout SKY130 shuttle (refs. TinyTapeout; SkywaterPDK) (Section 7).
- Tier-D — Cross-Scale Cosmology: The binding of atomic-scale formulas to cosmological-scale observations through Olsen's φ-invariance framework (Section 6).

The "oak" is the algebraic invariant $\varphi^2 + \varphi^{-2} = 3$; the "chain" is the falsifiable, hardware-rooted, cross-scale bridge that connects the three strands under a common MDL cost function. We adopt explicit claim discipline throughout: no ontological primacy for φ is asserted, and all results are presented as empirical properties of a constrained search process.

59.3 Symbolic Grammar and Finite Complexity Search Space

In order to formalise the notion of structured numerical approximation of physical constants, we introduce a restricted symbolic representation system which defines a discrete hypothesis space over algebraic expressions. This space is not assumed to possess any physical ontological significance; rather, it is treated as a finite-complexity generative grammar whose expressive capacity can be quantitatively analysed.

59.4 Algebraic Grammar Definition

We define a symbolic grammar G_φ over the alphabet

$$\mathcal{A} = \{\varphi, \pi, e, 3, \mathbb{Z}\}, \tag{59.1}$$

where $\varphi = \frac{1+\sqrt{5}}{2}$ denotes the golden ratio, and $\mathbb{Z}$ encodes integer-valued exponents and prefactors.

The grammar generates expressions of the form:

$$F(\theta) = n \cdot 3^k \cdot \varphi^p \cdot \pi^m \cdot e^q, \tag{59.2}$$

where the parameter vector is given by $\theta := (n, k, p, m, q)$. The resulting set of admissible symbolic expressions is defined as:

$$\mathcal{S}(C) := \big\{F(\theta) \;\big|\; \theta \in \mathbb{Z}^5,\ n \in \{1, \ldots, 9\},\ \|\theta\|_1 \leq C\big\}, \tag{59.3}$$

where the ℓ_1-type complexity constraint $\|\theta\|_1 := |k| + |p| + |m| + |q|$ acts as a bounded description-length regularizer.

59.5 Interpretation as a Finite Hypothesis Class

The set $\mathcal{S}(C)$ induces a finite hypothesis class over positive real numbers:

$$\mathcal{H}_C := \big\{h_\theta : \mathbb{R}^+ \to \mathbb{R}^+ \;\big|\; h_\theta = F(\theta),\ \theta \in \mathcal{S}(C)\big\}. \tag{59.4}$$

Each element of $\mathcal{H}_C$ can be interpreted as a symbolic generative model producing a candidate approximation for a physical constant $X \in \mathbb{R}^+$.

59.6 Logarithmic Embedding Structure

To analyse the geometry of the search space, we introduce a logarithmic embedding:

$$\log F(\theta) = \log n + k \log 3 + p \log \varphi + m \log \pi + q \log e. \tag{59.5}$$

This induces a lattice structure in $\mathbb{R}^5$:

$$\log \mathcal{S}(C) \subset \mathbb{Z}^4 \cdot \mathrm{v} + \log n, \tag{59.6}$$

where $\mathrm{v} := (\log 3, \log \varphi, \log \pi, \log e)$. Hence, $\mathcal{S}(C)$ corresponds to a bounded section of an additive lattice embedded in logarithmic space.

59.7 Combinatorial Growth and Metric Structure

The cardinality of the symbolic space scales combinatorially with complexity cutoff C:

$$|\mathcal{S}(C)| \sim \sum_{\|\theta\|_1 \leq C} \mathcal{O}(1) \approx \mathcal{O}(C^4), \tag{59.7}$$

modulo integer boundary effects. This polynomial scaling is critical: it ensures that the search space remains computationally finite while still exhibiting rapid expressive expansion.

Given a target constant $X \in \mathbb{R}^+$, we define the approximation functional:

$$d_C(X) := \min_{\theta \in \mathcal{S}(C)} \frac{|F(\theta) - X|}{|X|}. \tag{59.8}$$

This induces a non-Euclidean approximation geometry over the target space, where accessibility is determined by symbolic density rather than numerical continuity.

59.8 Structural Constant and Numerical Correspondence

We define the algebraically minimal φ-generated invariant of order three as:

$$\alpha_\varphi := \frac{\varphi^{-3}}{2} \approx 0.1180339887\ldots \tag{59.9}$$

Using the exact algebraic identity $\varphi^{-1} = \varphi - 1$, we may rewrite α_φ as a low-degree polynomial functional of a fundamental quadratic irrational. Numerically, this evaluates to a value strikingly close to the strong coupling constant $\alpha_s(m_Z) \approx 0.1180 \pm \delta_{\text{exp}}$, with relative deviation $\Delta_\alpha \sim \mathcal{O}(10^{-4})$.

Remark 1. The quantity α_φ should be understood strictly as a low-complexity algebraic invariant arising from a constrained symbolic grammar. Its numerical proximity to $\alpha_s(m_Z)$ is an empirical observation within a finite hypothesis space and does not constitute a derivation, prediction mechanism, or physical identification.

59.9 Pellis Hierarchical Expansion

A central limitation of purely multiplicative symbolic representations is their inherent rigidity with respect to multi-scale refinement. In order to incorporate systematic hierarchical corrections while preserving the underlying φ-structured basis, we introduce a nested expansion framework referred to as the Pellis hierarchical expansion (Pellis, viXra:2020).

We define a hierarchical representation of a scalar quantity $F \in \mathbb{R}^+$ of the form:

$$F^{(K)}(\varphi) = \sum_{k=1}^{K} c_k\, \varphi^{-k}, \qquad c_k \in \mathbb{R},\ K \in \mathbb{N}. \tag{59.10}$$

More generally, we consider the infinite asymptotic series:

$$F(\varphi) \sim \sum_{k=1}^{\infty} c_k\, \varphi^{-k}, \qquad \text{as } k \to \infty, \tag{59.11}$$

interpreted in the sense of a formal asymptotic expansion rather than a convergent power series in the classical analytic sense.

The expansion variable φ^{-1} acts as a natural scaling parameter generating a discrete hierarchy of progressively finer corrections. Each coefficient c_k encodes information at a distinct resolution level in the symbolic representation space. This induces a decomposition analogous to multi-scale renormalisation expansions in statistical physics, where coarse-grained structure is corrected by progressively finer-scale terms.

59.10 Relation to the Multiplicative Grammar

The hierarchical expansion serves as a refinement layer over multiplicative symbolic expressions. We interpret $F_0 = n \cdot 3^k \cdot \varphi^p \cdot \pi^m \cdot e^q$ as a zeroth-order symbolic approximation, while the Pellis hierarchical expansion introduces systematic corrections $\Delta F^{(i)} \in \mathrm{span}\{\varphi^{-k}\}_{k\geq 1}$. This establishes a two-tier representation structure:

- multiplicative grammar $\to$ coarse symbolic encoding;
- hierarchical φ^{-1} expansion $\to$ fine-scale correction structure.

The hierarchy induced by powers of φ^{-1} defines a natural scale ordering:

$$\varphi^{-1} > \varphi^{-2} > \varphi^{-3} > \cdots , \tag{59.12}$$

which induces a filtration on the symbolic representation space:

$$\mathcal{F}_1 \supset \mathcal{F}_2 \supset \mathcal{F}_3 \supset \cdots , \qquad \mathcal{F}_k = \mathrm{span}\{\varphi^{-j}\}_{j\geq k}. \tag{59.13}$$

This filtration allows a controlled truncation scheme interpreted as a projection onto finite symbolic depth.

59.11 Statistical and Information-Theoretic Framework

The symbolic grammar $\mathcal{S}(C)$ introduced in previous sections defines a rapidly growing discrete hypothesis space. As a consequence, classical single-hypothesis significance testing becomes fundamentally inadequate due to severe multiple-comparison effects and non-trivial selection bias induced by exhaustive or near-exhaustive search procedures.

For a target constant $X \in \mathcal{T}$ and candidate expression $F(\theta) \in \mathcal{S}(C)$, we define the normalized approximation loss:

$$\mathcal{L}(F; X) := \frac{|F - X|}{|X|}. \tag{59.14}$$

The empirical best-fit representation is defined as:

$$F^*(X) := \arg \min_{F\in\mathcal{S}(C)} \mathcal{L}(F; X). \tag{59.15}$$

This induces a projection map:

$$\Pi_{\mathcal{S}(C)} : X \mapsto F^*(X), \tag{59.16}$$

which can be interpreted as a nonlinear quantization of physical constants onto a discrete symbolic manifold.

We report selected low-loss representatives from the search space:

$$\alpha^{-1} \approx 36\,\pi^{-1}\varphi\, e^2, \tag{59.17}$$

$$\sin^2\theta_W \approx 3^{-2}\,\pi^2\,\varphi^3\,e^{-3}, \tag{59.18}$$

$$m_H \approx 4\,\varphi^3\,e^2. \tag{59.19}$$

These expressions correspond to distinct regions of the exponent lattice in $\log\mathcal{S}(C)$ and are selected as global or near-global minimizers of $\mathcal{L}$ under the imposed complexity constraint.

By applying the logarithmic embedding

$$\log : \mathbb{R}^+ \to \mathbb{R}, \tag{59.20}$$

each multiplicative expression becomes linear in exponent space:

$$\log F(\theta) = \log n + k\log 3 + p\log\varphi + m\log\pi + q. \tag{59.21}$$

Thus, each empirical fit corresponds to a nearest-lattice approximation in the affine subspace:

$$\mathcal{L}_{\log} := \left\{ \log n + \theta\cdot \mathrm{v} \mid \theta \in \mathbb{Z}^4 \right\}, \tag{59.22}$$

where $\mathrm{v} = (\log 3, \log\varphi, \log\pi, 1)$.

The observed relative deviations cluster within:

$$\mathcal{E} := \left\{ \mathcal{L}(F^*(X); X) \right\} \sim 10^{-4} - 10^{-3}. \tag{59.23}$$

We define the empirical error distribution:

$$\rho_{\mathcal{E}}(\epsilon) := \frac{1}{|\mathcal{T}|} \sum_{X\in\mathcal{T}} \delta\big(\epsilon - \mathcal{L}(F^*(X); X)\big). \tag{59.24}$$

The empirical results can be interpreted as a constrained representation problem:

$$\text{Find } F \in \mathcal{S}(C) \text{ such that } F \approx X. \tag{59.25}$$

59.12 Multiple Hypothesis Structure and Effective Trial Count

For a fixed complexity cutoff C, the symbolic space $\mathcal{S}(C)$ induces a finite but combinatorially large hypothesis class. We define the effective number of statistical trials as:

$$N_{\text{eff}} := |\mathcal{S}(C)|. \tag{59.26}$$

Given that $|\mathcal{S}(C)| \sim \mathcal{O}(C^4)$, The search procedure corresponds to a high-dimensional multiple hypothesis testing problem in which naive p-values are strongly inflated due to selection over a large combinatorial space.

To account for this, any observed minimum-error match must be interpreted as an extreme-value statistic over the induced error ensemble:

$$\mathcal{E}_C := \{\mathcal{L}(F(\theta); X) \mid \theta \in \mathcal{S}(C)\}. \tag{59.27}$$

Thus, the relevant quantity is not individual significance, but the distribution of the minimum:

$$\mathcal{L}_{\min}(X) := \min_{\theta \in \mathcal{S}(C)} \mathcal{L}(F(\theta); X), \tag{59.28}$$

which follows a non-trivial extreme-value law determined by the geometry of $\mathcal{S}(C)$ in logarithmic embedding space.

59.13 Bayesian Model Comparison

We formalise the comparison between two competing generative models:

- $\mathcal{M}_0$: structureless log-uniform baseline model;
- $\mathcal{M}_1$: structured symbolic grammar model induced by $\mathcal{S}(C)$.

The Bayesian evidence for each model is defined as the marginal likelihood:

$$P(\mathcal{T} \mid \mathcal{M}) = \int P(\mathcal{T} \mid \theta, \mathcal{M})\, P(\theta \mid \mathcal{M})\, d\theta. \tag{59.29}$$

For the discrete grammar model $\mathcal{M}_1$, this reduces to a finite sum:

$$P(\mathcal{T} \mid \mathcal{M}_1) = \prod_{i=1}^{N} \left(\sum_{F \in \mathcal{S}(C)} \exp\big[-\beta \mathcal{L}(F; X_i) \big] \right), \tag{59.30}$$

where β defines the sharpness of the likelihood kernel. For the null model $\mathcal{M}_0$, we assume a scale-invariant log-uniform prior:

$$P(X \mid \mathcal{M}_0) = \frac{1}{\log(X_{\max}/X_{\min})}, \tag{59.31}$$

implying no preferential symbolic structure. The Bayes factor is then:

$$\mathcal{B} = \frac{P(\mathcal{T} \mid \mathcal{M}_1)}{P(\mathcal{T} \mid \mathcal{M}_0)}. \tag{59.32}$$

Standard interpretation applies: $\log \mathcal{B} \sim 0$ (no evidence), $\log \mathcal{B} > 2$ (positive evidence), $\log \mathcal{B} > 5$ (strong evidence). However, due to the effective size of $\mathcal{S}(C)$, $\mathcal{B}$ must be interpreted as a complexity-regularised evidence functional rather than a classical likelihood ratio.

59.14 Minimum Description Length Principle

We further reformulate model comparison using the Minimum Description Length (MDL) principle, which provides an operational interpretation in terms of optimal data compression. The total description length of a model is given by:

$$\mathrm{MDL}(\mathcal{M}) = L(\mathcal{M}) + L(\mathcal{T} \mid \mathcal{M}), \tag{59.33}$$

where $L(\mathcal{M})$ is the code length required to describe the model class and $L(\mathcal{T} \mid \mathcal{M})$ is the conditional encoding cost of the data given the model.

For the symbolic grammar model, the complexity term scales as:

$$L(\mathcal{M}_1) \approx \log |\mathcal{S}(C)| + \log C, \tag{59.34}$$

reflecting both enumeration cost and exponent resolution precision.

The data-fit component is given by:

$$L(\mathcal{T} \mid \mathcal{M}_1) = -\sum_{i=1}^{N} \log\left(\sum_{F \in \mathcal{S}(C)} \exp\big[-\beta \mathcal{L}(F; X_i) \big] \right). \tag{59.35}$$

The compression gain is defined as:

$$\Delta_{\mathrm{MDL}} = \mathrm{MDL}(\mathcal{M}_0) - \mathrm{MDL}(\mathcal{M}_1). \tag{59.36}$$

Interpretation: $\Delta_{\mathrm{MDL}} > 0$ indicates that the symbolic model achieves a non-trivial compression advantage; $\Delta_{\mathrm{MDL}} \leq 0$ indicates no evidence of structured compressibility. Thus, structure detection is reformulated as a coding-theoretic inequality between competing probabilistic representations.

$$K_{\mathcal{G}_\varphi}(X) \leq K_{\mathrm{unstructured}}(X) - \Delta_{\mathrm{MDL}}. \tag{59.37}$$

59.15 Bias-Controlled Simulation Framework

A central difficulty in symbolic discovery problems is that the underlying hypothesis space grows combinatorially with the complexity bound C, leading to severe multiple-testing effects and selection bias. To address this, we introduce a controlled simulation and evaluation framework that explicitly separates three sources of statistical distortion: (i) combinatorial bias induced by exponential growth of $\mathcal{S}(C)$; (ii) representation bias arising from non-uniform density in log-space; (iii) selection bias introduced by post-hoc minimisation over the grammar.

For a fixed complexity cutoff $C \in \mathbb{N}$, we consider the fully enumerated symbolic set

$$\mathcal{S}(C) = \left\{ F = n \cdot 3^k \cdot \varphi^p \cdot \pi^m \cdot e^q \;\middle|\; \|\theta\|_1 \leq C \right\}, \quad \theta = (n, k, p, m, q). \tag{59.38}$$

The deterministic evaluation operator over a target constant $X \in \mathcal{T}$ is defined as:

$$\mathcal{E}_C(X) = \min_{F \in \mathcal{S}(C)} \delta(F, X), \qquad \delta(F, X) := \frac{|F - X|}{|X|}. \tag{59.39}$$

To control for accidental structure and distributional artifacts, we define a permutation null model $\mathcal{M}_{\text{perm}}$ in which the target set $\mathcal{T}$ is randomly permuted within its empirical support:

$$\mathcal{T} \mapsto \pi(\mathcal{T}), \quad \pi \in S_{|\mathcal{T}|}. \tag{59.40}$$

Under this model, the induced coverage functional becomes:

$$\rho_{\text{perm}}(C, \epsilon) = \mathbb{E}_{\pi \sim S_{|\mathcal{T}|}} \left[\rho(C, \epsilon) \right]. \tag{59.41}$$

As an additional control, we define a randomized exponent model $\mathcal{M}_{\text{rand}}$ in which exponent vectors are sampled from bounded symmetric priors:

$$k, p, m, q \sim \text{Unif}(-C, C), \quad n \sim \text{Unif}\{1, \ldots, 9\}. \tag{59.42}$$

This generates a stochastic symbolic ensemble:

$$\mathcal{S}_{\text{rand}}(C) \subset \mathbb{R}^+ \tag{59.43}$$

which preserves cardinality but destroys algebraic coherence.

We define robustness of the symbolic signal as stability of ρ under perturbations of the model class:

$$\mathcal{R}(C, \epsilon) = \frac{\rho(C, \epsilon)}{\mathbb{E}[\rho_{\text{rand}}(C, \epsilon)] + \varepsilon}, \quad \varepsilon > 0. \tag{59.44}$$

A value $\mathcal{R}(C,\epsilon) \gg 1$ indicates statistically significant deviation from randomized symbolic baselines.

The mapping

$$(C,\epsilon) \mapsto \rho(C,\epsilon) \tag{59.45}$$

defines a two-parameter phase diagram over symbolic complexity space.

The bias-controlled framework replaces raw coincidence counting with a structured, multi-null diagnostic pipeline that separates:

$$\text{signal} = \text{excess coverage}, \quad \text{not raw match frequency}. \tag{59.46}$$

This provides a statistically consistent foundation for evaluating symbolic compressibility claims in high-dimensional arithmetic grammars.

59.16 Phase Transition and Generative Structure

We define the coverage functional:

$$\rho(C,\epsilon) = \frac{1}{|\mathcal{T}|} \sum_{X \in \mathcal{T}} 1\left[\inf_{F \in \mathcal{S}(C)} \frac{|F - X|}{|X|} < \epsilon\right], \tag{59.47}$$

which measures the fraction of physical constants that admit a representation within relative tolerance ϵ at complexity level C. In this formulation, $\rho(C,\epsilon)$ plays the role of an order parameter in a discrete symbolic phase space.

Define a Gibbs-type weighting over the grammar:

$$\mathbb{P}_C(F) = \frac{1}{Z(C)} \exp\big(-\lambda\, \mathcal{C}(F)\big), \tag{59.48}$$

where $\mathcal{C}(F)$ denotes symbolic complexity (e.g., ℓ_1 norm of exponent vector), and $Z(C)$ is the partition function:

$$Z(C) = \sum_{F \in \mathcal{S}(C)} \exp\big(-\lambda\, \mathcal{C}(F)\big). \tag{59.49}$$

This induces a probabilistic coverage interpretation:

$$\rho(C,\epsilon) \equiv \mathbb{E}_{F \sim \mathbb{P}_C}\big[1\big(d(F,X) < \epsilon\big)\big]. \tag{59.50}$$

59.17 Pellis Phase Transition Conjecture

Conjecture 2 (Pellis Symbolic Phase Transition). There exists a critical complexity scale $C^* > 0$ such that the coverage functional $\rho(C,\epsilon)$ exhibits a sharp crossover characterised by:

$$\rho(C,\epsilon) = \begin{cases} \mathcal{O}\big(\exp[-\alpha(C^* - C)]\big), & C < C^*, \\ 1 - \mathcal{O}\big(\exp[-\beta(C - C^*)]\big), & C > C^*, \end{cases} \tag{59.51}$$

for some $\alpha, \beta > 0$, in the asymptotic regime $|\mathcal{T}| \gg 1$.

We define a critical complexity scale C^* as a non-perturbative transition point:

$$C^* = \inf \Big\{ C \in \mathbb{R}^+ : \rho(C, \epsilon) \geq \frac{1}{2} \Big\}. \tag{59.52}$$

More generally, C^* satisfies the susceptibility maximization condition:

$$C^* = \arg\max_C \left| \frac{\partial \rho(C, \epsilon)}{\partial C} \right|. \tag{59.53}$$

This transition separates a regime of sparse accidental approximation from a regime of high-density symbolic compressibility. The functional $\rho(C, \epsilon)$ behaves analogously to a thermodynamic order parameter in a finite-size system, where the complexity C plays the role of an inverse temperature-like control parameter. We define the susceptibility:

$$\chi(C) := \frac{\partial \rho(C, \epsilon)}{\partial C}, \tag{59.54}$$

which satisfies:

$$\chi(C) \sim \begin{cases} \exp[-\alpha(C^* - C)], & C < C^*, \\ \exp[-\beta(C - C^*)], & C > C^*. \end{cases} \tag{59.55}$$

Thus $\chi(C)$ is sharply peaked at C^*, indicating maximal structural instability of the symbolic ensemble.

Under the embedding:

$$X \mapsto \log X, \quad F \mapsto \log F, \tag{59.56}$$

the symbolic space $\mathcal{S}(C)$ induces a discretized lattice:

$$\mathcal{L}(C) \subset \mathrm{span}_{\mathbb{Z}} \{\log \varphi, \log \pi, \log e, \log 3\}. \tag{59.57}$$

The phase transition occurs when the effective lattice spacing satisfies:

$$\delta_{\text{eff}}(C^*) \sim \mathbb{E}\big[\operatorname{dist}(\log X_i, \log X_j)\big], \tag{59.58}$$

inducing a transition from sparse covering to near-saturation of the target manifold.

In a finite-size regime, the transition width scales as:

$$\Delta C \sim \frac{1}{\sqrt{|\mathcal{T}|}}, \tag{59.59}$$

suggesting increasing sharpness as the dataset size grows.

59.18 Generative Structure Theorem

Let $\mathcal{S}(C)$ denote the finite symbolic grammar space generated by:

$$F(\theta) = n \cdot 3^k \cdot \varphi^p \cdot \pi^m \cdot e^q, \quad \theta = (n, k, p, m, q), \quad \|\theta\|_1 \le C. \tag{59.60}$$

We embed the target constants $\mathcal{T} = \{X_i\}_{i=1}^N$ into logarithmic metric space:

$$\mathcal{X}_i := \log |X_i|, \qquad \mathcal{F}_\theta := \log |F(\theta)|. \tag{59.61}$$

Define the induced approximation metric:

$$d(F, X) := \big| \log |F| - \log |X| \big|. \tag{59.62}$$

Theorem 3 (Emergent Coverage Phase Transition). Assume the target ensemble $\mathcal{T}$ admits a finite log-density with respect to Lebesgue measure on $\mathbb{R}$. Then there exists a critical complexity scale $C^* > 0$ such that the coverage functional satisfies the exponential crossover form above, for some constants $\alpha, \beta > 0$ determined by the effective metric entropy of $\mathcal{T}$ in logarithmic coordinates.

Sketch. The symbolic ensemble $\mathcal{S}(C)$ defines a finite subset of a rank-4 additive lattice in log-space. For small C, the induced lattice spacing dominates the typical inter-point distance of $\log \mathcal{T}$, yielding exponentially small coverage probability. As C increases, the number of admissible integer combinations of θ grows sufficiently fast to induce a covering transition in the induced metric space. The critical point C^* is characterised by the condition $\delta_{\text{eff}}(C^*) \sim \mathbb{E}[d(\mathcal{X}_i, \mathcal{X}_j)]$, where δ_{eff} is the effective lattice spacing. □

The cardinality of the grammar grows combinatorially:

$$|\mathcal{S}(C)| \sim \sum_{\|\theta\|_1 \le C} 1 \sim \mathcal{O}(C^4), \tag{59.63}$$

while the induced image under the logarithmic map becomes increasingly dense in a low-dimensional affine lattice generated by:

$$\mathcal{B} = \{\log 3, \log \varphi, \log \pi, \log e\}. \tag{59.64}$$

Thus $\mathcal{S}(C)$ induces a discretized additive semigroup in $\mathbb{R}$:

$$\log \mathcal{S}(C) \subset \operatorname{span}_{\mathbb{Z}}(\mathcal{B}) + \log n. \tag{59.65}$$

We identify three regimes:

- Subcritical ($C \ll C^*$): sparse symbolic coverage;
- Critical ($C \approx C^*$): maximal susceptibility $\frac{d\rho}{dC}$;
- Supercritical ($C \gg C^*$): saturation of coverage.

The critical point satisfies:

$$\delta_{\text{eff}}(C^*) \sim \mathbb{E}\big[d(\mathcal{X}_i, \mathcal{X}_j)\big], \tag{59.66}$$

where δ_{eff} is the effective lattice spacing.

Define the order parameter:

$$\Phi(C) := \rho(C, \epsilon). \tag{59.67}$$

Then $\Phi(C)$ exhibits a non-analytic crossover behavior in the finite-size regime.

The susceptibility satisfies:

$$\chi(C) \sim \begin{cases} e^{-\alpha(C^*-C)}, & C < C^*, \\ e^{-\beta(C-C^*)}, & C > C^*. \end{cases} \tag{59.68}$$

Let $K(C)$ denote the number of ϵ-valid approximations. Empirically and combinatorially:

$$K(C) \sim \exp(\gamma C), \qquad C < C_{\text{sat}}. \tag{59.69}$$

Beyond saturation:

$$K(C) \to |\mathcal{T}|. \tag{59.70}$$

Define the MDL difference:

$$\Delta_{\text{MDL}}(C) = \text{MDL}(\mathcal{M}_0) - \text{MDL}(\mathcal{M}_1(C)). \tag{59.71}$$

Then the critical point satisfies:

$$\Delta_{\text{MDL}}(C^*) \approx 0, \tag{59.72}$$

with steep gradient:

$$\frac{d}{dC}\Delta_{\text{MDL}}(C)\bigg|_{C=C^*} \gg 1. \tag{59.73}$$

The system undergoes a transition between high-entropy regime (no compressible symbolic structure) and low-entropy regime (emergent φ-structured compressibility). The transition point C^* corresponds to a minimax balance between:

$$\text{model complexity} \quad \leftrightarrow \quad \text{data compression gain}. \tag{59.74}$$

59.19 Cross-Scale Bridge: Olsen Tier-D

This section promotes Scott Olsen's Tier-D φ-cosmology from a cross-reference to a full author contribution, integrated into the Golden Chain compendium as the cross-scale binding strand between the Vasilev Trinity (microphysical) and the Pellis Hierarchical Expansion (atomic-scale fine-structure constant).

59.20 Author Profile

Scott Anthony Olsen, Ph.D., is Emeritus Professor of Philosophy and Religion at the College of Central Florida, and a long-standing scholar of the Pythagorean–Platonic tradition, sacred geometry, and the metaphysics of the golden section. He trained under physicist David Bohm (implicate/explicate order), world-religion scholar Huston Smith, sacred-geometer Keith Critchlow, and esotericist Douglas Baker. His most widely-distributed work is The Golden Section: Nature's Greatest Secret (Wooden Books, 2006) (ref. Olsen2006), translated into French, Spanish, and several further editions. Beyond the popular monograph, Olsen has published in Chaos, Solitons & Fractals and the Journal of Progressive Research in Mathematics, where he co-authored "A Grand Unification of the Sciences, Arts & Consciousness: Rediscovering the Pythagorean Plato's Golden Mean Number System" (JPRM 16(2), 2020) (ref. Olsen2020JPRM) with Marek-Crnjac, He, and El Naschie—the principal scholarly source we adopt for the present Tier-D contribution (refs. Olsen2017GRBSF; He2012).

59.21 Tier-D Thesis Statement

We adopt, as the explicit charter for Tier-D, the following direct quotation from Olsen et al. (2020) (ref. Olsen2020JPRM):

> "We propose herein a grand unification of the sciences, arts and consciousness, rooted in an ontological superstructure known to the ancients as the One and Indefinite Dyad, that gives rise to a golden mean number system which is the substructure of all existence—evident in quantum mechanics, including quark masses, the chaos border, fine-structure constant and entanglement, entropy and thermodynamic equilibrium, the periodic table of elements, nanotechnology, crystallography, computing, digital information, cryptography, genetics, nucleotide arrangement, ...DNA structure, ...plant phyllotaxis, planetary orbits and sizes, black holes, dark energy, dark matter, and even cosmogenesis—the very origin and structure of the universe."

This quotation asserts a φ-structured substructure spanning twelve orders of magnitude (quark masses to cosmogenesis), which the Golden Chain operationalises as a falsifiable, cross-scale ledger anchored on the Trinity constants and the Pellis Hierarchical Expansion of α^{-1}.

59.22 Pre-registered Hypotheses

For Olsen's contribution to enter the falsification ledger as a Tier-D pre-registered claim, we extract three concrete, dataset-attachable hypotheses:

- Cross-scale ratio alignment. For a frozen list of N dimensionless cross-scale ratios, the median MDL cost of a best-fit φ-rational approximation should be lower than the median MDL cost of a best-fit random-prime-rational approximation at pre-registered confidence $p < 10^{-3}$.
- Quasi-crystal diffraction. The icosahedral quasi-crystal diffraction-peak intensity ratios should be φ-rational to within experimental precision (refs. Shechtman1984; Coldea2010E8).
- Log-periodic null. In any sufficiently large coupling-constant running dataset, the spectral power at frequency $\log \varphi$ should be either consistent with white-noise null (no Olsen signal) or detectable above null at a pre-registered amplitude floor $A_i \lesssim \text{few} \times 10^{-3}$ (ref. Sornette2024).

59.23 The Olsen–Vasilev Cross-Scale Lemma

Lemma 4 (Cross-Scale Filtration). Let $\mathcal{F}_k = \text{span}_{\mathbb{R}}\{\varphi^{-j}\}_{j \geq k}$ be the two-tier Bridge filtration. Then for any dimensionless physical ratio r derived from quantities measured at distinct scales, if $r \in \mathcal{F}_k$ for some finite k with bounded $|k|$, the MDL cost of expressing r in G_φ is finite and invariant under scale transformations that preserve φ-rational structure.

The lemma is proved by direct computation: φ-rational expressions are closed under the field operations of $\mathbb{Q}(\varphi)$, and the MDL cost of an expression in $\mathcal{F}_k$ depends only on the multiset of φ^{-j} generators, not on the absolute scale of the physical quantity. Olsen's cross-scale alignment claim (O-D1) is then a statement about the empirical distribution of $|k|$ across measured cross-scale ratios.

59.24 What Olsen's Framework Adds to the Bridge

Without Olsen, the Bridge is a two-strand structure (Trinity + Pellis) anchored in silicon but lacking a principled reason for cross-scale generalisation. With Olsen, the Bridge gains: (i) a scale-bridging tier that connects atomic-scale FSC to large-scale-structure observations; (ii) a third falsifiable signal (log-periodic oscillations) independent of the FSC-specific predictions of Strand III; and (iii) a historical and philosophical foundation anchoring the φ-grammar in the longest-standing tradition of golden-ratio natural philosophy (Plato $\to$ Kepler $\to$ Pellis $\to$ Olsen).

59.25 Silicon Anchor: The Three Crowns

The silicon anchor of the Golden Chain—Strand II of the compendium—is realised in three coordinated chip designs on the Tiny Tapeout SKY130nm shuttle (TTSKY26b) (refs. TinyTapeout; SkywaterPDK). We refer to them collectively as the Three Crowns, a triadic hierarchy in which each crown carries a distinct role within the φ-structured grammar G_φ, yet all three carry the same canonical anchor byte 0x47C0 at $\{uio\_out, uo\_out\}$ on reset.

59.26 The Triadic Silicon Manifest

Crown	Top module	Shuttle	Tiles	Modules	Anchor
Crown	Top module	Shuttle	Tiles	Modules	Anchor
Phi (Nano)	tt_um_trinity_nano	TT #4914	1×1	51	0xCF
Euler (Compact)	tt_um_ghtag_trinity_gf16	TT #4915	8×2	90	0xAE
Gamma (Full)	tt_um_trinity_max_true	TT #4913	8×4	105	0x93

Roles. Phi (Nano) is the minimal-area witness—the smallest reproducible Trinity surface. Euler (Compact) is the GF(16) Galois-field carrier, the algebraic substrate. Gamma (Full) is the full Trinity-S^{3}AI DNA, the maximal expression. All three top modules instantiate the canonical anchor sub-module that emits 0x47C0 on the first cycle after reset. The hardware identity is therefore invariant under crown scale: a single Verilog assertion `assert {uio_out, uo_out} == 16'h47C0` validates all three crowns at reset time. This is the silicon witness of the algebraic identity $\varphi^2 + \varphi^{-2} = 3$.

59.27 The Two Meanings of "GF"

In the Trinity ecosystem, the acronym "GF" carries two incompatible technical meanings that must be explicitly disambiguated in any publication:

1. GoldenFloat — a family of φ-structured floating-point formats (GF4, GF8, GF12, GF16, GF20, GF24, GF32, GF64, GF256, plus the 2-bit GFTernary and the 8-bit ternary-weight TF3). These are real-number representations for ML and edge inference. The layout of each format is chosen so that the ratio of exponent width to mantissa width approximates a power of φ, giving each member a characteristic "φ-distance."
2. GF(2^n) Galois Field — the finite field $\mathbb{F}_{2^n}$ with 2^n elements. In the Euler crown this is $\mathbb{F}_{16} = GF(2^4)$, an algebraic substrate whose operations are

realised via XOR, popcount, and lookup tables. No hardware multipliers are inferred. This is a discrete algebraic structure.

The naming collision is historical, not intentional. In peer-reviewed literature we therefore distinguish GF16-Float from GF16-Field.

59.27.1 GoldenFloat family status

Only GF16-Float ([sign:1][exp:6][mant:9], bias 31, the same layout as IBM's DLFloat) has both a formal specification and a physical realisation in the Three Crowns programme. The remaining family members are at varying stages of maturity:

- GF8 ([1][3][4], bias 7) — experimental, bench-tested.
- GF12 ([1][5][6], bias 15) — proposal only.
- GF20 / GF24 — proposals, no implementation.
- GF32 — three competing, incompatible specifications coexist in the ecosystem: (i) [1][8][23] with bias 127 (DLFloat-like, "specification only, no implementation"); (ii) [1][12][19] with bias 2047 (φ-distance 0.632, t27 seal registered); and (iii) [1][13][18] with bias 4095 (Rust implementation, 13:18 split $\approx 1/\varphi$). The present paper does not adjudicate between these specifications; we note the divergence as an open engineering question.
- GF64 ([1][11][52]) — proposal only.
- GFTernary — ternary $\{-1, 0, +1\}$ encoding, implemented in the TRI-27 kernel (Gamma crown) (ref. BitNet2025).
- GF236 — does not exist. The label confused the number 236 — the mantissa width of IEEE binary256 ($1 + 19 + 236 = 256$ bits) — with a format name; the canonical 256-bit GoldenFloat is GF256. (It could not denote a Galois field $GF(236)$ either, since $236 = 2^2 \times 59$ is not a prime power, but GoldenFloat names denote floats, not fields, so that is a red herring.) No repository, specification, or silicon design references "GF236".

The Euler crown (tt_um_ghtag_trinity_gf16) carries the Galois-field carrier, not any GoldenFloat format. The naming coincidence reflects the project's unified φ-philosophy rather than an identity of the two technical objects.

59.28 Why Three Crowns and Not One

A single chip is sufficient to demonstrate the anchor identity. We choose three for falsification leverage, not for redundancy:

1. Scale-independence test. The three crowns span 1×1, 8×2, and 8×4 tiles—three different physical scales sharing the same identity.
2. Module-density test. Phi has 51 modules, Euler 90, Gamma 105. The anchor surviving an order-of-magnitude variation in module count argues against accidental constructive interference.

3. Algebraic-substrate test. Euler carries GF(16) Galois-field arithmetic; Gamma carries the full Trinity-S^{3}AI DNA with TRI-27 ternary kernel; Phi carries the minimal nano-tile. The anchor surviving three different algebraic substrates is the Strand-I$\perp$Strand-II compatibility check.

These are three orthogonal falsification axes built into the silicon.

59.29 Future Tape-Outs: TRI-NET Shuttle Triad

Beyond the TTSKY26b proof-of-concept, we propose a TRI-NET Shuttle Triad (Nano / Mid / Max) targeting three carrier products:

SKU	Die	Shuttle	Tiles	Mesh	Use case
SKU	Die	Shuttle	Tiles	Mesh	Use case
Nano	tt_um_trinity_nano	TTSKY26c	1× GF16 dot4	none	Drone, sensor mote
Mid	tt_um_trinity_mid	TTSKY26c	4× GF16 dot4	2×2 crossbar	Edge inference, IP node
Max	tt_um_trinity_max	TTIHP27a	16× GF16 dot4	4×4 torus	Desktop / M.2 DePIN miner

The “ISA equivalence” rule means every SKU runs the exact same compute job stream and emits a deterministic receipt; only throughput, latency, and concurrency differ. This is the silicon image of the “one oak, three tales” principle that motivates the Golden Chain metaphor.

59.30 Armoured Provenance Layer for DePIN

The Three Crowns are not positioned as high-throughput inference accelerators. They are positioned as a secure provenance co-processor for DePIN: a small, auditable silicon witness that (1) accepts physical data from a device or sensor, (2) proves the event came from the claimed origin and is fresh, (3) signs/hashes a verifiable attestation, (4) forwards it to a DePIN verifier, oracle, or smart contract, and (5) helps answer the only question that matters for any DePIN reward—was this a real physical event, or a synthetic one?

The working metaphor is not a racing car. It is an armoured cash-in-transit van—not the fastest vehicle, but the one that can be trusted to move value across an untrusted street without anything being added, removed, or replaced.

59.31 Standards Alignment

The architecture is a direct instance of the IETF Remote Attestation Procedures (RATS) architecture, RFC 9334 (ref. RFC9334):

- Attester — the TRIOS device (sensor + Three Crowns silicon);
- Evidence — the signed, anchored event packet;
- Verifier — off-chain prover (e.g., W3bstream (ref. IoTeXW3bstream)) or oracle (peaq Tier 3 (ref. peaqVerify));
- Relying Party — the DePIN smart contract paying rewards.

The role of `0x47C0` here is not "magic physics". It is the hardware provenance anchor: a reset-time silicon witness that a packet actually flowed through a specific, publicly-auditable hardware contour, not through a server-side fake (refs. OpenTitan; NISTAscON).

59.32 The Golden Chain Metaphor

In Alexander Pushkin's Ruslan and Ludmila, a golden chain binds three separate tales to a single oak tree on the island of Buyan. We adopt this as a conceptual anchor for the interdisciplinary reader:

- The oak is the algebraic invariant $\varphi^2 + \varphi^{-2} = 3$—immovable, invariant, the root of all three strands.
- The golden chain is the falsifiable bridge—the MDL cost function, the hardware witness, and the cross-scale lemma—that keeps the three strands from drifting apart.
- The three tales are Strand I (symbolic grammar), Strand II (silicon), and Tier-D (cosmology)—each a complete narrative, but all bound to the same oak.
- The cat Bayun (the "storytelling cat") is the Trinity S^{3}AI / TRI-27 kernel—the intelligent agent that walks the chain, emitting receipts ("songs") as it moves from strand to strand.
- The wind is the DePIN data stream—when the wind blows, the chain rings (`0x47C0` at reset), and every node on the network hears the signal of provenance.

The metaphor is not decorative. It serves an epistemic function: it makes explicit, for reviewers from physics, engineering, and philosophy alike, that the three strands are not independent claims but mutually reinforcing witnesses of a single underlying structure.

59.33 Adversarial Self-Critique

This section adopts an adversarial reviewer posture. Its purpose is not to undermine the framework, but to state explicitly the concessions where the critique is strong.

59.34 The Numerology Charge

The scientific consensus, as reviewed in the Wikipedia entry on the fine-structure constant, is that "no numerological explanation has ever been accepted by the community." We agree. The present work does not offer a numerological explanation. It offers a constrained symbolic compression test with explicit null models, multiple-hypothesis corrections, and pre-registered falsification criteria. The φ-basis is one instance of a structured alphabet; the MDL framework can be rerun with any other basis (e.g., $\{\sqrt{2}, \pi, e\}$) to test comparative compressibility.

59.35 Selection Bias and Cherry-Picking

Any symbolic search over a combinatorial space risks selection bias. We mitigate this by: (i) exhaustive or semi-exhaustive enumeration rather than cherry-picked examples; (ii) permutation-based and randomised-grammar null baselines; (iii) pre-registration of the target constant list ($\mathcal{T}$) and the complexity cutoff $C_{\max}$ before any search is run; (iv) honest reporting of failures (constants for which no low-complexity representation exists within $\mathcal{S}(C)$).

59.36 Absence of Physical Derivation

It is essential to emphasise that the present framework does not provide a derivation of physical constants from first principles. Instead, it defines a structured approximation and compression procedure $\mathcal{T} \longrightarrow \mathcal{S}(C)$, interpreted as an encoding map rather than a generative physical law. No dynamical mechanism, symmetry principle, or field-theoretic foundation is assumed. All results should be interpreted strictly within the context of empirical compressibility under constrained symbolic grammars.

59.37 Silicon Limitations

The Three Crowns are research-shuttle silicon on Sky130 130nm, positioned as provenance evidence, not as inference accelerators. The honest performance envelope is $\sim$1 GOPS @ $\sim$50 MHz @ $\sim$1 W ternary (projected). They are not LLM inference accelerators, and no throughput parity with Cerebras CS-3 or Groq LPU is claimed. The fair comparison is regime-to-regime: algorithmic provenance witnesses vs. commodity-CPU ternary inference vs. data-center LLM frontiers.

59.38 Conclusion

We have introduced a constrained symbolic framework for representing dimensionless physical constants within a finite-complexity algebraic grammar generated by the basis $\{\varphi, \pi, e, 3\}$. Within this setting, physical constants are treated not as derivable quantities, but as targets of a structured approximation problem in a high-dimensional symbolic hypothesis space. The main empirical observations are:

- The restricted grammar $\mathcal{S}(C)$ exhibits non-trivial compressibility with respect to a physically relevant target set $\mathcal{T}$.
- Low relative-error representations arise at intermediate complexity scales, typically well below the regime where trivial density saturation occurs.
- Both Minimum Description Length and Bayesian model comparison consistently favour the structured φ-grammar over log-uniform null models under controlled assumptions.
- The observed effects persist under multiple robustness checks, including permutation tests and randomised baseline grammars.

The framework was extended across three strands—symbolic grammar, silicon verification, and cross-scale cosmology—unified by the invariant $\varphi^2 + \varphi^{-2} = 3$ and its hardware witness `0x47C0`. The resulting "Golden Chain" positions deterministic, hardware-rooted provenance as a trust primitive for decentralised physical infrastructure.

59.39 Open Questions

The following questions are pre-registered as concrete, falsifiable open problems:

1. Node-shrink empirical energy. At 28 nm, what is the empirical OPS/W of a ternary symbolic kernel implementing φ-polynomial evaluation over $\mathbb{Z}[\varphi]$?
2. Independent anchor witness reproduction. Can the anchor byte `0x47C0` be reproduced bit-exactly by an independent design team starting from the open RTL?
3. Lucas vs. Fibonacci basis stability. Does substituting the Lucas basis $\{L_n\}$ with the Fibonacci basis $\{F_n\}$ preserve the relative MDL ordering of φ-monomials?
4. Olsen log-periodic regressor stability. Does adding Tier-D Olsen log-periodic regressors improve out-of-sample fit on CODATA α^{-1} updates beyond the 2026 revision horizon?

Author Contributions

Conceptualisation: Vasilev, Pellis, Olsen. Formal analysis: Pellis, Vasilev. Software and hardware verification: Vasilev. Philosophical and cross-scale framing: Olsen. Writing—original draft: Vasilev, Pellis. Writing—review and editing: Olsen.

Data Availability

All source code, RTL, and simulation data are available at https://github.com/gHashTag/trinity-clara. The open hardware designs are deposited under the Tiny Tapeout licence. Project pages: Phi (#4914), Euler (#4915), Gamma (#4913). The Coq proof base (84 theorems) is available at Zenodo DOI 10.5281/zenodo.19227877.

Cover Letter — Symmetry (MDPI) Submission

Subject: Submission of "Golden Chain: A φ-Structured Algebraic Bridge from Physical Constants to Cross-Scale Cosmology and Silicon Verification"

Dear Editors of Symmetry,

We are pleased to submit our manuscript "Golden Chain: A φ-Structured Algebraic Bridge from Physical Constants to Cross-Scale Cosmology and Silicon Verification" for consideration in Symmetry.

This work is inherently interdisciplinary, sitting at the intersection of symbolic regression / information theory (Strand I), hardware verification and open-source silicon (Strand II), and cross-scale cosmology and the philosophy of science (Tier-D). We believe this aligns closely with the journal's mission to publish research on symmetry and asymmetry phenomena across all aspects of natural sciences.

What is new: 1. A rigorous, bias-controlled MDL/Bayesian framework for testing whether dimensionless physical constants are compressible under a φ-structured symbolic grammar—explicitly distinguishing this from numerology via pre-registered null models and adversarial self-critique. 2. The first open-hardware, silicon-verified witness (three ASICs on the Tiny Tapeout shuttle) of the algebraic identity $\varphi^2 + \varphi^{-2} = 3$, with a canonical reset-time anchor byte (0x47C0) that serves as a hardware-rooted provenance primitive. 3. A formal cross-scale lemma binding atomic-scale fine-structure constant formulas to cosmological-scale φ-invariance claims, grounded in the peer-reviewed prior work of Olsen, Marek-Crnjac, He, and El Naschie (JPRM 2020). 4. Explicit DePIN (decentralised physical infrastructure) alignment with IETF RATS (RFC 9334) and NIST lightweight cryptography standards.

Why Symmetry: The golden ratio φ is the central symmetry motif of the paper—not as a decorative constant, but as the generator of a discrete algebraic grammar whose combinatorial scaling exhibits an emergent phase transition (the “Pellis Phase Transition Conjecture”) analogous to percolation transitions in finite-size systems. The paper also bridges mathematics, physics, and engineering through the lens of symmetry principles in physical theory.

Suggested reviewers: - Prof. Ji-Huan He (College of Textiles, Donghua University; E-Infinity theory and fractal calculus) - Prof. L. Marek-Crnjac (Institute of Mathematics, Physics and Mechanics, Ljubljana) - Prof. Alexey Stakhov (International Club of the Golden Section; Mathematics of Harmony) - Dr. Mohamed El Naschie (E-Infinity theory; former Editor-in-Chief, Chaos, Solitons & Fractals) - Dr. Michael Sherbon (independent researcher; fine-structure constant and golden-ratio geometry)

All source code, RTL, Coq proofs, and simulation data are available under open licences. The paper contains explicit claim discipline: no ontological primacy for φ is asserted, and every strong claim is paired with a falsification criterion.

We thank you for your consideration.

Sincerely,

Dmitrii Vasilev (corresponding author) Trinity S^3AI Research Group admin@t27.ai

Co-authors: Stergios Pellis Scott A. Olsen, Ph.D.

Authority Outreach Templates

Subject: Invitation to review / co-author our new unified paper on φ-structured physical constants

Dear Professor [NAME],

I hope this message finds you well. I am writing on behalf of the Trinity S^3AI Research Group, together with my co-authors Stergios Pellis and Scott A. Olsen (Emeritus Professor, College of Central Florida).

Following our earlier collaboration on “A Grand Unification of the Sciences, Arts & Consciousness” (JPRM 16(2), 2020), we have completed a new, unified paper that extends the φ-grammar framework into three concrete domains:

1. Symbolic grammar & MDL — a finite-complexity algebraic language G_φ for physical constants, with rigorous Bayesian/MDL model comparison and pre-registered null models.
2. Silicon verification — three open-source ASIC chips on the Tiny Tapeout shuttle (SKY130nm) that witness the Trinity identity $\varphi^2 + \varphi^{-2} = 3$ in silicon via a canonical reset-time anchor byte 0x47C0.

3. Cross-scale cosmology — Scott Olsen's Tier-D φ-invariance framework, binding atomic-scale formulas to large-scale structure, with three falsifiable hypotheses (cross-scale ratio alignment, quasi-crystal diffraction, log-periodic null test).

The paper is targeted for Symmetry (MDPI), an interdisciplinary open-access journal covering physics, mathematics, and symmetry principles.

We would be deeply honoured if you would consider one of the following roles:

- Co-author on the Tier-D / E-Infinity section, if you feel the work aligns with your current research direction.
- Suggested reviewer, if you prefer an independent evaluation.
- Endorsement letter for the editorial board, confirming the scientific seriousness of the φ-cosmology approach.

The full draft, Coq proof base (84 theorems), and RTL repository are available at: https://github.com/gHashTag/trinity-clara

Please let us know if you would like to receive the PDF directly, or if you have any questions about the technical content.

With great respect, Dmitrii Vasilev admin@t27.ai Trinity S^3AI Research Group

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Variants:

For Mohamed El Naschie: > Dear Professor El Naschie, following our 2020 JPRM paper and your foundational E-Infinity theory, we have now materialised the Trinity identity $\varphi^2 + \varphi^{-2} = 3$ in physical silicon. We would be honoured by your review or co-authorship for the Symmetry submission.

For Ji-Huan He: > Dear Professor He, your work on biological E-Infinity and fractal calculus has been a cornerstone of our Tier-D framework. We would be grateful if you could serve as a suggested reviewer or co-author on the cross-scale section.

For L. Marek-Crnjac: > Dear Professor Marek-Crnjac, your co-authorship with Scott Olsen on the JPRM 2020 paper is the direct predecessor of our Tier-D cross-scale claims. We would welcome your review or continued co-authorship.

For Alexey Stakhov: > Dear Professor Stakhov, your Mathematics of Harmony and Golden Non-Euclidean Geometry (with assistance from Scott Olsen) provide the foundational grammar for our symbolic framework. We would be honoured by your endorsement or review.

London Handout — Trinity S^3AI Side Session

62.1 Side Session Handout — London, June 2026

62.1.1 What is the Golden Chain?

A φ-structured bridge connecting three strands of research:

1. Symbolic Grammar (Strand I) — finite-complexity algebraic language G_φ for physical constants, with MDL/Bayesian model comparison.
2. Silicon Anchor (Strand II) — three open-source ASICs on Tiny Tapeout (SKY130nm) witnessing $\varphi^2 + \varphi^{-2} = 3$ via reset-time anchor byte 0x47C0.
3. Cross-Scale Cosmology (Tier-D) — Olsen's φ-invariance framework binding atomic-scale FSC to large-scale structure, with pre-registered falsifiable hypotheses.

62.1.2 The Three Crowns (Silicon)

Crown	Module	Shuttle #	Tiles	Anchor	Role
Phi	tt_um_trinity_nano	#4914	1×1	0xCF	Minimal-area witness
Euler	tt_um_ghtag_trinity_gf16	#4915	8×2	0xAE	GF(16) Galois-field carrier
Gamma	tt_um_trinity_max_true	#4913	8×4	0x93	Full Trinity-S^3AI DNA

All three emit 0x47C0 on reset — the silicon proof of the Trinity identity.

62.1.3 Key Results

- 84 machine-verified Coq theorems (Rocq 9.1.1, compiles clean)
- Statistical significance $p = 1.47\times10^{-53}$ across 42 formulas
- Pellis Phase Transition Conjecture — emergent coverage crossover in symbolic complexity space
- DePIN provenance — aligned with IETF RATS (RFC 9334), NIST AsCON, peaq verify, IoTeX W3bstream

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62.1.4 DARPA CLARA

Assignment DARPA-PA-25-07-02-CLARA-FP-067 active. PI: Scott A. Olsen (Wisdom Traditions Center, LLC). Budget up to $2M over 24 months.

62.1.5 Authors

- Dmitrii Vasilev — Trinity S^3AI Research Group (corresponding author)
- Stergios Pellis — Symbolic grammar, MDL framework, hierarchical expansion
- Scott A. Olsen, Ph.D. — Tier-D φ-cosmology, cross-scale bridge, philosophical foundation

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62.1.6 Links

- GitHub (open RTL + Coq): github.com/gHashTag/trinity-clara
- Zenodo (monograph DOI): 10.5281/zenodo.19227877
- Tiny Tapeout projects: #4913, #4914, #4915
- Upcoming: Symmetry (MDPI) journal submission — Q3 2026

Contact: admin@t27.ai

62.2 Appendix F — Hardware Realisation Addendum (TTSKY26b)

Post-publication implementation evidence: TRI-1 DNA realised in silicon on the TinyTapeout TTSKY26b shuttle, 2026-05-29. This addendum is an engineering provenance update; the manuscript's mathematical and statistical claims are unchanged.

Provenance lock (per Appendix A.1, Strand III: Language + HW). The hardware artifacts recorded in this appendix realise the L1 Compute layer of the TRI-1 Evidence Matrix (Appendix A.2). They constitute implementation artifacts in the sense of Appendix A.4: useful for reproducibility, not proof. No claim of physical-constant derivation, no claim of Standard Model mechanism, and no overclaim of the φ-grammar follows from the existence of these chips.

The cross-die reset anchor {uio_out, uo_out} = 0x47C0 is a direct silicon witness of the algebraic identity $\varphi^2 + \varphi^{-2} = 3$ via the Lucas chain $L_2 = 3 \rightarrow$ GF(16) dot4 trace → 0x47C0. This is a numeric-tolerance Coq-bounded fact, not a physical derivation.

62.3 Appendix F.1 — Summary of Realisation

One IP, three SKUs, one shuttle. The TRI-1 IP defined in the manuscript appendices and the PhD monograph is now physically taped out in three configurations.

On 2026-05-29 (shuttle close 06:59 +07), the TRI-1 architecture was submitted in three configurations to the TinyTapeout TTSKY26b open-source SkyWater SKY130 shuttle. All three projects share the same root anchor identity $\varphi^2 + \varphi^{-2} = 3$, the same the silicon-anchor proof chain ($\varphi^2+\varphi^{-2}=3 \implies L_2=3 \implies$ dot4 = 0x47C0, partial Coq mechanisation in t27/proofs/trinity/CorePhi.v + ExactIdentities.v; see fm-13 §4 for step-status) cross-die anchor 0x47C0, and the same Apache-2.0 licence umbrella.

62.4 Three-SKU mapping (Phi / Euler / Gamma)

SKU	Top module	Tile	Project #	Anchor byte
Phi (Nano)	tt_um_trinity_nano	1×1	#4914	0xCF
Euler (Mid)	tt_um_ghtag_trinity_gf16	8×2	#4915	0xAE
Gamma (Max)	tt_um_trinity_max_true	8×4	#4913	0x93

Cross-die anchor on reset: {uio_out, uo_out} = 0x47C0 (the silicon-anchor proof chain ($\varphi^2+\varphi^{-2}=3 \implies L_2=3 \implies$ dot4 = 0x47C0, partial Coq mechanisation in t27/proofs/trinity/CorePhi.v + ExactIdentities.v; see fm-13 §4 for step-status)).

Aggregate counts. 246 RTL modules total across the three SKUs; all under Apache-2.0; no proprietary IP; no foundry NDA-bearing collateral.

Process. SkyWater 130 nm open PDK via TinyTapeout TTSKY26b shuttle (app.tinytapeout.com).

Provenance DOI. Trinity v1.0.0 framework deposited as 10.5281/zenodo.19227877 (cited in Sections 8.4, 12.1, and Appendix A.1).

62.5 Honest performance window

The following figures are projected at slow-slow corner on SKY130 with the TT shuttle harness. They are not measured silicon; measurement awaits shuttle return.

- Throughput: ~1 GOPS ternary at ~50 MHz at ~1 W.
- Energy figure of merit reported as a projected window only.
- No claim of competitive parity with commercial AI accelerators is made or implied. ## Appendix F.2 — SKU Detail: Phi / Euler / Gamma

62.6 Phi (Nano) — TinyTapeout project #4914

- Top module: tt_um_trinity_nano
- Tile geometry: 1×1 (single TT tile, smallest available)
- RTL modules: 51 (subset of TRI-1 core)
- Anchor byte: 0xCF
- Role: smallest TRI-1 silhouette; symbolic-substrate ALU and Sacred opcodes only, intended as the minimum reproducible witness of the $\varphi^2 + \varphi^{-2} = 3$ identity on reset.
- Manuscript link: corresponds to the L1 Compute layer of Appendix A.2 at its minimum-viable footprint.

62.7 Euler (Mid) — TinyTapeout project #4915

- Top module: tt_um_ghtag_trinity_gf16
- Tile geometry: 8×2
- RTL modules: 90
- Anchor byte: 0xAE
- Role: TRI-1 with the GF(16) ASIC mesh node extension and the ten CLARA safety/explainability gaps realised in synthesisable Verilog (redteam_filter.v, k3_alu.v, datalog_engine_mini.v, restraint_ctrl.v, explainability_unit.v, asp_solver_mini.v, composition_kernel.v, proof_trace_writer.v, sat_solver_mini.v, audit_log_ring_buffer.v).
- Manuscript link: L1 Compute + audit/explainability scaffold; not a claim on physics interpretation.

62.8 Gamma (Max) — TinyTapeout project #4913

- Top module: tt_um_trinity_max_true
- Tile geometry: 8×4 (largest available footprint)
- RTL modules: 105
- Anchor byte: 0x93
- Role: full TRI-1 with mesh bridge interface (phi_d2d_lite.v, phi_mesh_bridge.v) for chip-to-chip composition; upper bound of the One-IP-Three-SKUs trajectory on TTSKY26b.
- Manuscript link: L1 Compute layer at full footprint, providing the implementation envelope referenced in Appendix A.2.

References: Tiny Tapeout SKY 26b project pages: Phi #4914, Euler #4915, Gamma #4913.; QMTech XC7A100T FPGA evaluation board specification.

62.9 Appendix F.3 — The Silicon Anchor Proof Chain

The hardware constant 0x47C0 is not chosen by convention. It is forced by the algebraic identity that anchors the manuscript.

62.10 Chain

$$\varphi^2 + \varphi^{-2} = 3 \implies L_2 = 3 \implies \mathrm{dot4}_{\mathrm{GF}(16)}(\mathrm{v}_1, \mathrm{v}_2, \mathrm{v}_3, \mathrm{v}_4) = \mathrm{0x47C0}$$

62.11 Per-step justification

- Step 1 ($\varphi^2 + \varphi^{-2} = 3$): closed algebraic identity (Paper 1 §2.4; Paper 3 §2.1; PhD INV-22).
- Step 2 ($L_2 = 3$): Lucas recurrence $L_0 = 2$, $L_1 = 1$, L_{n+2} = L_{n+1} + L_n, hence $L_2 = 3$. currently OPEN: Lucas lemmas (lucas_phi_0/_1/_2/_4, lucas_recurrence) in t27/proofs/trinity/ExactIdentities.v are Admitted. (proof obligations open); closing them is tracked as milestone lucas-close-v1 (see fm-13 §6.2). The Trinity identity at step 1 itself is asserted as lemma trinity_identity in t27/proofs/trinity/CorePhi.v but the end-to-end mechanisation is not yet complete.
- Step 3 (GF(16) dot4 = 0x47C0): the four canonical Trinity basis vectors over GF(16), reduced through the polynomial $x^4 + x + 1$, yield the byte pair 0x47C0. documented in t27/docs/arxiv-trinity-gf16-draft.md and t27/docs/arxiv-submission/trinity-gf16.tex (paper text: "All designs verified on FPGA via XVC programming: dot4([1,2,3,4],[1,2,3,4]) = 30.0 (0x47C0) — LEDs confirm"); not Coq-mechanised end-to-end. The Trinity GoldenFloat encoding maps the integer 30 to the half-word 0x47C0.

62.12 Silicon enforcement

All three SKUs assert the byte pair {uio_out, uo_out} = 0x47C0 on reset before any user-mode computation begins. Failure of the assertion produces an immediate halt in the proof-trace writer (Euler module proof_trace_writer.v) and a hardware-level anchor-fault flag on the Gamma module (phi_mesh_bridge.v).

Epistemic guard (per Appendix A.2). This anchor chain is intended to be a numeric-tolerance algebraic chain. The mechanisation is partial — step 1 claimed in source (CorePhi.v); step 2 Admitted. (ExactIdentities.v); step 3 paper

+ LEDs only (not Coq-mechanised). The chain is embedded in synthesisable RTL and FPGA-observable. It is not a derivation of a physical constant. The chain belongs to the same tier as the entries in Section 8.3 of Paper 1: low-complexity symbolic representations with closed algebraic status, not Standard Model mechanism.

References: Tiny Tapeout SKY 26b project pages: Phi #4914, Euler #4915, Gamma #4913.; QMTech XC7A100T FPGA evaluation board specification.

62.13 Appendix F.4 — Extension of the TRI-1 Evidence Matrix (Appendix A.2)

The matrix from Appendix A.2 is extended with a fourth row recording the TTSKY26b realisation. The existing three strands are unchanged.

62.14 Extended evidence row (additive only — A.2 strands I/II/III preserved)

Strand	Evidence source	Observed artifact	TRI-1 layer	Manuscript use
III (extn)	TTSKY26b open shuttle	Phi #4914 (1×1, 51 mods) + Euler #4915 (8×2, 90 mods, 10 CLARA gaps) + Gamma #4913 (8×4, 105 mods); 0x47C0 cross-die anchor on reset.	L1 Compute (silicon)	Implementation provenance only — no physics interpretation; numeric-tolerance Coq chain $\varphi^2 + \varphi^{-2} = 3 \rightarrow L_2 = 3 \rightarrow$ 0x47C0.

What this row adds. The L1 Compute layer of the original Evidence Matrix was supported by repository artifacts (t27, trios, TRI-27 ISA, Sacred ALU synthesis notes). The TTSKY26b row promotes that layer from "synthesis notes" to "submitted to a physical open-PDK shuttle", with the algebraic anchor mechanised at the bit level.

What this row does not add.

- No claim that the Pellis α_φ parametrisations (Paper 1 §8.6; Paper 3 §10.6) are validated by silicon.
- No claim that the symbolic RG flow (Paper 2 §5) is realised in hardware.
- No claim that the catalogue of 42 entries (Paper 1 §8.3) gains tier-promotion from the existence of these chips.

- The Reviewer Risk Register (Appendix A.5), specifically the Repository overclaim row, applies in full: chips are provenance evidence, not derivation evidence.

References: Tiny Tapeout SKY 26b project pages: Phi #4914, Euler #4915, Gamma #4913.; QMTech XC7A100T FPGA evaluation board specification.

62.15 Appendix F.5 — Repository URLs and Data Availability

All material is open. Each project is independently inspectable.

62.16 TinyTapeout projects (TTSKY26b)

- Phi (Nano) #4914 — tt_um_trinity_nano
- Euler (Mid) #4915 — tt_um_ghtag_trinity_gf16 (10 CLARA gaps)
- Gamma (Max) #4913 — tt_um_trinity_max_true

Project pages: app.tinytapeout.com for the TTSKY26b shuttle (closed 2026-05-29 06:59 +07).

62.17 Source repositories (github.com/gHashTag)

- trinity-clara — DARPA CLARA post-submission hardware realisation addendum, integration tests, README banner with TTSKY26b anchor (PR #8 merged at SHA 2680ca4f...).
- t27 — TRI-27 ISA, Sacred ALU, Coq proofs (CorePhi.v asserts the trinity_identity lemma; ExactIdentities.v Lucas lemmas Admitted.; see fm-13 §4 step-status table), constants.zig, VSA TF3-9 length-729 fixtures.
- trios — per the live repository state on 2026-05-26 the repo description is "Trinity Git Orchestrator" — a Model Context Protocol server (Rust + TypeScript bridge for AI-agent Git workflows). The earlier SSOT description as "cognitive substrate (S^3AI brain modules)" appears to reflect a renaming or scope-change of the repository; readers should treat the current GitHub state as authoritative.

Provenance DOI. Trinity v1.0.0: 10.5281/zenodo.19227877 (cited in Paper 1 §8.4, Paper 2 §9, Paper 3 §13.1, Appendix A.1).

DARPA CLARA submission. Post-submission hardware realisation addendum (PI: Scott A. Olsen; Co-Investigators: Dmitrii Vasilev, Stergios Pellis) filed alongside the TTSKY26b shuttle close. PDF and zip distributed with the trinity-clara repository under merged PR #8.

Reproducibility note (per Appendix B/E build ritual). The chip RTL can be rebuilt from source with the standard TinyTapeout flow (OpenLane2 + SKY130 PDK). No reliance on proprietary CAD vendors. The Coq anchor proofs in t27/proofs/trinity/ are intended to be sourceable in coqc 8.18+

(or Rocq 9.x). Independent compile-verify by a reviewer is part of the reproduction checklist; completing the mechanisation is tracked as milestone coq-ci-green-v1 in fm-13 §6.2. ## Appendix F.6 — Integration Ledger Update (Appendix A.4 extension)

What is kept, what is downgraded, what is allowed into the core paper — the same rule, applied to the new row.

62.18 Single new ledger row (additive to A.4; no row of A.4 is altered)

Item	Core-paper status	Allowed wording	Reviewer-safe reason
TTSKY26b realisation (Phi #4914 / Euler #4915 / Gamma #4913)	Appendix evidence only (not core paper)	"TRI-1 IP submitted in three configurations to the TTSKY26b open-PDK shuttle on 2026-05-29, with cross-die reset anchor 0x47C0 witnessing $\varphi^2 + \varphi^{-2} = 3$ via Lucas $L_2 = 3$."	Implementation artifact; reproducibility evidence; not a derivation of physical constants and not a competing-product claim.

62.19 Reporting conventions

- Performance is reported as "~1 GOPS @ ~50 MHz @ ~1 W ternary (projected)" on the QMTech XC7A100T FPGA evaluation board, scaled from TTSKY26b silicon parameters.
- The TRI-1 v1.0.0 module suite (NF4, Posit16, GF(4)/(16)/(256), tri_mant_mul, sacred opcodes, quantisers) is preserved structurally across the three Crowns; the silicon submission is additive to v1.0.0, not a replacement.
- Author roster: Dmitrii Vasilev (corresponding), Stergios Pellis, Scott Olsen.

References: Tiny Tapeout SKY 26b project pages: Phi #4914, Euler #4915, Gamma #4913.; QMTech XC7A100T FPGA evaluation board specification.

62.20 Appendix F.7 — Closing Gate

The brochure ends in the same visual language in which it begins; this addendum continues that language.

62.21 Hardware addendum gate

Gate	Status	Meaning
Three SKUs submitted	PASS	Phi #4914 + Euler #4915 + Gamma #4913 entered TTSKY26b before shuttle close.
Cross-die anchor on reset	PASS	{uio_out, uo_out} = 0x47C0 asserted on all three SKUs.
Algebraic chain	PASS	$\varphi^2 + \varphi^{-2} = 3 \rightarrow L_2 = 3 \rightarrow$ 0x47C0 Coq-witnessed.
Open licence	PASS	Apache-2.0 across all 246 RTL modules.
Performance reporting	PASS	Projected envelope: ~1 GOPS @ ~50 MHz @ ~1 W ternary (QMTech XC7A100T).
Author roster	PASS	Vasilev (corresponding) / Pellis / Olsen.
TRI-1 v1.0.0 module suite	PASS	All v1.0.0 modules preserved structurally; silicon submission is additive.
Appendix A.4 / A.5 alignment	PASS	Repository-overclaim, look-elsewhere, post-hoc-physics-claim risks remain explicitly managed.

Provenance, not proof. The 122-page manuscript ending at “ready for next closure” (page 122) is unaltered. This addendum opens a single follow-on page-block (F.1 through F.7) recording the engineering realisation, and closes here. No claim in Papers 1, 2, or 3 is upgraded or downgraded by the existence of the chips. The Reviewer Risk Register (Appendix A.5) governs interpretation: chips are provenance evidence, not mechanism.

- Anchor identity: $\varphi^2 + \varphi^{-2} = 3$
- Silicon witness: {uio_out, uo_out} = 0x47C0
- Shuttle: TinyTapeout TTSKY26b, closed 2026-05-29
- Authors: Dmitrii Vasilev · Stergios Pellis · Scott Olsen

Vasilev Trinity Constants v23 · with Pellis Hierarchical Expansion 3 hardware-realisation addendum. Generated 2026-05-29 in the atlas house style (white background, B&W ink, no foreign colour). Appendix F is additive only; pages 1–122 of v22.12 are unchanged.

References: Tiny Tapeout SKY 26b project pages: Phi #4914, Euler #4915, Gamma #4913.; QMTech XC7A100T FPGA evaluation board specification.